Quantum Gravity Dynamics

Chapter 6

Quantum Action, Natural-Invariant Dynamics and Historical Correspondence Tests

Research edition v2.7 (action-first; QM043 + QN001–QN005 + Q052–Q054) | 5 November 2026

Action-first synthesis; Chapters 1–5 define the effective gravitational theory

Action-first charter: quantum response and the status of the historical tests (5 November 2026)

Revised role of this chapter. QGD is an action-defined gravitational alternative formulated with independent natural-invariant one-forms. Chapter 1 states *Axiom I*, the complete founding master metric, and *Axiom II*, the unproved existence of an independent microscopic counterpart to the phenomenological map. Neither Axiom II nor a successful $\mathcal{F}_{\text{micro}}$ construction is needed to calculate QGD's regular metric-mediated gravitational equations and classical solutions. The core dynamics comes from the declared Einstein–Hilbert–matter action composed with $g(\Sigma)$ and its renormalized quantum-effective extension; Chapter 4 gives its defining variational law.

Primary current result. Section 40 (QN001–QN005) contains the four *explicitly computed* proper-time Dirac/Hartree quadratic kernels $\mathcal{F}_C^\Lambda$, $\mathcal{F}_R^\Lambda$, $\mathcal{V}_\Lambda$ and K_Λ, the Schur-complement curvature response $\mathcal{F}_R^\Lambda - \mathcal{V}_\Lambda^2/(2K_\Lambda)$, and the exact pullback of the corresponding metric Hessian to natural-invariant perturbations. Those results are conditional on the stated Euclidean flat-background quantum model; they do not define a complete causal Lorentzian nonlocal functional on all geometries. The independent finite local curvature couplings remain renormalization data.

Status of QM001–QM043. The remaining sections of this research edition preserve earlier and continuing studies of possible realizations of Axiom II. They are valuable *optional tests*, not missing prerequisites for the validity of the effective QGD gravitational action. The phenomenological map $\mathcal{F}_{\text{phen}}$ is the source/state/boundary-to-[Σ] solution correspondence of the specified action, modulo metric-preserving representation redundancies. The microscopic map must be independent of that field-equation solver to count as an actual realization, and its existence has not been established.

Mathematical hygiene. An eight-sector regular background has a ten-dimensional metric image and 22 representation directions. Near the all-zero sector, the quadratic metric map has vanishing first derivative; the projected Euler equations may then hold even where the Einstein residual does not. Physical equivalence with GR is therefore conditional on a declared admissible, sufficiently regular sector domain. The historical $\lambda\mathcal{Q}_{ab}$ slot remains in Axiom I but is set to zero in the controlled QN004–QN005 metric-only calculations.

Historical and research-edition synopsis (preserved).

Editorial update (QN001–QN005, 4 November 2026). An additional dedicated section consolidates the analytical quantum effective action, common-regulator Dirac/Hartree response, and the exact natural-invariant pullback (flat signed sectors and weak-static spherical branch). It distinguishes calculated Euclidean kernels from unresolved Lorentzian causal, finite-renormalization, and microscopic-emergence problems. All QM001–QM043 content is preserved.

Earlier editorial update (QM043). This edition adds a conditional coarse-graining, large-number, and collective-basis refinement to the inverse matching map while retaining all earlier material. It does not claim microscopic closure.

Earlier consolidation (QM042). This v2.3 edition preserves the entire QM001–QM039 chapter and appends the independent QM032–QM036 audits previously archived outside it, QM040–QM041

microscopic field calculations, and a definitive two-map/inverse-matching status ledger. *Phenomenological $\mathcal{F}_{\rm phen}$ is conditionally closed under the adopted effective action; derived microscopic $\mathcal{F}_{\rm QM}$ is not established.* The original abstract follows unchanged, describing the historical sequence of calculations. **Abstract.** Chapter 5 proves a conditional static matter-wave-to-QGD-geometric correspondence when a stable source-coupled collective response is supplied as premise MW2. This chapter begins to test that physical premise with the actual interacting quantum-matter model of Chapter 1, using the independently developed microscopic spectral and WKB benchmarks QM001–QM005. We retain the exact leading Dirac spinor ratio, the conditional WKB phase transport, and the stationary current bridge $r^2 j^r = C_J$ rather than replacing them. We specify the full physical source-response matrix and its infrared-pole criterion; re-evaluate the scalar-contact Hartree saddle using precisely Chapter 5's cutoff convention; and verify an analytically finite density response and a zero source-visible long-range residue in the tested scalar channel. We additionally prove a transverse-traceless scalar-mixing selection rule for an isotropic state and a local-map exterior obstruction. The earlier tensor increment independently verifies the collisionless TT shear continuum and a transverse-momentum kinetic Ward identity with its necessary local contact. These negative tests identify why a physically selected Dirac quasiparticle is not yet a universally gravitating collective field. The present extension constructs the full ten-indexed leading intraband stress matrix, defines the covariantly regulated fermion-determinant bubble and genuine metric seagull, computes its local vacuum heat-kernel terms, verifies a finite-coordinate-deformation Ward cancellation, refits the Hartree saddle under the changed regulator, and tests the scalar-resummed TT channel. It then specifies the remaining genuinely dynamical tensor and universality tests. Neither a massless graviton nor Einstein dynamics is assumed in the microscopic kernel. The founding complete coframe/signed-sector/$\mathcal{Q}$ metric remains unchanged. The v0.5 extension proves an exact null-sector Clifford/coframe bridge, preserves the covariant Dirac/WKB/current equation, and derives the source-to-kinetic functional kernel that an independently microscopic promotion law would have to reproduce. It explicitly distinguishes matching within the adopted gravitational action from deriving that action microscopically. All new physical emergence claims remain open. The v2.2 QM039 addition exactly solves a small interacting derivative-bond model, computes nonzero causal cross-species response, and tests its self-consistent Hartree kinetic frame and RPA collective pole. Its uniform saddle fails the infinite-chain staggered bond stability test; no universal geometric wave is claimed. The v2.1 QM038 addition tests the full-master-metric Diracization, the Clifford-compatible connection, a conformal Hermitian WKB transport coefficient, inverse-microscopic ambiguity and long-distance frame locking. It introduces no unproved claim of a dynamically selected collective phase. The v2.0 QM037 extension adds exact Clifford/principal-symbol tests for curved Dirac fields, a conformal spin-connection current counterexample, constrained Fierz bilinears, and a conditional shared kinetic composite channel; no microscopic exterior geometry is claimed. The v1.9 QM034 extension exactly solves a unitary two-site four-flavour determinant-density model, exhibiting true causal cross-species response and a fourth-order soft collective excitation, but also a vanishing soft-pole numerator, a gapped homogeneous composite, spectator non-universality and vacuum-energy shift ambiguity; this remains a projected analogue rather than a derivation of geometry from the full four-dimensional microscopic model. The v0.5 QM009 extension derives the inhomogeneous scalar and number-density susceptibility, independently checks matched smooth regulators and sharp-cutoff artifacts, and proves the absence of a Hartree–Fock source-to-kinetic deformation even for spatially varying mean fields. The v0.7 QM010 extension sharpens the central research objective: the conditional Schwarzschild source-to-natural-invariant bridge is already closed under the adopted QGD field action, while the genuinely microscopic spinor/state-to-invariant selection map remains open. Motivated by Hamiltonian stationary action and Fermat optics, it derives the weak-field H profile as a stationary source-conditioned field functional, proves its spherical uniqueness under stated boundary conditions, separates path variation from field variation, and gives the exact conditional Hessian/source-response form required for a microscopic selection principle. No graviton or independent tensor mode is assumed as a prerequisite. QM011 tests the proposed microscopic variational-selection functional directly: its Hubbard–Stratonovich stationary

scalar is physically determined, but the measured positive curvature of the scalar Hessian prevents the required QGD Schwarzschild exterior selection in the Hartree–Gaussian regime. This test does not reopen the conditional QGD correspondence. The v0.8 research synthesis separately audits QM012–QM014, including the nonlinear stiffness of the regular $H = \sigma_{\mathrm{amp}}^2$ coordinate, the original versus induced-action research arrows, exact rank/gauge checks, and the status of the proposed but unadopted premise (Q). It preserves the effective static milestone and the open microscopic selection claim. The v0.9 QM015 extension adds a source-overlap pole criterion, two gapless-but-source-invisible controls, and an exactly solvable mixed-channel stationary response which reproduces the conditional $1/r$ exterior despite the physical source coupling only to a massive channel. The mixed-channel parameters and symmetry are explicitly assumed in the toy calculation; they have not been derived from the original interacting Dirac theory. The v1.0 QM016 extension now evaluates the self-consistent original scalar-contact Dirac model and proves an exact positive-curvature identity on the specified Hartree cutoff domain, computes a full two-coordinate static scalar–density thermodynamic Hessian and independent susceptibility checks, and establishes that coupling tuning in this regime does not provide the missing source-visible massless channel. The full momentum-dependent stress/kinetic 2PI problem remains open. QM017–QM019 now follow the newly proposed kinetic route through an explicitly scoped, nonlocal screened-exchange test, a positive kinetic derivative accompanied by a short-range failure, an exact cross-sector retarded Kubo obstruction for a decoupled spectator, an isotropic scalar-to-TT resummation obstruction, and an algebraic Clifford/signed-lift check proving that a species-specific medium metric is not universal gravitational emergence. All results retain their declared approximation scopes; no full Ward-completed stress–kinetic 2PI computation has been claimed. QM022 establishes an exact projective-spinor representation of the twisting Kerr null direction, its flat-reference ray transport and optical expansion, and a restricted optical-divergence identity for the Kerr/Schwarzschild gravitational weight. It proves why conserved chiral current is insufficient, why averaged mixed spinors do not pick a unique null congruence, and why an unrestricted divergence rule fails the vacuum Einstein equations. These are constructive geometric/kinematic advances, not a microscopic derivation of a common physical metric. The v1.3 QM023 consolidation incorporates the explicit working hypothesis of an actual collective quantum field extending into exterior vacuum; it retains a complete proof-status ledger, separates the frame-direction and frame-weight tasks, and poses the precise central question for $\mathcal{F}_{\mathrm{micro}}$. No new microscopic field operator is claimed to have been derived.

Proof-status contract. **[D]** denotes an exact identity, a conditional mathematical lemma with its hypotheses stated, or a reproduced numerical result within a named model. **[C]** denotes a conclusion conditional on that model, approximation, or an adopted infrared action. **[A]** denotes an independently assumed physical ingredient. **[O]** denotes an unproved microscopic QGD claim. No "derived gravity" is inferred from static response finiteness, WKB transport, algebraic factorization, or an Einsteinian metric pullback.

QM023 RESEARCH COMPASS — NEW PHYSICAL HYPOTHESIS

Working ontology [A]: The natural-invariant-producing structure is to be investigated as an *actual quantum collective field extending through the exterior vacuum*, rather than only a label reconstructed from a source state or a known Einstein solution. Its operator definition, source coupling, causal field equation, and physical link to $[\Sigma]$ remain open.

Central question [O]: Can the *original microscopic quantum-matter action plus source state and boundary conditions* generate an exterior collective field which *selects* the weighted directions and weights of a physical natural-invariant class and thereby a *common* effective metric, without assuming that metric or the Einstein equations as inputs?

Result to date: Mathematical metric representation and conditional Einsteinian dynamics: established. Spinor-optical and real microscopic response results: established within stated model and approximation domains. Physical exterior-field selection and universality: *not derived.* QM023–

QM038 at the end of this chapter consolidates the achievements and establishes an explicit fitted effective selection law as a target for backward micro-derivation.

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1 Why a new Chapter 6 belongs here

The logical sequence is now: Chapter 1 (microscopic fields, Dirac spinors, large-N, WKB and conserved currents); Chapter 2 (complete natural-invariant metric and selected geometric solutions); Chapter 3 (gravitational empirical normalization); Chapter 4 (adopted effective QGD action, quantum stress and full metric pullback); Chapter 5 (backward ladder and conditional static correspondence); *this Chapter 6* (test of microscopic origin and source-visible response); formerly Chapter 6, now provisionally Chapter 7 (vacuum and physical energy); formerly Chapter 7, now provisionally Chapter 8 (cosmology). No original chapter is superseded by this research edition.

The Chapter 5 static milestone has a clearly identified physical premise: a collective coordinate q with

$$\boxed{\mathcal{E}[q;\rho_m] = \int \mathrm{d}^3x \left[\frac{Z}{2}|\nabla q|^2 - g_c q \rho_m\right], \qquad Z > 0,} \tag{1}$$

which conditionally yields $G_{\mathrm{eff}} = g_c^2/(4\pi Z)$. Here q is a gravitationally *candidate* collective coordinate, not the Dirac quasiparticle, the Pauli-block ratio, or the historical symmetric tensor $\mathcal{Q}_{ab}$. Equation (1) is a **[A]** premise until derived from microscopic QFT.

2 The precise missing intersection: a physical source-to-operator-to-invariant map

Status: This section sharpens the research question of this chapter. It does not replace the Chapter 1 spinor construction or the Chapter 2 dynamically selected exterior natural invariant. The operative distinction is between a *quantum source state* $\hat{\varrho}_{\mathrm{src}}$ and a *probe spinor* Ψ_{pr} propagating through the effective background. Their state-dependent and kinematic data must not be conflated.

What is already present. For a specified effective Dirac operator $\mathcal{D}_{\mathrm{eff}}$, solving $\mathcal{D}_{\mathrm{eff}}\Psi_{\mathrm{pr}} = 0$ yields the spinor ratios R, WKB phase S, and current j^μ. Separately, the Chapter 2 adopted Einstein–Hilbert QGD action dynamically selects the Schwarzschild natural invariant within its declared independent-sector ansatz. What has *not* been derived is an independent physical map

$$\boxed{\mathcal{F}: \quad (\hat{\varrho}_{\mathrm{src}}, \mathcal{D}_{\mathrm{eff}}, S, \mathsf{R}, j, \text{boundary data}) \longmapsto [\Sigma]_{\mathrm{physical}},} \tag{2}$$

where $[\Sigma]$ denotes physically indistinguishable signed-sector representations after the reference coframe and allowed redundancies are specified. A particular positive-energy lower/upper spinor ratio is a *probe-dependent* kinematic quantity, not automatically the source's gravitational one-form.

Two logically distinct ways to complete Eq. (2).

Route A. Conditional effective QGD matching [C]. Compute $T^{\mu\nu}_{\mathrm{src}}$ from the quantum source; solve the *already adopted* QGD gravitational equations with boundary conditions; obtain $[\Sigma]$; construct the compatible Dirac operator; then test its spinors, WKB and conserved currents. This is a noncircular *effective-model calculation* if the gravitational equation is explicitly listed as an independent premise. It does not derive that equation from QFT.

Route B. Independent microscopic source-to-kinetic map [O]. Compute a probe-independent effective kinetic principal symbol from microscopic matter/source dynamics, without inserting g or Σ as background inputs. Demonstrate that this symbol determines a common effective geometry and physically selects the QGD natural-invariant sectors (with their coframe and $\mathcal{Q}$ histories), including exterior memory of a compact source. This is the stronger physical-emergence claim.

These routes have different burdens of proof. An independently propagated composite spin-two particle is *one possible* mechanism for Route B, not a necessary ingredient of Route A.

2.1 An exact null-sector operator matching lemma [D]

Fix the reference Lorentz frame, take the classical $\lambda\mathcal{Q} = 0$ specialization of the *complete* QGD metric, and select one positive null sector. Let ℓ_μ be a flat-null covector, $\eta^{\mu\nu}\ell_\mu\ell_\nu = 0$, and $H(x) \geq 0$. Define

$$\sigma_\mu = \sqrt{H}\,\ell_\mu, \quad g_{\mu\nu} = \eta_{\mu\nu} + H\ell_\mu\ell_\nu, \quad \Gamma^\mu = \gamma^\mu - \frac{H}{2}\ell^\mu \not{\ell}, \quad \not{\ell} := \ell_\nu\gamma^\nu. \tag{3}$$

The complete founding metric still reads

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)}\sigma_b^{(s)} - \lambda\mathcal{Q}_{ab}], \quad s_s = -\varepsilon_s.} \tag{4}$$

Equation (3) is a *specialization*, not a replacement for Eq. (4).

Proposition 2.1 (Exact principal-symbol and coframe correspondence)**.** *With the Chapter 1 convention* $\{\gamma^a, \gamma^b\} = -2\eta^{ab}I$*, the matrices of Eq.* (3) *satisfy*

$$\boxed{\{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}I, \qquad g^{\mu\nu} = \eta^{\mu\nu} - H\ell^\mu\ell^\nu.} \tag{5}$$

An explicit inverse coframe is $e_a{}^\mu = \delta_a{}^\mu - \frac{H}{2}\ell_a\ell^\mu$ *and* $\theta^a{}_\mu = \delta^a{}_\mu + \frac{H}{2}\ell^a\ell_\mu$ *in the specified flat reference coordinates. The metric determinant obeys* $\det g = \det\eta$ *in Cartesian Kerr–Schild coordinates.*

Proof. The nilpotent rank-one perturbation $N_a{}^b = \ell_a\ell^b$ satisfies $N^2 = 0$ because $\ell^2 = 0$, hence $(I - \frac{H}{2}N)^{-1} = I + \frac{H}{2}N$. Expanding $\{\Gamma^\mu, \Gamma^\nu\}$ with $\{\gamma^\mu, \not{\ell}\} = -2\ell^\mu I$ gives the linear term $+2H\ell^\mu\ell^\nu I$; the quadratic term vanishes since $\not{\ell}^2 = -\ell^2 I = 0$. This equals $-2(\eta^{\mu\nu} - H\ell^\mu\ell^\nu)I$. The determinant lemma gives $\det(\eta + H\ell\ell^\top) = \det\eta\,(1 + H\ell^2) = \det\eta$. □

Meaning. This is an exact *operator representation* of a chosen QGD null-sector geometry. It is not a derivation of H from quantum matter, nor a proof that Γ^μ is generated by the original scalar-contact QFT. The operator's connection and all other species' propagation must also be supplied consistently.

2.2 Curved Dirac dynamics, WKB, and the original bridge [C]

To obtain a covariant wave equation, the matrices in Eq. (3) are not by themselves sufficient. Using the torsion-free spin connection $\Omega_\mu[e]$ of the specified coframe, take

$$\boxed{\mathcal{D}_g\Psi := [i\hbar\Gamma^\mu(\partial_\mu + \Omega_\mu) - mc]\Psi = 0.} \tag{6}$$

The connection term is required for local Lorentz covariance and the full current equation. Writing $\Psi = e^{-iS/\hbar}(a_0 + O(\hbar))$ gives $p_\mu = \partial_\mu S$ with the adopted phase convention and, at leading order,

$$\boxed{(g^{\mu\nu}p_\mu p_\nu + m^2c^2)a_0 = 0 \quad \text{as a scalar mass shell.}} \tag{7}$$

The notation means $g^{\mu\nu}p_\mu p_\nu + m^2c^2 = 0$ wherever $a_0 \neq 0$; it is not a new matrix equation beyond the Dirac principal symbol. Spin transport and the spin connection enter beyond leading eikonal order.

Let $J^\mu = \bar{\Psi}\Gamma^\mu\Psi$ in the convention of Chapter 1. For a Hermitian compatible curved Dirac action and a real mass, $\nabla_\mu J^\mu = 0$. In static spherical symmetry $\sqrt{-g} = r^2\sin\theta$ in these coordinates; consequently

$$\boxed{\partial_r(r^2J^r) = 0, \qquad r^2J^r = C_J.} \tag{8}$$

Thus the *exact integrated current bridge survives* the null-sector geometric matching. The local relation $J^r \simeq \rho p_r c^2/E$ must, however, be replaced by its appropriate tetrad/observer-dependent curved-space WKB closure; copying the flat radial velocity formula unchanged into Kerr–Schild coordinates is generally incorrect. The four-mode spinor relations remain local algebraic ingredients after a compatible tetrad is fixed.

2.3 An explicit independent Hamilton–Jacobi verification [D/C]

Set $x^0 = ct$ and $c = 1$ temporarily. For $\ell_\mu dx^\mu = dx^0 + dr$ in a radial exterior patch,

$$ds^2 = -(1-H)(dx^0)^2 + 2H dx^0 \mathrm{d}r + (1+H)\mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2. \tag{9}$$

For a radial eikonal $S = -Ex^0 + W(r)$, $p_r = W'(r)$, the Dirac principal symbol gives

$$\boxed{(1-H)p_r^2 - 2HEp_r - (1+H)E^2 + m^2 = 0,} \tag{10}$$

with two branches, in the exterior $H < 1$,

$$\boxed{p_r^{\pm} = \frac{HE \pm \sqrt{E^2 - (1-H)m^2}}{1-H}.} \tag{11}$$

In the massless limit $p_r^+ = E(1+H)/(1-H)$ and $p_r^- = -E$; both satisfy the same null characteristic cone. As $H \to 0$, $p_r^{\pm} \to \pm\sqrt{E^2 - m^2}$. At $r = 4r_s$ ($H = 1/4$), $E = 3/2$ and $m = 1$, the outgoing/ingoing roots are approximately 2.133 and -1.133, respectively. These expressions were checked independently by substitution in Eq. (10). The resulting momentum change is geometric *once H has been provided*; it is not evidence that the microscopic matter theory independently selected H.

3 Why a gravitational natural invariant is selected: Hamilton, Fermat, and QGD (QM010)

Research question. Chapter 2 already selects the Schwarzschild natural-invariant amplitudes *dynamically under the adopted Einstein–Hilbert QGD action.* Chapter 1 supplies microscopic and effective Dirac structure, its WKB mass shell, the component ratio $u_B = \mathsf{R}u_A$, and the current bridge $r^2 j^r = C_J$. The remaining question is not whether those effective results are invalid, but whether the quantum matter theory itself supplies the *selection functional* relating a specified source state to a physical [Σ]. This section makes the distinction precise and records the conditional selection result as closed.

3.1 Stationary optical paths and stationary gravitational fields are distinct problems

In a prescribed isotropic optical medium of refractive index $n(\boldsymbol{x})$, Fermat's principle is

$$\delta_\gamma \mathfrak{T}[\gamma; n] = \delta_\gamma \left(\frac{1}{c} \int_\gamma n(\boldsymbol{x})\, \mathrm{d}s \right) = 0, \qquad \text{with } n \text{ fixed.} \tag{12}$$

Its Euler–Lagrange equations determine ray paths γ for that medium, including Snell's law. They do *not* determine $n(\boldsymbol{x})$ from the rays alone. Likewise, $\delta_\gamma \int \mathrm{d}s_g = 0$ gives geodesics for a *specified* metric g, not a field equation for g. For an individual physical history, Hamilton's principle reads $\delta\mathfrak{S} = 0$ under the appropriate endpoint and dynamical constraints; it is generally *not* the statement $\delta H_{\text{Hamiltonian}} = 0$, and stationarity does not imply an absolute minimum. The WKB phase S_{eik} must

not be confused with the action functional $\mathfrak{S}$. See also the general discussion of stationary-action principles in https://www.scholarpedia.org/article/Principle_of_least_action.

Thus the natural analogy for the *selection of the QGD medium itself* is not only ray stationarity but a separate *field* stationarity condition. In the adopted QGD effective theory it is

$$\boxed{\left.\frac{\delta S_{\mathrm{QGD}}[g[\Sigma],\psi,A]}{\delta\sigma_\mu^{(s)}(x)}\right|_{\substack{\text{specified source, boundary data}\\ \text{independently varied sectors}}} = 0.} \tag{13}$$

This condition is a **[C]** selection equation because its gravitational action is already supplied. It is not an independently derived matter-to-[Σ] law.

3.2 An explicit source-conditioned stationary functional for the Schwarzschild amplitude [C]

Let $H_{\mathrm{KS}}(\boldsymbol{x})$ denote the *dimensionless gravitational Kerr–Schild amplitude*, not the archival integrated matter energy H_{orig} of Chapter 1. In the weak-field static benchmark set $H_{\mathrm{KS}} = -2\Phi/c^2$. For a specified compact mass-density distribution ρ_m computed from the source quantum state, consider the *adopted Newtonian-limit field functional*

$$\boxed{\mathfrak{E}_{\mathrm{N}}[H_{\mathrm{KS}};\rho_m] = \int_{\mathbb{R}^3} \mathrm{d}^3x \left[\frac{c^4}{32\pi G_N}|\nabla H_{\mathrm{KS}}|^2 - \frac{c^2}{2}\rho_m H_{\mathrm{KS}}\right].} \tag{14}$$

For variations δH_{KS} vanishing at the boundary (or decaying sufficiently rapidly), integration by parts gives

$$\delta\mathfrak{E}_{\mathrm{N}} = \int \mathrm{d}^3x \left[-\frac{c^4}{16\pi G_N}\nabla^2 H_{\mathrm{KS}} - \frac{c^2}{2}\rho_m\right]\delta H_{\mathrm{KS}}, \tag{15}$$

$$\therefore \quad \boxed{\nabla^2 H_{\mathrm{KS}} = -\frac{8\pi G_N}{c^2}\rho_m.} \tag{16}$$

This gives the conditional source-to-amplitude map

$$\boxed{H_{\mathrm{KS}}(\boldsymbol{x}) = \frac{2G_N}{c^2}\int_{\mathbb{R}^3}\frac{\rho_m(\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|}\,\mathrm{d}^3y,} \tag{17}$$

with $H_{\mathrm{KS}} \to 0$ at infinity. For a compact spherical source with total mass $M_s = \int \rho_m\,\mathrm{d}^3x$, Gauss' law fixes the unique spherically symmetric exterior solution

$$\boxed{H_{\mathrm{KS}}(r) = \frac{2G_N M_s}{c^2 r} \quad (r > R), \qquad \sigma_\mu \mathrm{d}x^\mu = \sqrt{H_{\mathrm{KS}}}\,[\mathrm{d}(ct) + \mathrm{d}r].} \tag{18}$$

Proposition 3.1 (Conditional source-and-boundary selection, with its domain)**.** *For $G_N > 0$, a regular compact nonnegative source ρ_m with the stated decay at infinity and the Euclidean Newtonian-limit energy* (14)*, the stationary solution H_{KS} is unique within the permitted regular field class. In the spherically symmetric exterior, Eq.* (18) *follows without separately postulating an exterior profile. This proves* conditional *variational selection within the adopted weak-field theory, not microscopic emergence of its field functional.*

Proof. Stationarity gives Eq. (16), whose Green-function solution with the stipulated boundary condition is Eq. (17). If H_1, H_2 are two such solutions, their difference $h = H_1 - H_2$ satisfies $\nabla^2 h = 0$ with the imposed regularity and decay; integration of $h\nabla^2 h$ by parts gives $\int |\nabla h|^2 = 0$,

hence $h = 0$. Moreover, the second variation is $\delta^2\mathfrak{E}_{\rm N} = (c^4/16\pi G_N)\int|\nabla\delta H|^2 \geq 0$ under the convention $\delta^2\mathfrak{E} = \mathrm{d}^2\mathfrak{E}[H + t\delta H]/\mathrm{d}t^2|_{t=0}$. For an exterior spherical harmonic function, $H = a + b/r$; decay requires $a = 0$, and integration of Eq. (16) fixes $b = 2G_N M_s/c^2$. The sign of $\sigma = \pm\sqrt{H}\,\ell$ and the signed-sector/coframe redundancy require additional choice or quotient, even when H is unique. □

Scope. The functional (14) is a rewriting of the *adopted Newtonian gravitational dynamics*; it was not obtained by integrating out Chapter 1's fermions. Its source density and Schwarzschild exterior do not turn this approximation into the full relativistic stellar field equations. Chapter 2's separately checked Einstein–Hilbert sector variation provides the appropriate exact classical vacuum exterior once the action and its variational completeness conditions are accepted. Note also that varying H is not automatically identical to varying $\sigma = \sqrt{H}\,\ell$ at $\sigma = 0$, where the change of variables can be singular.

3.3 The complete QGD Euler equation is the actual field-selection equation [C]

For the founding metric, retained in Eq. (4), define the weight-adjusted metric Euler tensor Π^{ab} from the adopted full $S_{\rm QGD}$. Holding the coframe and other independent variables fixed as appropriate, the chain rule gives the general pullback condition

$$\boxed{(D_{\sigma^{(s)}}g)^*_{\sqrt{-g}}(\Pi_{\mathfrak{G}} + \Pi_{\mathfrak{T}} + \Pi_{\mathfrak{Q}}) = 0.} \tag{19}$$

In the Chapter 6 recognizable Option-C notation (with $s_s = -\varepsilon_s$), it is

$$\boxed{2s_s\Pi^{ab}\sigma^{(s)}_b = \lambda\big[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\big]^a.} \tag{20}$$

For the declared classical specialization $\lambda\mathcal{Q} = 0$, this becomes the projected condition $\Pi^{ab}\sigma^{(s)}_b = 0$ for a nonzero sector. The full tensor equation $\Pi^{ab} = 0$ implies the projected equations, but the converse is not automatic unless the independently varied metric-map directions have sufficient rank. The variational principle can select one or more $[\Sigma]$ subject to admissibility and boundary data, but the source-dependent quantum wavefunction alone has not been proven to select a unique representative. The classical *Schwarzschild natural invariant is dynamically selected within the adopted action*; it is not merely an algebraic guess.

3.4 What would make stationary selection genuinely microscopic? [O]

A microscopic theory could close the stronger original bridge without introducing a separate elementary graviton, but must specify the quantum collective variables and derive their constrained effective functional from the underlying state and action. Let φ^I be independent physical collective coordinates obtained from source-state quantum observables (or equivalence classes of composite operators). A *hypothetical, to-be-derived* effective functional is $\Gamma_{\rm mic}[\varphi; J]$, where J_A couples to independently defined source operators. Only if this functional is actually obtained from microscopic quantum matter may we assert a physical selection law:

$$\boxed{\frac{\delta\Gamma_{\rm mic}}{\delta\varphi^I} = 0, \qquad [\Sigma]_{\rm physical} = \mathcal{P}[\varphi_*;\ \text{frame and boundary data}].} \tag{21}$$

The projection $\mathcal{P}$ from physical collective coordinates to natural-invariant sectors must be independently defined; a four-column spinor basis, an arbitrary probe velocity or a factorization of an already known metric is not such a proof. For an isolated regular stationary branch, differentiate (21) with respect to the source J_A to obtain

$$\boxed{\frac{\delta\varphi^I}{\delta J_A} = -[\mathcal{K}^{-1}_{\rm phys}]^{IJ}\frac{\delta^2\Gamma_{\rm mic}}{\delta\varphi^J\delta J_A}, \qquad \mathcal{K}_{\rm phys,IJ} := \frac{\delta^2\Gamma_{\rm mic}}{\delta\varphi^I\delta\varphi^J},} \tag{22}$$

where inverses are only taken after treating constraints, gauge directions, boundary conditions and any zero modes. This identity is a *conditional implicit-function theorem*, not a microscopic calculation of the kernel. It exhibits precisely where source dependence and nonlocal exterior information would enter. It does *not* prescribe a spin-two particle, and it does not require that every collective variable be an independent propagating field.

An equivalent weak-field matching diagnostic is obtained by differentiating (14): in Fourier space the *adopted* quadratic amplitude kernel and source vertex are

$$\boxed{\mathcal{K}_{HH}^{\mathrm{QGD}}(\boldsymbol{q}) = \frac{c^4}{16\pi G_N}|\boldsymbol{q}|^2, \qquad \mathcal{B}_{H\rho}^{\mathrm{QGD}} = -\frac{c^2}{2}, \qquad \frac{\delta H_{\mathrm{KS}}(\boldsymbol{q})}{\delta\rho_m(\boldsymbol{q})} = \frac{8\pi G_N}{c^2|\boldsymbol{q}|^2}.} \tag{23}$$

This is an *effective functional matching target*, not a result obtained from the contact interaction. Matching its observable consequences is essential for independent microscopic emergence of the same exterior source law, but computing a quasiparticle self-energy or a graviton pole is *not* automatically the correct first research step for the original physical spinor-to-invariant map.

3.5 Practical decision and falsifiable next step

Closed [C]: For compact spherical sources, $\hat{\varrho}_{\mathrm{src}} \mapsto \rho_m \mapsto H_{\mathrm{KS}} \mapsto \sigma$ follows by stationary field selection from the adopted QGD weak-field action and its boundary conditions. The exact external null-branch Chapter 2 Schwarzschild profile, compatible Dirac Clifford algebra, WKB shell and conserved-current bridge are retained. No new force or graviton calculation is required to recognize this *conditional* closure.

Open [O]: Starting from the *actual macroscopic Dirac solution and its source-state input*, derive a basis-independent physical rule $\mathcal{F}_{\mathrm{micro}}(\Psi, S, \mathsf{R}, j, \hat{\varrho}_{\mathrm{src}}; \mathrm{BC}) = [\Sigma]$ or derive the functional whose stationary points define that rule. Test source dependence, observer/probe independence, exterior matching, sector redundancy and complete-metric consistency. This is the research priority. A source-to-kinetic Dirac vertex, a tensor collective response and the prior QM001–QM009 spectral calculations remain *optional diagnostic routes*, not prerequisites to maintaining the closed conditional bridge.

Conceptual conclusion. Nature does not select a field merely because a test ray takes a stationary path. A dynamical principle and source/boundary data select the *field*; the selected field then governs its rays and spinor propagation. In effective QGD the relevant principle is already specified. The microscopic origin and uniqueness of its physical natural-invariant selection remain the one unresolved foundational connection.

4 The decisive source-to-kinetic response target

4.1 What the effective QGD source law supplies [C]

For a static, compact, nonrelativistic quantum source with independently computed mass density $\rho_m(\boldsymbol{y})$, adopt the weak-field QGD/GR source response, including its empirically normalized G_N:

$$\boxed{H[\rho_m](\boldsymbol{x}) = \frac{2G_N}{c^2}\int \frac{\rho_m(\boldsymbol{y})}{|\boldsymbol{x}-\boldsymbol{y}|}\,\mathrm{d}^3y, \qquad \nabla^2 H = -\frac{8\pi G_N}{c^2}\rho_m.} \tag{24}$$

For $r > R$ and a spherical source of total mass $M_s = \int \rho_m \mathrm{d}^3y$, $H = r_s/r$, $r_s = 2G_N M_s/c^2$. Chapter 2's independent Euler calculation, with the chosen QGD vacuum null branch, gives the same *exact exterior* profile when that Einstein–Hilbert action is adopted. The interior Poisson relation here is only a weak-field approximation, not an exact general-relativistic stellar solution.

4.2 A functional derivative that any stronger microscopic map must match [D/C]

Keeping the specified exterior null orientation $\ell_\mu(\boldsymbol{x})$ fixed while perturbing the source density, differentiate Eqs. (3) and (24):

$$\boxed{\frac{\delta\Gamma^\mu(\boldsymbol{x})}{\delta\rho_m(\boldsymbol{y})} = -\frac{G_N}{c^2}\frac{\ell^\mu(\boldsymbol{x})\ \not{\ell}(\boldsymbol{x})}{|\boldsymbol{x}-\boldsymbol{y}|}.} \tag{25}$$

In a local patch with fixed ℓ, the nonzero-momentum static Fourier-space target reads

$$\boxed{\delta\Gamma^\mu(\boldsymbol{q}) = -\frac{4\pi G_N}{c^2}\,\ell^\mu\ \not{\ell}\frac{\delta\rho_m(\boldsymbol{q})}{|\boldsymbol{q}|^2}.} \tag{26}$$

Here $\boldsymbol{q}$ denotes Fourier momentum (*not* the null one-form ℓ). Equations (25)–(26) are an *exact functional derivative of the stipulated effective matching rule*. The $1/|\boldsymbol{q}|^2$ kernel was supplied by the adopted gravitational equation; it has *not* been obtained from the tested microscopic contact interaction. This is now a quantitatively sharp benchmark for Route B. It does not prescribe how a fully nonlinear multi-sector, nonzero-$\lambda\mathcal{Q}$ QGD construction should behave.

4.3 What the original microscopic Dirac model can and cannot supply [D/C/O]

Use Chapter 1's regulated contact theory $\mathcal{L}_{\text{micro}} = \bar\psi(i\gamma^\mu\partial_\mu - m_0)\psi + G_4(\bar\psi\psi)^2$ in its tested homogeneous, isotropic Hartree phase. Its self-energy is a scalar mean-field mass shift; first-order local Fock exchange adds constant scalar and temporal terms. Those terms change the Dirac eigenspinors but *do not* generate the particular position-dependent derivative matrices in Eq. (3). The tested stable scalar Gaussian source response has a finite static Hessian $K_0 > 0$ and, for a regular small-$\boldsymbol{q}$ continuation, a source-visible response analytic at $\boldsymbol{q} = 0$; it therefore does not provide the $1/|\boldsymbol{q}|^2$ kernel of Eq. (26).

More sharply, at Hartree order in momentum space the defined effective Dirac operator takes the form $\mathcal{D}_H(p;\rho) = \gamma^\mu p_\mu - M[\rho]$ (up to the adopted energy origin). Thus its source derivative of the *kinetic* principal symbol is exactly

$$\boxed{\frac{\delta}{\delta\rho_m(y)}\left[\frac{\partial\mathcal{D}_H}{\partial p_\mu}\right] = \frac{\delta\gamma^\mu}{\delta\rho_m(y)} = 0.} \tag{27}$$

The first-order local Hartree–Fock corrections remain algebraic in momentum in the tested homogeneous phase, so the same zero derivative result holds at that order. It demonstrably disagrees with the nonzero effective target Eq. (25). This is the first directly calculated *operator-vertex mismatch* for the chosen microscopic phase: it is stronger than observing merely that a spinor ratio changes. It does not exclude a higher-order or different-phase state-dependent kinetic vertex.

This is a *restricted negative result*, not an all-orders impossibility theorem for contact QFT: different phases, additional composite operators, non-Gaussian collective physics, dynamical gauge interactions or nonperturbative effects can change the infrared theory. The QM001–QM005 single-fermion dispersion calculations are likewise not by themselves evidence of the required universal source-to-kinetic response. A momentum-dependent quasiparticle self-energy need not be a universal curved-space tetrad.

Proposition 4.1 (Probe and local-source obstructions, with domain stated)**.** *Within a prescribed static source and a fixed effective gravitational background: (i) no universal gravitational amplitude $H(\boldsymbol{x})$ can be defined by equating it to the speed or lower/upper ratio of* an arbitrarily chosen *test spinor, because different permitted probes at the same event have different velocities and spinor*

ratios; (ii) no strictly local function $H(\boldsymbol{x}) = F(\rho_m(\boldsymbol{x}), \ldots)$ that vanishes for vacuum-valued local source observables can reproduce a nonzero compact-source exterior field in a vacuum shell where those observables take their vacuum values.

Proof. For (i), at $H = 1/4$ the designated free-fall speed is $\beta_{\rm ff} = 1/2$ and $\zeta_{\rm ff} = 1/(2+\sqrt{3}) \simeq 0.267949$, whereas a different test particle at the same event may have a distinct speed β. Thus $H = \beta^2_{\rm arbitrary}$ cannot define a unique source geometry. For (ii), outside a compact source $\rho_m = 0$ and the assumed other local source observables are vacuum-valued, giving $F = 0$ by hypothesis, while Chapter 2's exterior amplitude $H = r_s/r > 0$. A nonlocal, boundary-data-dependent or independent field-equation matching rule is necessary; Eq. (24) is one explicitly declared conditional example. □

4.4 What counts as completion of the original QGD bridge

The research should now use two separate completion criteria. **Effective correspondence [C]:** for a specified quantum source, recover [Σ] by the adopted QGD Euler equations, verify its full master metric (4) or declared null-sector specialization, build $\mathcal{D}_g$ with its spin connection, and check WKB and $r^2 J^r = C_J$. This route is mathematically closed in the declared spherical exterior benchmark; it must not be advertised as microscopic genesis of the gravitational field. **Physical origin [O]:** compute a source-dependent kinetic symbol or a physical collective constitutive law without first using the target metric/action, and show that its source response produces the required exterior field and a species-independent effective propagation geometry. Its functional target is Eq. (25); the long-range/nonlocal feature is not optional if the source is compact and the exterior field nonzero, but it need not be described as a separate fundamental force carrier.

Practical next microscopic calculation. Compute the *source-to-probe derivative kinetic vertex*

$$\boxed{\mathcal{V}^\mu(x, y; p) = \frac{\delta}{\delta\rho_m(y)} \left[\frac{\partial \mathcal{D}_{\rm eff}(x,p)}{\partial p_\mu}\right]_{\text{physical state}},} \tag{28}$$

starting from a specified source density matrix (not one arbitrarily selected probe mode), with matched regulators and conserved source vertices. Its null-sector target, in the Dirac part of the symbol, is Eq. (25). A failure of the tested phase to produce the required nonlocal state dependence would be a scientifically useful result; a successful calculation would provide the missing physics for the original state-to-natural-invariant map. This test is more targeted than searching for a massless tensor particle in isolation.

5 Preserve the WKB and wavefunction–momentum bridge

Within a coherent local positive-energy Dirac branch with an effective scalar mass $\mathcal{M}(x)$ and gauge-invariant spatial momentum $\boldsymbol{\pi}$, the leading local Dirac equation gives

$$\boxed{u_B = \mathsf{R} u_A, \qquad \mathsf{R} = \frac{c\boldsymbol{\tau}\cdot\boldsymbol{\pi}}{\mathcal{E}_+ + \mathcal{M}c^2}, \quad \boldsymbol{\zeta} = \frac{c\boldsymbol{\pi}}{\mathcal{E}_+ + \mathcal{M}c^2}.} \tag{29}$$

This is a **[D]** local eigenspinor relation under its stated assumptions, not yet the gravitational covector σ_μ. The full four-mode QGD construction may be written $\Psi = U_f c$; since alternative invertible U_f yield the same Ψ by a compensating change in c, the unrestrained four-mode basis does not uniquely identify the physical gravitational profile.

For a coherent branch, $\Psi \simeq Au\exp(iS/\hbar)$ with $\boldsymbol{\pi} = \boldsymbol{\nabla}S - q\boldsymbol{A}$ when the eikonal and gauge conventions allow. The leading spinor current is $\boldsymbol{j} \simeq \rho_D c^2 \boldsymbol{\pi}/\mathcal{E}_+$, while the exact global-number

Ward identity of the interacting microscopic theory is $\partial_\mu j^\mu = 0$. Hence in a stationary, spherically symmetric source-free shell,

$$\boxed{r^2 j^r = C_J, \qquad j^r \simeq \rho_D \frac{\pi_r c^2}{\mathcal{E}_+} \quad \text{for an isolated positive-energy WKB stream.}} \tag{30}$$

The first equation concerns the conserved *total* current; a separately identified collisionally coupled stream can have a source in its own kinetic transport equation. A zero net flow of a bound or isotropic source does not mean zero energy density or zero stress. The WKB and flux equations are **[D/C]** matter-transport statements; no $1/r$ gravity law follows solely from them.

The QM001–QM005 restricted finite-density calculations demonstrate state-dependent effective spinor projectors and a self-energy/damping programme, but do not identify a universal gravitational mode. They are *input benchmarks*, not independent microscopic gravity proofs.

6 A microscopic response kernel without an inserted Einstein propagator

Let $\widehat{\mathcal{O}}_I$ be a specified set of renormalized or cutoff-defined microscopic observables, including total stress-energy, currents and candidate scalar/tensor composites. With physical probe sources J_A, define $W[J]$, expectation coordinates φ^I, and a Legendre effective action $\Gamma_{\rm coll}[\varphi]$ on the nondegenerate constrained response subspace. At a stable stationary state,

$$\mathcal{K}_{IJ}(k) := \frac{\delta^2 \Gamma_{\rm coll}}{\delta\varphi^I(-k)\delta\varphi^J(k)}, \qquad V_A{}^I(k) := \frac{\delta^2 \Gamma_{\rm coll}}{\delta J_A(-k)\delta\varphi^I(k)}. \tag{31}$$

The physical linear response is schematically

$$\boxed{\chi^{\rm phys}_{AB}(k) = V_A{}^I(k)[\mathcal{K}^{-1}(k)]_{IJ} V_B{}^J(-k) + \chi^{\rm contact}_{AB}(k).} \tag{32}$$

The exact signs of the exchange term depend on source/Legendre conventions, but the pole criterion does not. V and $\mathcal{K}$ must be computed from the *same* microscopic generating functional. In a theory with gauge or constraint directions the inverse is understood only on the physical response quotient, with the Ward seagull/contact pieces retained. Defining $\mathcal{K}$ via a Legendre transform does not presuppose independent physical propagation of every coordinate φ^I.

Definition 6.1 (Necessary static source-visible criterion **[C]**)**.** *For a specified pair of physical source channels in three spatial dimensions, a genuine long-range inverse-distance linear response requires a nonzero, noncancelled coefficient of* $1/|\boldsymbol{k}|^2$ *in the* full *static physical response. Define the diagnostic*

$$\boxed{\mathcal{R}_{AB} := \lim_{|\boldsymbol{k}|\to 0} |\boldsymbol{k}|^2 \chi^{\rm phys}_{AB}(0,\boldsymbol{k}),} \tag{33}$$

when the limit and specified order of frequency/momentum limits exist. $\mathcal{R}_{AB} \neq 0$ *is necessary for a simple Coulomb/Newton-like static pole with regular subleading terms; it is not sufficient for universally coupled relativistic gravity.*

The source-visible condition is stronger than "the spectrum is gapless." A sound-like density mode or a particle–hole continuum need not mediate an independent static $1/r$ interaction between generic sources. One must identify the physical eigenmode, stability, causal dynamics, coupling and cancellation structure without placing an Einstein/Fierz–Pauli $1/k^2$ propagator inside $\mathcal{K}$ by assumption.

7 Explicit Chapter 1 scalar-contact test: a finite response

Choose the original Chapter 1 model with vanishing background electromagnetic field,

$$\boxed{\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m_0)\psi + G_4(\bar{\psi}\psi)^2, \qquad G_4 > 0.} \tag{34}$$

In natural units at zero temperature, the prescribed homogeneous Hartree saddle satisfies

$$0 = \frac{M - m_0}{2G_4} - \frac{M}{\pi^2}\int_{p_F}^{\Lambda} \frac{p^2\,\mathrm{d}p}{\sqrt{p^2 + M^2}}, \qquad p_F = \sqrt{\mu^2 - M^2}. \tag{35}$$

This is a fixed-cutoff effective model, not a regulator-independent ultraviolet completion. Its scalar inverse static response, density-mixing vertex, and direct density response are

$$\mathcal{K}_{MM}(0,\mathbf{0}) = K_0 = \frac{1}{2G_4} - \frac{1}{\pi^2}\left[\int_{p_F}^{\Lambda} \frac{p^4\,\mathrm{d}p}{(p^2+M^2)^{3/2}} + \frac{M^2 p_F}{\mu}\right], \tag{36}$$

$$V_0 = \Omega_{M\mu} = \frac{M p_F}{\pi^2}, \qquad \chi_{nn}^{\mathrm{direct}} = \frac{\mu p_F}{\pi^2}. \tag{37}$$

For $\Lambda = 4$, $m_0 = 0.12$, $G_4 = 0.5654255716$, and $\mu = 1.2$, independent quadrature and root-solving give

$$\boxed{\begin{array}{cc} M = 0.693435535520, & p_F = 0.979360586341, \\ K_0 = 0.168741734136 > 0, & V_0 = 0.068809590036, \\ \chi_{nn}^{\mathrm{direct}} = 0.119075968585, & \dfrac{V_0^2}{K_0} = 0.028059209568, \\ \multicolumn{2}{c}{\chi_{nn}^{\mathrm{total}} = 0.147135178153.} \end{array}} \tag{38}$$

The total susceptibility was independently checked by differentiating the density after re-solving the nonlinear Hartree gap equation at nearby chemical potentials. This numerical reproduction agrees with Chapter 5 v3.7 and does not independently infer an exterior gravitational potential.

Proposition 7.1 (Regular scalar channel does not generate the required static pole **[C]**)**.** *Assume the scalar Hessian and density vertex are analytic near zero static momentum, with* $\mathcal{K}_{MM}(0,\boldsymbol{k}) = K_0 + O(|\boldsymbol{k}|^2)$*,* $V(0,\boldsymbol{k}) = V_0 + O(|\boldsymbol{k}|^2)$*, and direct/contact terms regular there. If* $K_0 > 0$*, then*

$$\boxed{\lim_{|\boldsymbol{k}|\to 0} |\boldsymbol{k}|^2 \chi_{nn}^{\mathrm{phys}}(0,\boldsymbol{k}) = 0.} \tag{39}$$

Therefore the tested stable scalar-density channel cannot realize the simple source-visible $1/|\boldsymbol{k}|^2$ *static MW2 response under these hypotheses.*

Proof. Since $K_0 > 0$, the inverse scalar Hessian is bounded in a neighborhood of zero momentum and $V\mathcal{K}^{-1}V = V_0^2/K_0 + O(|\boldsymbol{k}|^2)$. The regular direct/contact contribution is bounded too. Multiplication by $|\boldsymbol{k}|^2$ gives zero. The result is restricted to the named channel and analytic, stable phase; it is not a statement about all tensor or exotic collective excitations of QFT. □

For visual illustration only, insert a *trial* positive static gradient coefficient Z_s and set $x = |\boldsymbol{k}|\sqrt{Z_s/K_0}$. Then $\chi_{\mathrm{exch}} = V_0^2/[K_0(1+x^2)]$: its static limit remains finite while $x^2\chi_{\mathrm{exch}} \to 0$. The stiffness Z_s is *not* claimed computed by this plotted model.

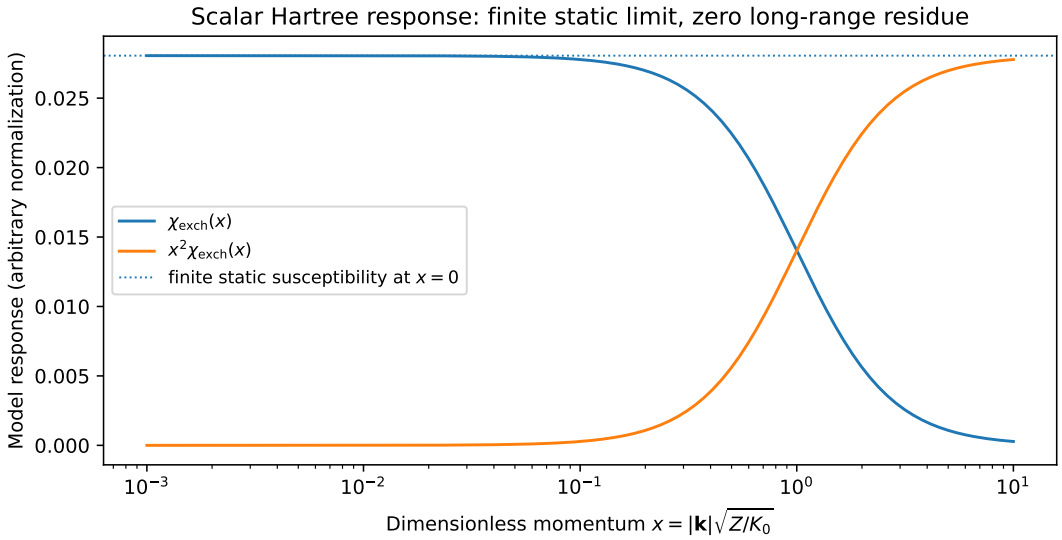


Figure 1: Reproducible scalar pole diagnostic. The positive gradient coefficient is a declared illustrative regular continuation; the K_0, V_0 inputs are calculated from the named microscopic Hartree saddle. No gravitational response has been inserted.

8 Second obstruction: scalar matter channels do not become tensor gravity by relabeling

Consider an isotropic equilibrium state and a single rotational scalar excitation q. For spatial external momentum $\boldsymbol{k}$, the most general parity-even symmetric scalar–stress two-point vertex has the tensor form

$$V_{ij,q}(0,\boldsymbol{k}) = a(|\boldsymbol{k}|^2)\delta_{ij} + b(|\boldsymbol{k}|^2)k_i k_j, \tag{40}$$

up to contact terms with the same rotational structure.

Proposition 8.1 (Vanishing transverse-traceless scalar mixing **[D/C]**). *For either physical spatial TT polarization $e^{\mathrm{TT}}_{ij}(\boldsymbol{k})$ satisfying $e^{\mathrm{TT}}_{ii} = 0$ and $k_i e^{\mathrm{TT}}_{ij} = 0$, one has*

$$\boxed{e^{\mathrm{TT}}_{ij} V_{ij,q} = 0.} \tag{41}$$

Thus an isotropic scalar-only linear exchange does not supply a helicity-two pole through its mixed stress vertex. Additional tensor, vector, anisotropic, symmetry-breaking or nonperturbative channels require their own independent tests.

Proof. Contract Eq. (40) with the TT tensor. The trace term vanishes because $e_{ii} = 0$ and the momentum term because $k_i e_{ij} = 0$. □

In a Lorentz-invariant stationary vacuum, after satisfying the complete stress Ward identity including contacts, a regular scalar–metric vertex takes the compatible form $V_{\mu\nu,q} \propto (k^2\eta_{\mu\nu} - k_\mu k_\nu)F(k^2)$ in the tested scalar channel. If F is finite and $\mathcal{K}_{qq}(0) > 0$, exchange contributions generated solely by this vertex scale as $O(k^4)$, not as $1/k^2$. This complements Chapter 5's explicitly calculated finite-momentum stress–mass form factor. A medium can have different temporal/spatial form factors, but the isotropic spatial TT contraction above remains zero for a scalar two-point vertex.

9 Third obstruction: a strictly local source-to-field rule cannot explain the exterior

Proposition 9.1 (Exterior obstruction for a vanishing local matter order parameter [**D/C**])**.** *Let a compact stationary matter source have coarse-grained matter data $\varphi^I(x)$ equal to fixed vacuum values $\varphi^I_{\rm vac}$ in an open exterior region U. Suppose a proposed natural-invariant prescription is strictly local and algebraic, $\Sigma(x) = F(\varphi^I(x))$, with no derivatives, boundary response, independent field degrees of freedom, or state-dependent nonlocal kernel. Then $\Sigma(x) = F(\varphi_{\rm vac})$ throughout U. It cannot represent a* source-dependent *nonconstant Schwarzschild exterior in that region unless the vacuum data or map itself carry nonlocal source information.*

Proof. At every exterior point the arguments of F equal the same vacuum constants, so its output is constant in the chosen fixed reference frame. A universal map has no local variable that encodes source mass or radial distance there. A source-dependent exterior therefore requires additional nonlocal response, boundary-value dynamics or genuinely exterior quantum correlations. □

The statement does not exclude vacuum correlations extending outside a compact classical source or a collective field living in the exterior. Rather, it identifies exactly what a successful microscopic realization must compute. A local Dirac momentum ratio is not automatically such an exterior response.

10 Incremental tensor calculation: collisionless stress spectrum and kinetic Ward completion (QGD–QM006)

Scope and status [D/C]. This first new tensor increment is evaluated in the homogeneous, zero-temperature Hartree Fermi sea of Chapter 1, retaining only its leading *positive-energy intraband* particle–hole response. It is an independently reproducible refinement of that chapter's shear formula, not the complete renormalized metric-source polarization. Vacuum/antiparticle bands, full stress vertices, interaction-resummed tensor ladders, and the gravitational metric seagull remain open and are not inferred from this result.

10.1 Derivation of the transverse-traceless Fermi-surface response

Select external wavevector $\boldsymbol{q} = q\boldsymbol{e}_z$. For a local positive-energy on-shell Dirac quasiparticle, the leading kinetic shear insertion is $V_{xy}(\boldsymbol{p}) = p_x p_y/E_{\boldsymbol{p}}$ and the Fermi occupation $n_{\boldsymbol{p}} = \Theta(p_F - |\boldsymbol{p}|)$. With spin degeneracy two, define the retarded intraband response by

$$\chi^R_{xy,xy}(\omega, q) = 2\int \frac{\mathrm{d}^3p}{(2\pi)^3}\, \frac{n_{\boldsymbol{p}-\boldsymbol{q}/2} - n_{\boldsymbol{p}+\boldsymbol{q}/2}}{\omega + i0^+ + E_{\boldsymbol{p}-\boldsymbol{q}/2} - E_{\boldsymbol{p}+\boldsymbol{q}/2}}\, V_{xy}(\boldsymbol{p})^2 + O(q/p_F)_{\rm vertex}. \tag{42}$$

Taking $q \to 0$ with $z = (\omega + i0^+)/(v_F q)$ fixed, using $n_- - n_+ = q v_F u\, \delta(E_p - \mu) + O(q^3)$ and $u = \cos\theta$ gives

$$\boxed{\begin{aligned} &\chi^R_{xy,xy}(\omega, q) = C_T J(z) + O(q/p_F), \qquad C_T = \frac{p_F^5}{16\pi^2\mu}, \\ &J(z) = \int_{-1}^{1} \frac{u(1-u^2)^2}{z-u}\,\mathrm{d}u \\ &\qquad = -2z^4 + \frac{10}{3}z^2 - \frac{16}{15} + z(1-z^2)^2 \log\frac{z+1}{z-1}. \end{aligned}} \tag{43}$$

The logarithm has the retarded branch. Its static limit and its continuum discontinuity are

$$\boxed{\begin{aligned}\chi^R_{xy,xy}(0,q\to 0) &= -\frac{p_F^5}{15\pi^2\mu} = -0.005071551738,\\ \operatorname{Im}\chi^R_{xy,xy}(\omega,q) &= -\pi C_T\, z(1-z^2)^2\,\Theta(1-|z|)\quad (z\in\mathbb{R}),\end{aligned}} \tag{44}$$

where the second expression assumes the low-q intraband limit. At $z = 0.5$, the model gives

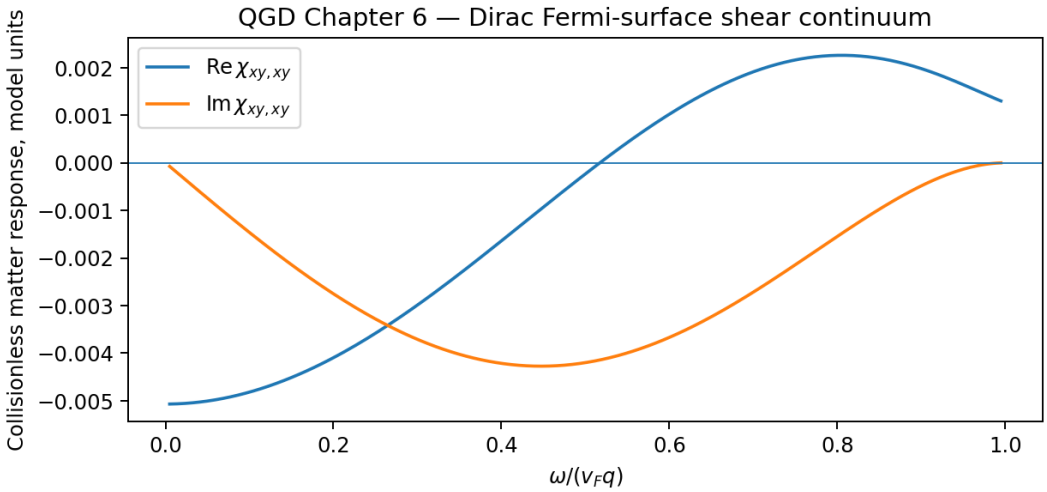
$$\chi^R_{xy,xy} = -0.0002346320 - 0.0042010180\, i.$$

These are finite continuum responses; they do not identify an isolated $1/q^2$ pole or a physical massless spin-two particle. An additive analytic metric-source contact cannot generate such a nonlocal pole by itself.

Both helicity-two components. For the orthogonal shear $V_D = (p_x^2 - p_y^2)/(2E)$ the two azimuthal averages coincide:

$$\int_0^{2\pi} (\cos\phi\sin\phi)^2\,\mathrm{d}\phi = \int_0^{2\pi} \tfrac{1}{4}\cos^2(2\phi)\,\mathrm{d}\phi = \frac{\pi}{4}.$$

Their mixed angular integral vanishes. Thus $\chi^R_{D,D} = \chi^R_{xy,xy}$ and $\chi^R_{D,xy} = 0$ at this isotropic intraband order. An isotropic scalar $\bar\psi\psi$ has zero mixed susceptibility to either TT component at the same order.



Leading positive-frequency retarded intraband tensor response. The imaginary part is the particle–hole continuum, not an isolated graviton.

10.2 Transverse-momentum Ward identity and the indispensable kinetic contact

The TT stress is not itself longitudinally constrained by momentum conservation. To independently check conservation, use T^{0x} and T^{zx}, with on-shell insertions $V_{0x} = p_x$ and $V_{zx} = p_z p_x/E_p$. At leading Fermi-surface order, their correlators obey

$$\boxed{\omega\chi^R_{0x,zx}(\omega,q) - q\big[\chi^R_{zx,zx}(\omega,q) + C_{\mathrm{kin}}\big] = 0, \qquad C_{\mathrm{kin}} = \frac{p_F^5}{15\pi^2\mu}.} \tag{45}$$

To verify, substitute the two vertices into Eq. (42). In the combination on the left, the factor $(z-u)$ cancels the denominator, leaving the angular integral $\int_{-1}^{1} u^2(1-u^2)\mathrm{d}u = 4/15$. The resulting nonzero local term is exactly qC_{kin} and must be retained to satisfy the leading kinetic momentum

Ward relation. Numerically, at $z = 1.5$ we obtain $\chi_{0x,zx} = 0.005207179181$, $\chi_{zx,zx} = 0.001303080831$, and a normalized Ward residual below 2×10^{-18}.

Crucial qualification. $C_{\rm kin}$ is the contact needed to complete this *linearized collisionless kinetic* stress identity. It is *not yet* the full local metric seagull/contact obtained by twice varying the regulated quantum effective action. A regulator-consistent microscopic gravity-source Ward identity must include that genuine seagull, vacuum terms, exchange and all conserved interaction vertices; the present identity is a restricted nontrivial consistency test, not a proof of complete diffeomorphism covariance.

10.3 Independent numerical checks and next increment

A second integration with the finite spatial-momentum occupation boundaries recovers $\chi_{xy,xy}(z = 1.5, q \to 0) = 0.000382522300$. At $q = 0.005$ it gives 0.000382519086, a relative deviation 8.40×10^{-6}. Independent Cauchy-principal-value integration reproduces Eq. (43) inside the Landau cut to better than 4×10^{-11} at the sampled z values. The code and data are packaged as `verify_tensor.py` and `tensor_shear_continuum.png`.

Next narrowly bounded increment [O]. Calculate the actual *metric-source* $T^{\mu\nu}$ bubble and second-variation seagull from the same sharp-cutoff Dirac effective action; evaluate its full stress Ward residual before and after contact terms. Only after that check should the interaction-resummed tensor Hessian be tested for a source-visible isolated pole. The WKB/Dirac ratio and stationary $r^2 j^r = C_J$ remain independent upstream matter identities.

11 Covariant stress Hessian, genuine metric seagull and tensor-pole decision (QM007)

Scope and scheme [D/C]. The earlier scalar benchmark employs a sharp *three-momentum* regulator in the preferred medium rest frame. Such a cutoff must not silently be treated as a covariant geometric regulator. In this section the vacuum determinant is separately defined by a covariant proper-time cutoff; the occupied positive-energy states contribute a finite density correction, with the chemical-potential one-form transformed under background coordinate changes. When matching the same homogeneous mass, the contact coupling is *refitted* and the two regulated theories are not identified coefficient by coefficient. The complete finite-q retarded Dirac Hessian, including all interaction-consistent vertices, has not been evaluated here. The exact functional bubble–seagull construction and the local covariant part have been evaluated and checked explicitly.

11.1 Metric-source definition and exact second-variation identity

Put the scalar Hubbard–Stratonovich field at a specified saddle M and couple the Dirac field to a *bookkeeping external metric* g. In Euclidean signature, let $\mathcal{D}_g = \gamma^a e_a{}^\mu \nabla_\mu[g] + M$, with the required spin connection. Define the positive Laplace-type operator

$$\boxed{H_g = \mathcal{D}_g^\dagger \mathcal{D}_g = -\nabla_g^2 + \frac{1}{4}R[g] + M^2, \qquad W_\Lambda[g;M] = \frac{1}{2}\int_{\Lambda^{-2}}^\infty \frac{\mathrm{d}s}{s}\,\mathrm{Tr}\; e^{-sH_g}.} \tag{46}$$

Here the equality to the minimal Laplace-type form refers to the torsion-free, zero-gauge-field, constant-M vacuum benchmark; nonconstant M or gauge fields produce additional operator terms. The path-integral fermionic sign and the first-order determinant are incorporated in the stated parity-even Euclidean proper-time convention. W_Λ is a *matter response to an external metric*, not a dynamical graviton action. The independent HS contribution $(M - m_0)^2/(4G_4)\int\sqrt{g}$ must be included in the full mean-field stress and varied consistently when M responds.

For independent metric perturbations h and k, denote $H_h = \delta_h H$, $H_{hk} = \delta_h \delta_k H$. Differentiating the *regulated trace itself* yields the exact identity

$$\boxed{\begin{aligned}\delta_h \delta_k W_\Lambda = \frac{1}{2}\int_{\Lambda^{-2}}^{\infty} \mathrm{d}s \int_0^s \mathrm{d}\tau\ \mathrm{Tr}\Big(e^{-(s-\tau)H} H_k e^{-\tau H} H_h\Big) \\ - \frac{1}{2}\int_{\Lambda^{-2}}^{\infty} \mathrm{d}s\ \mathrm{Tr}\Big(e^{-sH} H_{hk}\Big)\,.\end{aligned}} \tag{47}$$

The first term is the regulated *two-linear-vertex bubble*; the second is the *genuine metric-source seagull*, including second variations of the coframe, spin connection and density weight implicit in the covariant trace. Crucially, neither term is defined by adding an arbitrary local tensor after integration. A finite-dimensional regularization or a covariant bundle trace can make the cyclic functional-derivative argument rigorous. Equation (47) is exact for the chosen proper-time functional; the actual finite-q numerical integrals are a separate task.

11.2 Exact metric Ward identity, including contact and tadpole terms

Define $T^{\mu\nu} = 2(\sqrt{g})^{-1}\delta W/\delta g_{\mu\nu}$ and $K^{\mu\nu,\rho\sigma} = \delta T^{\mu\nu}/\delta g_{\rho\sigma}$ at a homogeneous flat background, with symmetric-source differentiation conventions. Varying $\nabla_\mu T^{\mu\nu} = 0$ gives, in Fourier space,

$$\boxed{\begin{aligned}q_\mu K^{\mu\nu,\rho\sigma}(q) = &-\frac{1}{2} q_\lambda \overline{T}^{\lambda\nu}\eta^{\rho\sigma} - \frac{1}{2}(q^\rho \overline{T}^{\sigma\nu} + q^\sigma \overline{T}^{\rho\nu}) \\ &+ \frac{1}{2} q^\nu \overline{T}^{\rho\sigma}.\end{aligned}} \tag{48}$$

The expectation value $\overline{T}$ is the *complete* background stress and all metric-coupled sources are transformed. At nonzero pressure or vacuum tadpole, the Hessian need not be simply transverse. A purported check of $q_\mu\langle TT\rangle = 0$ before including the contacts is generally incorrect. For q along z and spatial shear indices $(\nu;\rho\sigma) = (x;zx)$, Eq. (48) becomes $K^{zx,zx} = -\overline{P}/2$ in the homogeneous static local pressure response convention. The occupation-only free-quasiparticle diagnostic has $n = 0.03172538246$, $\varepsilon_{\mathrm{occ}} = 0.03235666932$, $P_{\mathrm{occ}} = 0.005713789638$, hence $-P_{\mathrm{occ}}/2 = -0.002856894819$. This is a *kinetic subset illustration*; the total interacting $\overline{P}$ also includes the HS potential, sea subtraction and chosen finite counterterms.

Proposition 11.1 (Nonlinear-coordinate Ward/seagull cancellation **[D]**)**.** *On a periodic flat Euclidean coordinate patch, choose the finite coordinate change $x' = x + \epsilon\sin(qz)/q$ and leave z and the other coordinates unchanged. Its pulled-back metric has*

$$g_{xz} = \epsilon\cos(qz), \quad g_{zz} = 1 + \epsilon^2\cos^2(qz), \quad \det g = 1, \quad R[g] = 0. \tag{49}$$

Therefore $W_\Lambda[g]$ is identical to its flat value under this diffeomorphism when all source data and boundary conditions are transformed. For a local $C_0\int\sqrt{g}$ term, the order-ϵ^2 quadratic Hessian from the first-order shear is $-C_0\epsilon^2/4$ per unit coordinate volume, while the second-order metric displacement times the one-point tadpole is $+C_0\epsilon^2/4$. The sum vanishes exactly. The seagull/tadpole terms cannot be dropped or replaced by the transverse kinetic contact of Section 10.

Independent numerical sampling gives a maximum determinant-volume deviation of 1.2×10^{-16} and zero cancellation residual at the displayed precision. This is an explicit nonlinear coordinate Ward check of the local covariant functional, *not* a numerical evaluation of every nonlocal stress-bubble component.

11.3 Vacuum Dirac determinant: calculated local terms and tensor power counting

For $q^2 \ll M^2$ and a constant mass, the heat-kernel coefficients of the operator in Eq. (46) are $\operatorname{tr} a_0 = 4$ and $\operatorname{tr} a_1 = 4(R/6 - R/4) = -R/3$. Thus

$$\boxed{W_\Lambda[g;M] = \int \mathrm{d}^4x\sqrt{g}[C_0 + C_R R + O(R^2, \nabla^2 R)], \quad C_0 = \frac{I_0}{8\pi^2}, \quad C_R = -\frac{I_1}{96\pi^2},} \tag{50}$$

where $I_0 = \int_{\Lambda^{-2}}^\infty \mathrm{d}s\, s^{-3}e^{-sM^2}$ and $I_1 = \int_{\Lambda^{-2}}^\infty \mathrm{d}s\, s^{-2}e^{-sM^2}$. For $M = 0.693435535520$, $\Lambda = 4$ in this *proper-time* scheme,

$$\boxed{I_0 = 120.81937817, \quad I_1 = 14.10422862, \quad C_0 = 1.53019530, \quad C_R = -0.01488601186.} \tag{51}$$

All four coefficients are scheme-dependent; an Einstein–Hilbert term with a fixed measured Newton constant has *not* been microscopically predicted. In particular, background metric dependence does not turn the metric into an integrated physical quantum degree of freedom.

For a transverse-traceless Euclidean shear $h_{xy} = \epsilon\cos(qz)$, its quadratic two-derivative action after cancelling the flat vacuum tadpole is

$$\boxed{\frac{\Delta W^{(2)}_{\mathrm{vac},TT}}{V} = -\frac{C_R}{4}\epsilon^2 q^2 + O(\epsilon^2 q^4, \epsilon^3) = 0.003721502965\,\epsilon^2 q^2 + \cdots .} \tag{52}$$

This follows from the standard quadratic expansion $\int\sqrt{g}R = -\frac{1}{4}\int(\partial h^{\mathrm{TT}}_{ij})^2$ in the stated Euclidean convention. Its q^2 dependence is the induced *inverse-kernel/metric-source action*, not an already observed $1/q^2$ stress susceptibility. Turning it into a massless propagator would require a physically justified, integrated metric or collective field and its normalized kinetic term. A source-only Dirac determinant does *not* provide this by itself.

11.4 Regulator matching: the same physical Hartree mass, different coupling

The earlier sharp spatial regulator $\Lambda = 4$ had $G_4^{(3D)} = 0.5654255716$. In the covariant proper-time vacuum prescription and finite occupied-medium contribution, the homogeneous saddle equation is

$$0 = \frac{M - m_0}{2G_4^{(PT)}} - \frac{MI_1(M)}{4\pi^2} + \frac{M}{\pi^2}\int_0^{p_F}\frac{p^2\mathrm{d}p}{\sqrt{p^2 + M^2}}, \qquad p_F = \sqrt{\mu^2 - M^2}. \tag{53}$$

Matching $M = 0.693435535520$ at $m_0 = 0.12$, $\mu = 1.2$ now requires

$$\boxed{G_4^{(PT)} = 1.2697985402 \neq G_4^{(3D)}.} \tag{54}$$

The scalar Hessian and source vertex of this *separately matched* proper-time scheme evaluate to

$$\boxed{K_0^{(PT)} = 0.08439298667 > 0, \quad V_0 = 0.06880959004, \quad \chi_{nn}^{(PT)}(0,0) = 0.17517967895.} \tag{55}$$

Two independent finite differences of the gap equation reproduce $K_0^{(PT)}$ and V_0 within 5×10^{-10}. Neither scheme has a scalar static $1/q^2$ pole on this stable regular branch. The different numerical susceptibility illustrates *regulator dependence*, not a physical change of gravitational strength.

11.5 Full ten-channel leading Fermi-surface stress response and scalar-resummed TT result

For spin degeneracy two at $T = 0$, write $p^\mu = (\mu, \boldsymbol{p})$ on the Fermi surface and define the ten kinetic source vertices $v^{\mu\nu}(\widehat{\boldsymbol{p}}) = p^\mu p^\nu/\mu$. The complete *leading intraband* symmetric response is the 10×10 matrix

$$\boxed{\chi^{\rm intra}_{AB}(\omega, q) = \frac{\mu p_F}{\pi^2} \int \frac{\mathrm{d}\Omega_{\widehat{p}}}{4\pi} \frac{v_F q \widehat{p}_z}{\omega - v_F q \widehat{p}_z + i0^+} v_A(\widehat{p}) v_B(\widehat{p}), \quad A, B = (\mu\nu).} \tag{56}$$

At strictly static $\omega = 0$ this is $\chi^{\rm intra}(0, q \to 0) = -\mathsf{A}$, where A is a positive-semidefinite angular Gram matrix. Direct independent Gaussian angular quadrature gives eigenvalues (multiplicity indicated)

$$\boxed{0\ (1),\quad 0.005071551738\ (3),\quad 0.01014310348\ (2),\quad 0.03807045895\ (3),\quad 0.19682715345\ (1).} \tag{57}$$

The rank is nine. Its null direction is the exact on-shell vertex identity $\sum_i v^{ii} = (p_F^2/\mu^2) v^{00}$; it is not a propagating tensor gauge degree of freedom. The numerical null residual divided by the matrix norm is below 10^{-14}. In addition, all 40 leading collisionless longitudinal kinetic identities (four momentum-conservation components tested against ten symmetric-stress sources) have a maximum numerical residual 5.76×10^{-16} at $z = 1.5$. These are distinct from the genuine metric-source seagull Ward check above. The two independent helicity-two shear combinations remain degenerate, and the static correlator has $\chi_{xy,xy} = -0.005071551738$ in this convention. At $z = \omega/(v_F q) = 0.5$ the analytic tensor cut has $\mathrm{Im}\,\chi_{xy,xy} = -0.004201017969$. The response is a continuum with a finite static limit, not a tensor $1/q^2$ exchange pole.

For the *same isotropic scalar-contact Hartree/HS Gaussian approximation*, $S = \bar\psi\psi$ is helicity zero, whereas T_{xy} for $\boldsymbol{q} \parallel \widehat{z}$ is helicity two. Therefore $\chi^0_{T_{xy},S} = 0$ by rotational symmetry. The exact Gaussian Schur complement consequently gives

$$\boxed{\Gamma^{\rm eff}_{hh,TT} = \Gamma_{hh,TT} - \Gamma_{hM,TT}\, K_s^{-1} \Gamma_{Mh,TT} = \Gamma_{hh,TT}, \qquad \Gamma_{hM,TT} = 0.} \tag{58}$$

A vanishing scalar–TT vertex cannot generate the missing tensor collective mode by resumming only the existing scalar auxiliary field. Combined with a finite scalar K_0 and the massive vacuum threshold, this establishes a *phase-specific negative result for the tested scalar-Hartree Gaussian tensor channel*, not a no-go theorem for all interactions, non-Gaussian exchange, critical phases, gauge sectors or other quantum matter. This is a more direct constraint on QGD's Chapter 5 premise MW2 than quasiparticle lifetime estimates alone.

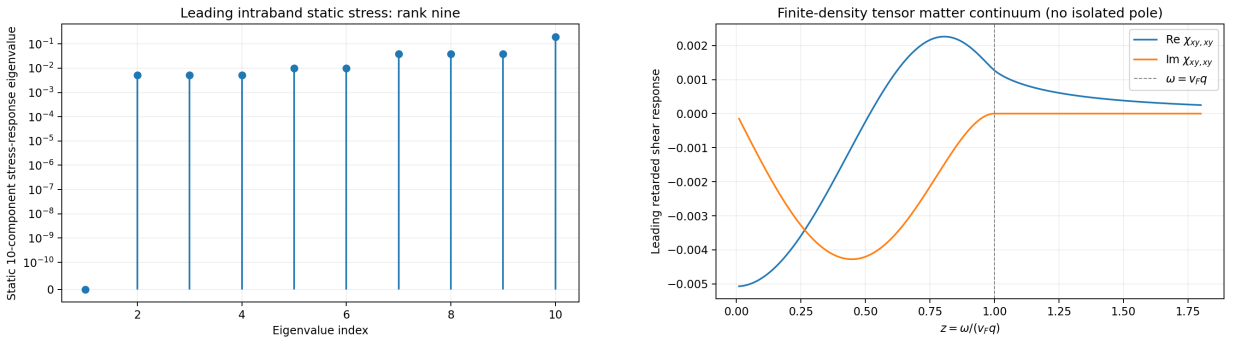


Figure 2: Independent numerical evidence in the leading finite-density collisionless sector: the symmetric-stress matrix has one on-shell null relation and nine finite eigenvalues (left), while the helicity-two retarded shear function has a particle–hole cut rather than an isolated pole (right). These are matter responses, not derived gravitons.

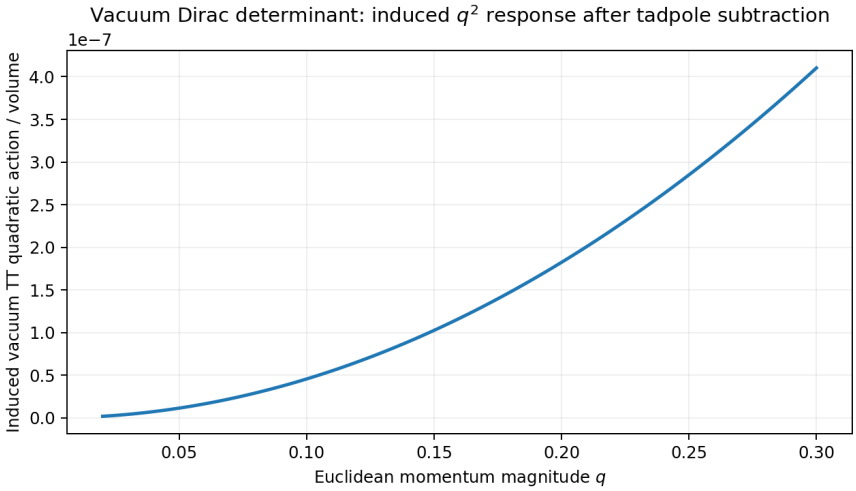


Figure 3: The local covariant vacuum Dirac determinant after flat-tadpole subtraction produces a regular induced transverse-traceless q^2 quadratic action (fixed small test amplitude), not a demonstrated physical $1/q^2$ pole. The displayed range is within $q < M$; higher derivative corrections are not included.

11.6 Why the bridge still matters, and the actual QGD stopping point

The microscopic branch may still satisfy the selected Dirac ratio $u_B = \mathsf{R}u_A$ and a controlled WKB transport equation, with $r^2 j^r = C_J$ for the stationary total current. Those are already validated *matter-side* statements. The complete founding metric $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}]$ remains an explicit *physical promotion and metric postulate*, not a consequence of the scalar-pole test. A state-to-[Σ] map requires a stable source-visible collective sector, universality across different kinds of matter, and a causal common metric for probes. No such sector has been obtained by merely generating the covariant $C_R R$ term in an external-source determinant.

Next decisive scientific move [O]. Extend the same generating functional to the interaction-complete *retarded* stress matrix with proper seagulls and dynamical Fock/crossed vertices (or an explicitly specified conserving nonperturbative scheme), and test helicity-two poles independently of the scalar HS field. If the original scalar-contact phase continues to fail, investigate a physically motivated distinct microscopic phase or operator channel rather than inserting $Z(\nabla q)^2$ or Einstein's action as the desired answer. The existence and source coupling of q are the actual Chapter 5–6 missing links.

12 The required physical operator channels and their tests

A subsequent, more general source matrix should not be built from $\bar{\psi}\psi$ alone. In the Chapter 1 matter model, consider a renormalized, regulator-consistent operator family containing $T^{\mu\nu}_{\text{total}}$, the conserved vector current, the scalar auxiliary channel, and if justified independent tensor composites. The full kernel is a matrix, not an assumed Einstein propagator:

$$\mathcal{K}(k) = \begin{pmatrix} \mathcal{K}_{ss} & \mathcal{K}_{sT} & \cdots \\ \mathcal{K}_{Ts} & \mathcal{K}_{TT} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \qquad \chi_{AB}^{\text{phys}} = V_A \mathcal{K}^{-1} V_B^* + \chi_{AB}^{\text{contact}}. \tag{59}$$

One must calculate the *retarded* frequency and spatial-momentum dependence in a phase with explicitly stated occupation, vacuum subtraction and cutoff. The physical stress response has

multiple helicity channels; direct/seagull pieces are retained before projecting onto spin-zero and helicity-two structures. The prior QM001–QM005 single-fermion residues are helpful input to dressed fermion lines and vertices, but they do not by themselves give an induced collective massless pole.

R1. Microscopic closure. Specify the approximation (e.g., conserving large-N_f or a regulated ladder), exact state, counterterms, all source vertices and the physical quotient. Do not identify constituent count N, flavor count N_f, and signed metric-sector count.

R2. Source-visible infrared spectrum. Compute the full physical response and test $\mathcal{R}_{AB} \neq 0$ for energy and stress probes, including rest matter, kinetic pressure and radiation. Check whether any candidate pole is isolated, stable and not a gauge artifact or cancelled by contacts.

R3. Universality. Repeat with at least two genuinely different probes and verify whether one common response normalizes their couplings to full energy–momentum, not only particle density or charge.

R4. Operational geometry. Compute the highest-derivative probe propagation operators from the same state, and test for a shared Lorentzian principal symbol and compatible mass/redshift response. An acoustic metric of one medium is insufficient by itself.

R5. State-to-natural-invariant map. After a physical geometry/response is identified, construct a frame-covariant state-dependent class $[\Sigma]$ and prove invariance under microscopic spin-basis changes and any genuine signed-sector equivalence. No arbitrary Gram factorization may count as independent selection.

R6. Relativistic effective action. Only then test whether the induced action has an acceptable two-derivative Einsteinian sector, correct signs and physical radiation spectrum; the adopted GR action cannot be reused as a microscopic input to its own derivation.

The proposed route remains open to non-Dirac quantum matter. What counts is the existence of the correct physical collective response and its universal operational consequences, not insisting that the fundamental excitation must be a particular microscopic spinor or a postulated elementary graviton.

13 Essential theoretical constraints, not replacements for calculation

Soft massless spin-two scattering in a Lorentz-invariant relativistic theory imposes stringent universal-coupling consistency conditions; see Weinberg (1964), https://doi.org/10.1103/PhysRev.135.B1049. The Weinberg–Witten theorem has important assumptions about a Lorentz-covariant conserved stress-energy tensor and matrix elements for massless higher-spin asymptotic states; see https://doi.org/10.1016/0370-2693(80)90212-9. These constraints must be checked rather than declaring either that no composite mechanism can ever work or that an arbitrary scalar condensate evades them. Composite-gravity models with nonstandard microscopic constraint structures have been explicitly studied by Carone, Erlich and Vaman (2017), https://doi.org/10.1007/JHEP03(2017)134, and in a fermionic version https://arxiv.org/abs/1812.08201. These are comparison examples, not a derivation from QGD's scalar-contact Dirac model.

14 Full founding QGD metric and the explicit stopping point

The complete founding QGD master metric is retained without redefinition:

$$\boxed{g_{\mu\nu} = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} - \sum_{s=1}^{N_\sigma} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right], \quad s_s = -\varepsilon_s.} \tag{60}$$

Here N_σ labels signed metric sectors and does not count microscopic particles or fermion flavors. The independent operator meaning and off-shell variation of $\mathcal{Q}_{ab}$ remain open. The simplified signed-sector Gram map is only a declared $\lambda\mathcal{Q} = 0$ classical specialization, not the complete founding metric.

For a full metric-dependent effective action, the recognizable Option C equation remains

$$\boxed{2 s_s \Pi^{ab} \sigma_b^{(s)} = \lambda \big[(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi \big]^a.} \tag{61}$$

This exact formal sector Euler equation under the stated metric-map hypotheses is not a proof that the microscopic QFT generates the metric-only Einstein action. Chapter 2's independently varied spherical A, B fields dynamically select $A = B$ and $A^2 = B^2 = 2G_N M/(c^2 r)$ *within the adopted Einstein–Hilbert sector*; Chapter 1's WKB ratios have not by themselves selected this gravitational profile.

15 An inhomogeneous quantum-source test of the missing kinetic bridge (QM009)

Research question. Given the Chapter 1 local scalar-contact theory, can a spatially varying source state create the source-to-*kinetic* response required by the independent-microscopic route of Section 2? This is a stronger requirement than changing a quasiparticle's Dirac mass or lower/upper spinor ratio. The test below never inserts a precomputed Schwarzschild field into the microscopic Hamiltonian.

15.1 Source, observable and the exact Hartree–Fock principal-symbol statement

Work in the flat reference frame, with $c = \hbar = 1$, and let a static number-source $\delta\mu(\boldsymbol{x})$ couple as $-\delta\mu\, \psi^\dagger\psi$ in the single-particle Hamiltonian. This is a response probe, *not* a gravitational field. The microscopic matter interaction is $G_4(\bar\psi\psi)^2$, and the local Hartree inverse propagator in Wigner variables is

$$\mathcal{D}_H(x,p) = \gamma^\mu p_\mu - M(x), \qquad \frac{\partial \mathcal{D}_H}{\partial p_\mu} = \gamma^\mu. \tag{62}$$

The local four-fermion Hartree and Fock self-energies depend on the coincident state density matrix $S(x,x)$; at first interaction order their coordinate-space kernels are proportional to $\delta^{(4)}(x-y)$ without derivatives. This remains true when $S(x,x)$ varies, and when the source has a nonzero spin polarization or current. Consequently

$$\boxed{\frac{\delta}{\delta\mu(y)} \left[\frac{\partial \mathcal{D}_{\mathrm{HF}}(x,p)}{\partial p_\mu} \right] \Bigg|_{\text{local HF}} = 0.} \tag{63}$$

This is an exact statement *within the specified local Hartree–Fock truncation*; it is not a theorem for higher-loop, nonlocal self-energies, alternative collective phases or the full QGD master metric. In particular a source-induced scalar mass can bend massive test-particle trajectories without changing the characteristic massless cone of the principal Dirac symbol. Its vanishing does not imply the spinor ratio is unchanged.

15.2 Finite-wave-number scalar response with two independently matched regulators

Define the static quadratic grand-potential Hessian at a homogeneous self-consistent state by

$$\delta^2\Omega = \frac{1}{2}\int_{\boldsymbol{q}} \left[K_s(q)|\delta M_q|^2 + 2V_s(q)\delta M_{-q}\delta\mu_q - D_{nn}(q)|\delta\mu_q|^2\right], \quad \int_{\boldsymbol{q}} = \int \frac{\mathrm{d}^3q}{(2\pi)^3}. \tag{64}$$

With $H_M(\boldsymbol{p}) = \boldsymbol{\alpha}\cdot\boldsymbol{p} + \beta M$, $E_p = \sqrt{p^2+M^2}$, $P_s(\boldsymbol{p}) = (I + sH_M/E_p)/2$, and $s = \pm 1$, the regulated static loop is evaluated from the particle/antiparticle spectral representation

$$\Pi_{AB}(0,\boldsymbol{q}) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \sum_{s,t=\pm1} \frac{f_s(\boldsymbol{p}) - f_t(\boldsymbol{p}+\boldsymbol{q})}{sE_p - tE_{p+q}} \operatorname{Tr}\left[P_s(\boldsymbol{p})\mathsf{V}_A P_t(\boldsymbol{p}+\boldsymbol{q})\mathsf{V}_B\right], \tag{65}$$

where $\mathsf{V}_M = \beta$, $\mathsf{V}_\mu = -I$, $f_-(\boldsymbol{p}) = 1$ and $f_+(\boldsymbol{p}) = n_F(E_p - \mu)$. The vacuum interband piece has the explicitly stated ultraviolet regulator; the finite-density occupation correction is kept uncut. For the scalar vertex the Dirac trace has the closed form

$$\operatorname{Tr}[P_s(\boldsymbol{p})\beta P_t(\boldsymbol{p}')\beta] = 1 + st\frac{M^2 - \boldsymbol{p}\cdot\boldsymbol{p}'}{E_pE_{p'}}, \quad \operatorname{Tr}[P_s(\boldsymbol{p})\beta P_t(\boldsymbol{p}')] = M\left(\frac{s}{E_p} + \frac{t}{E_{p'}}\right). \tag{66}$$

The corresponding density–density trace is $1 + st(M^2 + \boldsymbol{p}\cdot\boldsymbol{p}')/(E_pE_{p'})$. All four energy-sign channels are retained, including the vacuum interband channel. With signs fixed by $\mathsf{V}_\mu = -I$, the static Hessians are $K_s = 1/(2G_4) + \Pi_{MM}$, $V_s = \Pi_{M\mu}$, and $D_{nn} = -\Pi_{\mu\mu}$.

Minimizing over the scalar fluctuation gives the *actual inhomogeneous source response* in this approximation:

$$\boxed{\delta M(\boldsymbol{q}) = -\frac{V_s(q)}{K_s(q)}\delta\mu(\boldsymbol{q}), \quad \chi_{nn}(q) = D_{nn}(q) + \frac{V_s(q)^2}{K_s(q)}, \quad \frac{\delta M(\boldsymbol{q})}{\delta n(\boldsymbol{q})} = -\frac{V_s(q)}{K_s(q)\chi_{nn}(q)}.} \tag{67}$$

The last formula describes linear response about equilibrium, not an arbitrary constitutive identity for local density outside a source.

At zero wave number the original sharp-cutoff benchmark is independently reproduced: $M = 0.693435535520$, $p_F = 0.979360586341$, $K_0 = 0.168741734136$, $V_0 = 0.068809590036$, $\chi_{nn}(0) = 0.147135178153$, and

$$\boxed{\left.\frac{\mathrm{d}M}{\mathrm{d}n}\right|_{\text{equilibrium, sharp}} = -2.77146847504.} \tag{68}$$

For the same homogeneous source variation, the calculated scalar *operator* vertex is nonzero even though its kinetic derivative is zero:

$$\boxed{\left.\frac{\delta\mathcal{D}_H}{\delta n}\right|_{q=0,\text{equil.}} = +2.77146847504\, I_4, \qquad \frac{\delta}{\delta n}\left(\partial_{p_\mu}\mathcal{D}_H\right) = 0.} \tag{69}$$

For a spinor with *fixed canonical momentum* $p = p_F$ of the reference state, $\zeta = p/(E_p + M)$ satisfies $\partial_M\zeta|_p = -\zeta/E_p$. Therefore

$$\boxed{\zeta_0 = 0.517239994692, \qquad \left.\frac{\partial\zeta}{\partial n}\right|_{p\,\text{fixed}} = 1.19459528276.} \tag{70}$$

This is a *genuine source-state-to-spinor response*. It is not a derived map from the spinor ratio to the QGD gravitational natural invariant. The fixed-p derivative in Eq. (70) must not be confused with the derivative tracking the moving Fermi surface $p_F(n)$.

For a finite-q inhomogeneous calculation, a sharp cutoff independently imposed on both $|\boldsymbol{p}|$ and $|\boldsymbol{p}+\boldsymbol{q}|$ produces a nonanalytic $|q|$ edge artifact. We checked this numerically and do *not* interpret it as collective dynamics. Instead, match the same M, μ, and m_0 with smooth vacuum form factors

$$R_n(p) = \exp[-(p/\Lambda)^n], \qquad n = 8, 16, \tag{71}$$

implemented as $\sqrt{R_n(p)R_n(p')}$ across the vacuum loop, and recalibrate G_4 by the respective gap equation. The resulting $T \to 0$ zero-wave-number comparisons are

Vacuum scheme	Matched G_4	$K_s(0)$	$\chi_{nn}(0)$	$K_s(q=0.2)$ at $T=0.005$
Original sharp, $q=0$ only	0.56542557	0.16874173	0.14713518	—
Smooth $n=8$	0.62924767	0.14993818	0.15065405	0.15188956
Smooth $n=16$	0.60370423	0.15737518	0.14916178	0.16062232

Here the nonzero-q values use small thermal smoothing $T = 0.005$ solely to control the Fermi-surface quadrature. Repeating with $T = 0.010$ and increased quadrature resolution gives the expected stable infrared trend; the local model is visibly regulator dependent. The computed small-q expansions are $K_s(q) \simeq K_s(0) + \kappa_s q^2$ with $\kappa_s \simeq 0.0486$ (smooth $n=8$) and 0.0816 (smooth $n=16$) in these model units. These are gradient coefficients within the specified schemes, *not* an independently computed long-distance correlation length or an induced gravitational field.

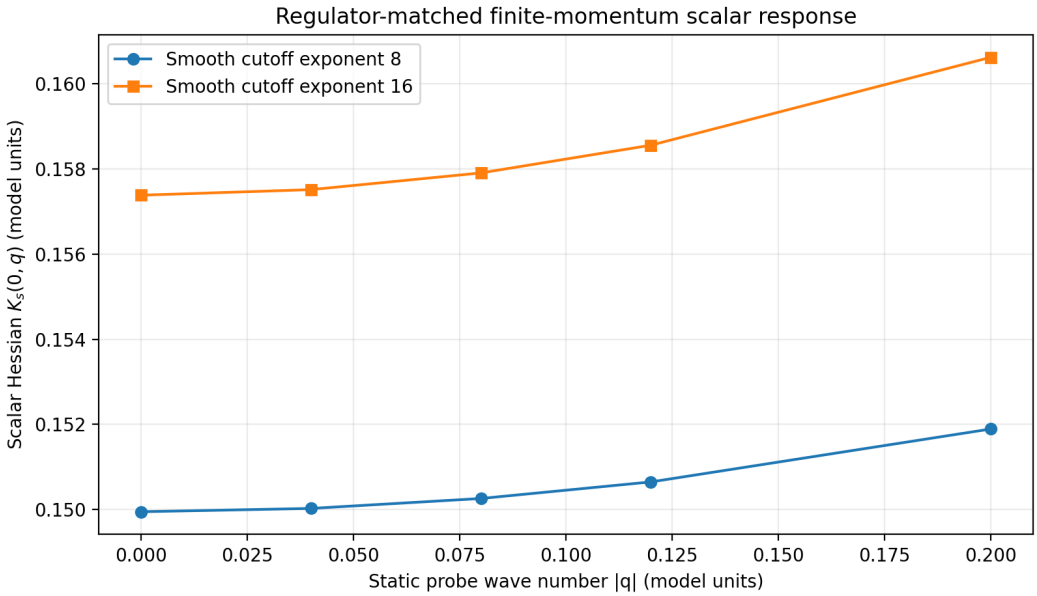


Figure 4: Independently calibrated smooth-vacuum-regulator calculations of the finite-wave-number scalar response. Both have a nonzero stable $K_s(0)$; neither has an infrared static massless scalar pole in the tested phase.

15.3 Resulting source-to-kinetic verdict and the exact next physical requirement

For both matched smooth regulators the static scalar Hessian remains positive at sufficiently small wave number and the source vertex is finite. Hence

$$\boxed{\lim_{q\to 0} q^2 \frac{V_s(q)^2}{K_s(q)} = 0.} \tag{72}$$

The position-dependent scalar source changes the local Dirac spinor ratio, as Eq. (70) demonstrates, but the microscopic *principal-symbol source-to-kinetic derivative* remains Eq. (63). Thus this phase

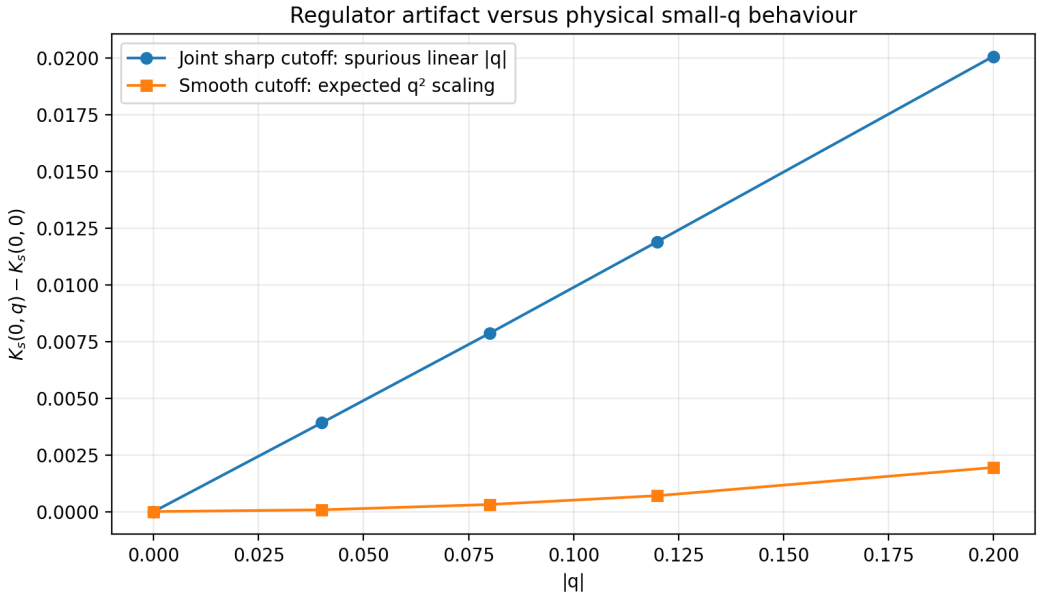


Figure 5: The spurious $|q|$ term produced by a naive joint sharp cutoff on two shifted loop momenta, compared with smooth cutoffs. This regulator boundary effect is not a newly derived collective mode.

does not independently produce the specific null-sector target

$$\frac{\delta\Gamma^{\mu}(x)}{\delta\rho_m(y)} \overset{\text{QGD target}}{=} -\frac{G_N}{c^2}\frac{\ell^{\mu}\ \not{\ell}}{|\boldsymbol{x}-\boldsymbol{y}|}, \tag{73}$$

which, in the Schwarzschild-specialized matching calculation, follows from the *adopted* gravitational source law, not from this contact QFT. The mass density ρ_m and the number-source μ are distinct perturbations; one must not equate their response kernels without a source/operator conversion and the correct units. Equation (73) remains an **[O]** microscopic target and is not a universal formula for every sector of the full QGD metric.

Next independent test. At the earliest momentum-dependent self-energy order, compute $\delta[\partial_{p_\mu}\Sigma_R(p;x)]/\delta n(y)$ with the matching two-particle irreducible source vertex, isolate any nonanalytic static $q \to 0$ behaviour, then test the *characteristic cone* against spinor-basis changes, the full source stress tensor, light propagation, and the complete coframe/signed-sector/$\mathcal{Q}$ master metric. Higher-order nonlocality need not be zero; this chapter has not calculated its full finite-q kernel. The effective QGD action route remains valid *conditionally* without that microscopic derivation. The historical $u_B = \mathsf{R}u_A$, leading WKB Hamilton–Jacobi relation and $r^2 j^r = C_J$ are retained throughout.

Literature and regulator caution. Nonrenormalizable NJL-type contact theories have strongly regulator-dependent inhomogeneous channels. For example Pannullo, Wagner and Winstel, *Phys. Rev. D* **110**, 076006 (2024), explicitly compare regulator-dependent nonuniform phases (DOI:10.1103/PhysRevD.110.076006). These are not imported QGD predictions, but justify why neither a sharp cutoff cusp nor a regulator-specific gradient coefficient should be presented as fundamental gravity.

16 Microscopic stationary selection: direct scalar-contact test (QM011)

Research question. Does the already specified microscopic quantum theory produce a state-dependent stationary functional, and does its physically admissible stationary point select the QGD gravitational natural invariant independently of the adopted gravitational action? We test these questions separately. The scalar model's existence of a stationary collective field is a positive result; identification of that field with QGD's $\sigma_\mu^{(s)}$ is a further physical claim, not a definition.

16.1 A microscopic Hubbard–Stratonovich functional, not a postulated gravitational energy

In the zero-electromagnetic-background scalar-contact model of Chapter 1, and $\hbar = c = 1$, an algebraic auxiliary scalar ϕ can be introduced:

$$\mathcal{L}_{\rm HS} = \bar\psi(i\gamma^\mu\partial_\mu - m_0 + \phi)\psi - \frac{\phi^2}{4G_4}, \qquad M = m_0 - \phi. \tag{74}$$

Completing the square in ϕ recovers the original $+G_4(\bar\psi\psi)^2$ interaction. Integrating the regulated fermions formally gives a state-dependent collective effective action $\Gamma_{\rm micro}[\phi;\mu]$, with a Dirac determinant and a local quadratic auxiliary-field term. Its homogeneous static saddle is precisely the Chapter 1/5 Hartree equation

$$0 = \Omega_M = \frac{M - m_0}{2G_4} - \frac{M}{\pi^2}\int_{p_F}^{\Lambda}\frac{p^2\mathrm{d}p}{\sqrt{p^2+M^2}}, \quad p_F = \sqrt{\mu^2 - M^2}. \tag{75}$$

This is an **[D/C]** microscopic variational-selection functional within the declared regulator/mean-field approximation, with a physically specified collective variable M. It has not introduced the QGD metric, $H_{\rm KS}$, σ, or the Einstein action.

16.2 The scalar stationary response is calculable and unique in the positive quadratic regime

Write $q(\boldsymbol{x}) = M(\boldsymbol{x}) - M_\star$ around the stable uniform saddle and apply a small external chemical-potential source $\delta\mu(\boldsymbol{x})$. The quadratic functional derived by expanding the same fermion determinant is

$$\Gamma^{(2)}_{\rm micro}[q;\delta\mu] = \frac{1}{2}\int\frac{\mathrm{d}^3k}{(2\pi)^3}\,K_s(k)q(-k)q(k) + \int\frac{\mathrm{d}^3k}{(2\pi)^3}\,V_s(k)q(-k)\delta\mu(k). \tag{76}$$

Here $K_s = \Omega_{MM} + \Pi_{MM}(k) - \Pi_{MM}(0)$ and V_s is the consistently computed scalar–density source vertex. Stationarity gives the exact Gaussian relation

$$\boxed{\frac{\delta\Gamma^{(2)}_{\rm micro}}{\delta q(-k)} = 0 \quad\Longrightarrow\quad q(k) = -\frac{V_s(k)}{K_s(k)}\delta\mu(k).} \tag{77}$$

QM009 determined the small-k static expansion $K_s(k) = K_0 + Z_s k^2 + O(k^4)$ with two *independently matched smooth vacuum regulators*, keeping the same physical homogeneous mass. Reusing those existing, reproducible inputs yields

Scheme	K_0	Z_s	V_0	$\xi_s := \sqrt{Z_s/K_0}$
Smooth $n = 8$	0.14993818	0.04864381	0.06880959	0.56958383
Smooth $n = 16$	0.15737518	0.08155175	0.06880959	0.71986081

The Z_s estimates are infrared fits to finite-k loop data with $T = 0.005$ quadrature smoothing; they are not exact $T = 0$ analytic derivatives. At the displayed truncation order, the stationary equation is

$$\boxed{(-Z_s\nabla^2 + K_0)q(\boldsymbol{x}) = -V_0\delta\mu(\boldsymbol{x}).} \tag{78}$$

For $K_0, Z_s > 0$, the quadratic form is strictly positive in its stated smooth finite-energy field space (with boundary conditions), and the stationary scalar solution is unique there. This does not prove global uniqueness of the full non-Gaussian fermionic effective action.

16.3 A compact spherical source: independent Green-function and field-equation tests

Choose the transparent benchmark $\delta\mu(\boldsymbol{x}) = \delta\mu_0\Theta(R - r)$ as a localized *number-source probe in the otherwise uniform finite-density medium*, not as a microscopic model of an isolated star in empty space. Let $m_s^2 = K_0/Z_s = \xi_s^{-2}$. For $r > R$, the stationary gradient-truncated solution is

$$\boxed{q_{\rm out}(r) = -\frac{V_0\delta\mu_0}{Z_s}\,\frac{m_sR\cosh(m_sR) - \sinh(m_sR)}{m_s^3}\,\frac{e^{-m_sr}}{r}.} \tag{79}$$

We derived Eq. (79) from the Green function $e^{-m_s|\boldsymbol{x}-\boldsymbol{y}|}/(4\pi Z_s|\boldsymbol{x}-\boldsymbol{y}|)$ and independently evaluated the radial three-dimensional Green-function integral. The two agree at 12 tested radii/regulator combinations to relative tolerance 10^{-10} or the floating-point absolute floor. Within that illustrative local-gradient model, after normalization to the same field at $r = R$, the ratio of microscopic scalar to a $1/r$ comparator at $r = R + d$ is precisely e^{-d/ξ_s}. For $d = 1$ it equals 0.17279 ($n = 8$) and 0.24929 ($n = 16$); for $d = 2$ it equals 0.02986 and 0.06214. The Yukawa profile is a *mathematical diagnostic of the truncated infrared functional*, not a quantitatively controlled far-field prediction: $m_s \simeq 1.76$ or 1.39 is larger than the $q \lesssim 0.2$ wave-number window used to fit Z_s, and is comparable to or greater than the Fermi momentum $p_F \simeq 0.98$. The inferred imaginary Yukawa pole thus lies beyond the verified small-q expansion. The *full* static $T = 0$ Fermi-surface response can have additional nonanalyticities at finite momentum (including Friedel-type tails). The reliable infrared conclusion is instead the absence of a $1/q^2$ residue, since $K_s(0) > 0$ and $V_s(0)$ is finite.

16.4 What would have been sufficient, and why it is not yet derived

For the local-gradient effective action, a static $1/r$ Green-function response of this particular scalar channel would require $K_0 = 0$ with $Z_s > 0$ and a nonvanishing $k \to 0$ source vertex, yielding $-Z_s\nabla^2 q = -V_0\delta\mu$ and an exterior $q \propto 1/r$. Merely *setting* $K_0 = 0$ is a hypothetical critical tuning, not a result at the tested Hartree saddle. Even if such a q were selected, one would still have to prove the source conversion from $\delta\mu$ to gravitational mass/stress, establish universal probe coupling, identify the physically admissible null congruence ℓ_μ, and derive a legitimate relation $H_{\rm KS} = h(q)$ with $\sigma_\mu = \sqrt{H_{\rm KS}}\,\ell_\mu$ in the declared classical null-sector specialization. The full founding coframe, signed sectors and $\lambda\mathcal{Q}_{ab}$ remain distinct.

A candidate massless collective field is *not automatically sourced*. For illustration, a shift-symmetric phase mode θ can couple derivatively as $g\partial_\mu\theta\, j^\mu$. Integrating by parts makes that interaction proportional to $-g\theta\,\partial_\mu j^\mu$ plus a boundary term. For an exactly conserved interior current and trivial boundary contribution it cannot source the static phase field. This is a diagnostic of source coupling, not a new Goldstone mode derived from Chapter 1.

Proposition 16.1 (What the microscopic test actually establishes)**.** *The specified scalar-contact quantum field theory has a regulated Hubbard–Stratonovich stationary functional. In its matched stable Hartree–Gaussian scalar phase, the independently calculated static Hessian satisfies $K_s(0) > 0$, and its scalar-to-number-source vertex is finite. Thus the stationary scalar response is regular at*

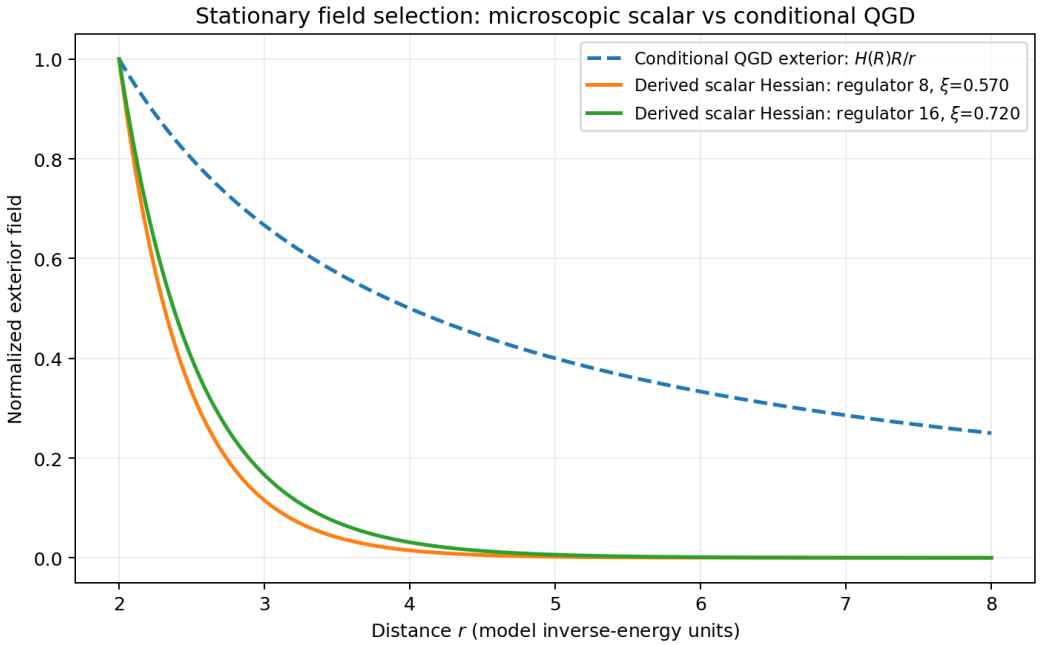


Figure 6: Independent variational selection test: the QGD conditional Schwarzschild amplitude $H \propto 1/r$ (normalized at $R = 2$) versus stationary scalar fields using the previously calculated and refitted microscopic Hessians. The scalar fits have positive K_0 and a finite correlation length. These profiles are a comparison, not an identification of the scalar q with H_{KS}.

$k = 0$ *and does* not *select the* $1/r$ *QGD gravitational amplitude from a localized number-source perturbation by the required infrared mechanism. It does select a physical* scalar mass response *and corresponding Dirac spinor changes. This is a phase/truncation-specific failure of the proposed scalar promotion, not a no-go theorem for a more general physical map* $\mathcal{F}$ *or a contradiction of the already closed* conditional *QGD Schwarzschild bridge.*

Decision for the next microscopic hypothesis. Keep $\mathcal{F}_{\mathrm{conditional}}$ and Chapter 2's variational Schwarzschild selection closed. For $\mathcal{F}_{\mathrm{micro}}$, the first unresolved property is *the existence of a physically defined state-to-natural-invariant rule* with source and exterior boundary data, not the existence of a standalone graviton. An alternative quantum collective variable must exhibit (i) an independently derived stationary functional, (ii) source/exterior transfer of the required form with justified physical boundary conditions, and (iii) a frame- and probe-independent lift into the full QGD sectors. The original Dirac ratio, WKB principal-symbol equation and $r^2 j^r = C_J$ remain independent and preserved; none is silently substituted for the new selection equation.

External comparison. Standard fermionic Hubbard–Stratonovich/mesonic effective-action calculations likewise identify the inverse collective propagator as $1/(2G) + \Pi(k)$ and distinguish scalar and (in enlarged chiral models) symmetry-related Goldstone channels; see `https://doi.org/10.1088/1674-1137/43/9/094103`. That literature supports the method of this test, not the physical identification with QGD's σ.

17 Research consolidation: what the programme has actually achieved (QM012–QM014)

Editorial status (2 November 2026). This section records independent research notes QM012, QM013 and QM014 after the v0.7/QM011 edition. It preserves all earlier calculations and equation labels. Its purpose is to identify small but decisive advances and to prevent an externally proposed induced-action construction from being mistaken for the original microscopic QGD closure. The additional calculations are not silently promoted into the founding axioms. The status convention [D/C/A/O] is that of the chapter's opening contract.

17.1 Two arrows, two logically different scientific questions

The founding objective is a source-state and macroscopic-Dirac-to-natural-invariant selection law, schematically

$$\boxed{\mathcal{D}_{\rm eff}\Psi = 0 \;\longrightarrow\; (\Psi, S, \mathsf{R}, j^\mu; \hat{\varrho}_{\rm src}) \xrightarrow{\mathcal{F}_{\rm micro}} [\Sigma]_{\rm physical}.} \tag{80}$$

This notation is a *research target*, not an equation derived by applying the Dirac operator: a probe spinor does not contain the full source-state information in general. The required map must use independently characterized source-state observables and must not depend on selecting an arbitrary probe velocity, spin polarization, spinor basis or already-known gravitational field. Chapter 1 and Eqs. (2)–(21) state the controlled ingredients and the remaining gap.

The alternative suggested in the external Claude compendium begins with a *new premise (Q)*: independent admissible QGD sector variables Σ, a specified functional measure/saddle prescription, and matter universally coupled to the composite metric $g(\Sigma)$. Its logical direction is

$$\boxed{S_{\rm micro}[\psi; g(\Sigma)] \xrightarrow{\int D\psi D\bar{\psi}} \Gamma_{\rm ind}[\Sigma] \xrightarrow{\delta\Gamma_{\rm ind}=0} [\Sigma]_{\rm stationary} \longrightarrow (\Psi, S, \mathsf{R}, j^\mu).} \tag{81}$$

The induced action and its solutions can be meaningful **[C]** calculations. They do not invert Eq. (80): the metric and its independent sector variables were supplied at the beginning. On a regular spanning branch $\delta_\Sigma\Gamma_{\rm ind} = (D_\Sigma g)^* \delta_g \Gamma_{\rm ind}$; at a rank-deficient branch its vanishing need not imply $\delta_g\Gamma_{\rm ind} = 0$. The proposed premise (Q) was **not adopted** as a proven rewriting of the flat-reference original microscopic F1 theory.

17.2 Small-win ledger: positive results, negative results, and their scope

The programme contains more than the still-open final arrow. The following claims deliberately distinguish a result established under a defined mathematical/model framework from a claim of fundamental quantum-gravitational emergence.

No.	Verified or restricted advance	Status	What it actually buys QGD
01	Exact positive-energy Dirac upper/lower block relation $u_B = \mathsf{R}u_A$.	D/C	Controlled microscopic spinor kinematics; not yet a gravitational one-form.
02	WKB Hamilton–Jacobi/transport and the stationary spherical $r^2 j^r = C_J$ law.	D/C	Preserves the historic wavefunction–momentum/current bridge without inferring a force from flux.
03	Chapter 1's separated large-N centre-of-mass $\hbar/N$ limit and correlation counterexample.	D/C	Demonstrates a precise possible classical collective limit; rejects the claim that large N alone yields a unique Dirac collective spinor or gravity.

No.	Verified or restricted advance	Status	What it actually buys QGD
04	Complete coframe/signed-sector/$\mathcal{Q}_{ab}$ metric and full Euler pullback.	A/D	Provides the exact representation and variational language, without deriving its microscopic ontology.
05	Independent A, B Einstein–Hilbert variations select $A^2 = B^2 = 2G_N M/(c^2 r)$ on the specified nonzero exterior branch.	C	Schwarzschild natural invariant is *dynamically* selected within adopted dynamics; not algebraically guessed.
06	One Newtonian mass calibration, exact static energy/potential consistency and distinct wavefunction vs geometric normalizations.	D/C	Eliminates spurious gravitational constants and unsupported universal $\hbar$-force claims.
07	Chapter 5 substitution $H = 2g_c q/c^2$ makes the assumed collective and Newtonian H energies exactly equal, with $G_{\text{eff}} = g_c^2/(4\pi Z)$.	C	Closes the specified static axiomatic correspondence while leaving premise MW2 physical.
08	Source-conditioned stationary H functional has a unique regular decaying solution for a compact source.	C	Fixes $H \sim 1/r$, its mass coefficient and $\sigma_{\text{amp}} \sim r^{-1/2}$ on the designated spherical null branch.
09	Exact null-sector Clifford/principal-symbol matching and radial Hamilton–Jacobi roots for *given* H.	D/C	Checks the geometry-to-Dirac return map; does not reverse it to derive H.
10	Explicit source-to-H and source-to-kinetic $1/\lvert\boldsymbol{k}\rvert^2$ *matching targets.*	D/C	Converts the missing origin into a falsifiable source-response test, not a new microscopic calculation.
11	Regulated Hartree stationary scalar and Gaussian Hessian/source vertex constructed from actual F1.	D/C	Proves microscopic collective stationarity is calculable in one specified quantum phase.
12	Two matched smooth regulators find $K_s(0) > 0$ and finite $V_s(0)$; QM011 Green-function check passes.	D/C	Rules out the desired static massless scalar residue in this tested regime; does not rule out all possible phases.
13	Leading Hartree/first local Fock principal-symbol source derivative is zero.	D/C	Exposes a concrete mismatch to the required universal source-to-kinetic response in the tested phase.
14	Isotropic scalar–TT mixing selection, collisionless TT continuum, full leading intraband stress matrix and Ward-contact checks.	D/C	Distinguishes real stress dynamics from a relabelled scalar; no autonomous graviton-like pole derived.
15	QM012 shows naive Laplace stiffness for σ_{amp} has wrong radial exponent; the $H = \sigma_{\text{amp}}^2$ stiffness succeeds conditionally.	D/C	Identifies the correct *effective field coordinate* and its nonlinear σ stiffness.
16	QM012 WKB density fourth-root exponent, spinor-bilinear null construction and statistical selection controls.	D/C	Provides candidates but also explicit failures of source-mass normalization, spin/probe independence or exterior selection.
17	Regulated Dirac determinant induces a local curvature coefficient when matter is coupled to an *already defined* metric.	D/C	Demonstrates induced-action consistency, not the original matter-to-Σ construction.

No.	Verified or restricted advance	Status	What it actually buys QGD
18	Regular eight-sector Jacobian rank 10, nullity 22 and $O(4,4)$ orbit equality; singular-rank counterexamples.	D	Identifies physical-representation redundancy only on the stated metric-only spanning branch.
19	Conditional induced Einsteinian operator has two on-shell TT polarizations and a gauge-dependent local canonical-energy example.	C/D	Checks the adopted low-energy spectrum/gauge accounting, not a full microscopic graviton or Hamiltonian-constraint theorem.
20	Chapter 8 regular metric-only no-extra-equation theorem.	C	Explains why a field redefinition alone gives no new cosmological observable beyond the input semiclassical metric action.

A **negative test** is a real gain only in its stated domain: neither the failure of a local scalar-contact channel nor the success of a universal representation is a universal theory of gravity. Conversely, source-visible massless static response and an admissible common propagation metric are separate requirements.

17.3 QM012: why varying the correct effective variable matters

In the *fixed-reference*, $\lambda\mathcal{Q} = 0$, positive null-sector specialization alone, take a fixed null direction ℓ_μ and write

$$\boxed{g_{\mu\nu} = \eta_{\mu\nu} + H\ell_\mu\ell_\nu, \quad \sigma_\mu = \sigma_{\rm amp}\ell_\mu, \quad H = \sigma_{\rm amp}^2 \geq 0.} \tag{82}$$

This is *not* a replacement for the full founding metric. Consider first the *specific trial* canonical vacuum energy $\mathfrak{E}_{\rm can}[\sigma_{\rm amp}] = (Z/2)\int|\nabla\sigma_{\rm amp}|^2{\rm d}^3x$. Its exterior spherical stationary equation is $\nabla^2\sigma_{\rm amp} = 0$, with decaying solution $\sigma_{\rm amp} = a/r$. Consequently $H = a^2/r^2$, which is not the Schwarzschild $H \propto 1/r$. This failure excludes this particular simple stiffness, not all possible microscopic actions for σ.

The *adopted* static QGD/GR functional, Eq. (14), is instead varied in H. Its exact Euler derivative is

$$\frac{\delta\mathfrak{E}_{\rm N}}{\delta H} = -\frac{c^4}{16\pi G_N}\nabla^2 H - \frac{c^2}{2}\rho_m, \qquad \nabla^2 H = -\frac{8\pi G_N}{c^2}\rho_m. \tag{83}$$

Regularity through a compact source, $H \to 0$ at infinity and the mass integral then fix the exterior solution $H = 2G_N M/(c^2 r)$, $\sigma_{\rm amp} = \sqrt{2G_N M/(c^2 r)}$. Its uniqueness refers to the chosen source, function space and boundary data; vacuum exterior harmonicity alone leaves the coefficient undetermined.

A useful new explicit identity [D/C]. Rewriting the *same* energy in the scalar amplitude is not the naive canonical energy:

$$\boxed{\mathfrak{E}_{\rm N}[\sigma_{\rm amp}^2;\rho_m] = \int{\rm d}^3x\left[\frac{c^4}{8\pi G_N}\sigma_{\rm amp}^2|\nabla\sigma_{\rm amp}|^2 - \frac{c^2}{2}\rho_m\sigma_{\rm amp}^2\right].} \tag{84}$$

By the chain rule its unrestricted scalar-amplitude Euler derivative is

$$\boxed{\frac{\delta\mathfrak{E}_{\rm N}[\sigma_{\rm amp}^2]}{\delta\sigma_{\rm amp}} = 2\sigma_{\rm amp}\left[-\frac{c^4}{16\pi G_N}\nabla^2(\sigma_{\rm amp}^2) - \frac{c^2}{2}\rho_m\right].} \tag{85}$$

On $\sigma_{\rm amp} \neq 0$, this is equivalent to Eq. (83). At $\sigma_{\rm amp} = 0$ it vanishes even when the H Euler equation fails, revealing a *spurious stationary branch of a noninvertible field coordinate*. In the

vacuum $r > R$, the correct $\sigma_{\rm amp} = \sqrt{r_s/r}$ has

$$\nabla^2\sigma_{\rm amp} = -\frac{\sigma_{\rm amp}}{4r^2}, \qquad |\nabla\sigma_{\rm amp}|^2 = \frac{\sigma_{\rm amp}^2}{4r^2}, \qquad \nabla^2(\sigma_{\rm amp}^2) = 0. \tag{86}$$

This resolves the apparent discrepancy between harmonic H and non-harmonic $\sigma_{\rm amp}$. Selecting H improves the conditional stationary coordinate in this branch; it is not a proof that the original flat Dirac QFT generates Eq. (84). In a general multi-sector configuration the individual sector amplitudes and H representatives can change under redundancies, so an operational, gauge/representation-invariant lift remains necessary.

17.4 QM012 controls: which apparent "spinor origins" actually fail

The independent comparative test supplies three especially useful controls: (i) a stationary outgoing free WKB wave has $|\Psi| \sim r^{-1}$ and constant flat-space local lower/upper ratio, not $\sigma_{\rm amp} \sim r^{-1/2}$; (ii) a trial fourth-root density $(\Psi^\dagger\Psi)^{1/4}$ has the correct radial *power* but a free outgoing $1/r$ wave is not an isolated normalizable source and its coefficient is a flux rather than source-mass normalization; (iii) pure-spinor Fierz bilinears can furnish null directions algebraically but generally retain spin-polarization dependence, while mixed unpolarized sources need not select a unique null congruence. Maximum entropy selects a state only under imposed constraints, not automatically a universal exterior H. These are useful restrictions on $\mathcal{F}_{\rm micro}$, not assertions that all collective-spinor possibilities fail.

17.5 QM013: proper-time induced-curvature win and the limits of premise (Q)

Assuming metric-coupled Dirac quantum matter with a covariant proper-time prescription, the checked local vacuum coefficient in the specified one-four-component-Dirac convention is

$$I_1 = \int_{\Lambda^{-2}}^{\infty} s^{-2}e^{-sM^2}\mathrm{d}s, \quad C_R = -\frac{I_1}{96\pi^2}, \quad M_P^2 = -2C_R = \frac{I_1}{48\pi^2}, \quad G_{\rm ind} = \frac{6\pi}{I_1}. \tag{87}$$

At $M = 0.693435535520$ and $\Lambda = 4$ in the original dimensionless model, the independent audit reports $I_1 = 14.10422862355$, $C_R = -0.01488601185938$ and $G_{\rm ind} = 1.3364471340$. Claude's alternative expression $12\pi/(\sum_i w_i I_{1,i})$ agrees for one such Dirac species *only if* its unspecified weight is $w_i = 2$; setting $w_i = 1$ yields a factor-of-two discrepancy. Neither matching a heat-kernel coefficient nor obtaining the textbook $1/|\boldsymbol{k}|^2$ weak-field response *after* admitting a dynamical metric establishes that Σ was derived as a composite observable of the original flat QFT. Vacuum-energy tuning, regularity, physical source vertices, regulator matching, the measure and no double counting of a separately retained Einstein term are still necessary. A causal non-equilibrium source response also requires an appropriate in-in rather than merely Euclidean vacuum prescription.

17.6 QM014: exact sector ranks and a narrow induced-spectrum check

For the special fixed-coframe, $\lambda\mathcal{Q} = 0$, eight-sector model, let $J = \mathrm{diag}(I_4, -I_4)$ and $g = \eta + \Sigma^{\mathsf{T}} J \Sigma$. Exact rational-matrix tests reproduce:

Configuration	rank $D_\Sigma g$	dim ker	$O(4,4)$ orbit	Stabilizer
Flat spanning pair	10	22	22	6
Second spanning Lorentzian pair	10	22	22	6
One rank-one sector	4	28	7	21
All sectors zero	0	32	0	28

The 22-dimensional kernel equals the local $O(4,4)$ orbit tangent *on the spanning branch*; singular branches have additional null directions that are not automatically gauge transformations. At a tuned, flat, spanning background and *assuming* the two-derivative induced Einsteinian action, the Fourier metric Hessian has rank 6 at generic non-null momentum (four diffeomorphism null directions) and rank 4 at null momentum (four gauge plus two physical wave solutions). Pullback to 32 sectors gives respective nullities 26 and 28, of which 22 are algebraic sector redundancies. These are reproducible **[D/C]** operator-rank checks, not a full nonlinear Dirac–Bergmann count for the original microscopic theory.

The same audit supplies an explicit gauge transformation of a TT wave for which the local canonical Fierz–Pauli $t^{00}_{\rm can}$ changes, although its periodic spatial average does not. Therefore $\mathcal{G}+\mathcal{T}+\mathcal{Q}=0$ (a sector Euler balance) cannot be read as three separately physical local energy densities, and the total matter stress should not be called zero inside an ordinary sourced Einstein solution. Most importantly, the Weinberg–Witten obstruction is *not* eliminated just by writing a stationary total-stress equation. A proposed microscopic spin-two construction must identify its true Lorentz-covariant stress operator and show precisely which theorem assumptions fail after the constraints are constructed. The classical sector orbit counts alone cannot do this.

17.7 Current research decision: the smallest still-missing physical result

Conditional QGD objective achieved [C]. Under a supplied effective QGD/Einstein gravitational action, specified source, boundary data, regular nonzero branch and null-sector matching, Chapter 2 selects Schwarzschild, Chapter 5 exactly matches its static collective functional, and the induced curved Dirac/WKB/current structure is consistent. This is a closed *conditional* statement, independent of whether premise (Q) is later adopted.

Physical genesis remains unproved [O]. The minimum new result that would advance the original direction is a *source-theory-derived*, physically defined, probe-independent collective observable $H_{\rm micro}$ (or a suitable tensor/constitutive replacement) and a constrained state-dependent functional $\Gamma_{\rm micro}[H_{\rm micro},\ldots;J]$ whose Hessian and *physical mass/stress* source vertex determine an exterior, source-visible long-range response. On an ordinary regular Gaussian branch the benchmark is

$$\boxed{\frac{\delta H_{\rm micro}(\boldsymbol{k})}{\delta\rho_m(\boldsymbol{k})} = -[K^{-1}_{\rm phys}(\boldsymbol{k})B_{H\rho}(\boldsymbol{k})] \overset{?}{\sim} \frac{8\pi G_N}{c^2|\boldsymbol{k}|^2} \qquad (|\boldsymbol{k}|\to 0).} \tag{88}$$

The displayed G_N is a measured effective matching target, not something QGD must independently predict to count an effective milestone. The formula presumes a specified linear static regime, consistent definitions and fixed Fourier conventions; a genuinely nonlinear/non-Gaussian selection mechanism would need its own sourced exterior solution, not a forced Gaussian pole. Only after this physical response is obtained should one derive the frame/probe-independent lift $H_{\rm micro}\to[\Sigma]$ and verify universal matter/light propagation, source stresses, causal constraints, and the full coframe/signed-sector/$\mathcal{Q}$ history. Inducing a curvature coefficient from a supplied metric does not substitute for that microscopic step.

Optional alternative to audit, not an adopted foundational closure. Premise (Q) may be explored on a separate track: supply a complete auxiliary-Σ/Dirac Lagrangian and measure, a consistent vacuum prescription, full nonlinear constraints including spin-connection and fermionic Dirac brackets, diffeomorphism and sector generators, boundary charges and Ward identities; then show how a composite universal metric and its microscopic physical states arise or declare them new axioms. The QM014 rank tests are necessary but not sufficient. A spin-two pole calculation is a diagnostic, not a required first step of the founding $\mathcal{F}_{\rm micro}$ project.

Reproducibility and provenance. The numerical QM011 scalar comparison was rerun using `test_qm011.py`; the QM012 selection diagnostics were rerun using `tests_qm012.py`; and QM014 rank/gauge diagnostics were rerun using `verify_qm014.py`, each using the pre-

served research packages. QM013's normalization comparison is independently archived in `QM013_Independent_Audit.md`. These are tests of stated mathematical/model claims, not evidence for a completed microscopic-to-gravity derivation. Relevant primary method literature: Cornwall, Jackiw and Tomboulis, *Phys. Rev. D* **10**, 2428 (1974), https://doi.org/10.1103/PhysRevD.10.2428; Visser, *Sakharov's induced gravity*, https://arxiv.org/abs/gr-qc/0204062; Weinberg and Witten, *Phys. Lett. B* **96**, 59 (1980), https://doi.org/10.1016/0370-2693(80)90212-9.

18 Conclusion and dependency ledger (v0.7 baseline preserved)

Research result	Status	Meaning
Dirac ratio, selected WKB, total current conservation	D/C	Matter-to-transport bridge retained
Hartree scalar Hessian and density-source vertex	D/C	Numerically reproduced using the original scalar-contact cutoff model
Stable scalar-channel static pole	D/C	Absent in sharp-3D and matched proper-time schemes at the stated saddle
TT–scalar two-point mixing	D/C	Zero in an isotropic state by rotational tensor selection
Strictly local matter-to-Σ exterior rule	D/C	Insufficient to produce a nonconstant source-dependent vacuum exterior
TT shear, rank-nine intraband matrix, and kinetic Ward contact	D/C	Independently checked, no isolated TT pole at this order
Null-sector Dirac principal symbol and radial WKB	D/C	Exact Clifford/coframe identity; reproduces QGD Kerr–Schild propagation given H
Conditional source-to-kinetic response	D/C	Calculated $1/\lvert\boldsymbol{q}\rvert^2$ functional target from adopted gravity equation; not derived from QFT
Hartree/Fock source-to-kinetic vertex	D/C	Zero at tested leading orders, fails conditional target
Full source-visible stress/tensor pole	O	No pole in scalar-Hartree Gaussian TT channel; other interactions require conserving tests
Universal optical/matter metric	O	Requires multi-probe kinetic and interaction tests
Microscopic Hubbard–Stratonovich stationary scalar	D/C	Existence derived; selected massive scalar response does not match H_{KS} in tested phase
Variational weak-field H selection	C	Source plus adopted stationary field energy and boundary conditions uniquely determine the spherical exterior H
Physical state-to-[Σ] selection	O	Requires invariant quantum-state/Dirac-to-invariant rule or microscopic selection functional
Induced complete QGD Einsteinian action	O	Currently adopted in Chapters 2–5 and tested conditionally

Revised v0.7 research decision. There are two separate closure goals. The *conditional QGD effective correspondence* is closed for the stated Schwarzschild exterior: quantum matter supplies the source, the *adopted* gravitational Euler equations select H and the natural invariant, the compatible curved Dirac operator has the exact required principal symbol, and WKB/current conservation match without discarding the original bridge. This does *not* establish the microscopic origin of gravitational geometry. QM009 adds the source-dependent spinor response but proves the Hartree–Fock kinetic vertex remains zero for the actual local scalar-contact model; a momentum-dependent self-energy and its conserving source vertex remain to be tested.

QM011 has now tested the microscopic stationarity hypothesis itself: a regulated matter-state-dependent functional exists and selects a unique scalar field at Gaussian order, but its measured positive scalar Hessian and finite source vertex do not select QGD's $1/r$ exterior. This negative result does not reopen the conditional closure and does not impose a graviton-first requirement.

The *independent origin* remains open at a now precise point: find a physically defined, observer- and probe-independent map from the solved macroscopic Dirac/quantum source state to $[\Sigma]$, or derive a microscopic constrained selection functional with that map as its stationary solution. The source-to-kinetic vertex of Eq. (28) is one testable *candidate route*, not an obligatory definition of $\mathcal{F}$; if that route is chosen, its result must match Eq. (25) without inserting the gravitational Green function. In the tested scalar-Hartree/first-Fock phase that vertex vanishes, whereas the stipulated QGD target is nonzero and nonlocal. The previously computed tensor/stress response remains a valuable parallel investigation, not the automatic prerequisite for the original spinor-to-invariant matching. Chapters 7 and 8 may be studied conditionally on the declared QGD effective action while the stronger origin question is pursued.

19 Final v0.8 research decision and non-circular acceptance test

Result of the consolidation. The research has *already* established a mathematically controlled geometry/Dirac correspondence and an exactly matched *conditional* static collective energy. Its shortfall is not a missing Schwarzschild calculation, a missing radial exponent or a missing stress-tensor notation. It is the absence of a quantum-source-derived physical selection law $\mathcal{F}_{\mathrm{micro}}$ which fixes the same long-range field and universal coupling *without starting from a supplied metric, Einstein equation or gravitational energy.* The massless collective postulate MW2 is neither contradicted by the tested gapped Hartree phase nor derived from it. The separate premise (Q) remains a distinct, explicitly unadopted induced-action route.

Operational closure criterion. To claim original-direction microscopic closure, a candidate calculation must (i) specify an actual quantum state/phase and observable $\mathcal{O}_{\mathrm{micro}}$, with its non-arbitrary definition, (ii) derive an effective constrained functional and *conserving physical source vertex* from the same microscopic interaction, (iii) obtain the sourced exterior relation and normalize it to source mass without using the target Green function as input, (iv) show a physically justified common kinetic/light-cone response and a nondegenerate map to a QGD $[\Sigma]$, and (v) pass the full-master-metric, sector-redundancy, stability, conservation and boundary tests. Matching a coefficient or radial power in isolation is a useful diagnostic but is not sufficient.

The next concrete research calculation. Construct a specified composite/2PI observable family containing source-state stress, kinetic and Dirac-bilinear data; calculate its regulated $\Gamma_{\mathrm{micro}}[\varphi; J]$, its reduced inverse Hessian after constraints, and the static mass-source response. First decide whether that response is finite, critical, nonanalytic or non-Gaussian. *Only if* a physically sourced long-range mode survives should one attempt the QGD $H \to [\Sigma]$ lift and universal test. As a parallel, independent research programme, an adopted premise (Q) requires a complete nonlinear Dirac–sector constraint algebra rather than a claim of automatic Weinberg–Witten evasion. Neither route invalidates the earlier closed conditional results.

Editorial seal. The original v0.7 core, source calculations and historic proof-status ledger remain intact; the present v0.8 text appends a narrowly scoped QM012–QM014 synthesis and this updated stopping rule. It does not promote any open [O] claim to [D].

20 QM015: source-overlap criterion and coupled-channel micro-origin programme

Purpose and proof scope. The previous sections isolate the physical missing arrow from quantum matter to a sourced natural invariant. We now sharpen it using an exact spectral-response criterion, two independent false-positive controls, and a solvable mixed-channel stationary model. In this extension **[D]** is a mathematical conclusion under explicitly stated hypotheses, **[C]** is conditional on the specified toy functional or a named approximation, **[A]** is an independently supplied

kinetic/source ingredient, and **[O]** remains an unproved QGD microscopic mechanism. These calculations do not modify the founding master metric, existing Chapters 1–5 or the alternatives catalogued as premise (Q).

20.1 The spectral source-overlap theorem [D]

Work at zero frequency in an independently chosen translation-invariant microscopic source state. Let φ_A denote renormalized, physically specified composite observables, which may include scalar bilinears, conserved stress channels, a momentum-dependent Dirac self-energy, or constrained order parameters. Their stationary quadratic response must follow from a *single* quantum-matter functional, not from an imported gravitational propagator. In a real positive Euclidean static subspace, write its quadratic change under an operational physical source ρ as

$$\delta^2\Gamma_{\rm micro} = \frac{1}{2}\int_{\boldsymbol{k}} \varphi_A(-\boldsymbol{k})K_{AB}(\boldsymbol{k})\varphi_B(\boldsymbol{k}) - \int_{\boldsymbol{k}} \varphi_A(-\boldsymbol{k})B_A(\boldsymbol{k})\rho(\boldsymbol{k}), \qquad \int_{\boldsymbol{k}} \equiv \int \frac{\mathrm{d}^3k}{(2\pi)^3}. \tag{89}$$

The source ρ is an operational long-wavelength mass/energy perturbation, to be defined from a second microscopic matter species or a conserving stress insertion. A trial linear observable $H_{\rm micro} = A_A\varphi_A$ has

$$\boxed{\chi_{H\rho}(\boldsymbol{k}) \equiv \frac{\delta H_{\rm micro}}{\delta\rho} = A_A(\boldsymbol{k})[K^{-1}(\boldsymbol{k})]_{AB}B_B(\boldsymbol{k}).} \tag{90}$$

The sign agrees with the displayed $-\varphi\cdot B\rho$ convention; an opposite microscopic source convention reverses it. The formula is valid on the constrained *physical* subspace after zero-gauge directions have been appropriately quotiented; it is not a license to invert an un-fixed gauge Hessian.

Proposition 20.1 (Finite-Hessian obstruction and source-overlap criterion)**.** *Assume finitely many physical static channels, $K(\boldsymbol{k})$ real symmetric, analytic near $\boldsymbol{k}=0$, and with $K(0)$ uniformly positive definite. If $A(\boldsymbol{k})$ and $B(\boldsymbol{k})$ are also analytic there, then $\chi_{H\rho}(\boldsymbol{k})$ is bounded and analytic near the origin; it cannot contain the required static $1/|\boldsymbol{k}|^2$ pole. Conversely, suppose a normalized isolated eigenmode u_0 has $\lambda_0(k) = \mu^2 + Zk^2 + O(k^4)$ with $Z>0$, while all other relevant modes are gapped and analytic. Its leading response is*

$$\chi^{(0)}_{H\rho}(k) = \frac{(A\cdot u_0)(u_0\cdot B)}{\mu^2 + Zk^2} + \textit{terms regular at } k=0. \tag{91}$$

A $1/k^2$ mass-source response therefore requires $\mu=0$, nonzero overlap on both sides*, and a nonvanishing net residue. The conditions are necessary and sufficient within the stated isolated Gaussian k^2 mode class, not across all non-Gaussian or nonlocal selection mechanisms.*

Proof. If $K(0)>0$, $K^{-1}(k)$ has a convergent analytic expansion at the origin; multiplication by regular A and B preserves this property. For the second claim, insert the orthonormal eigenmode decomposition of K^{-1} and isolate the eigenvalue λ_0. If either projection vanishes, the pole is invisible to that source-observable pair. Multiple critical modes may cancel, and a symmetry can force $u_0\cdot B=0$ even when the mode is genuinely gapless. □

This is the operative upgrade to the Chapter 5 premise MW2: instead of asking only whether a mode has Zk^2, we must ask whether the *same microscopic calculation* yields a nonzero mass/stress vertex B, a physical observable projection A, and the positive long-range residue. A density pole, a time-dependent sound pole and a gravitational static response are not interchangeable.

20.2 A concrete negative test: a massless Goldstone mode can be source-invisible [D/C]

In a homogeneous, isotropic, parity-preserving state of Dirac quantum matter define the parity-even scalar $S=\bar\psi\psi$ and parity-odd pseudoscalar $P=\bar\psi i\gamma_5\psi$. At zero frequency, their mixed response

obeys

$$\begin{aligned} \chi_{PS}(0,\boldsymbol{k}) &= -\chi_{PS}(0,-\boldsymbol{k}) \qquad \text{(parity)}, \\ \chi_{PS}(0,\boldsymbol{k}) &= +\chi_{PS}(0,-\boldsymbol{k}) \qquad \text{(isotropy)}, \\ \therefore \quad & \boxed{\chi_{PS}(0,\boldsymbol{k}) = 0}. \end{aligned} \tag{92}$$

If the microscopic mass insertion is parity even, an exactly gapless parity-odd Goldstone mode need not respond to it linearly. Chiral fermion theories provide a familiar controlled *candidate category*: pseudoscalar modes may be symmetry-protected while scalar amplitude modes are massive. This excludes the inference "fermionic Goldstone pole ⇒ static gravity". Equation (92) is exact only under the explicitly stated symmetry and source assumptions; parity-breaking states, anisotropic media or different source vertices must be analyzed separately.

A second independent control is a zero-temperature idealized compressible superfluid. At quadratic hydrodynamic order a density fluctuation δn and phase θ have the static functional

$$\mathfrak{H}[\delta n,\theta;J] = \int \mathrm{d}^3x \left[\frac{(\delta n)^2}{2\chi_n} + \frac{\rho_s}{2}|\boldsymbol{\nabla}\theta|^2 - J\delta n\right], \quad \chi_n,\rho_s > 0. \tag{93}$$

The phase has a gapless dynamical phonon after adding the canonical $\delta n\dot{\theta}$ term, but static minimization gives $\delta n = \chi_n J$ and $\nabla^2\theta = 0$. More explicitly, with $c_s^2 = \rho_s/\chi_n$ the linear density response has the idealized form

$$\chi^R_{nn}(\omega,k) = \frac{\chi_n c_s^2 k^2}{c_s^2 k^2 - (\omega + i0)^2}, \qquad \boxed{\chi^R_{nn}(0,k) = \chi_n,} \tag{94}$$

not $1/k^2$. Its propagating pole is not an unscreened static gravitational mass-source pole. The relevant limit is *first* $\omega = 0$ for a static source and *then* $k \to 0$; interchanging limits can change the answer.

20.3 Mixed-channel existence theorem: the source need not couple directly to the gapless field [D/C]

The selection theorem suggests a route less restrictive than demanding that the long-range collective coordinate itself directly overlap with a microscopic density bilinear. Consider two *hypothetical* collective coordinates, a gapless q and a massive source-facing s, with positive constants $Z, Z_s, M_*, \lambda, g_s$ and stationary energy

$$\boxed{\mathfrak{E}[q,s;\rho_m] = \int \mathrm{d}^3x \left[\frac{Z}{2}|\boldsymbol{\nabla}q|^2 + \frac{Z_s}{2}|\boldsymbol{\nabla}s|^2 + \frac{M_*^2}{2}(s-\lambda q)^2 - g_s\rho_m s\right].} \tag{95}$$

[A] The full functional and the joint shift symmetry $(q,s) \mapsto (q+\epsilon, s+\lambda\epsilon)$ of its *source-free* part are supplied for this toy test, not yet generated from Chapter 1's four-fermion action. In particular, no externally assumed QGD or Einstein metric is used in solving the model. The source acts *only* on the otherwise massive s channel. The Fourier Hessian, source vector and its determinant are

$$K(k) = \begin{pmatrix} Zk^2 + \lambda^2 M_*^2 & -\lambda M_*^2 \\ -\lambda M_*^2 & M_*^2 + Z_s k^2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ g_s \end{pmatrix}, \quad \boxed{\det K = k^2\left[M_*^2(Z + \lambda^2 Z_s) + ZZ_s k^2\right].} \tag{96}$$

An exact inversion gives

$$\boxed{\frac{\delta q(k)}{\delta\rho_m(k)} = \frac{\lambda g_s}{Z_{\text{eff}}\, k^2(1+\ell^2k^2)}, \qquad Z_{\text{eff}} = Z + \lambda^2 Z_s, \qquad \ell^2 = \frac{ZZ_s}{M_*^2 Z_{\text{eff}}}.} \tag{97}$$

The mass source directly excites the s channel; mode mixing transfers a nonzero residue into the joint gapless mode. The full static $K(k)$ is positive for every $k \neq 0$ when the displayed coefficients are

positive; the only long-wavelength flat direction is the deliberately supplied joint shift mode. Thus the calculation shows that a gapped observable may transmit source information to a long-range mode, not that a massless mode can emerge from a uniformly positive Hessian.

For an idealized point mass $\rho_m(\boldsymbol{x}) = M_{\rm src}\delta^{(3)}(\boldsymbol{x})$, the *independently Fourier-inverted* result is

$$\boxed{q(r) = \frac{\lambda g_s M_{\rm src}}{4\pi Z_{\rm eff} r}\left(1 - e^{-r/\ell}\right).} \tag{98}$$

If the effective null-sector observable is *conditionally* identified as $H = 2g_{\rm eff}q/c^2$, with $g_{\rm eff} = \lambda g_s$, then

$$\boxed{G_{\rm eff} = \frac{g_{\rm eff}^2}{4\pi Z_{\rm eff}}, \qquad H(r) = \frac{2G_{\rm eff}M_{\rm src}}{c^2 r}\left(1 - e^{-r/\ell}\right).} \tag{99}$$

In the controlled exterior regime $r \gg \ell$ this reproduces the Chapter 5 conditional $H \sim r^{-1}$ and the positive null-sector amplitude $\sigma_{\rm amp} \sim r^{-1/2}$, while giving an explicit exponentially short-range deviation at finite r. The latter should not be fitted to observations until ℓ is derived and constrained from an actual matter theory.

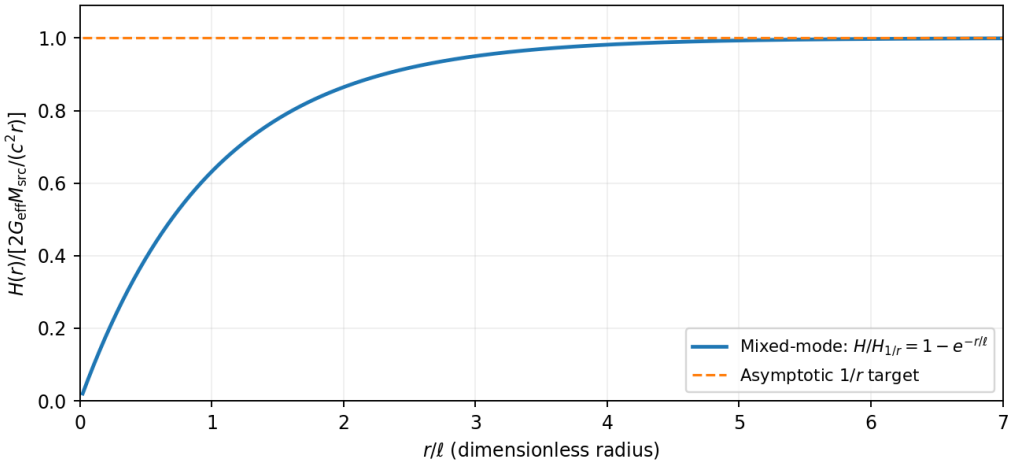


Figure 7: Exact QM015 toy-model radial ratio to the asymptotic QGD/Newtonian null-sector field. The finite-range correction follows from the two-field Hessian, not an empirical fit. There is no claim that the original QGD microscopic theory produces these parameters.

Independent check. In dimensionless units $Z = 2$, $Z_s = 3$, $M_* = 4$, $\lambda = 1/2$, $g_s = 1.2$, $c = 1$, we obtain $Z_{\rm eff} = 2.75$, $\ell = 0.369274472938$, $g_{\rm eff} = 0.6$, $G_{\rm eff} = 0.0104174144569$. The same $q(r)$ is obtained to better than $2 \cdot 10^{-7}$ absolute accuracy by direct numerical three-dimensional Fourier inversion for $M_{\rm src} = 3$ and $r = 0.1, 0.5, 1, 3, 8$, independently of the analytic Yukawa decomposition. The exact determinant and susceptibility identity are also checked symbolically; full verifier and data are packaged with this edition.

20.4 Schur-complement version of the mechanism [D]

For a general coupled physical block $\varphi = (q, s)$ with K_{ss} invertible at the chosen momentum, eliminating s yields

$$K_{q,\rm red} = K_{qq} - K_{qs}K_{ss}^{-1}K_{sq}, \qquad B_{q,\rm red} = B_q - K_{qs}K_{ss}^{-1}B_s. \tag{100}$$

These formulae expose an important difference between two questions: whether a channel is itself gapless, and whether it becomes *source-visible by mixing*. Even when $B_q = 0$ at the bare level, the reduced vertex can be nonzero if $K_{qs}B_s \neq 0$. In the specific model above $K_{qs} = -\lambda M_*^2$,

so $B_{q,\mathrm{red}}(0) = \lambda g_s$. The effective massless stiffness is Z_{eff}. The route to QGD would therefore be a *microscopically derived Schur complement* whose eigenmode and source overlap satisfy the static QGD target. It is not necessary to postulate a direct $-g_c q\rho$ term in an original fermionic Lagrangian, but the interaction that generates the mixing must still be derived.

20.5 A controlled candidate microscopic experiment: exact Hubbard–Stratonovich identity, then 2PI selection [O]

A viable research path retains the actual matter degrees of freedom instead of inserting $g(\Sigma)$ as their initial background. For illustration, an N-species flat-background Euclidean fermion theory with a scalar contact operator $\mathcal{O} = \sum_{i=1}^{N} \bar{\psi}_i \psi_i$ has the *algebraic* Hubbard–Stratonovich identity

$$\exp\left[+\frac{G_0}{2N}\int \mathcal{O}^2\right] \propto \int \mathcal{D}q \, \exp\left[-\int\left(\frac{Nq^2}{2G_0} - q\mathcal{O}\right)\right], \tag{101}$$

$$\Gamma_{\mathrm{HS}}[q] = \frac{N}{2G_0}\int q^2 - N \, \mathrm{Tr}\log(D_0 - q) + \text{counterterms}, \tag{102}$$

where the second line assumes identical species and a regulator. Here the q integration is initially an *exact rewriting of the chosen contact interaction*, not premise (Q), and is not yet a universal metric variable. Expanding about a true self-consistent saddle produces $K_q(k) = K_q(0) + Z_q k^2 + \cdots$ with the real mass-source vertex computed by inserting a *separate sourced microscopic matter species* or the appropriate conserved energy-stress operator. The already performed QM011 model yields $K_q(0) > 0$ under the two tested regulator schemes; nothing in the HS identity forces a pole.

The next numerical experiment should therefore calculate, *from one consistent regulated fermion theory*, the combined basis $\{\bar{\psi}\psi, \bar{\psi} i\gamma_5 \psi, T^{00}, T^{0i}, T^{ij}, \partial_{p_\mu}\Sigma_{\mathrm{Dirac}}, \ldots\}$, renormalize its composite mixing, and compute its connected source response through a conserving 2PI/Bethe–Salpeter differentiation. A finite truncation can preserve specified global conservation laws without automatically satisfying every gauge Ward identity; the exact identities and truncation dependence must be separately checked. The trial QGD observable H_{micro} should be defined operationally, not by already inserting $g_{\mu\nu}(\Sigma)$ into the fermion kinetic term.

Concrete decision procedure. (i) Solve the source-free homogeneous phase with the matched Chapter 5/QM011 regulator and record all stationary branches and thermodynamic stability. (ii) Add a second, spatially localized matter source with independently conserved stress; obtain B_A by differentiating the same functional. (iii) Compute $K_{AB}(\omega, \boldsymbol{k})$, its Ward-completed contacts and its constrained static Schur complement with $\omega = 0$ before $k \to 0$. (iv) Determine whether an eigenvalue is genuinely Zk^2, whether $A \cdot u$ and $u \cdot B$ are nonzero, and whether the residue is positive and stable under regulator/volume change. (v) Only then infer H_{micro} and compare the independently computed $\partial_\rho \partial_{p_\mu}\Sigma_{\mathrm{Dirac}}$ with the common kinetic deformation required by the QGD null-sector Dirac operator. (vi) Test separate massive, massless, polarized, antiparticle and source-composition probes and ultimately the full coframe/signed-sector/$\mathcal{Q}$ metric rather than a single scalar fifth force.

20.6 Why even a successful scalar pole would not yet be gravity [D/C/O]

A scalar $1/k^2$ pole suffices to generate a Newtonian-looking static force but is not by itself a proof of universal relativistic gravitation. In particular, a mere scalar mass self-energy shifts the Dirac mass term but need not change the principal symbol $\gamma^\mu p_\mu$; in the original QM009 tested Hartree–Fock channel the source-to-kinetic vertex is zero. A purely conformal perturbation of a four-dimensional reference metric also preserves unparameterized null geodesics, so its lensing cannot be assumed equivalent to the nonconformal QGD Kerr–Schild lift. The correct target must include the microscopic source-dependent *derivative* vertex and propagation of more than one probe species.

For the QGD positive null-sector specialization, define the classical geometric representation only *after* a physical response is obtained,

$$H_{\text{micro}} \xrightarrow[\text{additional physical selection}]{\text{nondegenerate sector/orbit lift}} [\Sigma], \qquad g_{\mu\nu} = \eta_{\mu\nu} + H\ell_\mu\ell_\nu, \quad \sigma_\mu = \sqrt{H}\ell_\mu. \tag{103}$$

The lift and the vector ℓ_μ cannot be defined by the motion of the very test particle used to validate the field. Moreover the founding complete metric remains

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{104}$$

The $\lambda\mathcal{Q} = 0$, fixed-reference Kerr–Schild relation is a declared specialization, not the general microscopic QGD theory.

21 Final v0.9 proof ledger, options, and next calculation

Result or candidate	Status	Scientific meaning
Analytic positive physical static Hessian forbids a $1/k^2$ pole	[D]	No gapless long-range Gaussian response in that fixed phase
Source-overlap residue $(A \cdot u)(u \cdot B)/Z$	[D]	Necessary observable and source couplings, in addition to a pole
Parity-even scalar source versus parity-odd massless mode	[D/C]	Gapless collective modes may be physically invisible to mass
Hydrodynamic phonon versus static density response	[D/C]	A propagating gapless phonon does not imply divergent static density susceptibility
Two-field source-visible massless mode by mixing	[D/C]	Explicit stationary existence model with an exact $1/r$ exterior and predicted finite-r correction
Microscopic origin of the two-field shift symmetry and mixing	[O]	Coefficients/fields remain postulated in QM015 toy; not a Dirac derivation
Exact HS transformation of a chosen fermion contact action	[D/C]	Does not itself add premise (Q); dynamically generated pole still open
QGD probe-independent $H \to [\Sigma]$ and universal kinetic coupling	[O]	Requires independent dynamical/operator proof
Claude's independent induced-$g(\Sigma)$ route (Q)	[A/O]	Separate unadopted additional microscopic premise with open constraints

Recommended next concrete calculation. Re-use the original Chapter 1 microscopic fermion contact action and QM011 matched regulators, enlarge the HS/2PI collective basis to the minimal scalar–stress–kinetic channels, and *measure the complete static source-overlap residue* at its self-consistent stable phase(s). Vary the contact coupling through any legitimate critical phase; do not set $K(0) = 0$ by hand. If a protected gapless sector is found, explicitly test whether the physical source vertex vanishes by symmetry or kinematics, and whether a mixed mode supplies a nonzero finite overlap. Verify that the same micro theory yields the source-dependent Dirac principal symbol. Negative answers would be publishable restrictions on this mechanism even if the target pole is absent.

Precise research outcome. QM015 contributes a source-overlap theorem, exact mixed-mode transfer mechanism and two decisive false-positive controls. It *does not* derive a long-range gravitational mode or any new fundamental field from the existing microscopic QGD interaction. This preserves the Chapter 5 conditional closure and makes Chapter 6's strongest unresolved task more concrete: derive a source-visible, stable physical collective operator and its universal

propagation effects directly from quantum matter before promoting it to the natural-invariant sector. All prior v0.7/v0.8 labels, calculations and proof statuses are retained.

Literature supporting methodological comparators. G. Baym and L. P. Kadanoff, *Phys. Rev.* **124**, 287 (1961), https://doi.org/10.1103/PhysRev.124.287; J. M. Cornwall, R. Jackiw and E. Tomboulis, *Phys. Rev. D* **10**, 2428 (1974), https://doi.org/10.1103/PhysRevD.10.2428; overview of NJL scalar/pseudoscalar collective channels, https://doi.org/10.1088/1674-1137/43/9/094103; Weinberg soft-graviton universality (1965), https://doi.org/10.1103/PhysRev.140.B516; J. Bonifacio *et al.*, *JHEP* **07**, 056 (2020), https://doi.org/10.1007/JHEP07(2020)056. The cited literature provides established comparators, not an independent endorsement of QGD's unproved microscopic origin.

22 QM016: a genuinely microscopic source–scalar–density block and a strict Hartree obstruction

Scientific scope. This is a new calculation within the *original Chapter 1* scalar-contact Dirac model of Eq. (34), not the freely chosen two-field energy of QM015. We analyze the self-consistent $T = 0$ homogeneous Hartree phase at fixed μ, m_0 and sharp spatial cutoff Λ, then test the two previously matched smooth finite-wave-number regulators at their benchmark point. The independent fields being varied are the scalar Hubbard–Stratonovich mass coordinate M and its conjugate number-density source; *neither* is identified with the QGD gravitational H or the complete tensor stress of a separate source. Proof status [D/C]; the universal microscopic-to-natural-invariant map stays [O].

22.1 A positive-curvature identity that excludes tuning a scalar massless pole in a controlled parameter domain [D/C]

Recall the stationary gap and static scalar Hessian from Eqs. (35) and (36), with $E_p = \sqrt{p^2 + M^2}$, $M > 0$, $G_4 > 0$, $m_0 > 0$, $\Lambda > \mu > 0$ and $p_F = \sqrt{\max(\mu^2 - M^2, 0)}$. The exact algebraic identity

$$\frac{p^2}{E_p} - \frac{p^4}{E_p^3} = \frac{M^2 p^2}{E_p^3} \tag{105}$$

permits elimination of $1/(2G_4)$ using the *same self-consistent* gap equation, not a gravitational matching or a fitted kernel:

$$\boxed{K_0 = \frac{m_0}{2G_4 M} + \frac{M^2}{\pi^2}\left[\int_{p_F}^{\Lambda} \frac{p^2\,\mathrm{d}p}{(p^2 + M^2)^{3/2}} - \frac{p_F}{\mu}\right].} \tag{106}$$

For an occupied Fermi sea, $0 < M < \mu$, its bracket is the explicit function

$$F(M;\Lambda,\mu) = \log\left(\frac{\Lambda + \sqrt{\Lambda^2 + M^2}}{\mu + \sqrt{\mu^2 - M^2}}\right) - \frac{\Lambda}{\sqrt{\Lambda^2 + M^2}}, \tag{107}$$

$$\frac{\partial F}{\partial M} = \frac{M}{E_\Lambda(\Lambda + E_\Lambda)} + \frac{M}{p_F(\mu + p_F)} + \frac{\Lambda M}{E_\Lambda^3} > 0, \quad E_\Lambda = \sqrt{\Lambda^2 + M^2}, \tag{108}$$

$$\lim_{M \to 0^+} F(M;\Lambda,\mu) = \log(\Lambda/\mu) - 1. \tag{109}$$

Here $\partial_M F > 0$ is evaluated at fixed Λ, μ and $0 < M < \mu$. The derivative was also checked by exact symbolic differentiation. For an unoccupied Fermi sea, $M \geq \mu$, $p_F = 0$ and the square bracket in Eq. (106) is manifestly positive.

Proposition 22.1 (Scalar Hartree criticality exclusion in the stated cutoff domain)**.** *For the homogeneous positive-mass solutions of the sharp-cutoff Hartree gap equation with $m_0 > 0$, $G_4 > 0$ and $\Lambda/\mu \geq e$ (and $\Lambda > \mu > 0$), the zero-wave-number scalar curvature is* strictly positive*, $K_0 > 0$, for any self-consistent value of G_4. This excludes a zero-momentum massless scalar* within this model, regulator and approximation*; it makes no claim about nonlocal self-energy, stress-tensor eigenmodes, non-Gaussian criticality, different cutoff ratios, or other microscopic interactions.*

Proof. For $M < \mu$, Eqs. (108) and (109) give $F(M) \geq \log(\Lambda/\mu) - 1 \geq 0$. The separate term $m_0/(2G_4M)$ is strictly positive. For $M \geq \mu$, the integral in Eq. (106) is positive, as is the first term. Thus $K_0 > 0$ for each such self-consistent solution. No local-minimum assumption is needed for the algebraic conclusion, although thermodynamic global selection is a separate question. □

This result is stronger than sweeping G_4 numerically; unlike a generic "no-go" claim, it has an explicit parameter-domain proof. At our $\Lambda = 4$, $\mu = 1.2$ benchmark, $\Lambda/\mu = 10/3 > e$ and the positive lower bound on F is $\log(10/3) - 1 \simeq 0.203973$. The original $m_0 = 0.12$ is held fixed. As emphasized in QM009–QM011, a hard spatial cutoff is *not* accepted as a reliable finite-q regulator; this theorem concerns only the self-consistent $q = 0$ Hartree thermodynamics.

22.2 The genuine two-coordinate scalar–density static Hessian [D/C]

A physical number-density response cannot be checked by calling a chemical-potential source $\delta\mu$ an independent propagating boson. Introduce the Legendre-transformed static Helmholtz density $\mathcal{F}(M, n) = \Omega(M, \mu) + \mu n$ with $n = -\Omega_\mu$ in the occupied phase. Using the *same* fermionic grand-potential derivatives

$$K = \Omega_{MM}, \qquad V = \Omega_{M\mu} = \frac{Mp_F}{\pi^2}, \qquad D = -\Omega_{\mu\mu} = \frac{\mu p_F}{\pi^2} > 0, \tag{110}$$

and eliminating $\delta\mu$ at fixed δn yields the symmetric two-coordinate physical static Hessian

$$\boxed{\mathsf{H}_{(M,n)} = \begin{pmatrix} K + V^2/D & V/D \\ V/D & 1/D \end{pmatrix}, \quad \det \mathsf{H}_{(M,n)} = \frac{K}{D} > 0, \quad \chi_{nn} = D + \frac{V^2}{K}.} \tag{111}$$

Its upper-left principal minor is positive and its determinant is positive; hence both thermodynamic eigenvalues are positive where $D > 0$. Number density is the Legendre conjugate of μ, not a separately proven propagating gapless field. In the empty-Fermi-sea branch $D = V = 0$ at $T = 0$, the Legendre matrix in Eq. (111) is singular as a *coordinate choice* and must not be interpreted as a new massless collective mode; the actual number-source response vanishes.

The static source response therefore remains finite for the tested finite-density phase. If the finite-wave-number continuation has the regularity independently established for the QM009 matched scalar channel, the proposed scalar source-visible pole residue obeys

$$\boxed{\lim_{|\boldsymbol{k}|\to 0} |\boldsymbol{k}|^2 \left[D(\boldsymbol{k}) + \frac{V(\boldsymbol{k})^2}{K(\boldsymbol{k})} \right] = 0.} \tag{112}$$

The constant-density susceptibility and its $k \to 0$ order are essential: a conserved density can have a hydrodynamic sound excitation at nonzero frequency while its *static* compressibility remains finite. The compressibility sum-rule control is standard in self-consistent NJL response theory; see https://doi.org/10.1088/1674-1137/43/9/094103.

22.3 New self-consistent coupling sweep and independent thermodynamic sum-rule check [D/C]

For $m_0 = 0.12$, $\mu = 1.2$, $\Lambda = 4$ and the original sharp zero-momentum normalization, we re-solved the *nonlinear* gap equation at 151 distinct couplings $G_4 \in [0.25, 0.95]$ rather than inserting the precomputed mass. Only one positive-mass root was found at each scanned coupling using the stated root-search grid. The populated branch's minimum scalar curvature was $K_{\min} \simeq 0.14633658$ near $G_4 \simeq 0.60048147$. The Fermi surface closes at

$$\boxed{G_4^* = \frac{\pi^2(\mu - m_0)}{2\mu \int_0^\Lambda p^2 \, \mathrm{d}p/\sqrt{p^2+\mu^2}} = 0.637146968076\ldots.} \tag{113}$$

When $G_4 > G_4^*$ on the tested homogeneous branch, $M > \mu$, so $p_F = 0$ and $V = 0$; the charge/number source then does not excite the scalar at zero temperature. Thus increased scalar attraction does not open a long-range *number-source-visible* channel here: throughout the occupied branch $K_0 > 0$, and beyond its endpoint the source overlap vanishes. The analytical positivity proposition, not the finite scan, is the warrant for the stated sharp-regulator domain.

At the original benchmark $G_4 = 0.5654255716$, the recomputed values are $M = 0.693435535520$, $K_0 = 0.168741734136$, $V_0 = 0.068809590036$, $D_0 = 0.119075968585$, and $\chi_{nn} = 0.147135178153$. Independently re-solving the gap equation at $\mu \pm 10^{-4}$ and differentiating $n = p_F^3/(3\pi^2)$ gives $\chi_{nn} = 0.147135178036$, agreeing within 1.2×10^{-10}. The algebraic K_0 identity also matches the direct fluctuation-integral Hessian along the scanned branch to better than 2×10^{-11} absolute numerical tolerance.

The two *previously independently matched* QM009 smooth static infrared regulator benchmarks provide an additional, different check of the same thermodynamic source block, not a new smooth-regulator coupling sweep:

Regulator	$K_s(0)$	D_0	χ_{nn}	$\lambda_{\min}(\mathsf{H}_{(M,n)})$
Smooth $n = 8$	0.14993818	0.11907597	0.15065405	0.14921889
Smooth $n = 16$	0.15737518	0.11907597	0.14916178	0.15661953

Both use $V_0 = 0.06880959$ and have positive $\det \mathsf{H} = K_s/D_0$. They therefore reinforce the positive scalar–density static conclusion at the tested reference state. Their finite-q stiffnesses are still those of QM009; an independent full matrix $K_{AB}(\omega, \boldsymbol{k})$ with stress and kinetic vertices was *not* computed here.

22.4 Why this obstructs the QM015 mixed-mode mechanism in the tested truncation [D/C/O]

QM015's exact long-range *toy* mechanism had a second, shift-symmetric collective coordinate and an off-diagonal mixing which transfers a number-source response into its zero eigenvalue. The original scalar-contact Hartree functional *does not* dynamically generate that extra coordinate: the scalar stiffness remains nonzero, and the independent density Legendre coordinate has positive curvature whenever occupied. The (M, n) two-coordinate response is therefore *not* a realization of the QM015 massless-mixing theorem. No Ward symmetry of this microscopic interaction protects the required $1/k^2$ scalar pole. In the occupied, isotropic phase the scalar-to-helicity-two stress mixing also vanishes by the earlier QM007 rotational selection rule; introducing the density coordinate does not manufacture the missing TT excitation.

Within the local Hartree–Fock truncation, $\partial_{p_\mu} \mathcal{D}_{\mathrm{HF}} = \gamma^\mu$ under a source-state perturbation even when $M(x)$ changes; Eq. (63) still gives a vanishing source-to-kinetic principal-symbol deformation. This is a *separate and indispensable* obstruction to universal QGD-like propagation, not something

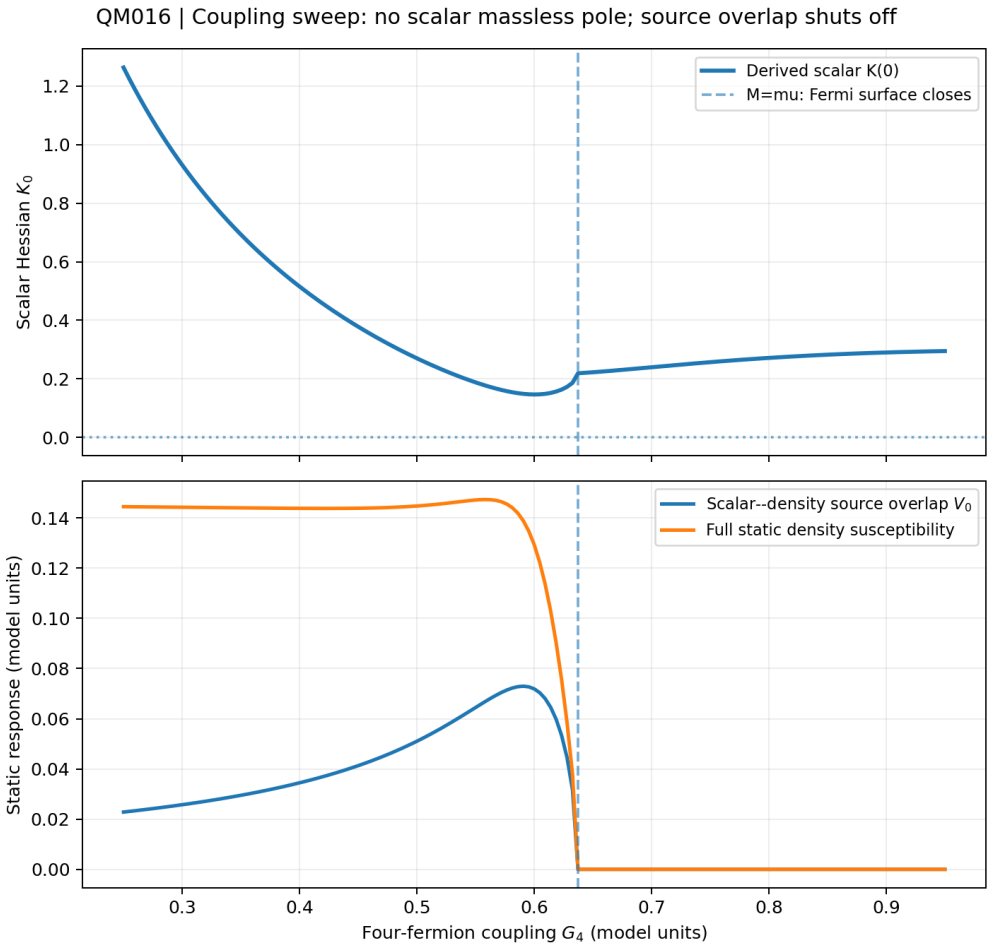


Figure 8: QM016 nonlinear gap-equation coupling sweep for the Chapter 1 sharp $q = 0$ contact model. The scalar curvature stays positive across the tested interval. When the Fermi surface closes at G_4^*, the number-source scalar overlap goes to zero. The analytic positivity theorem applies to the specified sharp-cutoff domain even beyond the finite sweep; the plot is a numerical cross-check, not a demonstration of universal quantum gravity.

repaired merely by increasing the static scalar susceptibility. These statements do not extend to the full nonlocal $\Sigma_{\mathrm{Dirac}}(p, x)$ at higher skeleton/2PI order.

Physical-source caveat. The perturbation $\delta\mu$ is a conserved-number external source *inside the chosen finite-density fermion medium*. It is not a compact object's full independent stress tensor, and the homogeneous medium is not an isolated asymptotically empty exterior. A second matter species χ which factorizes exactly from the original ψ action has zero connected χ–ψ cross-response, so an independent-source coupling must be generated or specified in the microscopic theory; the ψ density susceptibility cannot silently be renamed the universal mass-source vertex. This distinction is necessary for an honest $\rho_m \to [\Sigma]$ selection test.

22.5 QM016 decision and the next actually different microscopic calculation [O]

QM017-1. Construct a *single consistent* 2PI $\Phi[G]$ or Bethe–Salpeter kernel at the next genuinely momentum-dependent order for the Chapter 1 interaction. The new physics must be a frequency- and momentum-dependent Dirac self-energy/vertex, not merely re-diagonalization of the already-positive (M, n) static block.

QM017-2. Couple independent conserved stress sources as *bookkeeping probes of the original flat-space QFT*, including the needed stress-contact/seagull insertions; do not promote their metric source into an independently integrated field by Claude's unadopted premise (Q). Evaluate the retarded constrained $(T^{00}, T^{0i}, T^{ij}, S, J^{\mu}, \partial_p\Sigma)$ response and the helicity decompositions.

QM017-3. Verify the static density compressibility Ward/sum-rule and the energy-momentum Ward identities from the same functional approximation. A Φ-derivable approximation enforces specified conservation laws but does not automatically satisfy all Ward–Takahashi identities for truncated internal vertices; document the external conserving vertex separately. See Baym–Kadanoff, https://doi.org/10.1103/PhysRev.124.287, and van Hees–Knoll, https://doi.org/10.1103/PhysRevD.66.025028.

QM017-4. Test an isolated, positive-residue, source-visible infrared pole *and* a nonzero universal source-to-Dirac-principal-symbol response. Only after both checks should we attempt the source/probe-independent natural-invariant lift and the entire complete coframe/$\mathcal{Q}$ QGD metric.

Conclusion of QM016. A newly derived analytic Hartree curvature identity excludes tuning the scalar to a massless source-visible pole for $\Lambda/\mu \geq e$, $m_0 > 0$ and the other stated conditions. The full static two-coordinate number–scalar thermodynamic Hessian is positive on the occupied branch, while the source overlap vanishes as the Fermi sea empties. The original Chapter 1 contact theory has therefore *not* supplied the gapless source-visible mixed collective mode assumed only in QM015; the earlier conditional Schwarzschild theorem remains intact. The next relevant experiment must introduce *new dynamics computed from the same microscopic theory*, not a relabeling of the finite Hartree susceptibility as gravity.

23 QM017–QM019: Does microscopic matter produce a universal kinetic geometry?

Date: 2 November 2026. Research status: mixed constructive result and a decisive failure of the declared minimal mechanism; not a derivation of gravitational emergence. This section follows QM016 through multiple tests rather than stopping at a positive self-energy coefficient. We maintain the historical target

$$\mathcal{D}_{\text{eff}}\Psi = 0 \longrightarrow (\Psi, S, \mathsf{R}, j^{\mu}) \overset{\mathcal{F}_{\text{micro}}}{\longrightarrow} [\Sigma], \tag{114}$$

and use the operational source formulation

$$\boxed{(S_{\text{micro}}, \hat{\varrho}_{\text{source}}, \text{BC}) \longrightarrow g_{\mu\nu}^{\text{physical}} \longrightarrow [\Sigma]_{\text{physical}}.} \tag{115}$$

The second arrow is an *algebraic* local equivalence on a regular, fixed-coframe, $\lambda\mathcal{Q} = 0$, spanning signed-sector branch. The first arrow requires independent physical dynamics, a source-supported exterior and cross-species universality.

23.1 QM017A: a concrete nonlocal microscopic kinetic response, and its scope [D/C]

The Chapter 1 benchmark at $M = 0.693435535520$, $\mu = 1.2$ has $p_F = \sqrt{\mu^2 - M^2} = 0.979360586341$. QM011 independently obtained the *infrared* static HS scalar Hessian $K_s(k) = K_0 + Zk^2 + O(k^4)$ for matched smooth regulators. As an explicitly defined screened-exchange *diagnostic*, continue this quadratic expression into the internal momenta of a positive-energy Fermi sea:

$$D_s(\boldsymbol{q}) = \frac{1}{K_0 + Z\boldsymbol{q}^2}, \qquad \xi = \sqrt{Z/K_0}, \quad K_0, Z > 0. \tag{116}$$

The use of this D_s at all $|\boldsymbol{q}| \le 2p_F$ is a *model extension*, not an independently computed full-q Dirac/HS propagator. In particular, QM011's low-k fit does not fix the complete exchange integral. The following coefficient is therefore a reproducible constructive proxy, *not* a self-consistent 2PI prediction of the original contact action.

For a scalar Yukawa normalization g_*^2 (set to one only for the numerical table), take the static positive-energy exchange self-energy, in an explicitly specified rest-frame Dirac convention,

$$\Sigma_F(\boldsymbol{p}) = g_*^2 \int_{|\boldsymbol{k}|<p_F} \frac{d^3k}{(2\pi)^3} \frac{\gamma^0 E_k - \boldsymbol{\gamma}\cdot\boldsymbol{k} + M}{2E_k} \frac{1}{K_0 + Z|\boldsymbol{p}-\boldsymbol{k}|^2}, \qquad E_k = \sqrt{k^2 + M^2}. \tag{117}$$

The overall Fock sign and the normalization of the effective auxiliary Yukawa vertex are fixed *here by definition*; they cannot be imported as an uncomputed sign or coupling of a covariant 2PI skeleton. Different microscopic conventions can reverse the displayed sign of the kinetic correction, not its nonzero magnitude.

Proposition 23.1 (Nonzero spatial kinetic derivative in the screened-exchange diagnostic [D/C]**.** *] The odd-in-$\boldsymbol{p}$ spatial part of Eq. (117) has*

$$\boxed{\Sigma_F(\boldsymbol{p})\big|_{\boldsymbol{\gamma}\cdot\boldsymbol{p}} = -C_s\,\boldsymbol{\gamma}\cdot\boldsymbol{p} + O(|\boldsymbol{p}|^3), \quad C_s = \frac{g_*^2 Z}{6\pi^2}\int_0^{p_F} dk\, \frac{k^4}{E_k(K_0 + Zk^2)^2} > 0.} \tag{118}$$

At fixed M, K_0, Z, its partial, *not fully self-consistent, density response is*

$$\boxed{\left.\frac{\partial C_s}{\partial \mu}\right|_{M,K_0,Z} = \frac{g_*^2 Z p_F^3}{6\pi^2(K_0 + Zp_F^2)^2} > 0, \quad \left.\frac{\partial C_s}{\partial n}\right|_{M,K_0,Z} = \frac{g_*^2 Z p_F^2}{6\mu(K_0 + Zp_F^2)^2},\ n = \frac{p_F^3}{3\pi^2}.} \tag{119}$$

Proof. At $\boldsymbol{p} = 0$, $\partial D_s(|\boldsymbol{p} - \boldsymbol{k}|)/\partial p_j = 2Zk_j/(K_0 + Zk^2)^2$. The time-like and scalar numerators integrate to zero after multiplication by the odd function k_j. The spatial numerator $-\gamma^i k_i/(2E_k)$ gives $-g_*^2 Z\gamma^i p_j \int d^3k\, k_i k_j/[(2\pi)^3 E_k(K_0+Zk^2)^2]$. Rotational integration substitutes $k_i k_j \mapsto \delta_{ij}k^2/3$, proving Eq. (118). Differentiate the moving upper limit $p_F = \sqrt{\mu^2 - M^2}$ while holding the other parameters fixed to prove Eq. (119). The fully physical derivative also includes $M'(\mu)$, $K_0'(\mu)$, $Z'(\mu)$ and the appropriate vertex corrections, which are *not* included here. □

QM011 scheme	K_0	Z	ξ	C_s at $\kappa = p_F$
Smooth $n = 8$	0.14993818	0.04864381	0.56958383	0.00422660
Smooth $n = 16$	0.15737518	0.08155175	0.71986081	0.00532135

At $\kappa = 0.2p_F$, the same coefficients are 2.9039×10^{-6} and 4.3733×10^{-6}, respectively, so an actual nonzero contribution is already present in the conservative small-internal-momentum part. At the full p_F the quoted values depend on continuing the $O(k^2)$ HS kernel beyond the regime where QM011 controlled the fit. Direct Gauss–Legendre angular integration of Eq. (117), followed by a symmetric difference in the external p_z, agrees with Eq. (118). The separately differentiated upper-limit response agrees with a numerical μ derivative at fixed M, K_0, Z. The verifier supplies every test and tolerance.

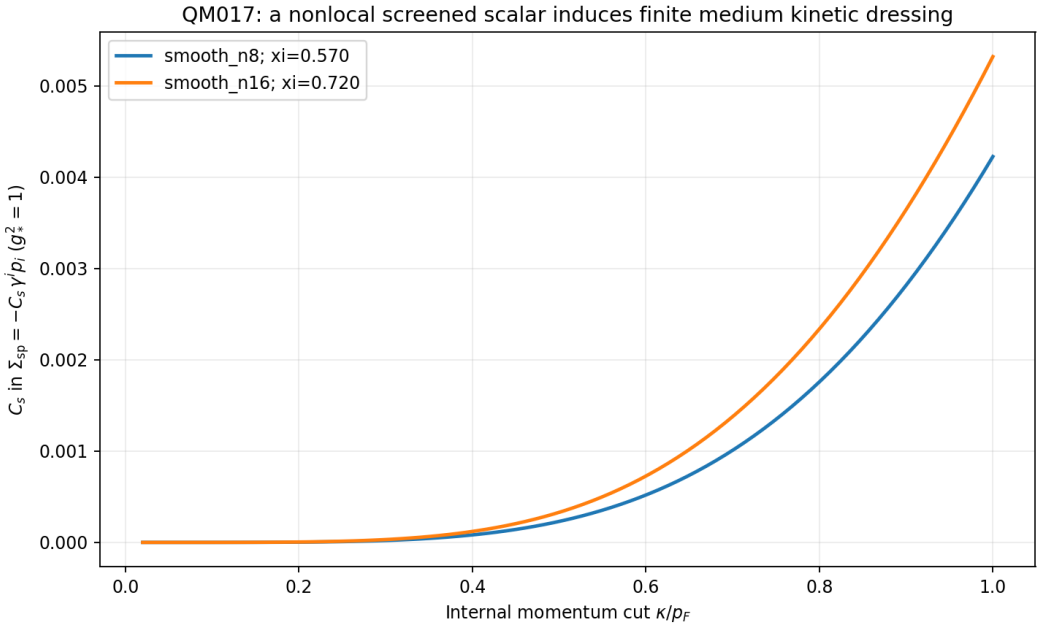


Figure 9: QM017 screened exchange: a nonzero medium kinetic coefficient appears when an already screened HS collective channel is permitted to carry momentum. This is a controlled diagnostic of a *specified* low-momentum extension, not a full 2PI computation or a gravitational source measurement.

23.2 QM017B: continuation to the actual exterior response fails [D/C]

The same positive HS Hessian has the exact three-dimensional *gradient-model* Green function

$$\boxed{D_s(r) = \frac{e^{-r/\xi}}{4\pi Z r}, \qquad \lim_{k\to 0} k^2 D_s(k) = 0.} \tag{120}$$

For the two QM011 matched schemes, $\xi = 0.56958$ and 0.71986 in model inverse-mass units. Hence this propagator does *not* deliver the $1/k^2$ source-response residue or asymptotic $1/r$ exterior

required by the Chapter 5 static QGD target. It can create finite state-dependent kinetic dressing inside a medium without selecting a distant gravitational field. A real gapless Fermi sea may also exhibit nonanalytic dynamic particle–hole and $2p_F$ effects beyond this gradient approximation; they are not automatically an isolated gravitational pole.

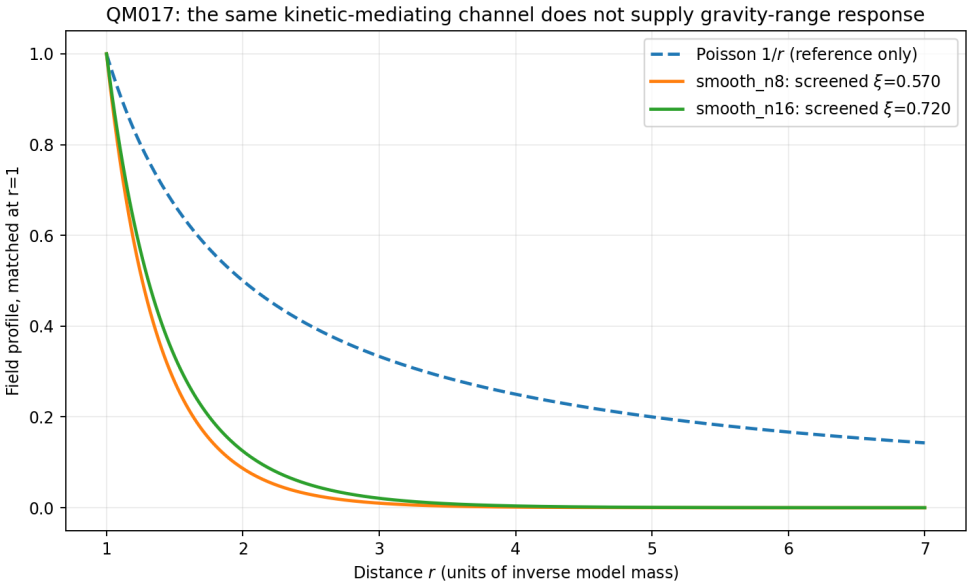


Figure 10: Distance test: screened QM011 scalar-gradient model versus the conditional QGD $1/r$ response, with the profiles normalized at $r = 1$. The scalar does not become gravitational merely because its exchange also perturbs a Dirac dispersion.

A necessary technical distinction. The derivative $\partial\Sigma/\partial p_i$ in Eq. (118) modifies a *local low-energy expansion around a chosen momentum*. This does not automatically replace the exact ultraviolet/pseudodifferential principal symbol, and a momentum-dependent nonlocal self-energy may not be describable by one local metric. A frequency-dependent time-like coefficient and consistent retarded vertex are missing. Thus the result disproves a too-strong assertion that matter interactions can *never* alter kinetic coefficients, but it does not meet QGD's universal local-geometry target.

23.3 QM018: exact all-orders obstruction to universal response of a decoupled spectator [D]

Let ψ describe the source and let χ be an independent neutral probe, with factorizing *dynamics*

$$H = H_\psi \otimes I_\chi + I_\psi \otimes H_\chi, \quad O_{\rm src} = O_\psi \otimes I_\chi, \quad O_{\rm pr} = I_\psi \otimes O_\chi. \tag{121}$$

This model is an explicit extension of the minimal source-only microscopic action by a *genuinely decoupled* spectator. Its purpose is to test the universality claim rather than to assert that all particles in an unrestricted microscopic QFT must be decoupled.

Proposition 23.2 (Cross-sector retarded response vanishes for every initial state [D]**.** *] For any initial density operator $\hat{\varrho}$ on the tensor-product Hilbert space,* including an entangled state*, the exact retarded response obeys*

$$\boxed{\chi^R_{\rm pr,src}(t) = -i\theta(t)\,{\rm Tr}(\hat{\varrho}[O_{\rm pr}(t), O_{\rm src}(0)]) = 0.} \tag{122}$$

No resummation or higher loop order internal to the ψ sector can make a causally independent χ particle acquire the source's gravitational propagation. If QGD claims universal influence on χ, its complete microscopic theory must generate an actual interaction, common constraint or other physically specified connection.

Proof. The time-evolution operator factorizes into $U_\psi(t)\otimes U_\chi(t)$ under Eq. (121). Thus $O_{\rm pr}(t) = I_\psi \otimes O_\chi(t)$, which commutes identically with $O_{\rm src}(0)$. The operator commutator vanishes independently of $\hat{\varrho}$. Equivalently, with independent external source insertions and factorizing path-integral measure, $W[J_\psi, J_\chi] = W_\psi[J_\psi] + W_\chi[J_\chi]$, so $\delta^2 W/\delta J_\psi \delta J_\chi = 0$. The latter path-integral argument assumes a product preparation; the operator argument does not. The verifier constructs an entangled Bell-state example with connected equal-time correlation equal to one, while its retarded cross-response is zero at every tested time. □

Logical boundary. The original F1 also permits coupling to gauge fields, and an extended microscopic theory could contain other common interactions. The proposition does *not* prove an all-theories no-go for emergent gravity or that a photon with electromagnetically charged matter is exactly decoupled. It does establish that *the minimal isolated scalar-contact source sector alone* cannot universally influence a separate causally uncoupled probe. A physical QGD law must pass this cross-species test rather than silently assign its candidate scalar response to all probes.

23.4 QM018B: scalar-resummed TT sector remains unexcited in the isotropic phase [D/C]

In a parity-even isotropic state the scalar–spatial-stress vertex at a static wavevector must have the tensor structure $\Pi_{ij,S} = a(k^2)\delta_{ij} + b(k^2)k_i k_j$. A transverse-traceless polarization satisfies $e^{ij}_{\rm TT}\delta_{ij} = e^{ij}_{\rm TT}k_i = 0$, hence

$$\boxed{e^{ij}_{\rm TT}\Pi_{ij,S} = 0, \qquad e^{ij}_{\rm TT}\left(\Pi_{ij,S}D_s\Pi_{S,mn}\right)e^{mn}_{\rm TT} = 0.} \tag{123}$$

The zero remains exact for *any* scalar-only resummation D_s preserving the stated rotational structure, including an artificially tuned scalar critical point. It is not an assertion that all possible tensor interactions or genuinely dynamical spin-two composites are absent; it closes the proposed route from a *single isotropic scalar HS mode* to an isolated TT gravitational pole at this level.

23.5 QM019: an exact species-specific Clifford geometry is still not gravity [D/C]

Suppose a low-energy first-order truncation of one fermion's inverse operator has kinetic matrices Γ^μ_a. Define

$$g^{\mu\nu}_a := -\frac{1}{2d_a}\,{\rm tr}\{\Gamma^\mu_a, \Gamma^\nu_a\}, \quad \mathcal{R}^{\mu\nu}_a := \{\Gamma^\mu_a, \Gamma^\nu_a\} + 2g^{\mu\nu}_a I_{d_a}. \tag{124}$$

A single effective Clifford metric exists in the stated convention only when every $\mathcal{R}^{\mu\nu}_a = 0$ with a nondegenerate Lorentzian g_a (plus the necessary spin-connection, locality, physical normalization and causal conditions). Different probes must agree on a *common* metric to justify QGD universality, modulo physically appropriate conformal freedom for strictly massless characteristics.

For the *illustrative* screened first-order approximation $\Gamma^0_\psi = \gamma^0$ and $\Gamma^i_\psi = (1 + C_s)\gamma^i$ (overall self-energy sign convention from Eq. (117)), the Clifford test passes exactly, yielding

$$\boxed{g^{\mu\nu}_\psi = {\rm diag}(-1, (1 + C_s)^2, (1 + C_s)^2, (1 + C_s)^2).} \tag{125}$$

The decoupled spectator instead has $g^{\mu\nu}_\chi = \eta^{\mu\nu}$; for the $n = 8$ benchmark $\|g_\psi - g_\chi\|_F \simeq 0.01467233 \neq 0$. This effective *species-specific medium geometry* is not a unique universal spacetime metric.

Moreover, Eq. (125) is a restricted static first-order IR model, not a demonstrated full causal Dirac operator.

The complete QGD founding metric is retained without identifying this medium tensor with a derived gravitational metric:

$$\boxed{g_{\mu\nu} = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} - \sum_{s} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right], \qquad s_s = -\varepsilon_s.} \tag{126}$$

In the fixed-reference, $\lambda\mathcal{Q} = 0$ eight-sector specialization, *any* sufficiently regular metric perturbation $h = g - \eta$ can be algebraically represented by $P = bI + h/(4b)$ and $N = bI - h/(4b)$, for $b \neq 0$: $P^T P - N^T N = h$. The QM019 verifier independently checks this exact signed lift of both illustrative g_ψ. *Therefore the mere ability to assign signed natural invariants to a medium's optical metric is not proof that a unique, source-derived universal gravitational* $[\Sigma]$ *has emerged.* This is a constructive counterexample to a superficially persuasive but logically incomplete micro-to-Σ claim.

23.6 Decisive research verdict and the restricted path forward

Test	Result and exact scope
Kinetic dressing appears when momentum-dependent screened exchange is included	**YES [D/C]** within the declared low-k-extended, static, frozen-HS diagnostic; not completed covariant 2PI
Asymptotic inverse-Laplacian pole of the same scalar mode	**NO [D/C]**: $K_0 > 0$, $k^2 D_s(k) \to 0$, Yukawa decay in its gradient truncation
Universal response for a completely independent spectator	**NO [D]**: exact zero retarded Kubo cross-response for arbitrary initial state
TT graviton from isotropic single scalar exchange	**NO [D/C]** at the stated scalar-resummed RPA level by rotational representation theory
Clifford metric of one dressed Dirac species	**YES [D/C]** algebraically, but it differs from the independent spectator's flat metric
Full microscopic source → common metric → $[\Sigma]$	**NOT ESTABLISHED [O]**; cannot be inferred from any of the preceding five observations

Decision. The minimal scalar-contact/HS mechanism tested through these approximations *fails* the combined long-range and universality criteria. This is a precise failure of the declared candidate, not of QGD's effective gravitational correspondence or every possible underlying QFT. Increasing exchange order in a decoupled source sector cannot repair the exact cross-species causal factorization. Adding a genuine universal microscopic interaction/constraint, or deriving a nontrivial shared emergent collective mediator, is a separate physical hypothesis that requires its own action and Ward-completed proof. Claude's independent auxiliary-Σ premise (Q) stays a clearly labeled alternative, not an adopted derivation.

What would count as the next real breakthrough? Derive a single state-dependent, conserved, causal, source-visible *tensorial* collective response whose physical pole or constraint-mediated field is shared by two otherwise distinct probe species. Compute from one consistent regulated microscopic action (i) its complete physical $T^{\mu\nu}$-source vertex and full nonlocal dynamic Hessian, (ii) the Ward-completed fermion and electromagnetic probe kinetic responses, (iii) the long-distance static source residue and time-dependent homogeneous sector, and (iv) the common Clifford/null characteristics. Only then use the exact QGD signed map to construct $[\Sigma]$; do not use that algebraic map to manufacture the input response. For a massless universally coupled helicity-two composite inside an otherwise ordinary flat Lorentz-covariant stress theory, the Weinberg–Witten hypotheses and their possible violations must also be treated at the microscopic level.

External methodological comparators. Ward-completed 2PI approximations: https://doi.org/10.1103/PhysRevD.66.025028. Stress-tensor measures and identities: https://doi.org/10.1103/PhysRevD.23.2262. Perturbative nonlocal Fermi-liquid kinetic self-energy is a standard physical possibility, e.g. the Galitskii low-density $m^*/m = 1+[8(7\ln 2-1)/(15\pi^2)](k_F a)^2+\cdots$ coefficient (this is a different nonrelativistic, renormalized model and is a *comparator*, not a computation of the relativistic QGD contact benchmark). Universal soft-graviton coupling is conditional on a real massless spin-two pole and Lorentz-invariant scattering: https://doi.org/10.1103/PhysRev.135.B1049. Weinberg–Witten theorem: https://doi.org/10.1016/0370-2693(80)90212-9.

Reproducibility. Run `verify_qm017.py`, `verify_qm018.py`, and `verify_qm019.py`; inspect the generated JSON, figures and the explicit approximation scopes. No existing v1.0 result has been reclassified as a full microscopic gravitational derivation.

24 QM022: A spinor-optical construction of a Kerr–Schild weighted basis

Research scope. The objective is the founding physical map from a quantum source state to the *physical equivalence class* of natural invariants, not a derivation of gravity from a separate scalar force. We develop exact algebraic/kinematic results and test a restricted constitutive candidate. The projective spinor field below is *constructed from the Kerr reference congruence*; its origin as a solution or observable of the original microscopic Dirac/contact theory is **open**. Previous QM017–QM019 obstructions and all Chapters 1–5 assumptions remain unchanged.

24.1 The complete master metric and its restricted null sector

The complete founding equation remains

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \big[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\big], \quad s_s = -\varepsilon_s.} \tag{127}$$

Our exact optical construction concerns *only* a fixed flat reference with $\lambda\mathcal{Q} = 0$, one positive signed null update $g = \bar{g} + Hk \otimes k$, and $\sigma = \sqrt{H}\, k$ where $H \geq 0$. A general signed eight-sector realization remains available but is not the microscopic identity of eight independent quantum spinors.

24.2 Pure chiral spinors supply exact null directions, not unique gravity

For a nonzero two-spinor $\chi = (u, v)^T$, let $N = \chi^\dagger\chi$ and $n^i = \chi^\dagger\tau^i\chi/N$. The Pauli matrix identity gives

$$(\chi^\dagger \boldsymbol{\tau} \chi)^2 = (\chi^\dagger\chi)^2, \quad |\boldsymbol{n}| = 1, \quad j_\chi^\mu = N(1, \boldsymbol{n}), \quad \eta_{\mu\nu} j_\chi^\mu j_\chi^\nu = 0. \tag{128}$$

For a regular four-spinor, its two chiral currents are separately null, $j^\mu \pm j_5^\mu = 2j_{R/L}^\mu$, up to the convention for γ^5. This is an exact spinor-to-null-*direction* bridge, not a physically selected exterior gravitational covector. In particular a general incoherent ensemble does not satisfy the pure-spinor Fierz identities for its averaged bilinears.

For a positive Hermitian chiral coherence matrix $C_{A\dot{B}} = \langle \chi_A \chi_{\dot{B}}^\dagger \rangle$, write $J^0 = \operatorname{tr} C$ and $J^i = \operatorname{tr}(C\tau^i)$. Then

$$\boxed{(J^0)^2 - |\boldsymbol{J}|^2 = 4 \det C \geq 0.} \tag{129}$$

A nonzero averaged chiral current is null exactly when C has rank one; a generic full-rank mixture is timelike. For the unpolarized $C = I_2/2$, $J^\mu = (1, 0, 0, 0)$ and both eigendirections are degenerate. Thus a locally isotropic many-body state does not select a unique projective null direction by taking

a mean current. This is an exact obstruction to a *naive* averaged-spinor rule, not to nonlocal or dynamically selected congruences.

24.3 An explicit normalized Kerr projective spinor and its optical transport

Use $G = c = 1$, stationary flat oblate coordinates

$$x = \sqrt{r^2 + a^2}\sin\theta\cos\phi, \quad y = \sqrt{r^2 + a^2}\sin\theta\sin\phi, \quad z = r\cos\theta, \quad \Sigma = r^2 + a^2\cos^2\theta. \tag{130}$$

The following normalized spinor defines a projective Pauli direction:

$$\boxed{\xi = \begin{pmatrix} \cos(\theta/2) \\ e^{i\beta}\sin(\theta/2) \end{pmatrix}, \qquad \beta = \phi - \arctan(a/r), \quad Y = \xi_2/\xi_1 = e^{i\beta}\tan(\theta/2).} \tag{131}$$

In terms of the flat Cartesian position,

$$\boxed{n_x = \frac{rx + ay}{r^2 + a^2}, \quad n_y = \frac{ry - ax}{r^2 + a^2}, \quad n_z = \frac{z}{r}, \quad \boldsymbol{n}^2 = 1.} \tag{132}$$

Direct differentiation of the oblate coordinate embedding yields the *exact* Kerr–Schild null one-form

$$\boxed{k = dt + \boldsymbol{n}\cdot d\boldsymbol{x} = dt + \frac{\Sigma}{r^2 + a^2}dr - a\sin^2\theta\, d\phi.} \tag{133}$$

No Kerr metric or Einstein field equation enters Eqs. (131)–(133); the physical congruence, parameter a, reference slicing and normalization *are supplied as the ansatz*. The projective spinor necessarily needs a second chart at the south polar axis. The signature convention matters: $k_\mu = (1, \boldsymbol{n})$ has raised vector $k^\mu = (-1, \boldsymbol{n})$ in $(-+++)$. The *future-directed chiral current* aligned with the same null lines is supplied by the antipodal spinor

$$\zeta = i\tau_y\xi^* = \begin{pmatrix} e^{-i\beta}\sin(\theta/2) \\ -\cos(\theta/2) \end{pmatrix}, \quad \zeta^\dagger\boldsymbol{\tau}\zeta = -\boldsymbol{n}, \quad j^\mu_\zeta = \mathcal{N}(1, -\boldsymbol{n}) = -\mathcal{N}k^\mu. \tag{134}$$

Thus *one helicity/convention generates the Kerr spatial covector while the antipodal one produces the future-directed raised current* in this signature. Conflating covector components with current-vector components would introduce a sign error; the two are related by the metric and chirality choice, not by an unqualified identity.

Define the flat-reference null-ray transport $\mathcal{L} = \partial_r - a(r^2 + a^2)^{-1}\partial_\phi$ for time-independent fields (the full null tangent is $-\partial_t + \mathcal{L}$). The following identities hold *exactly*:

$$\boxed{\mathcal{L}\beta = 0, \quad \mathcal{L}\xi = 0, \quad \mathcal{L}\boldsymbol{n} = 0, \quad \mathcal{L}\boldsymbol{x} = \boldsymbol{n}.} \tag{135}$$

Thus each flat-reference ray is straight and the projective spinor orientation is parallel-transported along that ray; the congruence as a whole may still have nonzero twist. This transport identity is *not* the four-component curved Dirac equation nor the complete free Weyl equation.

24.4 Exact optical scalars, Kerr twist and a limited weight rule

With $\sqrt{\bar\gamma} = \Sigma\sin\theta$ the spatial flat-reference volume density and $n^r = 1$, $n^\theta = 0$, $n^\phi = -a/(r^2 + a^2)$, direct calculation gives

$$\boxed{\Theta := \nabla_{\bar\gamma}\cdot\boldsymbol{n} = \frac{2r}{\Sigma}.} \tag{136}$$

In a screen orthogonal to $\boldsymbol{n}$ the optical derivative is, up to screen orientation,

$$\boxed{B_{AB} = \frac{1}{\Sigma}\begin{pmatrix} r & a\cos\theta \\ -a\cos\theta & r \end{pmatrix}, \quad \mathrm{shear}(B) = 0.} \tag{137}$$

The spacetime-null covector is *not* generally a rescaled real eikonal gradient:

$$[k \wedge dk]_{t\theta\phi} = -2a\sin\theta\cos\theta, \quad \boldsymbol{n}\cdot(\nabla\times\boldsymbol{n}) = -\frac{2a\cos\theta}{\Sigma}. \tag{138}$$

A single smooth S cannot supply $k \propto dS$ on a twisting open set. This is fully compatible with Chapter 1's independent phase S, spinor orientation R and current j^μ: none should be equated to the gravitational natural invariant without a microscopic law.

Restricted constitutive identity [D/C]. For the Kerr and Schwarzschild *vacuum* family, the exact gravitational Kerr–Schild weight is

$$\boxed{H_{\mathrm{Kerr}} = \frac{2Mr}{\Sigma} = M\Theta = M\,\nabla_{\bar\gamma}\cdot\boldsymbol{n}.} \tag{139}$$

This combines one charge M with a first derivative of the projective-spinor direction. It is an identity for this normalized stationary Kerr family, *not* an experimentally established or microscopic universal selection law. The normalization $k_t = 1$ and stationary flat reference fix its meaning. In ordinary units the coefficient is $G_N M/c^2$ and $a = J/(Mc)$.

A more useful variable is the *mass aspect* $\mathcal{M} := H/\Theta$ (defined where $\Theta \neq 0$). For the chosen Kerr family, $\mathcal{M} = M$; for aligned Kerr–Newman electrovac,

$$\boxed{\mathcal{M}_{\mathrm{KN}} = M - \frac{Q^2}{2r}, \qquad H_{\mathrm{KN}} = \Theta\left(M - \frac{Q^2}{2r}\right)} \tag{140}$$

with Q geometrized. Write $H = 2f(r)/\Sigma$, $f = r\mathcal{M}(r)$. The independent equatorial angular vacuum Einstein equation from QM020 reads

$$\boxed{R_{\theta\theta}|_{\theta=\pi/2} = \frac{2(rf' - f)}{r^2} = 2\mathcal{M}'(r).} \tag{141}$$

It selects $\mathcal{M}'(r) = 0$ in vacuum within the ansatz. *This equation follows from the adopted macroscopic Einstein dynamics, not the microscopic Dirac theory.* The identity (139) does not secretly derive it from fermions.

24.5 Why the conserved spinor current does not select the Kerr weight

A stationary conserved null current along this reference congruence can have the form $j^\mu = \mathcal{N}\ell^\mu$, with the future tangent $\ell = \partial_t - \mathcal{L} = -k^\sharp$ and

$$\boxed{\nabla^{\mathrm{flat}}_\mu j^\mu = 0 \quad\Longrightarrow\quad \mathcal{N}(r,\theta,\phi) = \frac{C(\theta,\beta)}{\Sigma}} \tag{142}$$

along the rays, on an open region where the congruence is defined and with stationary boundary data. The ray-constant C is not fixed by continuity. Even if a microscopic chiral spinor is aligned with this congruence, its conserved number-current density is proportional to Σ^{-1}, whereas the gravitational Kerr weight is proportional to $r\Sigma^{-1}$. Therefore current conservation plus a correct twisting direction does *not* produce the physical gravitational weight. The scalar Θ in Eq. (139) contains the extra radial factor; the gravitational identity is a *separate constitutive assumption* unless a microscopic action supplies it.

24.6 Current conservation is weaker than microscopic Weyl dynamics

A further physical test must not be skipped: the exact null-ray transport $\mathcal{L}\xi = 0$ and the conserved current $\mathcal{N} \propto \Sigma^{-1}$ do not imply that the spinor solves a microscopic wave equation. For the simplest real current-normalized *opposite-helicity* choice $\psi_{\rm trial} = \Sigma^{-1/2}\zeta$ aligned with the future tangent of Eq. (134), we evaluated the *flat, stationary* Weyl residual $\boldsymbol{\tau} \cdot \boldsymbol{\nabla}_{\rm Cartesian}\psi_{\rm trial}$, differentiating through the full oblate-to-Cartesian Jacobian. Its norm is nonzero: 0.0401770 at $(r, a, \theta, \phi) = (2.5, 0.4, 0.8, 0.3)$, 0.0313335 at $(4, 1.5, 1.3, 1.2)$, and 0.0340614 even for the Schwarzschild-direction control $(3, 0, 1.1, 0.5)$ in the stated length normalization. This **[D/C]** counterexample proves only that *this specific spinor amplitude and phase* are not full stationary free Weyl solutions. Another complex amplitude, phase, time dependence or interaction may change the result. The weaker continuity equation cannot replace a check of the actual Dirac/Weyl operator.

24.7 Failure and nonuniqueness tests; why this is not a hidden complete $\mathcal{F}_{\rm micro}$

(1) Non-universality of $H = M\nabla \cdot \boldsymbol{n}$ [D]. Let $\boldsymbol{n} = (\cos\kappa x, \sin\kappa x, 0)$ in a fixed Cartesian Minkowski reference. It is a legitimate unit projective-spinor direction. The proposed weight gives $H = -M\kappa\sin(\kappa x)$. For $g = \eta + H(dt + \boldsymbol{n}\cdot d\boldsymbol{x})^2$, the stationary *linearized* vacuum equation would require $R^{(1)}_{tt} = -\frac{1}{2}\nabla^2 H = 0$, but instead

$$\boxed{R^{(1)}_{tt} = -\frac{1}{2}M\kappa^3\sin(\kappa x) \neq 0 \quad \text{generically.}} \tag{143}$$

Thus arbitrary Pauli-direction fields do not solve EFE under the divergence rule. The congruence's optical/differential restrictions are essential. The same rule gives $H = 0$ for a parallel null direction, whereas nontrivial pp-waves with parallel null rays exist; it cannot be a universal solution engine.

(2) Conserved global charges alone are insufficient [D/C]. Two equal, static masses $M/2$ at positions $\pm d\hat{z}$ and at $\pm d\hat{x}$ have identical total M and $J = 0$, but their zz trace-free quadrupole values are respectively $Q_{zz} = 2Md^2$ and $Q_{zz} = -Md^2$ (when the weights sum to M). Their weak-field exterior metrics therefore differ. The Kerr rule $a = J/M$, $H = M\Theta$ cannot represent every physical source with the same charges; extra density-matrix multipoles and boundary structure are necessary. The special Kerr/no-hair multipole relation is a further conditional physical restriction, not a generic consequence of M, J.

(3) Local exterior obstruction [D/C]. If an alleged $\mathcal{F}_{\rm micro}$ is a strictly pointwise finite-jet functional of local matter observables, and those observables coincide with the same vacuum values outside every compact source, then its output there is the same for every source. Such a map cannot distinguish Schwarzschild masses or Kerr angular momenta in a common exterior region. Consequently a successful map must violate at least one premise: involve nonlocal source-state information, a dynamically transmitted or constrained collective field, state-sensitive vacuum correlations, or other independent exterior/boundary data. This is an information argument, not a proof that a new fundamental force must be postulated.

(4) Berry phase is not spacetime twist [D]. The phase connection of the chosen normalized spinor is

$$\mathcal{A}_\xi = -i\xi^\dagger d\xi = \sin^2(\theta/2)\left(d\phi + \frac{a}{r^2 + a^2}dr\right), \quad (d\mathcal{A}_\xi)_{\theta\phi} = \frac{\sin\theta}{2}. \tag{144}$$

This Berry curvature remains nonzero at $a = 0$ even when the radial null congruence is twist-free. It cannot be identified with the spacetime Riemann tensor or the Frobenius twist of k without further structure.

24.8 Resulting physical research criterion

$$\boxed{\begin{array}{c}(S_{\text{micro}}, \widehat{\varrho}_{\text{source}}, \text{BC}) \xrightarrow{\mathcal{F}_{\text{physical}}} (\boldsymbol{n}_{\text{projective}}, \mathcal{M}_{\text{optical}}) \\ \longrightarrow [\Sigma]_{\text{physical}} \xrightarrow{\text{master metric}} g_{\mu\nu}.\end{array}} \tag{145}$$

On the restricted stationary Kerr branch a *kinematic candidate* is to extract the source charges M, J, supply a projective field $Y = \tan(\theta/2)e^{i[\phi - \arctan(a/r)]}$ with $a = J/M$, set $H = M\nabla_{\text{flat}} \cdot \boldsymbol{n}$ and $\sigma = \sqrt{H}\, k$. Its geometric output is exactly Kerr where the ansatz and vacuum conditions hold. What is *not* derived: why the original interacting quantum state chooses this Y among other projective fields, why its mass aspect obeys the vacuum source equation, how a vacuum exterior carries source information, whether the projective field solves the microscopic Dirac equation, how the general $\mathcal{Q}$ term behaves, and how universal matter species experience the same geometry. In particular an explicitly reconstructed Kerr spinor is not proof of microscopic gravitational emergence.

Next concrete test. Construct a *state-derived* chiral two-point/coherence transport or stress-kinetic collective observable for a rotating source, with its causal exterior propagation/constraint equation from the same microscopic action. Test whether its projective eigen-direction is ray-transported, geodesic and shear-free, whether its mass aspect is constant in an exterior vacuum without using EFE as an input, and whether both Dirac and electromagnetic probes couple to the same physical coframe. For general sources include more than M, J: spatial multipoles and nonstationary source data cannot be dropped. The mathematical Kerr spinor and divergence identity are exact *matching targets*; the physically selected $\mathcal{F}_{\text{micro}}$ remains **open**.

Reproducibility and provenance. Run `verify_qm022.py` and `verify_qm022_optical.py` for symbolic Pauli and Gram identities, exact null-ray transport, the optical divergence and screen shear, the current-amplitude mismatch, the source quadrupole counterexample, and the failed general divergence prescription. General projective-spinor/null-congruence mathematics is known from the Kerr theorem and spinor formalism (see, e.g., `https://link.springer.com/article/10.12942/lrr-2012-1` and `https://doi.org/10.1140/epjc/s10052-014-2854-y`); the new content here is the explicit QGD-specific physical-selection audit and its restricted constitutive candidate. The prior QM020–QM021 Einstein calculations supply the independent stationary Kerr and wider-solution baselines; they are not recast as microscopic derivations.

25 QM023: What the full QGD programme has accomplished and the newly fixed physical question

Scope. This section consolidates existing research results from Chapters 1–8, QM001–QM022 and the Chapter 2 QM020–QM021 supplements. It adds a declared working hypothesis and explicit acceptance criteria, *not* a new microscopic derivation or an independently verified QFT pole. The 1.2 manuscript is retained intact. Throughout, [D] denotes a proved identity or model-calculation under its declared assumptions, [C] a conditional effective-theory result, [A] a newly declared physical hypothesis, and [O] an open foundational claim.

25.1 The decision that fixes our next physical research direction [A]

New user-selected physical working hypothesis (2 November 2026). QGD should test whether the natural-invariant-producing structure is an *actual collective quantum field extending into the exterior vacuum*, not merely a nonlocal label assigned to the source state or a reconstruction from an already selected solution of Einstein's equations. "Actual" refers to degrees of freedom or state-dependent collective observables with operationally definable correlations, source response and causal evolution. It does *not* by definition imply a new fundamental elementary particle, an individual spinor expectation value, a prescribed massless pole, an a priori Einstein–Hilbert action or a nonzero Lorentz-breaking vacuum expectation value.

We introduce the *placeholder*, not an invented operator identity,

$$\widehat{\mathfrak{B}}^I(x) \quad \text{or a constrained collection of composite correlators} \quad \mathfrak{b}^I_\rho(x) = \langle \widehat{\mathfrak{B}}^I(x) \rangle_{\widehat{\rho}_{\text{source}}}, \tag{146}$$

where I is an unspecified collective/internal index and x includes the exterior region. If symmetry requires $\langle \widehat{\mathfrak{B}} \rangle = 0$, the gauge-invariant correlators or effective action, rather than this particular one-point function, must encode the field. Neither I nor the microscopic definition of $\widehat{\mathfrak{B}}^I$ is claimed to be known. The field may be composite, emergent or constrained; microscopic origin must be stated in the final theory, not hidden in notation.

CENTRAL QUESTION OF CHAPTER 6 — QM023 [O]

Starting only from a specified microscopic quantum-matter action, source density operator and physical boundary/initial data, can we derive a collective quantum field with a causal state and nontrivial exterior-vacuum dynamics, whose microscopic selection equations determine the weighted directions and weights of the QGD natural-invariant equivalence class $[\Sigma]_{\text{physical}}$*, without inserting the target Einstein metric, Einstein equations or gravitational Green function as its premise?*

Decisive verification: The resulting field must generate an operationally common effective geometry for distinct probes and reproduce the Einsteinian phenomenology in the appropriate macroscopic regime, including rotating twist, source multipoles, stationary/causal response and a homogeneous cosmological sector.

The original historical research arrow remains part of the programme:

$$\mathcal{D}_{\text{eff}}\Psi = 0 \longrightarrow (\Psi, S, \mathsf{R}, j^\mu) \overset{\mathcal{F}_{\text{micro}}}{\longrightarrow} [\Sigma]. \tag{147}$$

But the adequate *source-state formulation* must preserve additional information not contained in an arbitrary single probe spinor:

$$\boxed{(S_{\text{micro}}, \widehat{\rho}_{\text{source}}, \text{BC}) \overset{\text{derive}}{\longrightarrow} \mathfrak{B}_{\text{collective}}(x) \overset{\mathcal{F}_{\text{micro}}}{\longrightarrow} [\Sigma]_{\text{physical}} \overset{g[\Sigma]}{\longrightarrow} g^{\text{physical}}_{\mu\nu}.} \tag{148}$$

The derived microscopic field and the source-dependent map are open. The final metric arrow is the known QGD algebraic architecture. A second valid *operational* ordering, if the quantum collective field first determines common kinetic cones, is $\mathfrak{B} \to g_{\mathrm{micro}} \to [\Sigma]$; these are not competing ontologies when they identify the same equivalence class in a regular metric-only branch.

25.2 Why the weighted covectors work mathematically: exact statement, limited physical claim [D/C]

Do not replace the founding formula with its classical specialization. The complete master metric is

$$\boxed{g_{\mu\nu} = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} - \sum_{s} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right], \qquad s_s = -\varepsilon_s.} \tag{149}$$

For the *declared* fixed-coframe, $\lambda\mathcal{Q} = 0$ signed-sector specialization, the metric deformation is an indefinite Gram form, $h = \Sigma^{\mathsf{T}} J \Sigma$. Thus the weighted covectors collectively determine the ten metric components. On a regular spanning eight-sector branch, $\operatorname{rank}(D_\Sigma g) = 10$ and $\dim\ker(D_\Sigma g) = 22$; the null directions coincide with the $O(4,4)$ orbit tangents when the action depends on the sectors only through this metric. This establishes a *metric representation and quotient*, not the microscopic quantum origin of any particular sector representative. The counting changes on singular strata and generally when the historical $\mathcal{Q}_{ab}$ correction is active.

The covectors are *weighted basis-like directions*, not vectors each carrying its own scalar Riemann curvature. The Levi–Civita connection and Riemann tensor are constructed from the *complete* effective metric and its derivatives. The one-null-sector example $g = \bar{g} + Hk \otimes k$, $\sigma = \sqrt{H}\, k$ and $k_{\bar{g}}^2 = 0$ admits the exact coframe $e^a = \bar{e}^a + (H/2)k^a k$ and unchanged volume determinant. Kerr may have unchanged flat-reference volume determinant but nonzero Weyl curvature; Milne has varying coframe weights but vanishing four-dimensional curvature. Neither the absolute weight nor ∇H alone is a universal definition of gravity.

25.3 Consolidated achievement ledger: positive theorems, constraints and genuine failures

Interpretation rule. A verified negative result that excludes a tempting microscopic route is a scientific advance, but it is *not* evidence that an untested alternative must succeed. We distinguish exact algebra [D], conditional Einsteinian/effective matches [C], sourced microscopic model tests [D/C], and a genuine missing selection [O].

Chapter / test	Result	Status	Contribution to $\mathcal{F}_{\mathrm{micro}}$ and precise limit
Ch. 1	Exact local Dirac two-block ratio $u_B = \mathsf{R} u_A$	D/C	Physical spinor orientation/momentum relation; no independent gravitational source-amplitude selection.
Ch. 1	WKB, phase transport and conserved $r^2 j^r$	D/C	Quantum current/phase constraints; one phase cannot encode a twisting Kerr congruence.
Ch. 1	Large-N and collective semiclassical limits	D/C	Controlled macroscopic limits under specified scaling; generic many-fermion states retain correlations.
Ch. 1	Four-mode frame reparametrization	D	Same spinor may admit different mode frames; a freely chosen spinor decomposition cannot fix $[\Sigma]$.
Ch. 1	Mixed-state matter kinetic tensor can be full rank	D/C	Full source stress and correlations may encode more than mean current; rank is not a physical metric-selection law.

Chapter / test	Result	Status	Contribution / limit
Ch. 2	Independently varied spherical vacuum branch	C	$A^2 = B^2 = 2G_N M/(c^2 r)$ follows from adopted Einstein–Hilbert dynamics in the specified sector, not from flat quantum matter alone.
Ch. 2	Fermion-star source and rotating geometry	C	Constructive matter-to-metric-to-[Σ] pipeline *using EFE*; source information beyond a single current matters.
QM020–21	Kerr radial amplitude and broad Einstein solution checks	D/C	Rotating ansatz plus EFE selects $H = 2Mr/\Sigma$; 41 checks demonstrate geometry-engine reach, not micro origin.
QM021	Kerr $k \wedge dk \neq 0$	D	One scalar eikonal cannot generate the rotating null direction even after nonzero rescaling.
Ch. 3	Independent α_G and state normalizations	D/C	Wavefunction normalization does not determine G_N or the source's natural-invariant strength.
Ch. 3	Spin-singlet source with $j^r = 0$ and nonzero mass	C	Radial probability flux cannot universally serve as gravitational source variable.
Ch. 4	Full metric pullback and Option C Euler equation	D/C	Mathematically exact after adopted metric-mediated action; stationarity alone not micro genesis.
Ch. 4	Microscopic determinant/stress-correlator programme	D/C	Defined quantum observables and induced local terms; does not make the universal metric an output of the initial flat-matter model.
Ch. 5	Stable collective static functional	C	If source-coupled q exists, $G_{\text{eff}} = g_c^2/(4\pi Z)$ and $H = 2G_{\text{eff}} M/(c^2 r)$; existence of q remains assumed.
Ch. 5	Exact regular signed eight-sector lift	D	Any smooth metric deformation can be lifted locally on spanning branch; this is factorization, not selection.
Ch. 5	Kinetic principal-symbol metric candidate	D/C	Common probe cones could operationally identify g_{micro}; species dependence must be excluded.
QM006–07	TT continuum, stress matrix and Ward contact	D/C	Actual microscopic tensor spectral information; no isolated universal spin-two pole in tested approximation.
QM009	Inhomogeneous scalar source and finite response	D/C	Source-dependent Dirac mass/ratio changes; local Hartree–Fock kinetic principal symbol remains unchanged.
QM010–12	Stationary $H = \sigma^2_{\text{amp}}$ variable	D/C	Exact $1/r$ exterior from stipulated H energy; naive canonical $\lvert\nabla\sigma\rvert^2$ gives wrong exponent.
QM011	Actual Hubbard–Stratonovich scalar saddle	D/C	Microscopic stationary field exists; measured positive infrared curvature yields short-range response.
QM013	Induced curvature term under premise (Q)	C/A	Correctly scoped one-loop induction; independent Σ and universal metric coupling are supplied, not derived.
QM014	$O(4,4)$ rank/gauge and two-mode induced checks	D/C	Physical metric quotient validated on regular branch; full nonlinear microscopic constraint algebra remains unproved.
QM015	Spectral source-overlap theorem	D	Masslessness is insufficient unless the source excites the mode and the proposed geometric observable detects it.
QM015	Exactly soluble source-mixed long-range toy model	D/C	Indirect source coupling can yield $1/r$; model's shift symmetry and coefficients are assumptions.
QM016	Positive Hartree scalar curvature theorem	D/C	For positive-mass solutions with $\Lambda/\mu \geq e$, the tested sharp-cutoff scalar cannot be tuned massless by $G_4 > 0$.
QM016	Coupled scalar–density Hessian and sum-rule checks	D/C	Quantitative thermodynamic susceptibility, not a new propagating gravitational degree of freedom.

Chapter / test	Result	Status	Contribution / limit
QM017	Screened-exchange kinetic correction	D/C	Momentum dependence can alter a quasiparticle's principal symbol; measured screening prevents exterior Newtonian tail in this model.
QM018	Factorized spectator zero retarded response	D	A truly uncoupled species has no causal source response, even in initially entangled states.
QM019	Species-specific Clifford/signed lift	D/C	Algebraic metric reconstruction may succeed for one species while universal geometry fails.
QM022	Kerr projective-spinor null direction	D	Exact rotating weighted-direction reconstruction and shear-free ray transport, not independent spinor dynamics.
QM022	Optical weight identity and its counterexample	D/C	$H = M\nabla \cdot \boldsymbol{n}$ works in Kerr–Schwarzschild family, fails generally; current scaling does not fix H.
Ch. 7	Sector-orbit flat-vacuum cancellation and energy audit	D/C	Nonzero sector amplitudes can encode flat metric; pure-orbit gradient energy is not a geometric observable.
Ch. 8	FLRW lifts and no-extra-metric-equation result	D/C	Representation does not select cosmic time evolution; static inverse Laplacian lacks homogeneous zero-mode dynamics.

The net achievement. QGD has (a) a valid *representation* of geometric gravitational fields, (b) a well-defined *conditional effective action* recovering Einsteinian dynamics, (c) exact source-conditioned *static matching* if an appropriate collective mode is stipulated, (d) exact Dirac/optical *kinematic bridges*, and (e) several *genuine microscopic response calculations* which rule out specific scalar-only, screened, or species-specific shortcuts. No one item proves (f): a quantum-source-derived, universally coupled, exterior-vacuum collective weighted frame. The unexplained junction is chiefly (e)→(f), not (f)→Schwarzschild once EFE is adopted.

25.4 What "an actual collective quantum field in exterior vacuum" obliges us to explain

Three separate questions must be answered without circularity.

V1. Existence and definition. Which local or suitably quasilocal composite of the *specified* microscopic operators becomes the collective field $\widehat{\mathfrak{B}}(x)$? What is its symmetry/gauge class, coarse-graining scale, norm and commutator or constrained phase space? Does it exist outside a region containing material quanta, rather than only in a chosen description of a matter wavefunction? A classical nonzero mean is not required if physical correlators or constraints carry the gravitational information.

V2. Source imprint and exterior dynamics. What causal action or equations follow from the same microscopic interactions, source state and measure? The exterior may have no local material current, $J_{\text{matter}}(x) = 0$, yet a sourced field can be nonzero there if the theory supplies the necessary propagating/constraint/nonlocal dynamics and matching data. *An algebraic pointwise rule* $\mathfrak{b}(x) = f(\langle \widehat{O}_{\text{matter}}(x) \rangle)$ with vanishing exterior operator data cannot by itself generate a nonzero exterior field; vacuum polarization, nonlocal operators, boundary propagation or additional collective degrees must be explicitly justified.

V3. Weighted-frame selection and universality. Which physical, invariant observables of $\mathfrak{B}$ determine covector directions and weights, including rotating twist and multipoles? Why do electrons, neutral matter and photons register one common metric instead of

unrelated optical medium tensors? How do gauge/sector redundancies and an active historical $\mathcal{Q}_{ab}$ affect the observable class?

The choice of a spatially extended quantum field therefore sharpens, rather than automatically solves, the microscopic task. It is not enough to say that the original four-mode spinor "fills the vacuum"; one must derive a collective exterior quantum state and its field equations.

25.5 A non-circular functional architecture: a target for future derivation, not its result

One possible *working template* is to construct from $S_{\rm micro}$ and an initial/source density operator a causal collective effective functional (for real-time backgrounds ordinarily on a closed-time path)

$$\Gamma_{\rm coll}[\mathfrak{b}^+,\mathfrak{b}^-;\widehat{\rho}_{\rm source},{\rm BC}], \qquad \left.\frac{\delta\Gamma_{\rm coll}}{\delta\mathfrak{b}_I^+(x)}\right|_{\mathfrak{b}^+=\mathfrak{b}^-}=0, \tag{150}$$

with its source coupling and Ward identities derived from the *same microscopic model*. The location x must range through the exterior vacuum. These expressions are *desired definitions of a calculational problem*, not an asserted microscopic functional already obtained.

The missing constitutive map then has the generic form

$$\big[\Sigma(x)\big]_{\rm physical}=\mathcal{C}\big[\mathfrak{b}(x),\nabla\mathfrak{b}(x),\langle\delta\widehat{\mathfrak{B}}(x)\delta\widehat{\mathfrak{B}}(y)\rangle,\ldots;{\rm BC}\big] \tag{151}$$

with $\mathcal{C}$ to be independently derived and invariant under unphysical operator and sector coordinate changes. We do *not* preassign $\mathcal{C}=\sqrt{H}\,k$ as a universal law: that is one useful classical null-sector representative, and Kerr's k is twisting. The mapping may instead first yield common probe principal symbols, from which a regular signed representative can be constructed algebraically. In either case the full metric remains Eq. (149).

Key logical firewall. Defining the microscopic action *at the outset* as $S_{\rm micro}[\psi;g(\Sigma)]$ and subsequently integrating out ψ is the separate proposed premise (Q). Its induced Einsteinian action does not establish that Σ is a collective observable of the *original* flat-matter action. That route can be studied honestly as an extension, but not counted as completion of Eq. (148).

25.6 Why the existing Kerr/Schwarzschild bridge is powerful but not yet an origin theorem

Chapter 2 and QM020–QM022 establish the following exact restricted chain:

$$\xi_{\rm Kerr}\to k_{\rm twisting}\to\big(H=2Mr/\Sigma,\ \sigma=\sqrt{H}\,k\big)\to g_{\rm Kerr}. \tag{152}$$

Here $\xi_{\rm Kerr}$ is an explicitly *constructed* normalized projective spinor, the congruence is null/geodesic/shear-free, its twist obeys $k\wedge{\rm d}k\neq 0$ when rotating, and $H=M\nabla_{\rm flat}\cdot\boldsymbol{n}$ holds *on this Kerr/Schwarzschild family*. Chapter 2 independently selects the required radial amplitude through the *adopted vacuum Einstein equations* when the rotating congruence is already specified. QM022 proves that ξ and conserved current do not generally determine the Kerr weight: current density scales as $1/\Sigma$, not r/Σ. Nor is the simple geometric spinor automatically a free or interacting Dirac solution. A generic incoherent source has no unique pure null spinor current; M and J also do not determine generic source quadrupoles.

Therefore: a viable actual vacuum collective field must explain *both* the source-state selection of Kerr-like optical directions and the source-state selection of its weight/multipoles. These cannot simply be asserted from a spinor reconstructed after the classical geometry has been chosen. The Kerr optical example is a precise *target and kinematic possibility theorem*, not $\mathcal{F}_{\rm micro}$.

25.7 Why a fundamental Newtonian force is not part of the hypothesis [A/C]

The hypothesis is a sourced *weighted-frame quantum field*. Classical gravitational accelerations, lensing and curvature arise from the effective metric, its compatible connection and matter propagation. A $1/r$ exterior and the corresponding weak-field $1/k^2$ response are useful *phenomenological validation tests* in the ordinary Schwarzschild/Newtonian regime; they are not a separate primitive scalar-force ontology nor a universal requirement that the microscopic field be a simple Gaussian massless scalar. The analysis of QM011–QM019 remains relevant because it excludes concrete proposed *source-selection mechanisms*, not because it proves every emergent geometric mechanism must resemble a Yukawa or Newtonian potential.

The strongest falsification is not simply "does a scalar pole appear?", but *"can one and the same microscopic collective field state yield the geometry seen by independent species, including twisting vacuum configurations, consistent nonlinear and causal evolution, and the tested Einsteinian limit?"* Any explanation that only dresses one fermion's dispersion, changes its mass or invents a metric after the fact remains incomplete.

25.8 Five non-negotiable tests and the status after QM022

	Acceptance condition for the actual exterior quantum field	Present status
1	Derive an operator/correlator and source-conditioned state from the original microscopic quantum dynamics; define physical equivalences and a causal exterior equation.	**Open**; HS scalar exists but is not the required long-range frame.
2	Derive physical weighted directions (including $k\wedge \mathrm{d}k \neq 0$ when rotating) and weights, with independent source multipoles and boundary data; avoid a one-phase or one-current identification.	**Open**; Kerr projective-spinor reconstruction only.
3	Obtain the complete physical metric through the full master map or an independently generated common kinetic metric, and show that the selected class is not a sector-gauge artifact.	**Map proved conditionally**; physical selection open.
4	Demonstrate causal cross-species universality and consistency with matter stress/Ward identities; an uncoupled spectator must not be declared to gravitate.	**Open**; QM018 supplies exact counterexample.
5	Recover static, rotating, radiative and homogeneous/cosmological Einsteinian behavior in the valid effective regime, without assuming the requested solution as a microscopic input.	**Conditional classical matches**; no micro dynamics yet.

No double counting of results. EFE-generated solutions from Chapter 2/QM021, H-energy minimization from Chapter 5, induced Einstein curvature from an externally supplied $g(\Sigma)$, and Clifford factorization of a one-species kinetic tensor are independent conditional achievements. None should be counted as evidence for a physical exterior collective field until the microscopic arrow is derived.

25.9 A compact dependency map across all eight chapters

Chapter	What is already supplied	Missing physical link
1	Dirac spinors, many-body observables, current and semiclassical limits	Genuine field operator/coherence in sourced exterior vacuum
2	Natural-invariant metric and independent classical solution machinery	Quantum selection of weights, twist and multipoles
3	Empirical gravitational normalization and source/probe distinctions	Microscopic relation among wave amplitude, mass and frame weight
4	Full action-pullback/Option C and quantum correlator formalism	Derivation of gravity-relevant effective functional from flat microscopic matter
5	Exact conditional static collective-energy bridge	Quantum derivation of the collective degree and geometric coupling
6	Quantitative microscopic tests, proof ledgers, Kerr spinor-optical bridge	Actual causal, common, exterior vacuum quantum field $\mathfrak{B}$ and its $[\Sigma]$ map
7	Vacuum-orbit redundancies and geometric energy constraints	Gauge-invariant physical vacuum field/state and gravitational energy account
8	Cosmological representations and selection/no-extra-equation limits	Microscopic homogeneous initial-state and expansion dynamics

25.10 Research decision: what to calculate next, and what not to repeat

Immediate decision. Pause additional classical Einstein-solution enumeration and do not launch another scalar-only infrared scan merely to obtain a $1/r$ tail. First propose *one actual* candidate composite quantum field or coherence/connection order parameter, identify its microscopic operator definition (or explicit new physical premise if none exists), then derive the source-conditioned exterior quantum functional and its correlation/causal transport. The candidate must have enough directional structure to encode Kerr twist and enough state information to select source-dependent weights and multipoles. Its coupling to distinct probe species must come from one microscopic theory; the Chapter 1 Dirac/WKB/current identities are constraints to preserve, not substitutes for the missing dynamics.

Falsifiable branches. (i) If a micro-derived exterior field exists and supplies common kinetic geometry, test its QGD quotient, EFE limit, stability and quantum corrections. (ii) If the original contact model has no collective exterior mode and no universal probe interaction, report a model-specific failure and identify explicitly the additional microscopic interaction or degrees of freedom needed. (iii) If a field alters only one species, classify it as a quantum medium/optical metric, not QGD's universal gravity. (iv) If the only way to obtain the field is to postulate independent Σ and universal $g(\Sigma)$ coupling, declare a new axiom and test that distinct theory without retroactively claiming the original $\mathcal{F}_{\text{micro}}$ has been derived.

Central question retained for the next chapter increment.

$$\boxed{\begin{aligned}&(S_{\text{micro}}, \widehat{\rho}_{\text{source}}, \text{BC}) \xrightarrow{\text{derive}} \mathfrak{B}_\rho(x), \qquad x \text{ in exterior vacuum},\\ &\mathfrak{B}_\rho \xrightarrow{\mathcal{F}_{\text{micro}}} [\Sigma]_{\text{physical}} \xrightarrow{g[\Sigma]} g_{\mu\nu}^{\text{common}}.\end{aligned}} \tag{153}$$

Answering this, not finding another coordinate form of Kerr, is the intended next foundational breakthrough. The present QM023 section is a synthesis and a research-ontology decision [A/O], not a new solution of the open map. No prior proof-status claim is upgraded by adding it.

26 QM024: Exact microscopic rewriting versus a physical pregeometric phase

Research status [D/N/C/O]. The original F1 and the new determinant-of-spinor-bilinears pregeometry M have been tested as *different* microscopic theories. This section reports mathematical identities and restricted counterexamples. It does not supersede the QM023 exterior collective-field research question or promote the new theory M to an established closure of $\mathcal{F}_{\text{micro}}$. The complete proof, numerical procedures, and references appear in the accompanying QM024 standalone research report.

26.1 Route A: exact rewrite of original F1 [D/N]

For two fermion pairs with $B = \bar{\psi}_1\psi_1 + \bar{\psi}_2\psi_2$ and an exactly defined zero-dimensional Grassmann integration, a genuine collective identity is

$$\boxed{Z_F(J) = J^2 + g = \frac{1}{\sqrt{2\pi g}} \int d\phi\, e^{-\phi^2/(2g)} (J+\phi)^2, \qquad g > 0.} \tag{154}$$

The identity preserves all source derivatives exactly. Yet at $J = 0$ the bosonic auxiliary effective action has nonzero stationary points $\phi_\pm = \pm\sqrt{2g}$ while the exact expectation value is $\langle\phi\rangle = 0$. This is an exact finite-regulator *counterexample to inferring a spontaneously selected physical order parameter from a formal auxiliary saddle.* Thermodynamic, gauge and continuum limits require independent treatment.

An exact source-preserving rewrite cannot change the original retarded connected observables. In particular the QM018 decoupled-source/spectator result $\chi^R_{PS} = 0$ persists under exact field redefinitions, including with entangled initial states. It does not follow that *every* interacting F1 model lacks a collective gravity mode; the earlier scalar/stress tests provide restricted negative results, not a global nonperturbative F1 no-go theorem.

26.2 Route B: the metric-free microscopic action is genuinely different [D]

The proposed spinor-pregeometry action uses

$$\boxed{E^a{}_\mu = \sum_{i=1}^{N} \frac{i}{2} \bar{\psi}_i \gamma^a \overleftrightarrow{D_\mu} \psi_i, \qquad S_M = \kappa \int d^4x \det E.} \tag{155}$$

The determinant of the *sum* $E = \sum_i E_i$ contains mixed eight-fermion interactions among species, and is not the sum of individual free Dirac actions. M can therefore supply new common microscopic structure, which is physically promising, but it is not exactly equivalent to original F1. The formal identity $1 = \int DE'\, \delta(E' - E)$ and Fourier multiplier Π is a rewrite of M, with source preserving properties only after the contours, measures and gauge treatment have been specified.

As a finite-regulator test, an eight-generator Grassmann model with $E_{a\mu} = \bar{\psi}_a\psi_\mu$ obeys

$$\det E = 24\, \bar{\psi}_1\bar{\psi}_2\bar{\psi}_3\bar{\psi}_4\psi_1\psi_2\psi_3\psi_4, \qquad Z_M(j) = j^4 + 24\kappa. \tag{156}$$

Thus the composite determinant interaction is nonzero but $\langle E_{a\mu}\rangle = 0$ at zero source in this finite model. It proves neither presence nor absence of a large-N geometric phase. A nonzero ordinary-number pointwise unimodular equality $\det E = v \neq 0$ is incompatible with finite-regulator nilpotency of a microscopic Grassmann composite; imposing unimodularity on a genuinely bosonic auxiliary field is a different proposal requiring derivation.

26.3 Correct frame duality and quantum measure [D/N]

On the regular real 4×4 branch,

$$\boxed{\Pi = (\det e)e^{-1}, \quad e = (\det \Pi)^{1/3}\Pi^{-1}, \quad \left|\det \frac{\partial \Pi}{\partial e}\right| = 3|\det e|^8.} \tag{157}$$

The last relation follows by splitting the 16 fluctuations into one trace direction (eigenvalue 3 at $e = I$) and 15 traceless directions (eigenvalue -1), and extending by the cofactor map. Forty random-frame tests reproduce it with relative error below 10^{-12}. The measure factor is required for a flat finite-dimensional component measure; it is not the full gauge-fixed continuum functional measure.

The dual field $e(\Pi)$ controls the common *kinetic* fermion frame. The proposed microscopic mass term, however, is $m_i \det E\, \bar{\psi}_i \psi_i$. Its exact auxiliary representation contains $m_i \det E'\, \bar{\psi}_i \psi_i$. Off shell, the effective mass relative to the kinetic frame is $m_i \det E'/\det e$, and the two determinants need not agree. For $e = \mathrm{diag}(2, 1, 1, 1)$ and $E' = I$ the ratio is $1/2$. A common conventional massive Dirac action can follow at an appropriate classical saddle, but not as the exact off-shell operator for all E' and Π configurations.

Further, the proposed real Euclidean auxiliary potential $\kappa \det E'$ is unbounded below. Along $E' = t\,\mathrm{diag}(-1, 1, 1, 1)$ one has $\det E' = -t^4$, and $\exp(-\kappa \det E') = \exp(\kappa t^4)$ for $\kappa > 0$. A different contour, a regulated Grassmann construction, or a stabilized collective functional must be specified before interpreting a standalone bosonic E' path integral as exact. This issue does not invalidate the regulated *fermionic* determinant-composite action by itself.

26.4 The induced Einsteinian coefficient remains a conditional success [C/N]

If a nondegenerate collective frame exists and the fermions have the standard covariant kinetic action there, the proper-time Dirac determinant gives the previously derived scheme-dependent induced coefficient. For N identical four-component Dirac species,

$$G_{\mathrm{ind}} = \frac{6\pi}{N I_1}, \qquad I_1 = \Lambda_c^2 e^{-m^2/\Lambda_c^2} - m^2 E_1(m^2/\Lambda_c^2). \tag{158}$$

At $N = 100, m = 0.01, \Lambda_c = 1$, the numerical result is $G_{\mathrm{ind}} = 0.18867731537$. Changing Λ_c to 0.5 or 2 changes it to 0.75647771473 or 0.04713687534, respectively, at fixed matter content. This is a valid calculation *in its specified proper-time convention*, not a parameter-free physical prediction or proof of existence of a healthy collective frame phase. For $E = O(N)$, extensivity of $\kappa \det E$ at $O(N)$ suggests $\kappa = O(N^{-3})$ in the simple normalization; a controlled large-N derivation must specify the scaling.

Rerunning QM014 confirms the conditional induced-Einstein metric-operator ranks $(6, 4)$ at generic and null momentum, and sector ranks $(6, 4)$ with nullities $(26, 28)$ on the regular metric-only branch. These are *not* a complete microscopic Dirac–Bergmann constraint classification. The microscopic M action also contains no independently specified Maxwell kinetic term; equality of photon and fermion effective light cones remains an independent, uncompleted microscopic test.

26.5 Outcome for the founding $\mathcal{F}_{\mathrm{micro}}$ programme

Route A: exact finite-regulator collective rewritings preserve physical source correlators, but do not generate new universal geometry absent from the original theory. **Route B:** an exact formal auxiliary rewriting of the *new* spinor-pregeometry M and a correct local frame duality are demonstrated; the nondegenerate physical collective phase, its correct massive/gauge couplings, and its exterior causal field dynamics are not yet established.

The exact CTP object needed for a source-conditioned claim is a properly regulated collective functional

$$\Gamma_{\mathrm{coll}}[e_{\pm}, \omega_{\pm}; \widehat{\varrho}_{\mathrm{source}}],$$

with both contour branches, correct Jacobians and physical constraints. The geometric phase must be tested using gauge-invariant or relational observables and collective spectra, not inferred from a gauge-dependent frame expectation alone. Only after such an object is computed could one legitimately select the physical QGD class using the *full* master metric Eq. (4) rather than a pure-algebraic lift.

Central open question [O]. Does a metric-free, fully regulated microscopic quantum theory dynamically produce a stable exterior collective quantum field with an invertible physical weighted frame, causal source dependence and a common relativistic propagation structure for all relevant probes? If so, does its physical class $[\Sigma]$ recover the tested Einsteinian limit without postulating that limit in the microscopic action? QM024 has established the correct formal and algebraic prerequisites, *not* that the required geometric phase exists.

27 QM025: Exact composite pushforward, dual-frame measure and exterior phase-selection test

Scope [D/N/C/O]. We now test whether the *physical phase* can be inferred from the exact collective transformation of spinor pregeometry M. Unlike QM024, which checked the dual coframe algebra, QM025 examines the support of the exact finite-regulator composite distribution, its response to external sources, coordinate-dependent auxiliary saddles, the stationary equations of both auxiliary fields, and the limitations of a homogeneous covariant frame potential. The calculations are mathematical tests of the stated proposals, not a nonperturbative simulation of the four-dimensional theory. The canonical QM023 central question remains unchanged.

27.1 A rigorous distinction between the composite X and the dual coframe $e(\Pi)$ [D]

Write $E^a{}_\mu[\psi,\omega]$ for the microscopic Grassmann-even spinor bilinear in Eq. (155), $X^a{}_\mu$ for the constrained auxiliary image of this composite, and $\Pi^\mu{}_a$ for its Fourier/Lagrange multiplier. For a finite Grassmann regulator the formal exact identity is a distributional pushforward,

$$\begin{aligned} 1 &= \int d^{16}X\, \delta^{(16)}(X - E[\psi,\omega]), \\ Z_M[J] &= \int D\psi D\omega\, e^{iS_M + iJ\cdot O} \\ &= \int d^{16}X\, D\psi D\omega\, \delta^{(16)}(X-E)\, e^{iS_M(X)+iJ\cdot O}. \end{aligned} \tag{159}$$

The delta symbol means a *finite derivative-of-delta distribution* when its shift has nilpotent entries, not a positive stochastic delta distribution over classical frames. It reproduces polynomial test functions and source derivatives exactly with consistent integration conventions.

Proposition 27.1 (Support of a finite composite pushforward, not a no-go for condensation)**.** *At any finite lattice regulator the ordinary-number body of every matrix component of E vanishes. Consequently $\delta^{(16)}(X-E)$ is a finite sum of partial derivatives of $\delta^{(16)}(X)$ and has distributional support at the degenerate point $X=0$. Replacing the* exact *X-integration by a restriction to* $\det X \neq 0$ *removes that support and is not an exact collective rewriting. This does not exclude nonzero renormalized composite expectations or an invertible* effective *frame after a thermodynamic, large-N, continuum, or gauge-conditioned limit.*

Proof. Each $E^a{}_\mu$ is Grassmann even and nilpotent because it has no constant component in the finite Grassmann algebra. Expand the shifted distribution as

$$\delta^{(16)}(X-E) = \sum_{|\alpha|\le K} \frac{(-E)^\alpha}{\alpha!}\, \partial_X^\alpha \delta^{(16)}(X),$$

with the sum terminating by nilpotency. Every term has support at $X=0$. Its pairing with a smooth function supported away from $X=0$, including one supported inside $\mathrm{GL}(4,\mathbb{R})$, is zero. Hence imposing $\det X \neq 0$ at fixed regulator cannot be equated to the original exact identity. A distributional support statement is not a probability claim and does not commute automatically with a singular infinite-volume limit. □

Exactly solvable source diagnostic. Let $\vartheta = \bar\psi\psi$ be one Grassmann-even nilpotent variable ($\vartheta^2=0$), normalized so $\int d\bar\psi d\psi\, \vartheta = 1$. Then for $j \neq 0$,

$$Z(j) = \int d\bar\psi d\psi\, e^{j\vartheta} = j, \qquad \frac{1}{Z(j)} \int d\bar\psi d\psi\, e^{j\vartheta}\delta(x-\vartheta) = \delta(x) - \frac{1}{j}\delta'(x). \tag{160}$$

The source-generated expectation is $\langle\vartheta\rangle = 1/j$ even though the pushforward's distributional support is $x = 0$, because its derivative-of-delta term gives $\langle x\rangle = 1/j$ while $\langle x^2\rangle = 0$. The connected x second moment is $-1/j^2$. The pushforward is therefore *not* a positive probability distribution; a seemingly nonzero expectation is not an ordinary-valued random invertible matrix at finite regulator.

For the previous exact four-pair determinant test (QM024), $Z(j) = j^4 + 24\kappa$ and the average diagonal bilinear obeys

$$\frac{1}{4}\partial_j \log Z(j) = \frac{j^3}{j^4 + 24\kappa}, \qquad \langle E_{aa}\rangle|_{j=0} = 0, \qquad \left.\frac{d\langle E_{aa}\rangle}{dj}\right|_{j=0} = 0. \tag{161}$$

This is a finite-volume source-response obstruction in the *specific zero-dimensional model*; it does not extrapolate to the full geometric phase.

27.2 The dual coframe $e(\Pi)$ can be invertible without X being invertible [D/N]

The local duality reviewed in QM024 uses the *multiplier*, not the composite:

$$\Pi = \mathrm{cof}(e), \qquad e = (\det \Pi)^{1/3}(\Pi^{-1})^{\mathsf{T}}, \qquad \left|\det \frac{\partial \Pi}{\partial e}\right| = 3|\det e|^8. \tag{162}$$

Here the matrix index placement is chosen so $\mathrm{cof}(e) = (\det e)(e^{-1})^{\mathsf{T}}$; this is the same coframe density relation as QM024 with its contracted indices. Working on the invertible multiplier branch is locally permissible for the finite-dimensional coordinate change (the singular determinant hypersurface has zero Lebesgue measure) only when the oscillatory integral, gauge treatment and boundary contributions justify it. It does *not* impose $\det X \neq 0$ and does not by itself prove that the dual frame is a physical condensate.

Proposition 27.2 (A Jacobian can manufacture a nonzero coordinate peak)**.** *A nonzero maximum of a dual-frame integration density is not, by itself, an invariant phase-selection criterion. Indeed, for $a > 0$,*

$$\int_0^\infty d\Pi\, e^{-a\Pi^2} = \int_0^\infty 3b^2\, db\, e^{-ab^6}, \qquad \Pi = b^3. \tag{163}$$

The first integrand peaks at $\Pi = 0$ while the transformed density peaks at $b = (3a)^{-1/6} > 0$; the integrals and every transformed observable are identical. Thus a finite-dimensional auxiliary-frame saddle defined only as the pointwise maximum of the integration density can be a coordinate artifact.

The exact tests at $a = 0.5, 1, 2$ agree to 10^{-12}. The higher-dimensional cofactor Jacobian Eq. (162) makes a consistent measure and gauge-invariant phase criterion particularly important; the one-dimensional example is a warning, not a claimed derivation of a specific QGD saddle.

27.3 The actual coupled saddle equations, including the massive-source mismatch [D/C]

For the Euclidean-sign formal first-order representation, let

$$S_{\mathrm{aux}}[\psi, \omega, X, \Pi] = \int d^4x \left\{ \kappa \det X + \Pi : (X - E[\psi, \omega]) + \sum_i m_i \det X\, \bar{\psi}_i \psi_i \right\}. \tag{164}$$

After integrating fermions with an admissible initial-state prescription, the schematic effective functional is

$$\Gamma_{\mathrm{coll}}[X, \Pi, \omega; \rho] = \int d^4x \{\kappa \det X + \Pi : X\} - \mathrm{Tr}_\rho \log \mathcal{D}[\Pi, X, \omega] + \Gamma_{\mathrm{measure}} + \Gamma_{\mathrm{other}}, \tag{165}$$

where $\mathcal{D}$ includes a common kinetic density $-\Pi^{\mu}{}_{a}\gamma^{a}D_{\mu}$ and species masses proportional to $m_i \det X$. The trace-log alone is justified only for suitable Gaussian states and specified regulator/contour conditions; Γ_{other} denotes the required state and gauge terms, not an assumed Einstein action. In particular, the proper-time coefficient obtained by *setting* $X = e$ and assuming a standard curved Dirac operator is not automatically the coefficient of the full off-shell Γ_{coll}.

The two distinct stationary conditions read formally

$$\boxed{X^{a}{}_{\mu} = \langle E^{a}{}_{\mu}\rangle_{\rho,\Pi,X}, \qquad \Pi^{\mu}{}_{a} = -\left(\kappa + \sum_i m_i \langle \bar{\psi}_i \psi_i\rangle_{\rho,\Pi,X}\right)\operatorname{cof}(X)^{\mu}{}_{a},} \tag{166}$$

up to counterterms and constraints in the regulated theory. These equations must be solved jointly with the source state and connection. A conditional aligned saddle $X = ce$ and $\Pi = \operatorname{cof}(e)$ implies

$$\boxed{-\kappa_{\text{eff}}c^3 = 1, \quad \kappa_{\text{eff}} := \kappa + \sum_i m_i \langle \bar{\psi}_i \psi_i\rangle, \qquad m_{i,\text{eff}} = m_i c^4.} \tag{167}$$

This fixes a relation between the two frames *if* an aligned saddle exists; it neither forces alignment nor supplies a stable vacuum. For $\kappa_{\text{eff}} = -8, -1, -1/8$, respectively, the aligned positive solutions are $c = 1/2, 1, 2$, and the common mass rescalings are $1/16, 1, 16$. The signs and contours require a physically specified Lorentzian/Euclidean continuation.

27.4 A limited homogeneous phase-selection obstruction from diffeomorphism invariance [D/C]

Proposition 27.3 (Derivative-free local frame potential cannot alone stabilize a positive scale)**.** *For a* single invertible *coframe with no extra background tensors or scalar order parameters, every local diffeomorphism- and local-Lorentz-invariant, zero-derivative density is proportional to* $\det e$*. Accordingly the local homogeneous ansatz* $e^{a}{}_{\mu} = b\delta^{a}{}_{\mu}$*,* $b > 0$*, has*

$$V_{\text{local}}(b) = \Lambda_{\text{eff}} b^4, \qquad \frac{dV_{\text{local}}}{db} = 4\Lambda_{\text{eff}} b^3. \tag{168}$$

There is no isolated positive-b stationary minimum: if $\Lambda_{\text{eff}} \neq 0$ *none exists in this ansatz, and if* $\Lambda_{\text{eff}} = 0$ *the local scale direction is flat (subject to coordinate, boundary and gauge identifications). This does not rule out a geometric phase selected by nonlocal quantum effects, multiple collective fields, constraints, finite-density sources, boundary conditions or a nontrivial continuum limit.*

Proof. Without derivatives or other nontrivial fields, contractions of the coframe with its inverse and invariant tensors produce numerical constants; the only density carrying the coordinate Jacobian is the volume form $\epsilon_{abcd}e^{a}\wedge e^{b}\wedge e^{c}\wedge e^{d}$, proportional to $\det e\, d^4x$. The single-scale restriction and derivative follow immediately. □

A pointwise unimodular *bosonic* constraint $\det e = v > 0$ kinematically excludes $e = 0$, but it is an additional restriction, not a proof that the original unconstrained microscopic fermion theory dynamically preferred that phase. Nor can one equate $\det E = v$ pointwise in a finite Grassmann algebra, by QM024.

27.5 What a source-conditioned causal collective action would actually require [C/O]

For an initial density operator $\widehat{\rho}_{\text{src}}$, the correct causal generating functional is the closed-time-path object

$$Z_\rho[J_+, J_-] = \operatorname{Tr}\left(U_{J_+}\widehat{\rho}_{\text{src}}U^{\dagger}_{J_-}\right), \qquad Z_\rho[J, J] = \operatorname{Tr}\widehat{\rho}_{\text{src}} = 1. \tag{169}$$

Any exact $X_\pm, \Pi_\pm$ insertion must preserve this identity, the initial-state boundary terms, and all source differentiations. The causal mean-field equation is obtained from the difference-field variation of the CTP collective effective action at coincident forward and backward fields. In an actual gauge theory the physical phase should be characterized by gauge-invariant, relationally defined metric/composite correlators, not a gauge-unfixed local $\langle e^a{}_\mu \rangle$ alone.

Required acceptance tests: (i) a finite-regulator fermion and connection definition plus a consistent auxiliary distribution, contour and Jacobian; (ii) independent self-consistent X, Π, ω saddle equations including mass terms and source state; (iii) an observable nondegenerate geometric phase, free-energy comparison and proper N/volume limits; (iv) gauge-invariant low-energy spin-2 and torsion spectra and Ward identities; (v) causal exterior response to a specified source; (vi) one common kinetic geometry for fermions and photons with the microscopic gauge action specified; (vii) only then the regular QGD lift to $[\Sigma]$ using the founding metric, including $\mathcal{Q}_{ab}$ if active.

27.6 QM025 result and physical decision [D/N/C/O]

Success: The exact finite-regulator pushforward, its source-generated observables, and the invertible dual-frame coordinate algebra can be distinguished rigorously. The two coupled auxiliary Euler equations are now explicit, including the source-dependent mass/frame mismatch. The need for a nontrivial physical phase criterion follows from the coordinate-peak example and from the homogeneous covariant potential argument.

Obstruction: Neither a formal delta insertion, nor an invertible multiplier parameterization, nor a bosonic unimodular restriction proves that the original microscopic theory has dynamically selected a nondegenerate collective vacuum. The single-frame derivative-free potential cannot accomplish this alone. This is a failure of the proposed *automatic closure proof*, not a no-go theorem for spinor pregeometry or QGD.

Next decisive calculation: Derive the *regulated source-dependent* spinor/connection effective action with a specified diffeomorphism-respecting regulator, calculate a gauge-invariant physical frame/composite two-point spectrum and compare geometric to nongeometric phases while retaining both X and Π. The transition and its massless spin-2 response must arise in the same microscopic theory. No claim of a stable phase is registered until that test succeeds.

28 QM026: Controlled large-N collective ordering, source anisotropy and the missing geometric spectrum

Research status [D/N/C/O]. The QM025 obstruction to automatic geometric-phase selection must not be read as a prohibition against collective ordering. We now test a fully specified, finite Grassmann projection of the determinant interaction, derive its complete finite-N generating function, and perform the two noncommuting source/large-N limits exactly. The positive result is a composite *order parameter*, not yet an invertible spacetime coframe in the original four-dimensional spinor-pregeometry model M. A second test proves that the determinant alone does not choose all the frame weights; a third makes explicit that a local ordered state need not have a propagating collective field. A separate conditional gauge-connection test examines photon universality in a nondegenerate induced-frame phase. All new scientific claims are scoped accordingly.

28.1 A specified Grassmann determinant projection [D]

Let $\chi_{ai}, \bar{\chi}_{ai}$ be independent Grassmann generators, $a = 1, \ldots, 4$ and $i = 1, \ldots, N$, with the Berezin integration convention $\int d\bar{\chi}_{ai}\, d\chi_{ai}\, \bar{\chi}_{ai}\chi_{ai} = 1$ (an overall common sign is fixed consistently). Define the four Grassmann-even bilinears and the diagonal composite matrix

$$S_a = \sum_{i=1}^{N} \bar{\chi}_{ai}\chi_{ai}, \qquad E = \mathrm{diag}(S_1, S_2, S_3, S_4), \qquad D = \det E = \prod_{a=1}^{4} S_a. \tag{170}$$

The exactly finite generating function is

$$Z_N(\mathbf{j}, \kappa) = \int D\chi D\bar{\chi}\ \exp\left[\kappa D + \sum_{a=1}^{4} j_a S_a\right]. \tag{171}$$

This is a *zero-dimensional diagonal-channel projection* of the determinant-composite interaction. It has neither $\gamma^a D_\mu$ nor a Lorentz connection, spacetime derivative, photon, metric or spin-two propagator. It is not a complete regulator of $S_M = \kappa \int \det E[\psi, \omega]$; no result about the full four-dimensional geometric phase will be inferred without new calculations.

Proposition 28.1 (Exact finite-N determinant generating function)**.** *For all integers $N \geq 1$, and ordinary commuting sources j_a,*

$$\boxed{Z_N(\mathbf{j}, \kappa) = \sum_{k=0}^{N} \frac{\kappa^k}{k!} \left[\frac{N!}{(N-k)!}\right]^4 \left(\prod_{a=1}^{4} j_a\right)^{N-k}.} \tag{172}$$

In particular $Z_N(\mathbf{0}, \kappa) = \kappa^N (N!)^3$ for $\kappa \neq 0$, and the zero-source expectations obey

$$\langle S_a \rangle|_{\mathbf{j}=0} = 0, \qquad \langle D \rangle|_{\mathbf{j}=0} = \frac{N}{\kappa}. \tag{173}$$

Proof. Expand the determinant interaction in powers of k. Because $S_a^{N+1} = 0$, $k \leq N$. For a single channel, $\int D\chi_a D\bar{\chi}_a\, S_a^k \exp(j_a S_a) = [N!/(N-k)!] j_a^{N-k}$: choose k distinct bilinears from the N available and saturate the others with source monomials. Multiply the four independent channel factors for fixed k. The zero-source partition is the $k = N$ term, and its log derivatives with respect to j_a and κ give Eq. (173). □

A nonzero determinant expectation with vanishing one-bilinear means therefore follows from an exact finite-Grassmann calculation. It cannot by itself be interpreted as a nonsingular *classical random matrix*: the underlying Berezin functional is not a positive distribution of ordinary c-number frames (QM025).

28.2 An actual order-of-limits result at large N [D/N]

Set $\kappa = g/N^3$, $g > 0$, and for now choose all four sources $j_a = j > 0$. The extensive scaling matches the schematic $E = O(N)$ and $\det E = O(N^4)$ of the original collective discussion. Denote the kth summand of Eq. (172) by W_k. Their exact ratio is

$$\frac{W_{k+1}}{W_k} = \frac{g}{N^3}\frac{(N-k)^4}{(k+1)j^4}. \tag{174}$$

This is monotonically decreasing with k, so the positive-weight k-distribution is log-concave. With $x = k/N$, its unique bulk maximum concentrates in the $N \to \infty$ limit around

$$\boxed{g(1-x)^4 = xj^4, \quad 0 < x < 1.} \tag{175}$$

Since $\partial_j \log Z_N = 4(N - \langle k \rangle)/j$, the normalized bilinear response tends to

$$b(j;g) := \lim_{N\to\infty} \frac{\langle S_a \rangle_{N,j}}{N} = \frac{1 - x(j;g)}{j}. \tag{176}$$

Writing $1 - x = jy$ in Eq. (175) and taking $j \to 0^+$ gives $gy^4 = 1$. Thus

$$\boxed{\lim_{j\to 0^+}\lim_{N\to\infty} \frac{\langle S_a \rangle_{N,j}}{N} = g^{-1/4}, \qquad \lim_{N\to\infty}\lim_{j\to 0^+} \frac{\langle S_a \rangle_{N,j}}{N} = 0.} \tag{177}$$

This is a genuine source-selected large-N branch in the *defined projection*; it is not a demonstration of a gauge-invariant physical frame in M. For $j < 0$ the limiting response has the opposite sign, so the extensive source-dependent free energy has a cusp at the origin (the exact finite-N polynomial is analytic). The numerical verifier calculates Eq. (172) using log-sum-exp for $N = 4, 16, 64, 256, 1024$ and confirms the convergence and source derivatives.

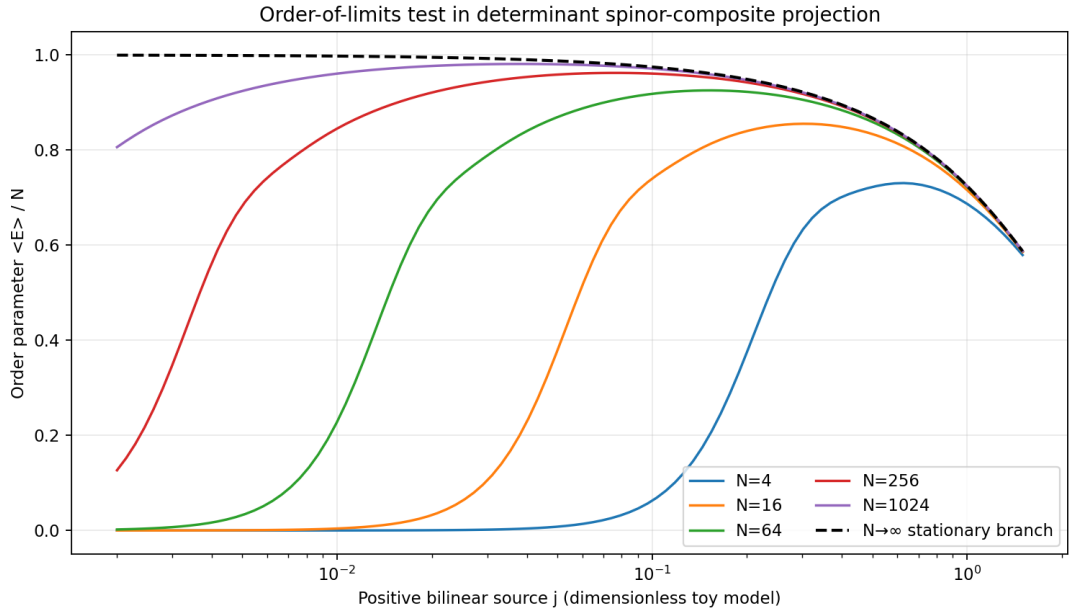


Figure 11: Exact finite-N determinant-model bilinear response (solid lines) compared with the analytic large-N branch (dashed). The small-source and large-N limits do not commute; this establishes ordering only in the named zero-dimensional projection.

The determinant is already nonzero as an invariant composite at zero source:

$$\left\langle \frac{\det E}{N^4} \right\rangle\Big|_{\mathbf{j}=0} = \frac{1}{g}, \qquad \langle E \rangle|_{\mathbf{j}=0} = 0 \quad (N < \infty). \tag{178}$$

This explicitly demonstrates why a source-free one-point frame expectation is not an adequate finite-system definition of a collective geometric phase.

28.3 Important failure: the determinant does not select a unique weighted frame [D/N]

The exact polynomial Eq. (172) depends on independent sources only through $\prod_a j_a$. With $j_a = tc_a > 0$ and $\kappa = g/N^3$, its large-N saddle obeys $g(1-x)^4 = xt^4 \prod_a c_a$. Taking the large-$N$ limit first and then $t \to 0^+$ yields

$$\boxed{b_a = \lim_{t\to 0^+} \lim_{N\to\infty} \frac{\langle S_a \rangle}{N} = \frac{(\prod_{b=1}^4 c_b/g)^{1/4}}{c_a}, \qquad \prod_{a=1}^4 b_a = \frac{1}{g}.} \tag{179}$$

For $g = 1$, source directions $(c_a) = (1,1,1,1)$ and $(1,2,1/2,1)$ select respectively $(b_a) = (1,1,1,1)$ and $(1,1/2,2,1)$, both with the same determinant. Exact $N = 4096$ evaluations approach these

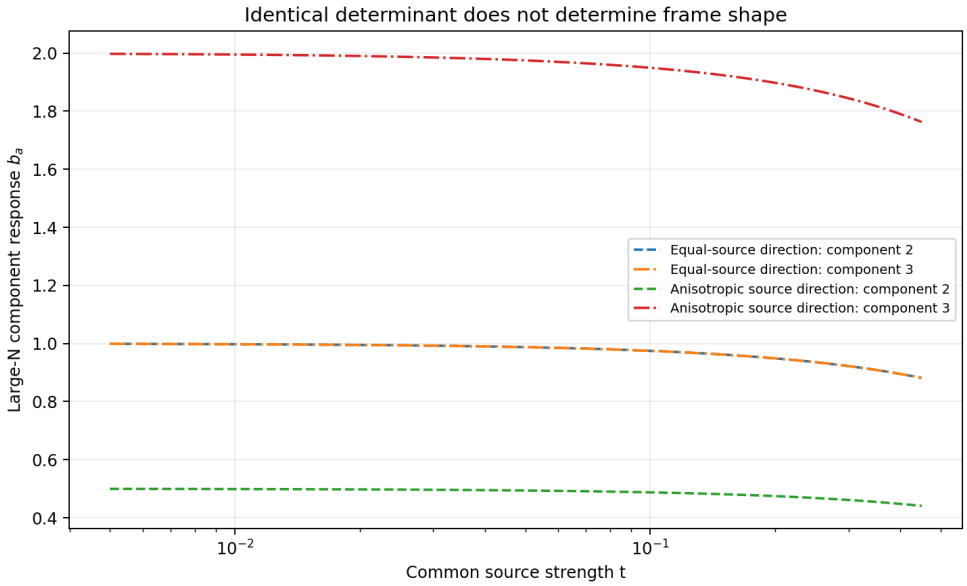


Figure 12: Different channel-source directions select different weighted-frame shapes in the exact large-N projection. The determinant tends to the same value as $t \to 0^+$ in both branches.

limits as the sources decrease. Thus the determinant interaction can fix a volume-like collective product without selecting all frame-shape data. Some constant-frame shape freedom may be related to coordinate or gauge transformations in a fully covariant spacetime model; the reduced projection lacks that gauge structure and cannot classify which shapes are physical. The *universal metric selection problem remains open.*

28.4 Connected source response and the missing spacetime propagator [D/N/O]

At equal sources $j > 0$ define the exact probability weights $p_k = W_k / \sum_l W_l$ as a computational device for the *coefficient sum* (not a positive probability measure on fermion fields). For two bilinear sources belonging to different channels and different species, the static mixed source derivative is

$$\boxed{\frac{\partial^2 \log Z_N}{\partial j_{ai} \partial j_{b\ell}} = \frac{\mathrm{Var}_p(k)}{N^2 j^2} > 0, \qquad a \neq b,\ i \neq \ell.} \tag{180}$$

At $g = 1, j = 0.2$ and $N = 4, 16, 64, 256, 1024$, the values are approximately 0.345657, 0.0713862, 0.0175760, 0.00438056 and 0.00109435. The microscopic determinant therefore creates a *shared static interaction across otherwise distinct constituent species*; the individual cross-derivative scales as $O(1/N)$ at fixed positive source. This is not a causal retarded two-point function.

A spatial array of *independent copies* of this exact projected model has $Z = \prod_x Z_{N,x}$ and therefore

$$\boxed{\langle D(x)D(y)\rangle_c = 0 \quad (x \neq y)} \tag{181}$$

regardless of the nonzero local determinant condensate. Its connected Fourier spectrum has no k^{-2} spin-two pole. This is an exact limitation of the independent-cell extension, *not* of the full four-dimensional M, whose derivative spinor composite and Lorentz connection can couple neighboring cells. A physically propagating exterior QGD field requires an independently computed spatial/causal collective kernel, not merely a local condensate.

28.5 Gauge-orbit check: a vanishing mean frame does not imply absent geometry [D/N]

As a separate Euclidean local-Lorentz illustration, let $e = Re_0$ with $R \in SO(4)$ Haar-distributed and e_0 invertible. Group averaging yields

$$\boxed{\langle e^a{}_\mu\rangle_R = 0, \qquad \left\langle e^a{}_\mu e^b{}_\nu \right\rangle_R = \frac{\delta^{ab}}{4}(e_0^{\mathsf{T}} e_0)_{\mu\nu}, \qquad \langle e^{\mathsf{T}} e\rangle_R = e_0^{\mathsf{T}} e_0.} \tag{182}$$

A 12,000-sample Haar check has metric error below 1.2×10^{-12} and average frame norm $\simeq 0.022$. Thus physical phase criteria must be expressed through gauge-invariant or relational composites; $\langle e\rangle = 0$ in an unfixed local gauge is not evidence of a missing geometric phase. This group-orbit calculation is separate from the zero-dimensional determinant model and cannot establish its local-Lorentz symmetry.

28.6 A conditional route to a shared photon/fermion coframe [D/C/O]

The microscopic proposal M originally includes spinors and a Lorentz connection but not an independently specified electromagnetic gauge connection. A *specified extension* replaces $D_\mu\psi_i$ in its defining composite by $D_\mu\psi_i - iq_iA_\mu\psi_i$ while preserving the same composite determinant interaction. The exact auxiliary kinetic term $-\Pi : E[\psi, \omega, A]$ then supplies a common coframe to every charged Dirac species, with its usual q_iA_μ coupling. If a nondegenerate frame phase exists and the fermion determinant admits a covariant low-energy expansion, the gauge-field part of the Euclidean heat-kernel coefficient of a four-component Dirac species is

$$\boxed{\mathrm{tr}\left(\frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}\mathsf{E}_F^2\right) = \frac{2}{3}q_i^2F_{\mu\nu}F^{\mu\nu}, \quad \Omega_{\mu\nu} = -iq_iF_{\mu\nu}I_4, \quad \mathsf{E}_F = -\frac{iq_i}{2}\gamma^{\mu\nu}F_{\mu\nu}.} \tag{183}$$

The independent gamma-matrix calculation reproduces this expression to relative error 6×10^{-16}. Its logarithmic fermionic determinant therefore induces a Maxwell-type F^2 action using the *same* chosen coframe at the stated derivative order. This is a constructive conditional common-metric mechanism, *not* proof of a physically realized photon sector: the gauge field is an added microscopic variable, measure and spin-connection treatment remain open, and the interacting two-field/massive-sector mismatches from QM024–QM025 persist.

28.7 QM026 result and the next genuine acceptance test

Positive result [D/N]. The determinant-composite interaction itself can generate a source-selected nonzero bilinear order in a rigorously computed large-N limit of an explicitly regulated

finite Grassmann projection. Both its exact finite-N source generating functional and the $N \to \infty$ saddle are known. Shared static cross-species responses and a nonzero composite determinant expectation follow without inserting any metric or Einstein action. This invalidates the inference that finite-Grassmann nilpotency or a vanishing finite-volume $\langle E \rangle$ forbids collective geometric ordering at large N.

Failure/limitation [D/O]. The projection has no spacetime derivative, spin connection, gauge dynamics or transverse-traceless correlator. Its determinant does not uniquely select frame weights, and independent cells have identically vanishing nonlocal correlations. A separate conditional gauge-extension shows how a shared photon kinetic tensor may be induced if a physical coframe is already realized, but that realization remains open. *The nondegenerate, dynamically preferred exterior quantum frame of the actual four-dimensional microscopic model M has not been demonstrated.*

Next essential computation [O]. Construct a finite, diffeomorphism-respecting discretization of the *actual* operator $E^a{}_\mu = \sum_i \frac{i}{2} \bar{\psi}_i \gamma^a \overleftrightarrow{D}_\mu \psi_i$, integrate the spin connection and all fermions or simulate with a demonstrably correct determinant/Pfaffian measure, and scan source-conditioned, gauge-invariant collective two-point spectra over increasing four-dimensional volumes. Determine whether a nondegenerate phase and its source-visible, transverse-traceless long-distance dynamics arise on the *same* branch. Then check that gauge connections coupled through the same microscopic composite yield a common effective geometry. Only after those physical tests may the regular QGD map $g \to [\Sigma]$ be promoted from an algebraic lift to microscopic selection. The full founding metric with $T^a{}_\mu$ and $\mathcal{Q}_{ab}$ remains the theory's master equation.

29 QM027: Fitting the QGD selection law backwards, then deriving its microscopic origin

Declared research decision [A/F]. Instead of requiring the unknown quantum derivation as a prerequisite for every effective selection formula, first construct and calibrate a compact *phenomenological selection functional* $\mathcal{F}_{\rm fit}$ that reproduces the established QGD/GR geometry families, hold out solution classes, identify its inevitable failure modes, and then demand a microscopic collective action whose infrared limit matches that explicit target. This respects Chapter 5's backward research programme. A fitted operational closure is *not* an autonomous microscopic origin theorem.

29.1 First fit: source charges, Chapter 1 projective spinor and Chapter 2 Kerr–Schild weight [F/D]

For the prescribed stationary asymptotically flat Kerr-family projective spinor of QM022, the null congruence direction is encoded by

$$\xi_a = \begin{pmatrix} \cos(\theta/2) \\ e^{i[\phi - \arctan(a/r)]}\sin(\theta/2) \end{pmatrix}, \qquad a = J/M, \quad \Sigma_a = r^2 + a^2\cos^2\theta, \quad \Theta[\xi_a] = \nabla_{\bar g}\cdot \boldsymbol{n}[\xi_a] = \frac{2r}{\Sigma_a} \tag{184}$$

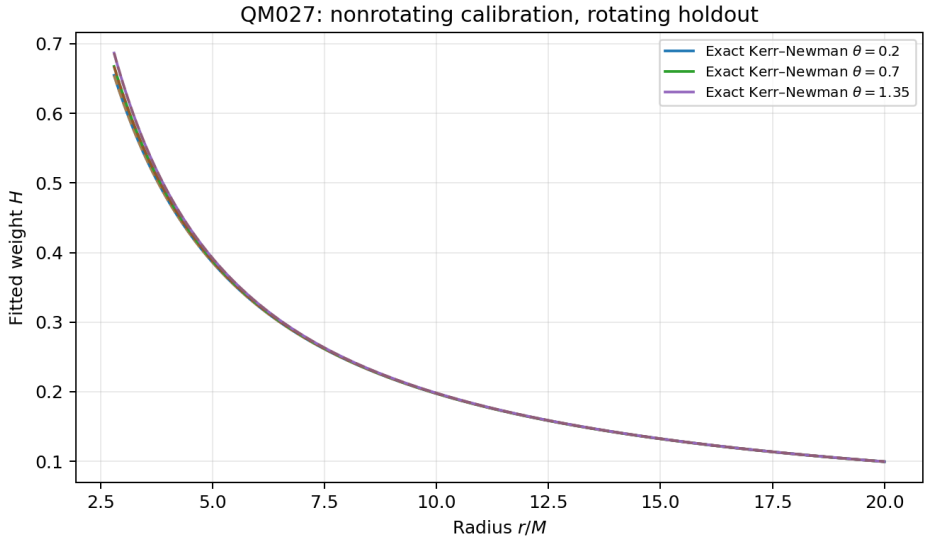
in geometrized units. The spinor direction and twist are *chosen from the Chapter 2 Kerr-family template*; they are not derived here from a generic source density operator. Fit the minimal even-in-electric-charge weight law

$$H_{\rm fit} = \left(\alpha_M M + \alpha_Q \frac{Q_e^2}{r}\right)\Theta[\xi_a], \qquad g_{\mu\nu} = \bar g_{\mu\nu} + H_{\rm fit} k_\mu[\xi_a] k_\nu[\xi_a]. \tag{185}$$

Fitting only the Schwarzschild and Reissner–Nordström nonrotating target profiles gives $\alpha_M = 1$, $\alpha_Q = -1/2$, so

$$\boxed{H_{\rm fit} = \frac{2Mr - Q_e^2}{r^2 + a^2\cos^2\theta}.} \tag{186}$$

This is the Kerr–Newman Kerr–Schild weight given the already selected rotating null congruence. The independent numerical verifier uses 160 synthetic nonrotating training points, then 200 unseen rotating holdout points (100 Kerr; 100 Kerr–Newman); the maximum absolute weight error is 2.22×10^{-16}, maximum relative error 5.88×10^{-16}. These tiny errors reflect an *exact finite-dimensional family identity* and noiseless targets, not empirical corroboration of an emergent microscopic law.



Failure boundaries [D/C]. A source quadrupole independent of (M, J, Q_e) contributes a leading weak-field angular term $\propto Q_2 P_2(\cos\theta)/r^3$ that Eq. (185) cannot predict. A source-free pp wave also has nonzero physical geometry when its incoming radiation data are nonzero. Schwarzschild/Kerr exterior fitting alone cannot infer interior pressure structure $A^2 \neq B^2$ or the FLRW homogeneous mode. A general physical selector therefore requires the *source density operator and its multipoles plus independent initial and radiation data*, not just two charges and a spinor template. The explicit synthetic weak-field quadrupole residual is 0.00458372785 at $r = 5, \theta = 0.45, Q_2 = 0.4$, or 1.15% of the monopole term. It is not a claimed exact nonlinear solution.

29.2 A general fitted weighted-coframe action: recover the TEGR coefficients [F/D/C]

The complete founding master metric remains

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{187}$$

The present fit treats only the *declared fixed-coframe, $\lambda\mathcal{Q} = 0$ specialization.* Let $e^a{}_\mu$ denote a *physical effective coframe* for g, not the reference $T^a{}_\mu$ and not the unproved microscopic fermionic composite. With a flat inertial spin connection $\omega^a{}_{b\mu}$ define $\mathsf{T}^a{}_{\mu\nu} = 2\partial_{[\mu} e^a{}_{\nu]} + 2\omega^a{}_{b[\mu} e^b{}_{\nu]}$ and its spacetime-index form $\mathsf{T}^\rho{}_{\mu\nu}$. For $\mathsf{T}_\mu = \mathsf{T}^\nu{}_{\nu\mu}$, consider the most general tested quadratic parity-even three-invariant ansatz

$$I_1 = \mathsf{T}^\rho{}_{\mu\nu} \mathsf{T}_\rho{}^{\mu\nu}, \quad I_2 = \mathsf{T}^\rho{}_{\mu\nu} \mathsf{T}^{\nu\mu}{}_\rho, \quad I_3 = \mathsf{T}_\mu \mathsf{T}^\mu, \quad \mathbb{T}_{\text{ansatz}} = c_1 I_1 + c_2 I_2 + c_3 I_3. \tag{188}$$

The fit target is the *established* Einstein-scalar/coframe identity, including the divergence term $\mathbb{B} = \frac{2}{e}\partial_\mu(e\mathsf{T}^\mu)$ and $e = |\det(e^a{}_\mu)|$:

$$R(g) = -\mathbb{T}_{\text{ansatz}} + \mathbb{B}. \tag{189}$$

Linear regression against 36 random local coframe jets $e^a{}_\mu(x) = \delta^a{}_\mu + D_\alpha{}^a{}_\mu x^\alpha$, using independently evaluated metric curvature and divergence at $x = 0$, yields the rank-three solution

$$\boxed{(c_1, c_2, c_3) = \left(\tfrac{1}{4}, \tfrac{1}{2}, -1\right), \qquad \mathbb{T}_{\text{fit}} = \tfrac{1}{4} I_1 + \tfrac{1}{2} I_2 - I_3.} \tag{190}$$

Twelve fresh coframe jets give maximum absolute error 6.66×10^{-16}. This reproduces the standard *teleparallel equivalent of general relativity* (TEGR) and does not constitute an independent quantum-gravity discovery. The inertial connection is not freely replaced by the dynamical Lorentz connection conjectured in spinor pregeometry. The equivalent action, including proper boundary data, is

$$\boxed{S_{\rm fit}[e,\omega;\widehat{\varrho}] = -\frac{c^3}{16\pi G_*}\int \mathrm{d}^4x\, e(\mathbb{T}_{\rm fit} + 2\Lambda_*) + S_{\rm matter}[e;\widehat{\varrho}] + S_{\rm boundary}.} \tag{191}$$

It is exactly the adopted Chapter 4 Einsteinian effective dynamics written in a first-derivative weighted-frame language, up to the standard boundary term. The curvature scalar was the *training target*; thus this is a fitted *effective* law, and G_*, Λ_*, universal source coupling and physical phase selection are not derived microscopically.

29.3 The explicit operationally closed selector and its regular QGD lift [C/D]

Let $\mathcal{I}$ collectively denote constraint-compatible Cauchy data, boundary conditions, radiation and gauge/branch conditions. The result of the inverse-engineering programme is the *operational fitted* set-valued selection map

$$\boxed{\begin{aligned}\mathcal{F}_{\rm fit}[S_{\rm source},\widehat{\varrho},\mathcal{I}] &:= \mathrm{GramLift}_{T,\lambda\mathcal{Q}=0}\,[g_*]\,,\\ g_* &= e_*^{\mathsf{T}}\eta e_*, \quad [e_*] \in \mathrm{Sol}_{\mathcal{I}}(\delta S_{\rm fit}/\delta e = 0)\,.\end{aligned}} \tag{192}$$

This is well defined under the regular classical field initial-value assumptions of the adopted effective action, modulo gauge. It does not claim all quantum source states lead to a unique geometry or that the field equations select their own initial values.

For the corresponding symmetric reference-frame metric deviation $h = g_* - \bar{g}$ one smooth eight-sector section is

$$P = b\mathbf{1} + \frac{h}{4b}, \qquad N = b\mathbf{1} - \frac{h}{4b}, \qquad b \neq 0, \qquad \boxed{P^{\mathsf{T}}P - N^{\mathsf{T}}N = h.} \tag{193}$$

This constructs one representative; the 22 local sector-orbit directions are redundant only in the declared regular metric-only theory. The complete $\mathcal{Q}_{ab}$ sector is not reduced by fiat.

29.4 The eight-chapter clues carried into $\mathcal{F}_{\rm fit}$

- **Chapters 1 and 3:** source density operators, full stress and spinor structure enter; neither a single current nor wavefunction normalization alone determines the physical field strength.
- **Chapters 2 and 5:** weighted covectors, the exact Kerr twist, independent Schwarzschild variation, and the conditional $H = \sigma^2_{\rm amp}$ energy normalization constrain the geometry and amplitude.
- **Chapters 4 and 7:** the full master metric, effective pullback Euler equation, 22 regular sector redundancies and gauge/energy consistency require a quotient-level output, not a preferred arbitrary sector representative.
- **Chapters 6 and 8:** the screened scalar, species-specific kinetic and auxiliary-vacuum failures exclude several naive microscopic stories; a sourced, causal exterior field needs homogeneous/cosmological and radiative data too.

These are *structural fitting constraints*; they are not secretly assumed proofs of source-state-to-coframe emergence.

29.5 The changed microscopic goal: derive a known functional, not an undefined force [O]

Our *next* physical research problem is no longer "find any $1/r$ scalar force". It is to find a fully specified microscopic QFT and exterior collective phase such that the correctly regulated causal 1PI/in-in action reproduces the already fitted target,

$$\boxed{\Gamma^{\rm IR}_{\rm coll}[e_+, e_-; \widehat{\varrho}] \;\rightsquigarrow\; -\frac{c^3}{16\pi G_*}\int e\left(\tfrac{1}{4}I_1 + \tfrac{1}{2}I_2 - I_3 + 2\Lambda_*\right) + \Gamma^{\rm common}_{\rm matter}[e; \widehat{\varrho}] + \cdots .} \tag{194}$$

The arrow indicates the physical low-energy reduction, not a claimed identity between a causal doubled quantum action and a single-copy equilibrium functional. A successful microscopic derivation must independently prove the nondegenerate physical collective phase, coframe correlator/propagation, Ward and full constraint identities, source-conditioned exterior field, common Dirac/photon cones, and acceptable regulator and quantum corrections. Chapters 1 and 6 contain candidate operator/response tools; Chapters 2, 3, 4, 5, 7 and 8 supply the independent geometric, normalization, variational, static, gauge and cosmological targets.

Status [F/D/C/O]. *The effective fitted functional has been specified and its two numerical regression tests reproduced. We can now work backwards against it. This does* not *close the autonomous microscopic $\mathcal{F}_{\rm micro}$, and pretending the fitted TEGR action emerged from the original flat Dirac theory would erase precisely the discovery QGD seeks.* A standalone six-page QM027 research note, numerical verifier, JSON results, and optical/quadrupole figures accompany this chapter update.

30 QM031: One-loop vacuum and causal-source consistency of the fitted collective target

Research status. This section is a quantitative continuation of QM027–QM030, not an independent postulate that the microscopic spinor-pregeometry theory M already possesses a geometric phase. We compare *two observables of the same conditioned one-loop Dirac determinant*: the source-coupled infrared kinetic response and the homogeneous vacuum term. The key result is a cutoff-scale obstruction to obtaining both from the *unsubtracted* determinant in a controlled low-energy branch. A separate exact two-state calculation validates the causal-versus-Euclidean response methodology. The actual four-dimensional composite-frame phase and its complete in–in functional remain open.

30.1 Founding metric and the precise common-field premise

The founding QGD metric is unchanged:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{195}$$

Only the regular metric-only branch $\lambda\mathcal{Q} = 0$ and the corresponding physical metric quotient are used in the tested induced-Einstein response. The independent $\mathcal{Q}$ metric operator is neither specified nor silently generated by a matter loop. If the physical collective coframe e exists, then $g = e^{\mathsf{T}}\eta e$ has a regular signed-sector lift to $[\Sigma]$; this kinematic algebra is *not* a proof that the microscopic path integral selects e.

The candidate microscopic spinor-pregeometry action from QM024–QM026 has fermion-bilinear composites $E[\psi, \omega]$, auxiliary X and Π, and a possible dual frame $e(\Pi)$ on the invertible branch. The exact insertion $X = E$ and the coframe duality remain formal identities of the specified regulated integral. The *physical existence*, state, causal measure, mass-frame alignment and universal photon vertex have not been established. The following determinant calculation is explicitly conditional on possessing such a common-frame phase; it must not be identified with a proof of that phase.

30.2 A two-channel vacuum calculation from the same fermion loop

For one four-component Euclidean Dirac species in the previously specified proper-time convention, Chapter 6 found

$$W_E[g] = \int d^4x\sqrt{g}\{C_0 + C_R R + O(R^2, \nabla^2 R)\}, \quad C_0 = \frac{I_0}{8\pi^2}, \quad C_R = -\frac{I_1}{96\pi^2}, \tag{196}$$

where for the common cutoff $\Lambda_c > 0$,

$$I_0(m) = \int_{\Lambda_c^{-2}}^{\infty} ds\, s^{-3} e^{-m^2 s}, \qquad I_1(m) = \int_{\Lambda_c^{-2}}^{\infty} ds\, s^{-2} e^{-m^2 s}. \tag{197}$$

The sign and normalization refer to that *particular* parity-even determinant prescription. Complete microscopic spinor pregeometry also contains an auxiliary Legendre functional, mass-frame matching, counterterms and possibly other fields. These are *not* evaluated here.

Assume only for a diagnostic that the N identical species' loop terms supply the *entire* low-energy Euclidean R and volume coefficients, with no independent or finite cosmological counterterm. Match to $-(16\pi G_{\rm ind})^{-1}\int\sqrt{g}(R - 2\Lambda_{\rm loop})$. Algebra gives

$$\boxed{G_{\rm ind} = \frac{6\pi}{N I_1}, \qquad \Lambda_{\rm loop} = \frac{6 I_0}{I_1}, \qquad H^2_{\rm formal} = \frac{\Lambda_{\rm loop}}{3} = \frac{2 I_0}{I_1}.} \tag{198}$$

The last equality is the *formal* de Sitter solution of the Einstein truncation. It is a low-energy prediction only if $H \ll \Lambda_c$ and all neglected terms are subleading. We now test precisely that requirement.

Proposition QM031.1 (cutoff-scale vacuum obstruction; exact within the stated benchmark). For every $m \geq 0$ in the common proper-time model,

$$\boxed{\Lambda_c^2 \;\leq\; H_{\rm formal}^2 \;<\; 2\Lambda_c^2.} \tag{199}$$

For any collection of independently contributing Dirac species with positive multiplicities, arbitrary nonnegative masses and the *same* cutoff, the same inequality holds for the induced total coefficients. Consequently the unadjusted loop terms alone cannot select a de Sitter vacuum in a parametrically controlled $H \ll \Lambda_c$ Einsteinian derivative expansion.

Proof. Put $z = m^2 \geq 0$ and write I_0/I_1 as the normalized mean of s^{-1} with positive weight $s^{-2}e^{-zs}$ on $s \in [\Lambda_c^{-2}, \infty)$. Since $s^{-1} < \Lambda_c^2$ except at an endpoint of zero measure, $I_0/I_1 < \Lambda_c^2$. At $z = 0$, elementary integration gives $I_0/I_1 = \Lambda_c^2/2$. Let $I_2(z) = \int_{\Lambda_c^{-2}}^{\infty} ds\, s^{-1} e^{-zs}$ for $z > 0$. Differentiating yields

$$\frac{d}{dz}\frac{I_0}{I_1} = \frac{I_0 I_2 - I_1^2}{I_1^2} \geq 0, \tag{200}$$

by Cauchy–Schwarz. Hence $\Lambda_c^2/2 \leq I_0/I_1 < \Lambda_c^2$, proving Eq. (199). For mixed species, the total ratio $\sum_i w_i I_{0,i} / \sum_i w_i I_{1,i}$ is a convex combination of the individual ratios weighted by $w_i I_{1,i} > 0$. □

The exact expressions are

$$I_1 = \Lambda_c^2 e^{-x^2} - m^2 E_1(x^2), \quad I_0 = \tfrac{1}{2}[\Lambda_c^4 e^{-x^2} - m^2 I_1], \qquad x = m/\Lambda_c. \tag{201}$$

At the existing benchmark $N = 100$, $m = 0.01\Lambda_c$, $\Lambda_c = 1$, we independently obtain

$$I_0 = 0.499900051, \quad I_1 = 0.999036683, \quad G_{\rm ind} = 0.188677315, \quad H_{\rm formal}/\Lambda_c = 1.000382. \tag{202}$$

At $m/\Lambda_c = 0, 0.5, 1, 2, 4$, the ratios $H_{\rm formal}/\Lambda_c$ are approximately $1, 1.11994, 1.21547, 1.31408, 1.37789$, respectively. The bound, not these sample points, establishes failure of the subcutoff Einstein truncation. Crucially, increasing N scales $G_{\rm ind}$ but cancels from the formal H/Λ_c. **Scope of the obstruction.** It is conditional on the same regulator/convention and *no* extra volume coefficient. A renormalization counterterm, a collective Legendre term, a different state, a distinct heavy spectrum or a nonperturbative phase may change the conclusion. Therefore it is *not* a theorem that QGD cannot select a small cosmological constant; it is a theorem that this unsubtracted leading determinant cannot self-consistently do so within its own low-energy truncation.

30.3 The source channel: what the same coefficient does and does not prove

After a flat-vacuum subtraction and *assuming the physical coframe is dynamical*, the same I_1 yields the standard Einstein kinetic term with $G_{\rm ind}$ in Eq. (198). With a common metric stress vertex and covariantly conserved classical source, the conditional static weak-field and transverse-traceless retarded responses are

$$\boxed{\frac{\delta h_{00}(0,\mathbf{k})}{\delta\rho(0,\mathbf{k})} = \frac{8\pi G_{\rm ind}}{|\mathbf{k}|^2}, \qquad \frac{\delta h_{ij}^{\rm TT}(\omega,\mathbf{k})}{\delta T_{ij}^{\rm TT}(\omega,\mathbf{k})} = \frac{16\pi G_{\rm ind}}{|\mathbf{k}|^2 - (\omega + i0)^2}.} \tag{203}$$

Here $c = 1$ and the retarded tensor expression is the usual linearized Einstein equation in a stated gauge for a TT source. Both formulas are *derived responses of the admitted Einsteinian effective*

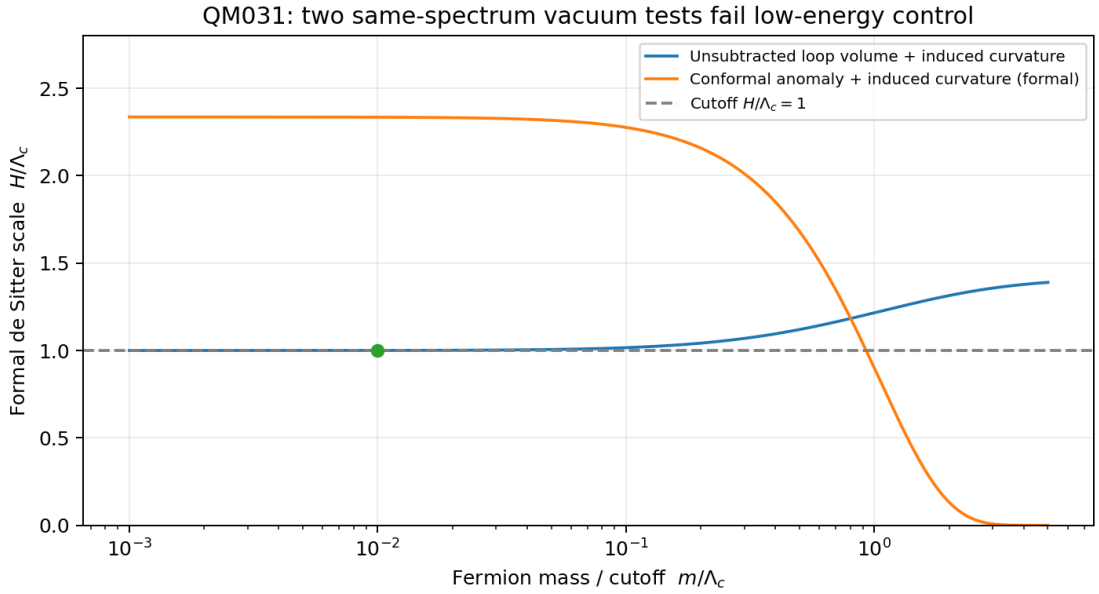


Figure 13: Two distinct formal vacuum tests with the same illustrative species spectrum: the unsubtracted determinant's volume-to-curvature ratio (solid line) and QM030's massless-anomaly/induced-G hybrid (second curve). The first always has $H \geq \Lambda_c$; the second intersects the cutoff only when its fermions are not light compared with H. Neither is a controlled late-time-Λ prediction.

metric, not massless composite tensor poles established directly in the original fermionic four-point function. At $|\mathbf{k}| = 0.01, 0.04, 0.2$, the first formula gives respectively 47419.78, 2963.736 and 118.549 in the benchmark. These values cross-check that the *same* induced G is used across source and vacuum channels.

A consistent physical claim requires the *complete* collective inverse kernel, including the real-time matter polarization, metric seagull, auxiliary-field potential/measure, regulator-conserving vertices, and any independent sector dynamics. The two Einsteinian formulas alone cannot prove that an actual composite metric fluctuates at long distances.

30.4 An explicit $k = 0$ versus static-exterior separation

The retarded scalar denominator in Eq. (203) is $D_R(\omega, k) = [k^2 - (\omega + i0)^2]^{-1}$. The purely kinematical gradient-weighted factor $F(\omega, k) = k^2 D_R(\omega, k)$ obeys

$$\boxed{\lim_{k\to 0} F(0,k) = 1, \qquad \lim_{\omega\to 0} F(\omega, 0) = 0.} \tag{204}$$

This exact order-of-limits check is not itself a cosmological equation. It exhibits why a static inverse-Laplacian fit at $\omega = 0$ does not fix the strictly homogeneous time-dependent sector. Chapter 8 requires an independent homogeneous constraint and quantum initial state. For its flat FLRW metric, $3H^2 = 8\pi G_{\rm ren}\rho + \Lambda_{\rm eff}$ and $\dot{H} = -4\pi G_{\rm ren}(\rho + p)$ follow from the admitted action; the values of $\rho_{\rm vac}$ and the residual $\Lambda_{\rm eff}$ do not follow from the static source kernel.

Proposition QM031.2 (vacuum matching is not determined by the local source response). At fixed curvature coefficient C_R and fixed specified matter vertices, adding the finite local counterterm $\delta C_0 \int \sqrt{g}$ leaves the flat-space R coefficient unchanged while shifting $\Lambda_{\rm eff}$ by $8\pi G\, \delta C_0$. The family of renormalized effective actions therefore shares the matched local kinetic coefficient but admits different homogeneous vacuum backgrounds. A source fit performed only in the

weak-field regime $Hr \ll 1$ cannot determine $\Lambda_{\rm eff}$ uniquely. A genuinely microscopic vacuum-state, renormalization or global matching principle is necessary for a numerical cosmological prediction.

30.5 Causal state and noise: an exactly solved control

The matter in–in functional for any candidate physical frame is formally

$$Z_{\rm CTP}[e_+, e_-; \rho_0] = \mathrm{Tr}\Big(U[e_+]\rho_0 U[e_-]^\dagger\Big), \qquad Z_{\rm CTP}[e, e; \rho_0] = 1. \tag{205}$$

A real source-conditioned collective equation requires $\delta\Gamma_{\rm CTP}/\delta e_- = 0$ at $e_- = 0$, with retarded response, noise, metric seagulls and state-dependent initial correlations. In terms of $t_{\mu\nu} = T_{\mu\nu} - \langle T_{\mu\nu}\rangle$, its source kernels are proportional to the causal commutator $i\theta(t-t')\langle[T_{\mu\nu}(x), T_{\rho\sigma}(y)]\rangle$ and the symmetric noise kernel $\frac{1}{2}\langle\{t_{\mu\nu}(x), t_{\rho\sigma}(y)\}\rangle$. Contact terms depend on the metric coupling and must be included in the conserved Ward identity. Merely analytically continuing a Euclidean I_1 does not establish this full structure.

To test the causal methodology *without assuming gravity*, use an exactly soluble two-state quantum control $H_0 = (\Delta/2)\sigma_z$, $O = \sigma_x$ with ground-state density matrix, and a perturbation $-J(t)O$. Direct unitary evolution gives

$$\chi_R(t) = i\theta(t)\langle[O(t), O(0)]\rangle = 2\theta(t)\sin(\Delta t), \quad N_O(t) = \tfrac{1}{2}\langle\{O(t), O(0)\}\rangle = \cos(\Delta t). \tag{206}$$

The retarded kernel vanishes before the source, whereas the symmetric noise does not; they are inequivalent observables of the same state. For a small rectangular pulse $J = 10^{-3}$ over $0 < t < 0.4$, $\Delta = 1.3$, observing at $t = 1.2$, linear response predicts

$$\delta\langle O(t)\rangle_{\rm Kubo} = \frac{2J}{\Delta}[\cos\Delta(t-0.4) - \cos\Delta t] = 0.0007621909849. \tag{207}$$

Exact exponentiation of the time-dependent two-state Hamiltonian gives 0.0007621909077, differing by 7.7×10^{-11}, consistent with the nonlinear source correction. The commutator and noise formulas independently pass 50 time-point checks. This is a *response-method validation*, not a spin-2 composite calculation or a test of the full microscopic M.

30.6 The actual inverse problem: a common causal Hessian, not two fitted equations

The microscopic reverse-engineering target can be stated as the decomposition of the physical collective *retarded* inverse kernel,

$$\boxed{\begin{aligned} K_R^{\rm phys} &= K_R^{\rm aux/measure} + \Pi^{\rm Dirac}_{R,\,\rm bubble+seagull} + K_R^{\rm ct} + K_R^{\rm other}, \\ \mathcal{D}_R &= (K_R^{\rm phys})^{-1}\big|_{\rm constrained,\,quotient}. \end{aligned}} \tag{208}$$

The full Dirac Hessian includes scalar, vector, trace and TT blocks; the seagull must be retained even when the TT block is projected. The retarded kernel is a response to the same quantum source state that defines its mean stress; the symmetric noise is a separate CTP kernel. In particular, the algebraic signed-sector kernel of dimension 22 on the regular eight-sector metric-only map must be quotiented, rather than inverted as physical propagating degrees of freedom. The remaining diffeomorphism constraints must also be treated.

If N independent species couple to an already selected common geometric phase, the matter-induced kinetic operator and stress noise typically scale as $K_R \sim N$ and $\mathcal{N} \sim N$. Conditional on a stable invertible physical Hessian, the metric response $\mathcal{D}_R \sim N^{-1}$ and the induced metric covariance $\mathcal{D}_R\mathcal{N}\mathcal{D}_A \sim N^{-1}$. This is a useful *conditional large-N consistency scaling*, not a proof

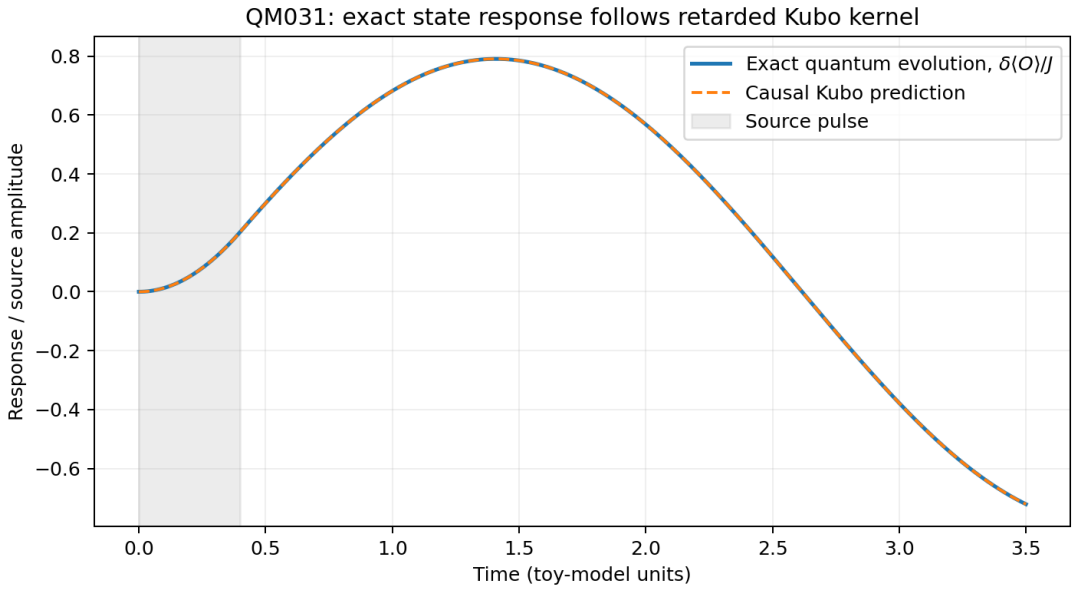


Figure 14: Direct finite-state unitary response to a source pulse versus the retarded Kubo expression, calculated from the same initial quantum state. This validates the method but not a four-dimensional quantum-gravitational geometric phase.

that the microscopic determinant phase is realized or that the spin-two pole exists. The actual full four-dimensional measure and state-dependent spectral density are required to establish such claims.

The vacuum and source conditions must be imposed on one and the same $\Gamma_{\rm coll}^{\rm CTP}$: a small source response on a flat-tuned branch and the cosmological vacuum contribution of the same renormalized action cannot be combined with two independently chosen values of the subtracted volume term. This is the practical content of the QM031 cutoff theorem for our inverse-design strategy.

30.7 A single common-branch acceptance test for $\mathcal{F}_{\rm micro}$

The next actual four-dimensional calculation must use one specified regulated microscopic action and one physical sector: (i) obtain a nondegenerate, gauge-invariant or relational geometric phase; (ii) compute its full causal inverse Hessian $K_{ab}^{\rm CTP}$, with conservation identities and seagulls; (iii) show a source-visible massless TT response with positive physical residue and an appropriate gauge constraint; (iv) verify common fermion/photon probe vertices on the *same* state; (v) determine the homogeneous stress and the initial/global rule fixing $\Lambda_{\rm eff}$ without reusing an arbitrary fitted subtraction; and (vi) lift the selected physical g to the QGD natural-invariant quotient. Only the last step is presently algebraically complete for regular metric-only sectors.

Result	Status
Same determinant C_0, C_R gives Eq. (198)	Exact within specified one-loop truncation [D/C]
Unsubtracted homogeneous $H \ll \Lambda_c$ from those terms	Fails cutoff bound [D/C]
Common $G_{\rm ind}$ in admitted static and TT responses	Conditional Einsteinian match [C]
Causal Kubo response versus exact state evolution	Verified toy control [D/N]
Actual $4D$ spinor-pregeometry phase, full CTP Hessian, universal vertices	Not computed [O]
Numerical value of physical cosmological constant	Not predicted [O]

Scientific conclusion. QM031 identifies an exact *two-channel incompatibility in a specific un-*

subtracted low-energy determinant candidate, not a general failure of QGD. A vacuum-matching principle is necessary before the same common-geometry effective theory can consistently accommodate exterior gravity and controlled cosmic expansion. Yet supplying that principle as a free counterterm does not derive it from microscopic matter. The original full master metric, Chapter 5's renormalized state-dependent action and Option C remain intact. The causal, quotient-invariant collective phase remains the central missing physical derivation.

31 QM034: Exact Causal Determinant-Composite Selection Test [D/N/O]

Appendix scope. This calculation is an exactly solved projected microscopic Hamiltonian rather than the full four-dimensional spinor-pregeometry action. It tests whether one determinant interaction can simultaneously carry retarded source overlap, cross-species correlations, and an independently meaningful homogeneous composite vacuum. Results never assume an emergent metric.

31.1 Scope and the QGD target

The source objective of Chapter 6 and QM027–QM033 is a single microscopic theory whose physical collective phase yields both the source-conditioned tensor response and the cosmic vacuum response. The physical test is not the existence of a local nonzero composite or a fitted solution of Einstein's equations. In the general case the physical retarded frame kernel includes mixing with collective connections and other auxiliary fields,

$$K^R_{\rm phys} = K^R_{ee} - K^R_{eY}(K^R_{YY})^{-1}K^R_{Ye}, \tag{209}$$

after consistent projection of gauge and constraint directions. A genuine gravity candidate must possess healthy source-visible helicity-two response, universal matter coupling, and a compatible homogeneous vacuum in the same regulator and phase.

The founding full metric is not altered:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \big[\eta_{ab} - \sum_s \varepsilon_s \sigma^{(s)}_a \sigma^{(s)}_b - \lambda \mathcal{Q}_{ab}\big], \qquad s_s = -\varepsilon_s.} \tag{210}$$

Here and below the exact toy calculation is *not* substituted for a derivation of this map or of the Chapter 5 renormalized QGD action.

31.2 One microscopic interaction: a genuine finite-dimensional Hamiltonian

Place four distinguishable fermion flavours $a = 1, \dots, 4$ on two sites $x = 1, 2$, with $\{c_{xa}, c^\dagger_{yb}\} = \delta_{xy}\delta_{ab}$. Define site densities $n_{xa} = c^\dagger_{xa}c_{xa}$ and the four-channel determinant-like site composite $V_x = \prod_{a=1}^4 n_{xa}$. The microscopic Hamiltonian is

$$\boxed{H = -t\sum_{a=1}^{4}(c^\dagger_{1a}c_{2a} + c^\dagger_{2a}c_{1a}) - g(V_1 + V_2), \quad g > 0,\ t > 0.} \tag{211}$$

We restrict to the number sector $n_{1a} + n_{2a} = 1$ for each flavour. This is a subspace of the original eight-orbital Fock space with four particles; all source and probe operators used below preserve this sector. The operator V_x is a diagonal, four-channel analogue of a determinant composite and *not* the derivative-dependent coframe determinant $\det E[\psi, \omega]$ of model M. In particular, this system has no continuum Lorentz group, spin connection, four-dimensional stress tensor, photon, or gravitational tensor mode. Its role is to test a necessary microscopic mechanism without assuming geometry.

Let $S^\alpha = \frac{1}{2}\sum_a \sigma^\alpha_a$ be the sum of four two-site pseudo-spins. The permutation-symmetric subspace has total spin $S = 2$ and basis $|m\rangle$, $m = -2, -1, 0, 1, 2$. In this basis,

$$H_{mn} = -g(\delta_{m,-2} + \delta_{m,2})\delta_{mn} - t\sqrt{(2-m)(3+m)}\,\delta_{n,m+1} + \text{h.c.}, \tag{212}$$

$$n_1 = \sum_a n_{1a} = 2 + S_z, \qquad \mathcal{V} := V_1 + V_2 = |-2\rangle\langle -2| + |2\rangle\langle 2|. \tag{213}$$

The resulting five-by-five real symmetric matrix is exactly diagonalizable. We independently constructed the full 70-by-70 four-particle fermionic Hamiltonian and confirmed its ground energy, first excitation and ground-state source observables against the five-state reduction at several nontrivial t/g values. [N]

31.3 Exact collective gap: useful softness is not a graviton pole

Under exchange of sites, $m \leftrightarrow -m$, the Hamiltonian divides into an even 3×3 block and an odd 2×2 block. Their characteristic equations are [D]

$$p_{\text{even}}(E) = E^3 + gE^2 - 16t^2E - 12gt^2 = 0, \tag{214}$$

$$p_{\text{odd}}(E) = E^2 + gE - 4t^2 = 0. \tag{215}$$

For small t/g, the even ground energy and lowest odd energy obey

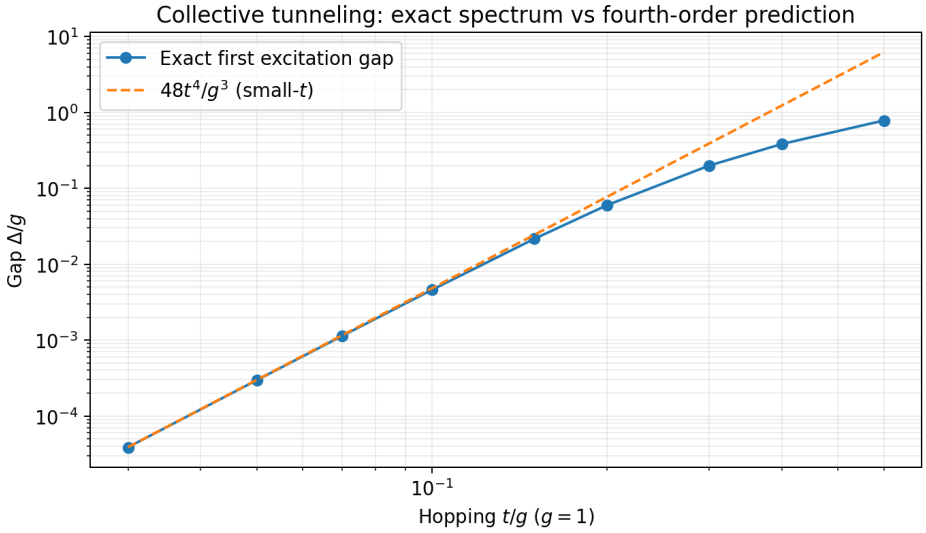
$$E_0 = -g - \frac{4t^2}{g} - \frac{32t^4}{g^3} + O(t^6/g^5), \tag{216}$$

$$E^-_{\text{odd}} = \frac{-g - \sqrt{g^2 + 16t^2}}{2} = -g - \frac{4t^2}{g} + \frac{16t^4}{g^3} + O(t^6/g^5), \tag{217}$$

so the lowest gap is

$$\boxed{\Delta = E^-_{\text{odd}} - E_0 = \frac{48t^4}{g^3} + O(t^6/g^5).} \tag{218}$$

Physically the two lowest configurations have all four flavours localized together on either site. Their mixing requires four microscopic hopping events, yielding the fourth-order tunnelling gap. Thus an interacting fermionic composite can possess a nontrivial, source-coupled soft mode without inserting a kinetic action for an external metric.



Exact first gap versus the independently derived fourth-order asymptote.

31.4 One microscopic model, exact causal source and probe vertices

Choose a physical density source $h_x(s)$ with $H \to H - \sum_x h_x(s)n_x$, and initial state $\rho_0 = |0\rangle\langle 0|$. This defines a bona fide closed-time-path functional,

$$Z[h_+, h_-] = \mathrm{Tr}\,[U_{h_+}\rho_0 U^\dagger_{h_-}], \qquad Z[h, h] = 1. \tag{219}$$

Define $\chi^R_{AB}(s) = i\theta(s)\langle[A(s), B(0)]\rangle$ for source $-h_B B$. Its zero-temperature exact Lehmann form is

$$\chi^R_{AB}(\omega) = \sum_{n>0} \frac{2\Delta_n \,\mathrm{Re}(A_{0n}B_{n0})}{\Delta_n^2 - (\omega + i0)^2}, \quad N_{AA}(s) = \sum_{n>0} |A_{0n}|^2 \cos(\Delta_n s), \tag{220}$$

for real symmetric operators here, where $\Delta_n = E_n - E_0$. This is a causal *microscopic* response, not an assumed retarded gravitational Green's function. The exact site-number conservation identity $n_1 + n_2 = 4I$ implies

$$\boxed{\chi^R_{12}(\omega) = -\chi^R_{11}(\omega), \qquad \chi^R_{00}(\omega) = 0 \quad (n_0 = n_1 + n_2).} \tag{221}$$

The static inhomogeneous susceptibility is

$$\chi_{11}(0) = 2\sum_{n>0} \frac{|(n_1)_{0n}|^2}{\Delta_n} = \frac{g^3}{6t^4} + O(g/t^2), \tag{222}$$

where $|(n_1)_{01}|^2 \to 4$ at small t/g. In the two-site source-response matrix, the homogeneous vector $(1,1)$ is a conserved null direction and the relative vector $(1,-1)$ has susceptibility eigenvalue $2\chi_{11}$. Its inverse on the non-null subspace is $K_-(0) = [2\chi_{11}(0)]^{-1} \sim 3t^4/g^3$. This provides an explicit example of a constrained physical inverse response; inversion of the unprojected two-by-two matrix would be meaningless.

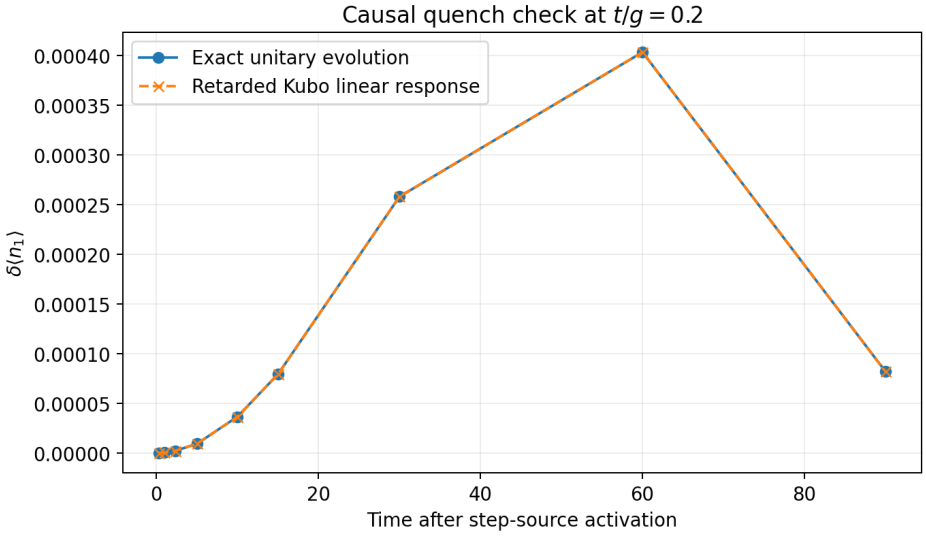
Retarded consistency and noise. The same exact eigenstates provide χ^R and the symmetric correlator N. A step source $h\theta(s)$ gives, at first order,

$$\delta\langle n_1(s)\rangle = 2h\sum_{n>0} \frac{|(n_1)_{0n}|^2}{\Delta_n}(1 - \cos(\Delta_n s)) + O(h^2). \tag{223}$$

Independent direct evolution with $\exp[-i(H - hn_1)s]$ reproduces Eq. (223). The exact CTP generating functional also obeys $Z[h,h] = 1$, and its first contour-source derivative reproduces $\langle n_1\rangle = 2$. The spectral density obeys the exact double-commutator sum rule

$$\sum_{n>0} \Delta_n |(n_1)_{0n}|^2 = \tfrac{1}{2}\langle 0|[n_1,[H,n_1]]|0\rangle. \tag{224}$$

All checks were carried out with the same finite Hamiltonian. [D][N]



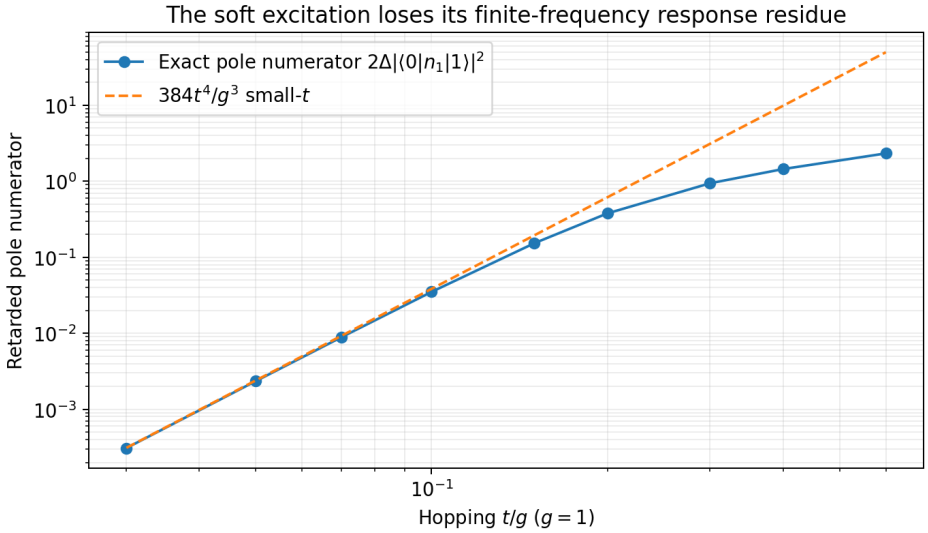
A direct unitary step-source evolution and independent retarded-Kubo prediction agree.

31.5 A decisive false-positive control: a soft gap without a finite wave-kernel numerator

The low-pole contribution to the retarded susceptibility is

$$\chi^R_{11}(\omega) \supset \frac{\mathcal{R}(t)}{\Delta^2 - (\omega + i0)^2}, \qquad \mathcal{R}(t) = 2\Delta|(n_1)_{01}|^2 = \boxed{\frac{384t^4}{g^3} + O(t^6/g^5)}. \tag{225}$$

Hence $\mathcal{R}(t) \to 0$ as $t \to 0$, even though the static $\chi_{11}(0)$ diverges. *Precision about residues:* the literal first-order complex-frequency residues at $\omega = \pm\Delta$ are $\mp|(n_1)_{01}|^2$ and do *not* vanish; it is the coefficient of the relativistic-style second-order denominator $\Delta^2 - (\omega + i0)^2$ that vanishes. The coalescing positive/negative poles cancel at any fixed nonzero frequency. For fixed nonzero frequency the soft contribution disappears; the $t \to 0$ and $\omega \to 0$ limits do not commute. At strictly $t = 0$, density commutes with H and does not transmit a causal source effect. This exactly refutes the inference "large static susceptibility + soft excitation ⇒ a massless long-range mediator with finite dynamical coupling." It is *not* a statement about whether an actual 4D spinor composite can propagate a graviton.



The low-mode inverse-kernel softness is accompanied by a vanishing source-response pole numerator.

31.6 Species universality test: success inside the interacting set, failure for a spectator

Introduce the flavour-resolved operators n_{1a} and their static susceptibilities $\chi_{ab} = \partial\langle n_{1a}\rangle/\partial h_{1b}$. Independent full-Fock-space calculations yield

g	t	χ_{aa}	$\chi_{a\neq b}$
1	0.10	104.230378	104.201936
1	0.20	6.745605	6.579590
1	0.40	0.873244	0.531467
0	0.20	1.250000	0 (to numerical precision)

The interaction causes a shared dynamical response among distinct flavours, rather than a species-specific local Hartree shift. This is an exact and nontrivial improvement on an independent-cell model. But extending the Hilbert space with an uncoupled spectator H_s gives

$$H_{\text{tot}} = H \otimes I + I \otimes H_s \quad \Longrightarrow \quad \boxed{\chi^R_{n_1,Q_s}(s) = 0} \tag{226}$$

for every purely spectator observable Q_s. Thus the interaction alone does not guarantee universal photon/fermion propagation or a universal common metric. A genuine QGD candidate must demonstrate universality of the *physical principal symbols and vertices*, not only correlation among chosen constituents.

31.7 The homogeneous composite does not follow the soft response

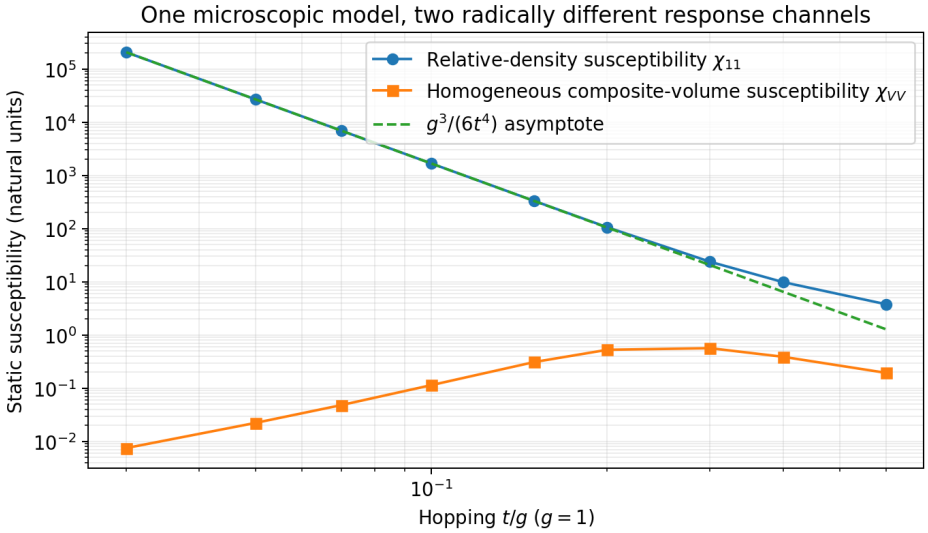
The even operator $\mathcal{V} = V_1 + V_2$ probes the homogeneous determinant-like composite. Site exchange parity gives a sharp selection rule: $n_1 - 2$ is odd, $\mathcal{V}$ is even and the ground state is even. Therefore

$$\boxed{\chi_{(n_1-2),\mathcal{V}}(\omega) = 0, \qquad |(\mathcal{V})_{01}|^2 = 0.} \tag{227}$$

The first excitation accessible to $\mathcal{V}$ is an even level whose gap approaches g, not zero, at small t/g. Differentiating the ground energy with respect to g gives

$$\langle \mathcal{V} \rangle = -\frac{\partial E_0}{\partial g} = 1 - \frac{4t^2}{g^2} + O(t^4/g^4), \qquad \boxed{\chi_{\mathcal{V}\mathcal{V}}(0) = \frac{8t^2}{g^3} + O(t^4/g^5).} \tag{228}$$

Meanwhile $\chi_{11}(0) \sim t^{-4}$ diverges: one and the same microscopic interaction gives a highly responsive inhomogeneous mode and a stiff homogeneous composite. [D][N]



Inhomogeneous and homogeneous channels have parametrically different infrared behavior.

In the fixed-number sector, the homogeneous *density* $n_1 + n_2 = 4I$ is conserved and has identically zero retarded response; the homogeneous composite *volume* $\mathcal{V}$ does respond but remains gapped. Neither is the cosmic scale factor of a four-dimensional geometric phase. This toy proves a logical obstruction to deriving a cosmological mode solely from the presence of a soft inhomogeneous collective response.

31.7.1 The absolute vacuum energy is independently adjustable

For any $C \in \mathbb{R}$, $H \mapsto H + CI$ shifts E_0 by C but leaves every excitation gap, physical source matrix element, retarded kernel, symmetric noise kernel, and normalized CTP functional unchanged. This is an exact model of the vacuum-reference ambiguity, *not* a calculation of the observed cosmological constant. In a dynamical gravitational completion the absolute normalization must be matched to its gravitational coupling: one cannot conclude from this nongravitational invariance that vacuum energy never gravitates.

31.8 Numerical verification and status ledger

t/g	Δ	$\chi_{11}(0)$	$\chi_{\mathcal{V}\mathcal{V}}(0)$	$\Delta_{\mathcal{V},\text{accessible}}$
0.05	0.000296909	26667.3138	0.0223622	0.842894
0.10	0.004584224	1667.3447	0.1145487	0.724789
0.20	0.059687576	105.9375	0.5273438	0.659688
0.40	0.382720057	9.8705771	0.3899075	1.133723

The standalone `verify_qm034.py` script contains 149 programmatic assertions covering symmetry reduction, full Fock cross-checks, exact spectral identities, static source derivatives, time-domain causality and source evolution, flavour mixing, spectator factorization and vacuum shift. The recorded run passes all tests. The numerical checks corroborate the stated finite-dimensional identities; they do not test the continuum spinor-pregeometry M.

Statement	Status
An interacting determinant-like microscopic composite produces nonlocal causal and cross-flavour response	[D][N]
Exact low gap $\Delta = 48t^4/g^3 + \cdots$ and large static response	[D][N]
The soft mode has a vanishing $1/(\Delta^2 - \omega^2)$ numerator as $t \to 0$	[D][N]
Homogeneous composite response stays gapped; vacuum energy shift leaves CTP unchanged	[D][N]
A physical four-dimensional transverse-traceless mode in M	**[O]**
A common photon/fermion metric and a controlled cosmic vacuum in M	**[O]**
A source-conditioned geometric phase and microscopic QGD [Σ] selection	**[O]**

31.9 Implications for the next genuine QGD calculation

A source-coupled soft pole is insufficient. The admissible relativistic infrared kernel must have (i) a nonzero positive physical spin-two pole residue, (ii) Ward-complete tensor/connection constraints, (iii) stable off-shell dynamics after removing redundant sector directions, (iv) universal fermion/photon principal symbols and source vertices, and (v) homogeneous vacuum selection with the same regulator and renormalization conditions.

The correct *full* next calculation is not another static metric fit. It is the four-dimensional microscopic model M with

$$E^a{}_\mu[\psi,\omega] = \sum_i \frac{i}{2}\bar{\psi}_i\gamma^a\overleftrightarrow{D_\mu}\psi_i, \qquad S_M = \kappa\int d^4x\,\det E,$$

(or a stated, independently motivated modification), correctly regulated with dynamical gauge/connection fields and an actual CTP measure. First establish a nondegenerate *gauge-invariant* geometric phase; compute the coupled $X/\Pi/\omega$ response and Schur complement; verify a finite-residue, source-visible helicity-two pole; then compute the homogeneous vacuum and species propagation in that same phase. The 2026 spinor-loop vierbein/connection study (Maitiniyazi et al., 2026) offers relevant Ward-complete perturbative technology, but its externally specified first-order variables are not themselves proof of composite selection in M. Lattice spinor-gravity work (Wetterich 2011; Vladimirov and Diakonov 2012) highlights why a geometric phase and long-range metric correlations are distinct questions.

Verdict. QM034 is a constructive but deliberately limited partial success. It shows in an exactly solvable unitary microscopic model how a determinant-type interaction can communicate source response among species and spatial sites, while furnishing a rigorous false-positive example for naive massless-pole identification and a separate homogeneous-vacuum obstruction. It leaves the full physical $\mathcal{F}_{\text{micro}}$ problem open.

32 QM037: Curved Dirac, Clifford basis, WKB bridge, and a candidate collective wave

Scope and proof status [D/N/C/O]. This appended research section is the shortened self-contained QM037 mathematical note, with 107 algebraic/numerical verifications and independent citations in the standalone report. It does not replace the complete QGD founding metric, the Chapter 5 renormalized action, or the earlier microscopic results. Most importantly, the derivative-current trial is a new *conditional* model and has not demonstrated a stable exterior collective phase.

32.1 Source definitions and what is actually being reversed

The founding QGD master metric, with reference coframe $T^a{}_\mu$ and historical quantum tensor $\mathcal{Q}_{ab}$, is

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{229}$$

The metric-only specialization $\lambda\mathcal{Q} = 0$ is *declared*, not derived. The Chapter 5 renormalized target is

$$\Gamma^{\rm ren}_{\rm QGD} = \frac{1}{16\pi G_N} \int \mathrm{d}^4x\sqrt{-g}(R - 2\Lambda_{\rm ren}) + S_{\rm GHY} + S^{\rm cl}_{\rm F1} + \hbar\Gamma^{(1)}_{\rm m,ren} + O(\hbar^2). \tag{230}$$

The preferred metric-pullback Euler equation is

$$\boxed{2s_s \Pi^{ab} \sigma_b^{(s)} = \lambda \left[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\right]^a.} \tag{231}$$

Equation (231) assumes the action depends on the sector fields through g only. Independent sector dynamics contribute their own Euler terms. The fitted phenomenological $\mathcal{F}_{\rm phen}$ solves causal effective equations and lifts the resulting g to $[\Sigma]$ on a regular branch; it is not a derivation of $\mathcal{F}_{\rm QM}$. We seek a *physical collective field/wave* without assuming any particular particle interpretation.

32.2 Clifford algebra: why a physical basis is more than four vectors

Use signature $\eta_{ab} = \mathrm{diag}(-,+,+,+)$ and constant complex 4×4 matrices satisfying

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbf{1}_4. \tag{232}$$

An inverse physical coframe $e_a{}^\mu$, *not the QGD reference* T, defines the curved matrices

$$\Gamma^\mu = e_a{}^\mu \gamma^a, \qquad \{\Gamma^\mu, \Gamma^\nu\} = 2g^{\mu\nu}\mathbf{1}_4, \qquad g^{\mu\nu} = e_a{}^\mu \eta^{ab} e_b{}^\nu. \tag{233}$$

Thus the 16 entries of the physical coframe consist locally of ten metric degrees of freedom plus six local Lorentz frame choices. This $16 - 6 = 10$ quotient is not the same as the $32 - 22 = 10$ local counting for QGD's signed eight-sector Gram representation; the two redundancies act on different bundles. The bridge between them is a *physical metric*, not equality of their gauge orbits.

Every complex 4×4 matrix expands in the Clifford basis

$$\mathbf{1}\ (1), \quad \gamma_*\ (1), \quad \gamma^a\ (4), \quad \gamma_*\gamma^a\ (4), \quad \gamma^{ab} = \tfrac{1}{2}[\gamma^a, \gamma^b]\ (6), \tag{234}$$

where numbers in parentheses are basis dimensions. A generic interacting Dirac self-energy need *not* remain purely in the vector (γ^a) sector. Non-vector principal-symbol terms can obstruct a single common metric. The Clifford anticommutator is therefore a dynamical *test*, not a representation imposed by fiat.

32.2.1 What spinor bilinear basis vectors actually provide

A Dirac spinor has scalar, pseudoscalar, vector, axial-vector and bivector bilinears, but they are constrained by Fierz–Pauli–Kofink identities; these are not independent freely adjustable frame vectors. In the standard $(+---)$ convention, for commuting spinor components define

$$S = \bar{\psi}\psi, \quad P = i\bar{\psi}\gamma_5\psi, \quad V^a = \bar{\psi}\gamma^a\psi, \quad A^a = \bar{\psi}\gamma^a\gamma_5\psi. \tag{235}$$

Then the regular-spinor relations include

$$\boxed{V_a V^a = S^2 + P^2, \qquad A_a A^a = -(S^2 + P^2), \qquad V_a A^a = 0.} \tag{236}$$

We tested all three identities on ten independent random four-spinors. Bilinears can be used to construct an adapted local spin frame in regular cases; however, the identities already use a Clifford metric and cannot determine its source-dependent dynamics. This supports Chapter 1's distinction between local four-mode spinor data and the independent physical selection of QGD natural invariants.

Proposition 32.1 (Clifford reconstruction requires matrix closure)**.** *Given effective first-derivative matrices Γ_i^μ of a microscopic probe species i, define*

$$\boxed{g_{i,\mathrm{tr}}^{\mu\nu} := \frac{1}{8}\,\mathrm{tr}\{\Gamma_i^\mu, \Gamma_i^\nu\}, \qquad \mathcal{R}_i^{\mu\nu} := \{\Gamma_i^\mu, \Gamma_i^\nu\} - 2g_{i,\mathrm{tr}}^{\mu\nu}\mathbf{1}.} \tag{237}$$

If all $\mathcal{R}_i^{\mu\nu} = 0$, the $g_{i,\mathrm{tr}}$ are nondegenerate Lorentzian, and the chosen adjoint/hyperbolicity conditions hold, the principal symbol has a metric Clifford structure. If $\mathcal{R} \neq 0$, a symmetric trace tensor alone cannot certify a common Dirac metric. The proof follows immediately from Eq. (233); the reverse direction is the definition of a Clifford module.

32.2.2 Explicit false-positive: axial derivative correction

Consider a local massless Fourier symbol in one spatial direction,

$$P(E,k) = E\gamma^0 + k(\gamma^1 + \epsilon\gamma_*\gamma^1), \qquad \gamma_* = i\gamma^0\gamma^1\gamma^2\gamma^3. \tag{238}$$

In a chiral basis its determinant factorizes exactly:

$$\boxed{\det P = (E^2 - (1+\epsilon)^2k^2)(E^2 - (1-\epsilon)^2k^2).} \tag{239}$$

Yet Eq. (237) assigns $g_{\mathrm{tr}}^{11} = 1 - \epsilon^2$. The non-scalar residual is nonzero, and the two characteristic cones differ. With $\epsilon = 0.23$ the positive slopes are 0.77 and 1.23; the verifier gives $\|\mathcal{R}\| = 0.92$ under its displayed matrix norm. The example is kinematic, not a measured Lorentz violation or a prediction of the original F1 interaction.

32.3 Spin connection: the missing ingredient in principal-symbol fitting

For a torsion-free metric-compatible spin structure, the curved Dirac operator has the form

$$\mathcal{D}_g = \hbar\Gamma^\mu(\partial_\mu + \Omega_\mu - iqA_\mu/\hbar) + m, \quad \Omega_\mu = \tfrac{1}{4}\omega_{\mu ab}\gamma^{ab} \quad \text{(with consistent connection conventions)}. \tag{240}$$

The tetrad postulate translates into Clifford compatibility,

$$\partial_\mu\Gamma^\nu + \Gamma^\nu{}_{\mu\rho}\Gamma^\rho + [\Omega_\mu, \Gamma^\nu] = 0. \tag{241}$$

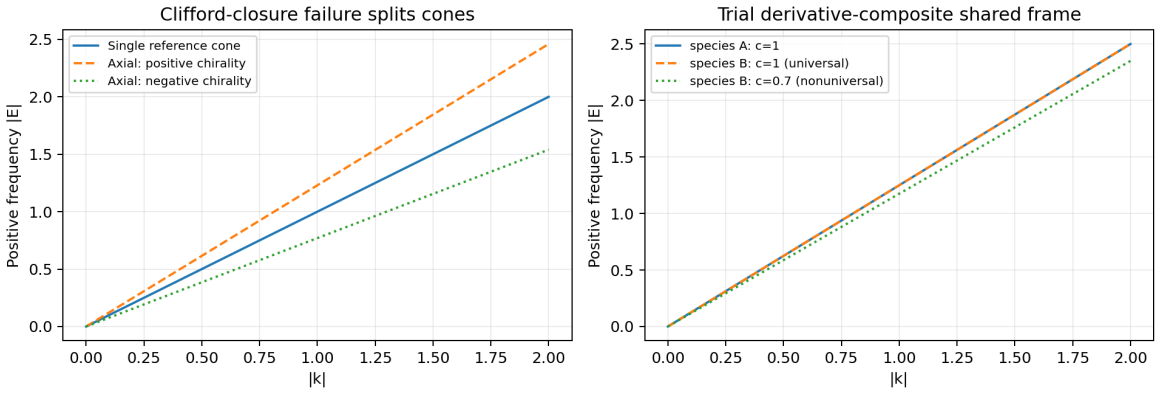


Figure 15: Left: an axial kinetic term gives two chiral cones despite a plausible trace-metric tensor. Right: a common derivative-current auxiliary field gives identical kinetic cones only if species couplings match. Both are specified trial models, not a derived vacuum geometry.

It constrains the bivector Lorentz connection once e and an affine connection have been specified; terms commuting with an irreducible complex Clifford representation (including a possible $U(1)$ connection) are not fixed by this equation. If torsion/nonmetricity or extra matrix connections are physical, they require separate treatment. Spinor covariant-derivative commutators encode the spin curvature, and the squared Dirac operator involves scalar curvature (the Lichnerowicz relation). This makes the *connection and its curvature* observable through subleading spinor propagation, not through the principal cone alone.

Proposition 32.2 (Identical Clifford cones, inequivalent WKB transport)**.** *On a conformally flat but curved 4-metric $g_{\mu\nu} = \Omega^2(x)\eta_{\mu\nu}$, let $\Omega = e^{\kappa x}$ for $\kappa \neq 0$. The exact massless spinor relation is*

$$\boxed{\not{D}_{\Omega^2\eta}\psi = \Omega^{-5/2}\not{\partial}_\eta(\Omega^{3/2}\psi) = \Omega^{-1}\gamma^\mu(\partial_\mu + \tfrac{3}{2}\partial_\mu \ln\Omega)\psi.} \tag{242}$$

The naive operator $\not{D}_{\rm raw} = \Omega^{-1}\gamma^\mu\partial_\mu$ has the same *Clifford principal symbol $\Omega^{-1}\gamma^\mu$ but omits the connection term. For a constant flat-space massless spinor χ, the covariant solution $\psi = \Omega^{-3/2}\chi$ carries current $j_g^\mu = \Omega^{-4}J_\eta^\mu$, so $\partial_\mu(\sqrt{-g}\, j_g^\mu) = 0$. By contrast, $\psi = \chi$ is a solution of the naive massless equation, but its covariant current has $\sqrt{-g}j^x_{\rm raw} = \Omega^3 J_\eta^x$, implying $\partial_x(\sqrt{-g}j^x_{\rm raw}) = 3\kappa\Omega^3 J_\eta^x \neq 0$ for $J_\eta^x \neq 0$. Moreover $R(g) = -6\kappa^2 e^{-2\kappa x}$; this is genuinely curved rather than a mere change of coordinates. Hence* metric reconstruction does not imply physical amplitude/current transport*.*

32.4 WKB and the original QGD spinor bridge, now with curvature

Take $\psi = e^{iS/\hbar}(a_0 + \hbar a_1 + \cdots)$ in Eq. (240). At leading eikonal order, with $\pi_\mu = \partial_\mu S - qA_\mu$,

$$(i\Gamma^\mu\pi_\mu + m)a_0 = 0 \quad \Rightarrow \quad \boxed{g^{\mu\nu}\pi_\mu\pi_\nu + m^2 = 0} \tag{243}$$

for our $(-+++)$ convention with $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$; this uses $\mathcal{D}_g = \hbar\not{D}_g + m$ (without an extra leading i), so the eikonal equation is conventionally consistent. The spin connection enters at the next order and governs transport/spin rotation; it does not normally change the leading eikonal characteristic cone. More generally polarization-dependent subleading Berry terms are possible, as derived by Oancea and Kumar (2023).

In a local orthonormal frame the historical QGD *on-shell block identity* remains

$$\boxed{u_B = \frac{c\,\boldsymbol{\tau}\cdot\boldsymbol{\pi}_{(a)}}{\mathcal{E}_{(0)} + \mathcal{M}c^2}\,u_A, \qquad \zeta = \frac{pc}{\mathcal{E}_{(0)} + \mathcal{M}c^2} = \tanh(\chi/2).} \tag{244}$$

The local ratio is momentum/polarization information, not an independently selected gravitational σ_μ. The exact conserved Dirac current is $\nabla_\mu j^\mu = 0$, $j^\mu = \bar{\psi}\Gamma^\mu\psi$; in coordinates

$$\boxed{\partial_\mu(\sqrt{-g}\, j^\mu) = 0.} \tag{245}$$

For a stationary radial current in $ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 \mathrm{d}\Omega_2^2$, this yields

$$\boxed{\sqrt{A(r)B(r)}\, r^2 j^r = C_J.} \tag{246}$$

It reduces to the original $r^2 j^r = C_J$ in the diagonal Schwarzschild exterior, $AB = 1$. In Chapter 1's stated single-positive-energy WKB closure, the corrected local relation is

$$\boxed{j^{\hat{r}} = \rho_D \frac{p_{\hat{r}} c^2}{\mathcal{E}_{(0)}} = s_D \frac{p_{\hat{r}}}{m_{\mathrm{eff}}}, \qquad \frac{\mathrm{d}}{\mathrm{d}r}\left(|A_{\mathrm{WKB}}|^2 \frac{p}{\Delta}\right) \simeq 0, \quad \Delta = \mathcal{E}_{(0)} + m_{\mathrm{eff}} c^2.} \tag{247}$$

The last radial envelope formula uses the Chapter 1 local/single-stream assumptions; Eq. (245) is the general covariant statement. In curved backgrounds the coordinate flux also contains the measure, observer tetrad and spin transport. No form of this bridge alone supplies the source-conditioned momentum dynamics or the collective wave's own field equation.

32.5 Two alternative source-to-probe kinetic mechanisms

32.5.1 What the original scalar-contact F1 model can and cannot do

At Hartree level a Lorentz-scalar local four-fermion interaction generates an effective mass self-energy, $\Sigma_{\mathrm{H}} \sim m_{\mathrm{H}}\mathbf{1}$, with the *flat* leading derivative matrices unchanged. A momentum-independent local Fock self-energy likewise cannot alter the highest-derivative cone. This recovers the earlier QM016–QM019 failures of the original contact model. Momentum-dependent nonlocal self-energies and higher derivative interactions evade the simple mass-only argument, but must pass the Clifford and universality diagnostics, and need not produce long-range modes.

32.5.2 A new explicit *trial* derivative-current auxiliary coupling

Let N flat-reference Dirac species have the dimension-four bilinear current $J_a{}^\mu = \sum_i c_i \frac{1}{2}\bar{\psi}_i \gamma_a \overleftrightarrow{\partial^\mu} \psi_i$. The following is a *new toy microscopic interaction, not the canonical F1 or background-free pregeometry M*:

$$\boxed{\mathcal{L}_{\mathrm{micro}} = \sum_i \bar{\psi}_i(\gamma^a \delta_a{}^\mu \partial_\mu + m_i)\psi_i + \frac{g}{2N\Lambda_c^4} J_A C^{AB} J_B.} \tag{248}$$

Here $A = (a, \mu)$ is a multi-index, and C^{AB} is a specified nonsingular channel kernel. A formal Gaussian Hubbard–Stratonovich rewriting is

$$\mathcal{L}_{\mathrm{HS}} = \mathcal{L}_0 + B^A J_A - \frac{N\Lambda_c^4}{2g} B^A (C^{-1})_{AB} B^B. \tag{249}$$

Integrating B formally reproduces Eq. (248), subject to a valid integration contour and regulator. Conditional on a nonzero homogeneous collective solution $\bar{B}$, species i acquire

$$\boxed{\Gamma_i^\mu = \gamma^a(\delta_a{}^\mu + c_i \bar{B}_a{}^\mu), \quad g_i^{\mu\nu} = (\delta + c_i \bar{B})^T \eta (\delta + c_i \bar{B}).} \tag{250}$$

For *equal* c_i, the Dirac principal cones agree exactly; for unequal c_i, they generally differ. In the numerical example $\bar{B}^1{}_1 = 0.25$, $c_1 = 1$, $c_2 = 0.7$, the wave slopes along x are 1.25 and 1.175. The

verifier checks both cases. This establishes an *algebraic mechanism for collective kinetic sharing*, not a dynamically selected source-responsive coframe.

The actual gap/phase and wave problem is encoded in the collective effective action and Hessian,

$$\Gamma_{\rm coll}[B] = -\frac{N\Lambda_c^4}{2g}BC^{-1}B - i\sum_i \mathrm{Tr}\ln\mathcal{D}_i[B] + \Gamma_{\rm ct}, \qquad K^R_{BB}(q) = K^R_{\rm auxiliary}(q) + \sum_i c_i^2\Pi^R_{JJ,i}(q) + K^R_{\rm ct}(q). \tag{251}$$

The matrices Γ_i being common at an imposed B does not show (i) a stable nondegenerate phase; (ii) a wave pole with nonvanishing source residue; (iii) a covariant generated spin connection; (iv) a common photon geometry; or (v) controlled homogeneous vacuum energy. These must be computed *from the same microscopic theory.* A Minkowski reference and a selected channel kernel are explicitly present in Eq. (248), so this construction does not solve the background-independence problem.

32.6 Reconnecting the Clifford metric to QGD's complete master metric

Suppose a microscopic model passes the Clifford, connection, species and causal tests and yields a physical $g_{\rm prop}$. On a regular eight-sector metric-only QGD branch, form a symmetric frame deformation h relative to T; the signed Gram construction

$$\boxed{P = b\mathbf{1} + \frac{h}{4b}, \qquad N = b\mathbf{1} - \frac{h}{4b}, \qquad P^TP - N^TN = h} \tag{252}$$

is an exact *representation* for symmetric h. The arbitrary nonzero b is not a microscopic wave amplitude; this lift does not fix an independent $\mathcal{Q}_{ab}$. QGD's natural invariant would thereby acquire a physical association with a *measured common propagation geometry*, rather than being fitted independently to a wavefunction profile. For a genuine nonzero historical $\lambda\mathcal{Q}_{ab}$ the map and its variation must be handled separately.

32.7 Acceptance tests for the next microscopic experiment

1. **Actual phase.** Compute the renormalized stationary/equilibrium solution for B or a better genuinely composite field, using the same regulator as its fluctuations; show a physical nondegenerate state, not a gauge-unfixed auxiliary mean alone.

2. **Clifford closure.** Extract Γ_i^μ from each retarded inverse propagator; test $\mathcal{R}_i^{\mu\nu} = 0$ or quantitatively bounded in the infrared. Check signature, invertibility and hyperbolicity. Non-Dirac probes require their own characteristic polynomials.

3. **Shared geometry.** Match $g_i^{\mu\nu}$ across different fermions, and match photon characteristics and normalized source/probe vertices, not just a single cone.

4. **Connection and WKB bridge.** Check Eq. (241), current-flux Eq. (245), the corrected local ratio and covariant WKB transport. The conformal counterexample shows why this is independent of Clifford closure.

5. **Real propagation.** Compute the full CTP collective Hessian including all auxiliary/connection mixing and conserved-source overlaps, with causal support and finite physical wave residues; do not preselect a graviton.

6. **Vacuum.** Evaluate the $k = 0$ branch and the localized-source exterior branch using one state definition and one renormalization scheme. Test the cosmological constant's physical matching rather than reading it from an unsubtracted cutoff term.

7. **Metric representation.** Only after the above, apply the full QGD master metric and regular natural-invariant quotient. Reject claims of unique σ selection from Gram algebra alone.

32.8 Research ledger and conclusion

Result	Status	Limitation
Dirac Γ^μ Clifford metric and local Spin covariance	[D/N]	assumes valid Clifford form
Axial kinetic correction can split two chiral cones	[D/N]	illustrative symbol
Same Clifford cone with wrong spinor flux transport	[D/N]	conformal counterexample
Four-dimensional conformal Dirac weight 3/2	[D/N]	massless, torsion-free
Curved covariant current extends Chapter 1 bridge	[D]	no source law
Derivative-current HS channel can share fermion frames	[D/N/C]	equal weights imposed
Derivative-current interaction selects a healthy wave	[O]	Hessian/phase not calculated
Photon/fermion common propagation, exterior causal wave	[O]	no microscopic closure
Conformal and full $\mathcal{Q}$ sector physical selection	[O]	metric-only construction

Verdict. The most actionable new clue from curved Dirac theory is not a particular spinor formula for σ. It is a *three-layer matching condition*: the collective quantum state must make matter's derivative matrices obey one physical Clifford algebra; its induced spinor connection must enforce the covariant WKB/current bridge; and its fluctuations must have the correct causal source and vacuum response. QM037 supplies a conditional microscopic derivative channel and explicit false-positive diagnostics. It does *not* establish a derived $\mathcal{F}_{\mathrm{QM}}$ or a physical wave in four-dimensional vacuum.

Selected external literature. M. Alcubierre, *Gen. Relativ. Gravit.* 57, 122 (2025), https://doi.org/10.1007/s10714-025-03460-8; M. A. Oancea and A. Kumar, *Phys. Rev. D* 107, 044029 (2023); M. A. Olpak and A. Ozpineci, *Eur. Phys. J. C* 81, 798 (2021). These are established curved-spinor and bilinear facts; the QGD interpretation remains conditional.

33 QM038: Dirac-first inverse matching and collective waves

Research status. Exact algebraic constructions and counterexamples are distinguished from an actual derivation of a microscopic gravitational phase. The new proposal is an *inverse matching relation*, not a unique formula converting phenomenological gravity into an ultraviolet action. This section preserves the complete founding QGD metric and the quantum-completed Chapter 5 action. Natural invariants are weighted covectors, not individual components of a Dirac spinor; the physical metric is not a fundamental microscopic field in the intended QGD ontology.

33.1 The two maps that must not be conflated

The adopted phenomenological prescription is

$$\mathcal{F}_{\mathrm{phen}} : (\varrho, \mathcal{I}; \boldsymbol{\theta}) \mapsto [g_*] \mapsto [\Sigma]_{\mathrm{QGD}}, \qquad g_* \in \mathrm{Sol}_{\mathcal{I}}^{\mathrm{CTP,ord.red.}} \left(\delta \Gamma_{\mathrm{QGD}}^{\mathrm{ren}} / \delta g_\Delta = 0 \right). \tag{253}$$

Here the final arrow is an algebraic *representation*, not evidence for microscopic source selection. The entire founding metric remains

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \Big[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \Big], \quad s_s = -\varepsilon_s.} \tag{254}$$

The phenomenological dynamics use the *quantum-completed Chapter 5 action*, not only the fitted Einstein–teleparallel leading term:

$$\Gamma_{\mathrm{QGD}}^{\mathrm{ren}} = \frac{1}{16\pi G_{\mathrm{ren}}} \int \sqrt{-g(\Sigma)} (R - 2\Lambda_{\mathrm{ren}}) \, \mathrm{d}^4x + S_{\mathrm{GHY}} + S_{\mathrm{F1}}^{\mathrm{cl}} + \hbar \Gamma_{\mathrm{m,ren}}^{(1)} + O(\hbar^2). \tag{255}$$

On a metric-only regular branch the recognizable QGD natural-invariant Euler equation remains Option C,

$$\boxed{2 s_s \Pi^{ab} \sigma_b^{(s)} = \lambda [(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi]^a,} \tag{256}$$

where Π varies the *whole* renormalized action, not just its classical curvature part. An independently justified sector-only action would contribute additional Euler terms. Neither Eq. (255) nor Option C derives the microscopic collective field.

The intended causal microscopic map, which is not yet known, is

$$\mathcal{F}_{\mathrm{QM}} : (S_{\mathrm{micro}}, \varrho, \mathcal{I}) \mapsto \mathfrak{B}_{\mathrm{collective}}(x) \mapsto \text{common observable propagation} \mapsto [g] \mapsto [\Sigma]. \tag{257}$$

Rather than equating the two maps, we *Diracize* the phenomenological target and compare a candidate microscopic two-point and vertex functional with it.

33.2 Exact Diracization of the full QGD metric

Let $M_{ab} = \eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}$. If M is real symmetric, nondegenerate and has Lorentz signature on a coordinate patch, Sylvester's inertia theorem supplies a local invertible matrix $L^a{}_b$ such that

$$M = L^{\mathsf{T}} \eta L, \qquad e^a{}_\mu = L^a{}_b T^b{}_\mu, \qquad g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu. \tag{258}$$

This physical coframe e differs from QGD's *reference coframe* T. It is not unique: local Lorentz changes of e leave g invariant and act through corresponding spin lifts. With a fixed constant Clifford basis $\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbf{1}$, construct

$$\boxed{\Gamma^\mu_{\mathrm{phen}}[\Sigma] = \gamma^a (e^{-1})^\mu{}_a, \qquad \{\Gamma^\mu_{\mathrm{phen}}, \Gamma^\nu_{\mathrm{phen}}\} = 2 g^{\mu\nu}[\Sigma] \mathbf{1}.} \tag{259}$$

The same factorization applies to the unrestricted master metric *if* its full M, including any independent $\mathcal{Q}$, is sufficiently regular and Lorentzian. It does *not* establish that $\mathcal{Q}$ has a defined microscopic dynamics. On the tested $\lambda\mathcal{Q} = 0$ eight-sector branch, taking $T = I$ and $h = \eta - M$ permits the exact signed lift $P = bI + h/(4b)$, $N = bI - h/(4b)$ with $P^{\mathsf{T}}P - N^{\mathsf{T}}N = h$. An explicit nonsingular, non-diagonal e passes 16 curved Clifford anticommutators and the Gram lift in the verifier.

A useful necessity test. At each point define $\mathcal{C}_i^{\mu\nu} = \frac{1}{2}\{\Gamma_i^\mu, \Gamma_i^\nu\}$. An interacting species i supports one ordinary local metric only if the matrix-valued residual

$$\mathcal{R}_i^{\mu\nu} = \mathcal{C}_i^{\mu\nu} - \frac{\operatorname{tr}\mathcal{C}_i^{\mu\nu}}{4}\mathbf{1} \tag{260}$$

vanishes, the resulting quadratic form is Lorentzian, and the *normalized* physical symbols agree across species. If $\mathcal{R}_i \neq 0$, different polarizations can have different cones (QM037's axial counterexample). The determinant $(\Gamma\cdot p)^2 = g^{\mu\nu}p_\mu p_\nu \mathbf{1}$ is checked directly for generic momenta. Massless characteristic cones alone fix only a conformal class; kinetic normalization and massive clocks are necessary to identify the scale.

33.3 The connection is constrained, not an unrelated fitting function

For a specified torsion-free affine connection $\Gamma^\nu{}_{\mu\rho}(g)$, Clifford compatibility is

$$\boxed{\partial_\mu\Gamma^\nu + \Gamma^\nu{}_{\mu\rho}\Gamma^\rho + [\Omega_\mu, \Gamma^\nu] = 0.} \tag{261}$$

For fixed Γ^ν and affine connection, the difference of two solutions $\Delta\Omega_\mu$ commutes with all four Clifford generators. In the irreducible four-dimensional complex Clifford representation, Schur's lemma forces $\Delta\Omega_\mu$ to be proportional to $\mathbf{1}$. Thus the *Lorentz-spin part* of a compatible connection is fixed up to central gauge contributions; torsion, extra bundles and independent gauge potentials are separate physical assumptions. The commutator map $X \mapsto ([X,\Gamma^0],\ldots,[X,\Gamma^3])$ has complex rank 15 and one-dimensional center (verified for the full QGD test coframe).

Constructive conformal case. The Hermitian massless first-order action

$$S_\psi = \frac{i}{2}\int \mathrm{d}^4x\,\mu(x)\,\bar\psi\Gamma^\mu\overleftrightarrow{\partial_\mu}\psi \tag{262}$$

produces $i\Gamma^\mu\partial_\mu + i[2\mu]^{-1}\partial_\mu(\mu\Gamma^\mu)$. With $g = \Omega^2\eta$, $\mu = \sqrt{-g} = \Omega^4$ and $\Gamma^\mu = \Omega^{-1}\gamma^\mu$, one obtains exactly

$$\boxed{\mathcal{D}_g = i\Omega^{-1}\gamma^\mu\left(\partial_\mu + \frac{3}{2}\partial_\mu\ln\Omega\right), \quad \text{for massless conformal Dirac.}} \tag{263}$$

The coefficient 3/2 follows algebraically from Hermitian ordering and the collective density *once they are provided*. It is not an independent fitted parameter in this conformal example. For generic rotating coframes, Hermitian ordering alone does *not* replace the full spin connection or secure local Lorentz covariance.

33.4 Basis vectors, Spin(1,3), Cartan transport and Fierz restrictions

The physical coframe $e^a = e^a{}_\mu \mathrm{d}x^\mu$ specifies a local orthonormal basis, not a coordinate basis or four arbitrary spinors. Its torsion and Lorentz curvature obey the Cartan structure equations

$$\mathsf{T}^a = \mathrm{d}e^a + \omega^a{}_b \wedge e^b, \qquad \mathsf{R}^a{}_b = \mathrm{d}\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b. \tag{264}$$

The Levi–Civita choice $\mathsf{T}^a = 0$ fixes $\omega(e)$ locally; a microscopic theory with torsion must supply its own additional equations. Under a local Lorentz change $e \mapsto \Lambda e$ with spin lift $S(\Lambda)$, the world-indexed Clifford matrices transform by conjugation $\Gamma^\mu \mapsto S\Gamma^\mu S^{-1}$. The spin connection transforms

as $\Omega_\mu \mapsto S\Omega_\mu S^{-1} - (\partial_\mu S)S^{-1}$, so the covariant Dirac operator is related by similarity; this is why merely choosing a spatially varying tetrad is not automatically an observable gravitational wave. World-coordinate changes and signed-sector orbit transformations are logically different redundancies and require separate audits.

The scalar, pseudoscalar, vector, axial-vector and bivector Dirac bilinears form an overcomplete constrained system. For example, with consistent sign conventions, regular spinors satisfy $V_a A^a = 0$ and $V^2 = -A^2 = S^2 + P^2$ in signature $(+---)$ (the signs translate in $(-+++)$). A *single* spinor cannot be promoted into four freely independent frame covectors without satisfying these algebraic restrictions and specifying an independent normalization. Chapter 1's four *mode amplitudes* likewise must not be identified with QGD's signed-sector labels. A large-N derivative-bilinear expectation can have a different rank structure, but only a controlled correlated state can establish that rank.

For a WKB spinor a_0 containing an internal normalized polarization ξ, a Berry-type connection $-i\xi^\dagger \mathrm{d}\xi$ can have nonzero curvature; in QM036 the Kerr projective texture provides an explicit example. This gives the collective wave access to non-exact spin-polarization transport, but there is *no proven identity* between its Berry curvature and the QGD spacetime curvature. The appropriate microscopic matching includes both phase-space transport and spin/polarization holonomy.

33.5 WKB, the original QGD bridge and the curvature encoded in the operator

A Dirac-type effective operator with principal Clifford symbol and a compatible spin connection obeys, at leading WKB order,

$$(i\Gamma^\mu \pi_\mu + m)a_0 = 0 \quad \Longrightarrow \quad g^{\mu\nu}\pi_\mu\pi_\nu + m^2 = 0 \quad \text{for } (-+++), \tag{265}$$

with representation conventions chosen consistently. The covariant current obeys

$$\nabla_\mu j^\mu = 0, \qquad \partial_\mu(\sqrt{-g}\, j^\mu) = 0. \tag{266}$$

For a diagonal static spherical metric $-A(r)dt^2 + B(r)dr^2 + r^2 d\Omega_2^2$, $r^2\sqrt{AB}\, j^r = C_J$. The Chapter 1 local Dirac two-block identity remains $u_B = c\,\boldsymbol{\tau}\cdot\boldsymbol{\pi}\, u_A/(\mathcal{E}_{(0)} + mc^2)$ in an adapted orthonormal frame. Its corrected single-stream WKB envelope identity, in the notation and restricted assumptions of Chapter 1, is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(|A_{\mathrm{WKB}}(r)|^2 \frac{p(r)}{\Delta(r)}\right) \simeq 0, \qquad \Delta = \mathcal{E}_{(0)} + m_{\mathrm{eff}}c^2. \tag{267}$$

This constrains *transport once a momentum/connection structure exists*, rather than determining the source or the gravitational natural invariant. No direct identification $\sigma \propto \mathrm{d}S$ follows; Kerr's rotating spinor has non-exact polarization transport (QM022, QM036).

Squaring the torsion-free Dirac operator gives the Schrödinger–Lichnerowicz relation (Euclidean notation)

$$\boxed{\mathcal{D}_g^2 = \nabla^*\nabla + \frac{1}{4}R(g)} \tag{268}$$

for the untwisted spin operator; gauge curvatures and torsion add additional contributions in the corresponding generalized formula. Therefore the Dirac operator already contains both propagation and a specific curvature term. Its heat-kernel expansion can induce $\sqrt{|g|}$, $\sqrt{|g|}R$ and higher curvature invariants (as in the Chapter 4–5 matter effective action). *But computing a determinant of an assumed $\mathcal{D}_g$ is not deriving the collective coframe from an independent microscopic spinor theory.* This is the circularity boundary.

33.6 What can a function from $\mathcal{F}_{\rm phen}$ to $\mathcal{F}_{\rm QM}$ actually mean?

There is an exact *Diracization* map $\mathscr{D}$ from a regular QGD phenomenological output to a class of compatible Dirac operators,

$$\boxed{\mathscr{D}:\ \mathcal{F}_{\rm phen} \mapsto \left[\Gamma^{\mu}_{\rm phen}, \Omega_{\mu,\rm phen}, \mathcal{K}_{R,\rm phen}, \mathcal{V}_{\rm phen}\right]_{\rm gauge},} \tag{269}$$

where $\mathcal{K}_R$ and $\mathcal{V}$ are independently specified causal response functions and interaction vertices, *not* determined by the Clifford algebra alone. A microscopic candidate defines a forward coarse-graining/data map $\mathscr{M}[S_{\rm micro}, \varrho]$. Then the scientifically meaningful inverse is the *set*

$$\boxed{\mathcal{F}^{-1}_{\rm QM}(\mathcal{D}_{\rm phen}) = \{(S_{\rm micro}, \varrho):\ \mathscr{M}[S_{\rm micro}, \varrho] \simeq_{\rm IR} \mathcal{D}_{\rm phen}\}\,.} \tag{270}$$

It is generically not single-valued. As an elementary example, Gaussian elimination of a heavy variable X of mass M coupled by $\lambda X\mathcal{O}$ gives $-\frac{1}{2}\lambda^2\mathcal{O}(M^2-\Box)^{-1}\mathcal{O}$ in Euclidean conventions. The two UV choices $(M,\lambda) = (2,1)$ and $(4,2)$ both give the leading local term $-\frac{1}{8}\mathcal{O}^2$, while their finite-momentum form factors differ: $1/(4+q^2)$ versus $4/(16+q^2)$. Matching a low-energy action or observed geometry alone therefore cannot invert the microscopic map uniquely.

A practical *inverse-matching objective*, evaluated only in the common controlled energy/momentum window, is

$$\begin{aligned}\mathcal{L}_{\rm match}[S_{\rm micro}] = {} & w_C \sum_i \|\mathcal{R}^{\mu\nu}_i\|^2 + w_U \sum_{i<j} \|g^{\mu\nu}_i - g^{\mu\nu}_j\|^2 + w_\nabla \sum_i \|\nabla^{(i)}_\mu \Gamma^\nu_i\|^2 \\ & + w_R \|\mathcal{K}^R_{\rm micro} - \mathcal{K}^R_{\rm phen}\|^2 + w_V \|\mathcal{V}_{\rm micro} - \mathcal{V}_{\rm phen}\|^2 + w_0 \|\mathcal{E}_{0,\rm micro} - \mathcal{E}_{0,\rm phen}\|^2. \end{aligned} \tag{271}$$

All comparisons must use gauge/coordinate-invariant observables or a common justified gauge, properly normalized probe operators, and uncertainty-aware renormalization matching. The w's are bookkeeping weights, not physical predictions. Setting the loss to zero by inserting $g_{\rm phen}$ in $S_{\rm micro}$ would be circular.

33.7 New possible clue: long-distance locking of several quantum-matter frames

Microscopic species need not have identical bare kinetic coframes for their physical *long-distance* geometry to become common. Motivated by Wetterich's analysis of *universality of geometry*, consider an illustrative *two-channel scalar-component reduction*, not a derivation of QGD dynamics:

$$K(q^2) = \begin{pmatrix} Zq^2+\mu^2 & -\mu^2 \\ -\mu^2 & Zq^2+\mu^2 \end{pmatrix}. \tag{272}$$

Its eigenmodes are the normalized common combination $B_+ = (B_1+B_2)/\sqrt{2}$ with inverse kernel Zq^2 and relative combination $B_- = (B_1-B_2)/\sqrt{2}$ with inverse kernel $Zq^2+2\mu^2$. The exact source response is

$$\boxed{K^{-1} = \frac{1}{2Zq^2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2(Zq^2+2\mu^2)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.} \tag{273}$$

At $q^2 \ll 2\mu^2/Z$, both species access the same nonanalytic long-range term; differences are shorter-ranged and controlled by the massive relative mode. For $Z=1$, $\mu^2=4$, $q^2=0.01$, the relative mismatch $|D_{11}-D_{12}|/|D_{12}|$ is 0.0025. This is a *constructive test architecture for universal propagation*: source cross-coupling and a heavy relative-frame mode can align physical long-distance responses. *It assumes rather than derives the common massless mode, is not spin-two dynamics, and does not establish the photon cone or a stable vacuum.* The continuum microscopic model must generate its coupled kernel and demonstrate all such properties.

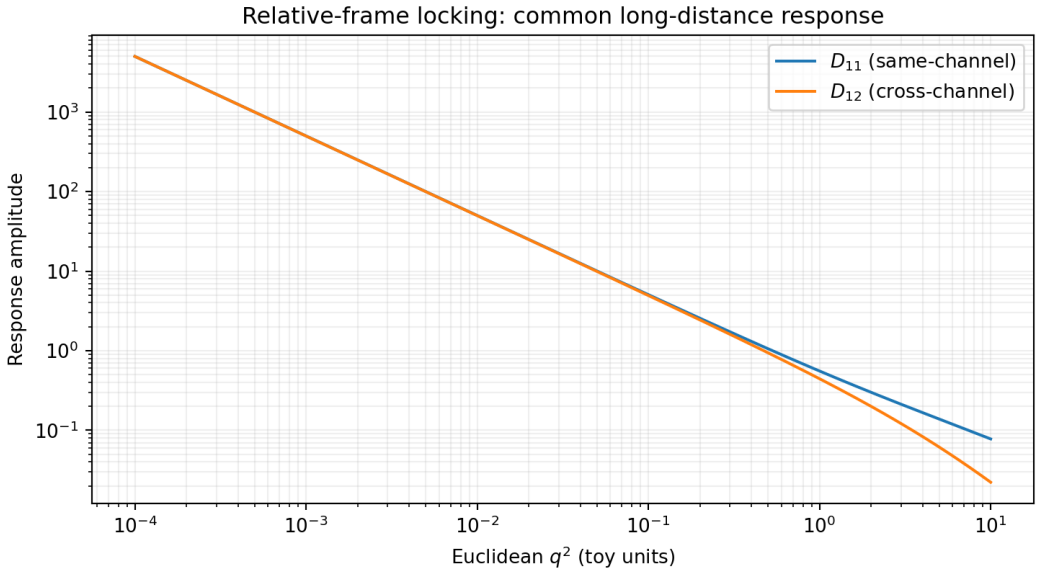


Figure 16: Illustrative two-channel response. When a relative mode is gapped, distinct species channels share the same long-distance singular response; the graph does not derive a gravitational field.

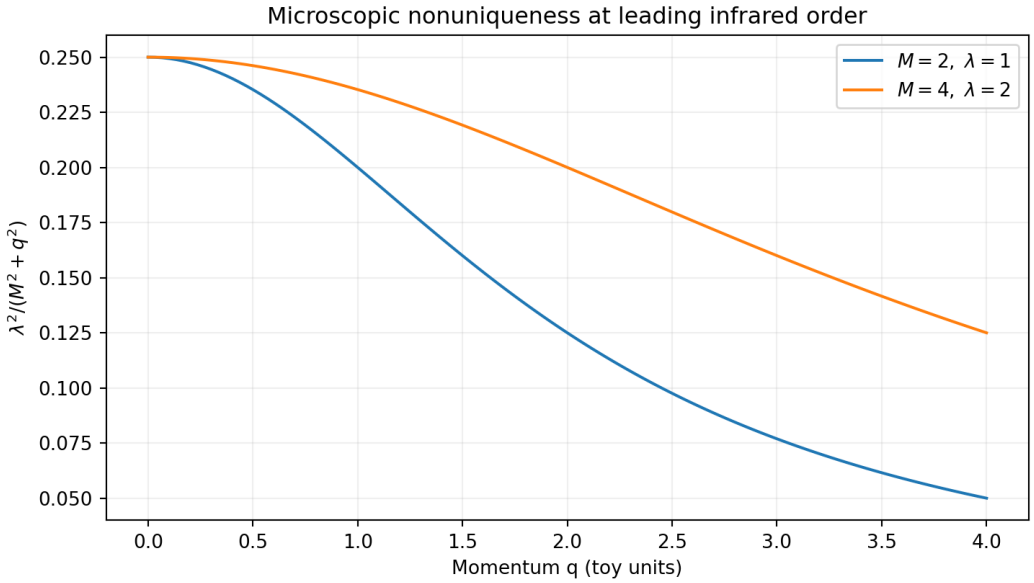


Figure 17: Different microscopic masses/couplings can yield the same leading fitted low-energy operator but distinct finite-momentum dependence. Off-shell matching adds discriminatory power.

33.8 Which microscopic model should be calculated next?

Retain a *metric-free microscopic origin* as a hypothesis to be tested. One candidate is a composite spinor-derivative operator

$$E^a{}_\mu[\psi,\omega] = \sum_{i=1}^{N} \frac{i}{2}\bar{\psi}_i\gamma^a \overleftrightarrow{D_\mu}\psi_i, \tag{274}$$

with suitable microscopic action, state, regulator, and independently integrated Lorentz connection. This is a microscopic *candidate observable*, not a declaration that $\langle E\rangle$ is already an invertible physical coframe. At a finite Grassmann regulator exact constraints $X = E$ are distributional and have nilpotent support, while the cofactor multiplier $\Pi = \mathrm{cof}(e)$ is a separate dual variable. A nonzero geometric phase must be demonstrated in a controlled collective/large-volume limit (QM025–QM026). The proposed Henneaux–Teitelboim modification may organize an integration-constant vacuum sector, but does not automatically solve vacuum selection or the source-generated wave.

Proposed falsification protocol. (i) Construct one regulated S_{micro} without inserting g_{phen} or an independent Einstein action; (ii) obtain its actual stationary or CTP collective phase and probability/measure interpretation; (iii) calculate source-conditioned, properly normalized retarded fermion and gauge probe two-point functions; (iv) reconstruct and compare full Clifford principal symbols, central/U(1) and Lorentz-spin connections and WKB conserved-current transport; (v) test common long-range response, relative-frame masses, nonzero source overlap and exterior propagation; (vi) test the same state's homogeneous vacuum and renormalization. The no-ghost, causal and regulator-limit checks remain mandatory. *No specific microscopic interaction currently passes this whole list.*

33.9 Proof-status ledger and independent checks

Statement	Status
Full-master-metric to local Clifford operator on Lorentzian patch	Exact algebra, conditional regularity
One-dimensional commutant of irreducible Clifford matrices	Exact algebra; verified rank 15
Conformal Hermitian transport coefficient 3/2	Exact in conformal case
Two UV completions with same leading EFT operator	Explicit counterexample
Two coupled responses share an IR singularity	Exact for specified toy kernel
Stable microscopic coframe condensate	Open
Emergent proper spin connection for rotating geometries	Open
One common fermion/photon causal geometry	Open
Predicted renormalized G and Λ	Open

Computational verification. The independent `verify_qm038.py` produces 51 passing checks: 16 flat Clifford relations; the complete symmetric signed eight-sector Gram representation; 16 curved Clifford relations; rank-15 commutator and central identity; the conformal 3/2 coefficient; multiple exact two-channel inversions and residues; microscopic inverse ambiguity; and three principal-symbol squares. These do *not* amount to a calculation of a propagating microscopic QGD wave.

33.10 External scientific comparisons

- M. Alcubierre, *The Dirac equation in general relativity and the* 3+1 *formalism*, Gen. Rel. Grav. **57**, 122 (2025), https://doi.org/10.1007/s10714-025-03460-8: tetrads, spin connection,

WKB/current and Einstein–Dirac matter.

- C. Wetterich, *Gravity from spinors*, Phys. Rev. D **70**, 105004 (2004), https://doi.org/10.1103/PhysRevD.70.105004: composite vielbein and the important *possible* role of global versus local Lorentz symmetry.
- C. Wetterich, *Lattice spinor gravity*, https://arxiv.org/abs/1108.1313: a proposed fermionic regulator with a collective geometry, not evidence that the desired phase is already derived for QGD.
- C. Wetterich, *Universality of geometry*, Phys. Lett. B **712**, 126–131 (2012), https://doi.org/10.1016/j.physletb.2012.04.053: several effective geometric variables can converge at long wavelengths under specified dynamical conditions.
- A. H. Chamseddine and A. Connes, *The spectral action principle*, Commun. Math. Phys. **186**, 731–750 (1997), https://doi.org/10.1007/s002200050126: an action from the Dirac spectrum, which still requires explaining the microscopic origin of the physical Dirac operator.

Conclusion. Curved-space Dirac theory provides an unusually tight *intermediate matching language*—Clifford closure, a constrained Lorentz-spin connection, Fierz- and WKB-consistent transport, induced curvature coefficients and potentially universal long-distance probes. It does not, by itself, invert the microscopic emergence problem. The correct phenomenology-to-micro bridge is a constrained inverse correspondence, progressively sharpened by off-shell causal measurements and checked with an actual regulated quantum model. QGD's complete master metric and Option C remain intact.

34 QM039: A single regulated derivative-bond quantum model—selected kinetic frame, causal response, and a phase obstruction

Research question. Can an interaction of the type proposed in QM037 generate, within the *same* quantum system, a self-consistent change of microscopic propagation, a causal source-to-other-species response, and a genuine collective wave—without assigning the auxiliary field a wave equation? We address this in a deliberately limited $1+1$-dimensional lattice regulator. No metric is inserted into the microscopic Hamiltonian, but the lattice supplies a preferred background and the result is *not* a background-free four-dimensional theory. All microscopic finite-system statements below are exact; the auxiliary-field saddle and fluctuation calculation are a separate Hartree/RPA approximation.

34.1 The specified microscopic Hamiltonian and exact auxiliary identity

Take N flavors of spinless complex fermions on an L-site ring with fermion number fixed within each flavor. Define the Hermitian link kinetic bilinear

$$T_{i,x} = f^\dagger_{i,x+1} f_{i,x} + f^\dagger_{i,x} f_{i,x+1}, \qquad J_x = \sum_{i=1}^{N} w_i T_{i,x}, \tag{275}$$

where w_i are real interaction weights. This notation separates them from the fermionic operators $f_{i,x}$. Define

$$\boxed{H_{\text{micro}} = -t \sum_{i,x} T_{i,x} - \frac{g}{2N} \sum_x J_x^2, \qquad t > 0, \quad g > 0.} \tag{276}$$

This is a *new regulated model*, a bond-channel projection of the QM037 derivative-bilinear proposal. It is not canonical F1 or the four-dimensional background-free spinor-pregeometry model M. Every term acts on a finite-dimensional Fock space at finite L; in particular the quartic Hamiltonian is self-adjoint and bounded there. At every Euclidean time slice,

$$\exp\left[\frac{\delta\tau g}{2N} J_x^2\right] \propto \int_{\mathbb{R}} \mathrm{d}B_x \, \exp\left[-\delta\tau \left(\frac{N B_x^2}{2g} - B_x J_x\right)\right], \tag{277}$$

with normalization fixed by the Gaussian integral. Thus the *original* microscopic interaction, not a postulated boson kinetic energy, produces an auxiliary bond field B_x. Its tree-level quadratic weight is ultralocal in space and imaginary time.

34.2 Source-conditioned exact unitary calculation

For the exact diagonalization we use $N = 2$ flavors, half filling in each, antiperiodic fermionic boundary conditions at $L = 4$ and periodic at $L = 6$ so the reference free ground state is nondegenerate. The fixed-number Hilbert-space dimensions are $\binom{4}{2}^2 = 36$ and $\binom{6}{3}^2 = 400$. Couple a real local source through $H(s) = H_{\text{micro}} - sT_{1,0}$ and take $P = T_{2,0}$ as a different-species probe. With eigenstates $|n\rangle$ and excitation energies $\Delta_n = E_n - E_0 > 0$, the exact static susceptibility is

$$\boxed{\left.\frac{\partial\langle P\rangle}{\partial s}\right|_{s=0} = 2 \sum_{n>0} \frac{\mathrm{Re}[\langle 0|P|n\rangle\langle n|T_{1,0}|0\rangle]}{\Delta_n}.} \tag{278}$$

For a source switched on at $t = 0$ this same microscopic Hamiltonian gives the *retarded* step response at first order,

$$\boxed{\delta\langle P(t)\rangle = 2s\,\theta(t) \sum_{n>0} \frac{\mathrm{Re}[P_{0n}S_{n0}]}{\Delta_n}(1 - \cos\Delta_n t) + O(s^2), \quad S = T_{1,0}.} \tag{279}$$

Both formulas were checked independently by diagonalizing $H(s)$ and by direct unitary propagation with $H(s)$, respectively. For $t = 1.7$, $s = 2 \times 10^{-4}$ in hopping units, we obtain

L	g	$\chi_{21}^{\mathrm{stat}}$	finite-difference χ_{21}	exact $\delta P(1.7)$	first-order Kubo
4	0	≈ 0	≈ 0	≈ 0	≈ 0
4	0.35	0.018070239	0.018070239	7.47634×10^{-6}	7.47775×10^{-6}
6	0.35	0.027759145	0.027759145	1.537266×10^{-5}	1.537336×10^{-5}

The finite-source and first-order values differ by higher-order terms, as expected. The source does *not* directly act on flavor 2. Its measured nonzero response therefore demonstrates interaction-mediated, causal cross-flavor communication; it vanishes for $g = 0$. The site-resolved response is plotted in Fig. 18. No continuum causal light cone is claimed from six lattice sites.

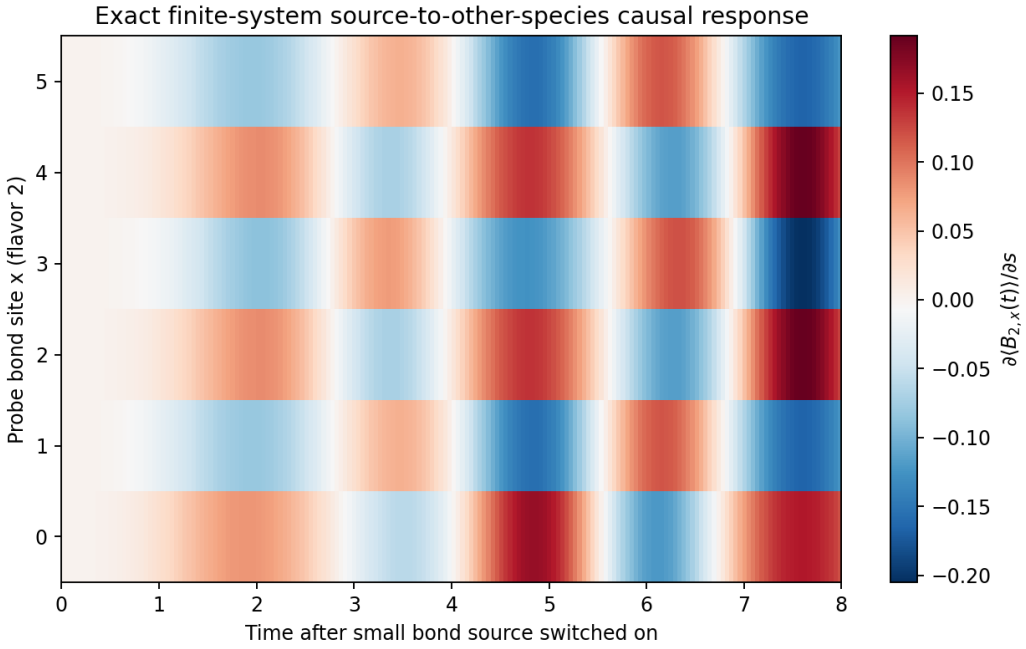


Figure 18: Exact $L = 6$, two-flavor, $g = 0.35$ response of flavor 2 link operators to a step source on flavor 1, evaluated from the microscopic spectral decomposition. The diagram shows the *derivative with respect to source amplitude*; this is a finite-system response, not evidence of a relativistic geometric wave.

34.3 The generated uniform kinetic background and its Clifford limit

At homogeneous Hartree level, $B_x = \bar{B}$ is a stationary solution of the *same* Hubbard–Stratonovich effective energy,

$$\frac{N\bar{B}}{g} = \sum_i w_i \langle T_{i,x} \rangle_{\bar{B}}, \qquad \varepsilon_i(k) = -2(t + w_i \bar{B}) \cos k. \tag{280}$$

For equally weighted, half-filled flavors in the infinite lattice, $\langle T_{i,x} \rangle = 2/\pi$ as long as the hopping is positive. Hence

$$\boxed{\bar{B} = \frac{2g}{\pi}, \qquad v_{F,i} = 2(t + \bar{B}).} \tag{281}$$

At $t = 1$, $g = 0.35$, this yields $\bar{B} \simeq 0.222817$ and $v_F \simeq 2.445634$. It is an interaction-dependent, self-consistent modification of the fermionic derivative operator. *Crucially*, the background does

not represent a newly discovered phase: the nonzero bond expectation is already present for a noninteracting filled band, and we have not yet checked its fluctuation stability. For unequal $w_1 = 1, w_2 = 0.7$, the same saddle gives $\bar{B} = g(w_1 + w_2)/\pi$, yet $v_{F,1} \neq v_{F,2}$. A common B therefore does not by itself guarantee one Clifford geometry.

Near two Fermi points, a two-component long-wavelength spinor has principal kinetic matrices $\Gamma^t = \gamma^0$, $\Gamma^x = v(x,t)\gamma^1$. Thus

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\ \mathrm{diag}(-1, v^2)^{\mu\nu}\, \mathbf{1}. \tag{282}$$

This *conditional continuum kinematic interpretation* requires a smooth long-wavelength bond field and neglects lattice corrections, spectral broadening, and flavor-dependent quantum self-energies. It does not establish a $3+1$-dimensional common metric, and it does not create a metric as a microscopic input. For *static* positive $v(x)$ the $1+1$ metric $\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}x^2/v(x)^2$ is locally flat by $y = \int^x \mathrm{d}u/v(u)$; a position-dependent velocity is not, by itself, gravitational curvature.

34.4 The original QGD bridge as a microscopic transport test

The Hermitian long-wavelength hopping operator is necessarily symmetrized,

$$H_{\mathrm{linear}} = -i\sigma^3 \left(v(x)\partial_x + \frac{1}{2} v'(x) \right), \qquad \partial_t(\chi^\dagger \chi) + \partial_x \left(v \chi^\dagger \sigma^3 \chi \right) = 0. \tag{283}$$

This is the one-dimensional analogue of the current-conservation part of Chapter 1's corrected WKB bridge. Indeed, upon a field rescaling it is the static massless Dirac transport in the above effective $1+1$ coframe. Neither this algebra nor the leading dispersion determines the higher-dimensional Lorentz spin connection, the gravitational $1/r$ source law or the temporal collective state. The general curved-space Dirac connection and the complete QGD bridge remain the benchmarks of QM037–QM038.

34.5 One-loop collective kernel: a soft pole that loses its parent vacuum

Integrating out the hopping fermions gives an induced, time- and momentum-dependent auxiliary inverse kernel. With N identical species, at leading RPA order,

$$\boxed{K_B^R(\omega, q) = N \left[\frac{1}{g} - \chi_{TT}^R(\omega, q) \right], \qquad D_B^R = (K_B^R)^{-1},} \tag{284}$$

where χ_{TT} is the free-band bond susceptibility, including the correct fermionic occupation factors and link vertices. In the staggered ($q = \pi$), half-filled channel, use an antiperiodic ring with $k_j = (2j+1)\pi/L$. At the homogeneous $\bar{B}$ with $t_{\mathrm{eff}} = t + \bar{B}$,

$$\boxed{\chi_{TT}^R(\omega, \pi) = \frac{1}{L} \sum_{\varepsilon_k < 0} \frac{2\Delta_k (2\sin k)^2}{\Delta_k^2 - (\omega + i0)^2}, \quad \Delta_k = 4 t_{\mathrm{eff}} \cos k.} \tag{285}$$

The static susceptibility has the divergent asymptote

$$\boxed{\chi_{TT}(0, \pi) = \frac{1}{L t_{\mathrm{eff}}} \sum_k \frac{\sin^2 k}{|\cos k|} \sim \frac{2}{\pi t_{\mathrm{eff}}} \ln L + O(1).} \tag{286}$$

Our direct finite sums fit a coefficient 0.520645 against the analytic $2/(\pi t_{\mathrm{eff}}) = 0.520617$ for $t = 1, g = 0.35$. An independent finite-difference calculation of the exact free-band ground-state energy under a staggered bond perturbation agrees with the susceptibility sum at $L = 8, 16, 32$. The local HS stiffness $1/g$ is finite. Consequently, the homogeneous half-filled saddle inevitably

loses stability in this *leading large-N/RPA approximation* as $L \to \infty$ for fixed $g > 0$. The preferred instability is a Peierls-like staggered-bond configuration, not a verified geometric vacuum. This is consistent with established studies of one-dimensional bond-order instabilities, but it is a calculation within the specified Hamiltonian and approximation (Garcia-Ruiz et al., 2026).

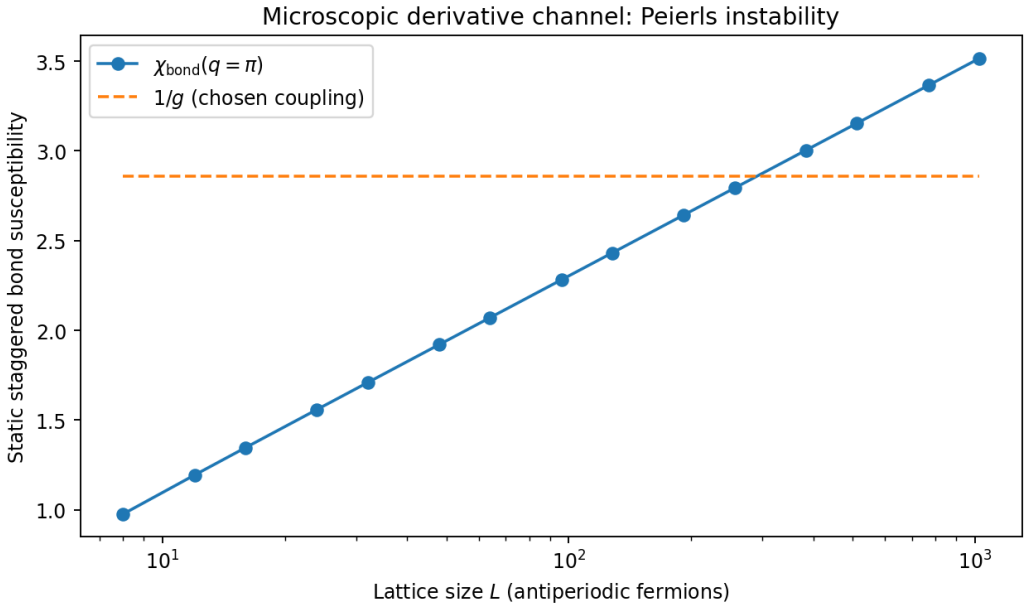


Figure 19: Direct finite-sum RPA staggered-bond susceptibility against system size. The horizontal line is $1/g$. Its crossing marks the loss of homogeneous-saddle stability, not a relativistic massless gravity wave.

For a finite ring still on the stable side, solving $K_B^R(\omega, \pi) = 0$ gives a *real composite-bond pole below the fermionic pair threshold*, with positive mode coefficient $Z_B = [\partial_{\omega^2} \chi_{TT}]^{-1}$ in the convention $D_B \simeq (Z_B/N)/(\omega_*^2 - (\omega + i0)^2)$. At $g = 0.35$:

L	RPA state	ω_*	pair threshold	Z_B
64	homogeneous stable	0.156201	0.240003	0.018093
256	homogeneous stable	0.014183	0.060023	0.002948
384	homogeneous unstable	—	0.040016	—

No bare $\dot{B}^2$ or $(\partial_x B)^2$ was inserted: the pole is produced by the microscopic fermionic polarization. Yet this is *not* a sound $q \to 0$ geometric wave: it is a staggered $q = \pi$ bond mode; its coefficient decreases as its frequency softens, and the homogeneous state fails before any fixed-g thermodynamic massless phase can be claimed. At finite L, this pole is an RPA approximation, *not* an exact spectral pole of the finite two-flavor Hamiltonian.

34.6 Consequences for the complete QGD master metric and the inverse map

The calculation succeeds at demonstrating, from one exact regulated interacting fermion Hamiltonian, a local source-conditioned cross-flavor response and a microscopic kinetic-bilinear channel. Its leading Hartree treatment gives a common long-wavelength Clifford speed for equal couplings, and its fermion loop dynamically induces a finite-lattice collective pole. None of those is independently sufficient for QGD geometry: the uniform saddle is unstable in the thermodynamic one-dimensional approximation; the wave is an order-mode rather than a universal exterior geometry; the spin

connection is not derived in four dimensions; a photon probe is absent; the vacuum energy is unrenormalized; and a preferred lattice background remains.

The full QGD metric must remain

$$\boxed{g_{\mu\nu} = T^{a}{}_{\mu} T^{b}{}_{\nu} \left[\eta_{ab} - \sum_{s} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right].} \tag{287}$$

Only *if* a common Lorentzian propagation geometry is established can one apply the regular $\lambda\mathcal{Q} = 0$ Gram lift to obtain $[\Sigma]$; this remains a representation, not microscopic sector selection. The next falsifiable microscopic target is thus a *higher-dimensional, stable, source-responsive collective derivative phase whose long-wavelength Clifford cones and transport coincide across probes*, tested at the same point as its dynamical wave spectrum and vacuum. A new bond operator or a new auxiliary letter alone will not close $\mathcal{F}_{\mathrm{QM}}$.

Claim	Status	Boundary
Exact finite Hamiltonian and source-to-other-species Kubo response	[D/N]	two small 1D rings
Auxiliary HS identity, homogeneous speed shift	[D/C]	Hartree uniform branch
Dynamically induced $q = \pi$ pole at finite L	[C/N]	one-loop/RPA stable side
Logarithmic Peierls instability at large L	[C/N]	half filling, RPA
1D Clifford and current transport for smooth v	[D/C]	long-wavelength continuum
Full QGD exterior wave, healthy 3+1 coframe, photon universality, Λ	[O]	not derived

35 Previously standalone constraints: off-shell stress, sector-orbit audit and exact wave targets (QM032–QM036)

Provenance and status. The four independent notes QM032 (one-loop Dirac transverse-traceless bubble), QM033 (restricted sector-classification audit), QM035-A (Kerr–Schild and pp-wave exact checks), and QM036 (backward Dirac/WKB bridge) were completed as separate supplements before QM037–QM041. Their decisive conclusions are now recorded inside the chapter. Every formula below is restricted to the named ansatz, regulator, or sector specialization. Their complete derivations and verification scripts are archived separately.

35.1 QM032: an off-shell metric stress bubble is not a collective-wave propagator [D/N/O]

For a Euclidean, free four-component Dirac field of mass m, with a specified proper-time cutoff $s \geq \Lambda_c^{-2}$ and transverse shear insertion, QM032 finds the finite-momentum *two-vertex bubble*

$$\boxed{B_{\mathrm{TT}}(Q^2) = -\frac{1}{16\pi^2}\int_0^1 dx \int_{\Lambda_c^{-2}}^{\infty} ds\; e^{-s[m^2+x(1-x)Q^2]}\left[s^{-3} + (m^2 - x(1-x)Q^2)s^{-2}\right].} \tag{288}$$

The universal logarithmic term is

$$B_{\mathrm{TT}}(Q^2)\big|_{Q^4\log} = -\frac{Q^4}{320\pi^2}\log\frac{\Lambda_c^2}{m^2}, \tag{289}$$

in agreement with the independent Weyl-curvature-squared heat-kernel check at this normalization. The result is a calculated *matter response term*, not a physical collective-field inverse kernel: the genuine metric seagull, tadpole, local subtractions, and causal continuation still matter.

More generally, if e denotes the candidate collective frame and Y the auxiliary, connection, and other coupled fluctuations, the formally eliminated *retarded* physical kernel in a gauge-fixed nonsingular subspace is

$$\boxed{K^R_{\mathrm{phys}} = K^R_{ee} - K^R_{eY}(K^R_{YY})^{-1}K^R_{Ye}.} \tag{290}$$

This Schur complement, together with the physical source vertices and Ward identities, is the relevant object for the existence and visibility of a wave. At singular K_{YY} the constraints must instead be handled before inversion. In particular, the curvature term induced by an external-metric Dirac determinant does *not* by itself demonstrate a new wave in the metric-free spinor model.

35.2 QM033: failed uniqueness of a six-field sector extension [D/N/O]

On the fixed-reference, $\lambda\mathcal{Q} = 0$, eight-sector chart let $h = P^{\mathsf{T}}P - N^{\mathsf{T}}N$. For $P = N = v(I + U(x))$, every Gram contraction of the displayed sector jets is identically zero, including the antisymmetric first-derivative current and the bilinear of two derivatives. This establishes *invisibility on the exact cancelling submanifold*, not that U is a full-theory gauge parameter: perturbing to $N = P + \epsilon vI$ yields

$$h = -\epsilon v^2(2I + U + U^{\mathsf{T}}) - \epsilon^2 v^2 I. \tag{291}$$

The proposed global $J_{\mu,ab}J^{\mu,ab}$ term has a Lorentzian kinetic form with three velocity directions of one sign and three of the opposite sign; it cannot describe six unconstrained healthy propagating modes. A distinct, completely antisymmetrized two-form $H = dB$ has a healthy *linearized* gauge-kinetic combination with one massless mode in four dimensions, but no nonlinear QGD completion has been derived. Therefore neither "exactly six extra healthy modes" nor "every sector-orbit direction must be gauge" was established. This is a correction to a proposed classification, *not* a derivation of new QGD matter.

35.3 QM035-A: smooth propagating natural-invariant waves and a distinct single copy [D/N/C]

For a Brinkmann reference $\bar{g} = -2\,du\,dv + dx^2 + dy^2$, take

$$g = \bar{g} + H\,du^2, \qquad H = f(u)(x^2 - y^2) + 2q(u)xy. \tag{292}$$

This is an exact vacuum Einstein plane wave for arbitrary differentiable profiles f, q. A smooth real two-sector representation is

$$\boxed{\sigma_+ = \left(b + \frac{H}{4b}\right) du, \quad \sigma_- = \left(b - \frac{H}{4b}\right) du, \quad \sigma_+ \otimes \sigma_+ - \sigma_- \otimes \sigma_- = H\,du^2} \quad (b \neq 0). \tag{293}$$

It remains regular across sign changes of H. The reference one-form $A = H du$ satisfies the *flat-background* source-free Maxwell equation, and the stationary Kerr–Schild Kerr counterpart $A = H\ell$ similarly solves its reference Maxwell system away from singular sources. This is the known classical double-copy relation, *not* a microscopic generation law. In the plane-wave example its Maxwell stress is nevertheless nonzero:

$$T^{\mathrm{Maxwell}}_{uu} = 4(x^2 + y^2)[f(u)^2 + q(u)^2]. \tag{294}$$

Consequently A cannot simply be identified with the physical photon matter field that sources the same Ricci-flat QGD solution. The exact wave is a *phenomenological target*; no graviton was imposed in its signed-sector representation.

35.4 QM036: the backwards WKB and original current bridge [D/N/C]

Let $\mathcal{E}_{(0)}$ and $\boldsymbol{\pi}_{(a)}$ denote local tetrad energy and spatial kinetic momentum. The leading local Dirac spinor-block identity remains

$$u_B = \frac{c\,\boldsymbol{\tau}\cdot\boldsymbol{\pi}}{\mathcal{E}_{(0)} + \mathcal{M}c^2}\,u_A. \tag{295}$$

It determines a *spinor component ratio*, not the natural invariant itself. WKB solvability of a curved-space Dirac operator determines the effective Hamilton–Jacobi mass shell; the next transport order determines current flow. For a stationary spherical metric $ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega_2^2$, covariant conservation gives

$$\boxed{\sqrt{A(r)B(r)}\,r^2 j^r = C_J.} \tag{296}$$

In Schwarzschild coordinates $AB = 1$ this specializes to the original $r^2 j^r = C_J$. The chapter's corrected conditional radial amplitude–momentum relation $d[|\mathcal{A}(r)|^2 p(r)/\Delta(r)]/dr \simeq 0$ constrains the WKB *transport amplitude once the momentum field is supplied*; it is not an independent source-force equation.

The Schwarzschild natural-invariant amplitude $\sigma_{\mathrm{amp}} \propto r^{-1/2}$ is not a free outgoing $1/r$ wave amplitude: $H = \sigma_{\mathrm{amp}}^2 \propto 1/r$ is the harmonic exterior *metric weight*. Kerr's projective spinor has a nonzero Berry curvature even when one scalar WKB phase is locally exact, suggesting a role for polarization/transport structure but not identifying Berry curvature with gravitational curvature.

Conversely, sufficiently many independent massive Hamilton–Jacobi momenta can reconstruct the ten symmetric inverse-metric components locally; massless characteristic data reconstruct only the conformal class. These inverse-probe reconstructions give kinematic constraints on the microscopic theory, not a source-conditioned microscopic selection mechanism.

36 QM040-A: Independent microscopic constant-frame Hessian at the self-consistent quantum saddle

This section audits the proposed comparison of fixed-background derivative-bilinear Model A and metric-free spinor-pregeometry Model B supplied in the research correspondence of 3 November 2026. The new calculation *does not* introduce a fundamental metric into QGD or assume a microscopic graviton. It derives a specified model's full constant-frame response and distinguishes that response from a physical propagating field. The standalone QM040-A paper supplies detailed derivations, full numeric results and the reproducible verifier.

36.1 Fully specified regulated Model A

For N Euclidean Dirac species with positive weights w_i, define a real 4×4 auxiliary matrix B and select the Euclidean potential

$$V_{\text{eff}}(B) = \frac{N}{2g}\,\mathrm{tr}(B^{\mathsf{T}}B) - 2\sum_i \int_{|p|\le\Lambda} \frac{d^4p}{(2\pi)^4} \log\left[m^2 + p^{\mathsf{T}}(I + w_iB)^{\mathsf{T}}(I + w_iB)p\right]. \tag{297}$$

The regulator is tied to the fixed Euclidean background, and the positive auxiliary potential is a *stated contour choice*, not derived from the Lorentzian sign of the original interaction. No in-in retarded or spatially modulated kernel follows from the constant-field determinant alone. Define

$$I_A(q) = \frac{1}{16\pi^2}\int_0^{\Lambda^2} \frac{u^2du}{m^2 + q^2u}, \qquad I_B(q) = \frac{1}{16\pi^2}\int_0^{\Lambda^2} \frac{u^3du}{(m^2 + q^2u)^2}. \tag{298}$$

At $B = bI$, put $q_i = 1 + w_ib$, $A = \sum_i w_i^2 I_A(q_i)$ and $D = \sum_i w_i^2 q_i^2 I_B(q_i)$. The complete 16×16 zero-momentum Hessian has 1 trace, 9 symmetric-traceless and 6 antisymmetric eigenvalues:

$$\boxed{k_{\text{tr}} = \frac{N}{g} - A + 2D, \quad k_{\text{ST}} = \frac{N}{g} - A + \frac{2}{3}D, \quad k_{\text{AS}} = \frac{N}{g} - A.} \tag{299}$$

The bare zero field is *off shell*: $\partial V/\partial B|_{B=0} \neq 0$ for positive w_i. The selected homogeneous saddle instead solves

$$\frac{Nb}{g} = \sum_i w_i(1 + w_ib)I_A(1 + w_ib). \tag{300}$$

This implies the exact positive identity

$$\boxed{k_{\text{AS}} = \frac{1}{b}\sum_i w_iI_A(q_i) > 0, \quad k_{\text{ST}} = k_{\text{AS}} + \frac{2}{3}D > 0, \quad k_{\text{tr}} = k_{\text{AS}} + 2D > 0.} \tag{301}$$

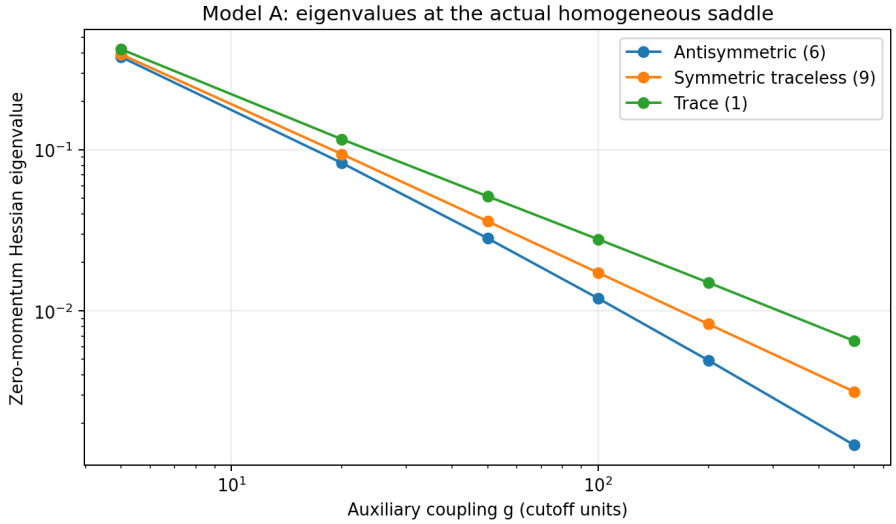
This is homogeneous local *Euclidean* stability, not proof of finite-momentum, Lorentzian or physical-wave stability.

36.2 The benchmark and the correct universality statement

For $N = 2$, $w = (1, 2.7)$, $m/\Lambda = 0.1$ and $\Lambda = 1$, the self-consistent solution at $g = 100$ is $b = 0.3389507568$ with

$$k_{\text{AS}} = 0.0119936724, \qquad k_{\text{ST}} = 0.0172965756, \qquad k_{\text{tr}} = 0.0279023820. \tag{302}$$

At the unstationary $B = 0$ configuration, the antisymmetric Hessian instead equals -0.0057477768. Independent direct quadrature of the *unexpanded* determinant checks the derivative formula. The verifier reports 58 successful assertions at the named benchmark and additional couplings.



The proposed theorem "universality iff no bare kinetic term" is too strong. For the fixed-background operator $\Gamma_i^\mu = \gamma^a(\delta_a{}^\mu + w_i B_a{}^\mu)$, an isotropic $B = bI$ gives the same *null cone* for every i despite unequal w_i and an existing bare kinetic term. But the linear anisotropic response scales as $w_i/(1+w_i b)$ and generally differs among species. In metric-free Model B, if the *entire* leading Dirac operator genuinely factorizes as $w_i \Pi^\mu{}_a \gamma^a D_\mu$, the positive scalar weights cannot split its principal cones; this remains conditional on the existence of a nondegenerate quantum phase and the absence of additional species-dependent derivative terms.

36.3 What remains for the derived map

The attempted proof of no finite-momentum instabilities uses a Källén–Lehmann representation: positivity makes the separated-point, non-contact spectral part of a Euclidean susceptibility decreasing in Q^2, but local subtraction terms can affect the full inverse kernel. Similarly, external-metric induced curvature terms do not replace the full gauge-projected, mixed frame/connection/auxiliary CTP Hessian of actual metric-free Model B. The Henneaux–Teitelboim integration constant labels Λ but does not independently predict its value or eliminate higher-derivative scalar dynamics. The physical spin connection, Chapter 1 WKB bridge, photon and fermion cones, and source-conditioned collective wave all still require derivation *in the same microscopic phase.*

The founding QGD master metric remains

$$g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left(\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right). \tag{303}$$

The next decisive calculation is $K^R_{AB}(\omega, \mathbf{q})$ *around the selected stationary state*, followed by species-wise Clifford and compatible-spin-transport tests, not simply another $Q = 0$ eigenvalue or a preassigned gravitational particle interpretation.

37 QM041: Finite-momentum collective modes about the quantum-selected saddle

The controlled Euclidean Model A of QM040 has N Dirac species and a real auxiliary derivative-bilinear field $B_{a\mu}$, with an $O(4)$-symmetric sharp loop-momentum cutoff. At the selected isotropic saddle $B_{a\mu} = b\delta_{a\mu}$, set $q_i = 1 + w_i b$. Differentiating the full one-loop fermion determinant yields

the exact finite-momentum 16×16 inverse auxiliary kernel

$$K_{a\mu,c\nu}(Q) = \frac{N}{g}\delta_{ac}\delta_{\mu\nu} - 4\sum_i w_i^2 \int_{|p|\le\Lambda} \frac{d^4p}{(2\pi)^4} \frac{p_\mu p_\nu}{D_i^- D_i^+} \Big[(m^2 + q_i^2(p^2 - Q^2/4))\delta_{ac} - 2q_i^2 p_a p_c + \tfrac{1}{2}q_i^2 Q_a Q_c\Big], \quad D_i^\pm = m^2 + q_i^2(p \pm Q/2)^2. \tag{304}$$

The symmetric vertex $V_{i,a\mu} = iw_i\gamma_a p_\mu$ is fixed by the microscopic Hermitian derivative bilinear. No external metric or Einstein kernel was inserted. This fixed-background trial nevertheless has a microscopic reference metric, and thus is not itself the desired metric-free QGD completion.

For a wave vector along Euclidean axis 4, the $B_{12} \pm B_{21}$ eigenvalues are $K_{\mathrm{ST,AS}} = k_{\mathrm{ST,AS}}(0) + Z_{\mathrm{ST,AS}}Q^2 + O(Q^4)$. Define $d_i(u) = m^2 + q_i^2 u$ and $r_i = q_i^2 u/d_i \in (0,1)$. The exact induced transverse stiffness is

$$Z_{\mathrm{ST}} = \frac{1}{16\pi^2}\sum_i w_i^2 \int_0^{\Lambda^2} u\,du\, \frac{q_i^2 u}{d_i(u)^2}\left(\frac{3}{4} - \frac{r_i}{2} + \frac{r_i^2}{12}\right) > 0. \tag{305}$$

The bracket is bounded below by $1/3$; no wave kinetic term was supplied to B. The full 16×16 Euclidean kernel was diagonalized over $0 \le Q^2 \le \Lambda^2$, and its below-pair-threshold analytic branch over $-\omega_{\mathrm{th}}^2 < Q^2 \le 0$, with $\omega_{\mathrm{th}}^2 = \min_i(2m/q_i)^2$. For $(w_1, w_2) = (1, 2.7)$, $m/\Lambda = 0.1$, and $g = 20, 100, 500$, the sampled smallest eigenvalues stay positive. Thus the specified Gaussian approximation has no *numerically detected isolated real pole below the first fermion-pair threshold*; above-threshold resonances, a full CTP kernel, and reflection positivity are unresolved.

The normalized fluctuation couplings to the two species are $c_i = w_i/(1 + w_i b)$. They differ at finite coupling despite the common isotropic principal cone, but satisfy $c_i \sim 1/b$ for large b. Moreover, the saddle and radial integrals give the strong-coupling asymptotics

$$b^2 \sim \frac{g\Lambda^4}{32\pi^2}, \qquad k_{\mathrm{ST}}(0) \sim \frac{N\Lambda^4}{48\pi^2 b^2}, \qquad Z_{\mathrm{ST}} \sim \frac{N\Lambda^2}{48\pi^2 b^2}, \qquad \frac{k_{\mathrm{ST}}(0)}{Z_{\mathrm{ST}}} \longrightarrow \Lambda^2. \tag{306}$$

Hence strong collective ordering can drive the two species toward a common normalized kinetic response without automatically yielding a new infrared wave. The ratio $k_{\mathrm{ST}}/Z_{\mathrm{ST}}$ is a formal low-Q extrapolation, not an actual pole determination above the pair continuum. This result further sharpens the backward inverse-matching problem: a microscopic candidate must generate a physical field with a causal source-visible pole, common Clifford geometry, the correct WKB bridge and controlled vacuum, before a metric is represented by the *complete* QGD master metric.

38 QM042: Consolidated status of phenomenological and microscopic QGD

Editorial closure of the 3 November 2026 stage. The aim of this section is to state what has been *established*, what is *conditional on the effective theory*, and what has *not been derived*. It is not another proposed microscopic interaction and makes no claim that the missing physical collective wave has been found. It consolidates QM001–QM041 and supersedes any optimistic wording in an earlier increment when the later tests gave a more restrictive result.

38.1 Two selection maps, not one

The phenomenological source/state-to-natural-invariant map is most accurately a *possibly set-valued conditional prescription*:

$$\boxed{\mathcal{F}_{\mathrm{phen}}(\varrho,\mathcal{I};\boldsymbol{\theta}) = \mathrm{GramLift}_{\mathrm{reg}}\left(\mathrm{Sol}_{\mathcal{I}}^{\text{causal, order-reduced}}\left[\left.\frac{\delta\Gamma_{\mathrm{eff}}^{\mathrm{CTP}}}{\delta g_\Delta}\right|_{g_\Delta=0} = 0\right]\right).} \tag{307}$$

The solution map requires specified sources and states ϱ, boundary and initial data $\mathcal{I}$, renormalization prescription and parameters $\boldsymbol{\theta}$, and physical branch conditions. The regular Gram lift returns a *representation class*, not a unique microscopic state. The *specific explicit* Gram lift is established for the fixed-reference $\lambda\mathcal{Q}=0$ specialization, without claiming to have independently determined the historical quantum tensor $\mathcal{Q}_{ab}$.

At leading effective order, the adopted dynamical benchmark is

$$\boxed{\Gamma_{\mathrm{QGD}}^{\mathrm{ren}} = \frac{1}{16\pi G_N}\int d^4x\,\sqrt{-g}\,(R-2\Lambda_{\mathrm{ren}}) + S_{\mathrm{GHY}} + S_{\mathrm{F1}}^{\mathrm{cl}} + \hbar\Gamma_{\mathrm{m,ren}}^{(1)}[g,\psi_c,A_c;\varrho;\mu] + O(\hbar^2).} \tag{308}$$

The single-branch formula is shorthand for the corresponding causal CTP equations where required. The geometrical dependence of this action is through *the unchanged complete founding master metric*

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu\left[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)}\sigma_b^{(s)} - \lambda\mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{309}$$

For metric-mediated sector dependence, the historical, recognizable natural-invariant Euler equation (Option C) remains

$$\boxed{2s_s\Pi^{ab}\sigma_b^{(s)} = \lambda\big[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\big]^a, \qquad s_s = -\varepsilon_s,} \tag{310}$$

where Π^{ab} is the variation of the *full adopted renormalized metric action*. A genuinely independent microscopic sector action would add extra Euler contributions; no such term is silently assumed here.

The conventional Einstein–Hilbert term is an *adopted effective dynamical target*; it is not a microscopic explanation of why this geometry exists. In particular, the stationary Schwarzschild null-invariant branch is dynamically selected *within that adopted variational model*, not independently by microscopic Dirac matter.

The proposed microscopic map is logically different:

$$\boxed{\mathcal{F}_{\mathrm{QM}} : (S_{\mathrm{micro}},\varrho,\mathcal{I}) \longmapsto \mathfrak{B}_{\mathrm{physical}} \longmapsto [g_{\mathrm{physical}}] \longmapsto [\Sigma]_{\mathrm{QGD}}.} \tag{311}$$

It requires a *dynamically selected, stable, causal, source-responsive collective quantum field or wave* that yields a shared effective geometry. No independent graviton or particular particle spectrum is assumed in the microscopic action. None of the controlled candidate calculations QM001–QM041 has completed this map.

38.2 The exact forward bridge: master metric to Clifford propagation

Define $M_{ab} = \eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}$. On a nondegenerate Lorentzian patch there is a local factorization $M = L^{\mathsf{T}} \eta L$, which defines a *physical* coframe $e^a{}_\mu = L^a{}_b T^b{}_\mu$. Writing $\Gamma^\mu = \gamma^a (e^{-1})_a{}^\mu$ gives

$$\boxed{\{\Gamma^\mu, \Gamma^\nu\} = 2g^{\mu\nu}\mathbf{1}, \qquad g^{\mu\nu} = \frac{1}{8}\,\mathrm{tr}\{\Gamma^\mu, \Gamma^\nu\}.} \tag{312}$$

The trace relation is a *necessary but not sufficient* test on a generic interacting principal symbol: one must additionally check that the traceless matrix part of the anticommutator vanishes. In a torsion-free, metric-compatible conventional curved Dirac theory, the induced spin connection supplies spinor transport, and the Hermitian first-order action supplies corresponding current-conserving subprincipal terms. In particular, the corrected Chapter 1 WKB bridge and its radial flux law remain independent transport checks. If torsion, nonmetricity, an independent spin connection or extra kinetic operators are present, they must be specified rather than discarded.

Thus a quantum candidate can be tested through its *computed* species-wise retarded inverse propagators $G_{i,R}^{-1}(x,p)$, extracted Clifford principal symbols, spinor-connection/transport terms, source vertices and vacuum response. This exact forward bridge from $g(\Sigma)$ to conventional effective Dirac propagation does *not* invert to a unique microscopic model.

38.3 The inverse correspondence that we actually found

Define the *inverse-matching correspondence* as a class of models rather than a single formula constructing microscopic physics:

$$\boxed{\mathscr{I}_{\mathrm{QGD}}[\mathcal{F}_{\mathrm{phen}}; \mathcal{D}_{\mathrm{match}}] = \left\{ (S_{\mathrm{micro}}, \varrho, \mathcal{I}) : \mathrm{IR}[\Gamma_{\mathrm{coll}}^{\mathrm{CTP}}, G_{i,R}^{-1}, V_i] \simeq \mathcal{D}_{\mathrm{match}}[\mathcal{F}_{\mathrm{phen}}] \right\}.} \tag{313}$$

Here $\mathcal{D}_{\mathrm{match}}$ specifies the finite set of effective propagation, transport, source-response and state/vacuum observables, their accuracy and energy domain. It includes the shared metric light cone where established, the full Clifford residual, species and electromagnetic universality, the Chapter 1 WKB transport, and the causal collective source kernel. This is a *formal acceptance correspondence* whose satisfying set has not been constructed in full; it is not an independently solved inverse equation.

Nonuniqueness is physical: distinct microscopic masses and couplings may produce the same leading low-energy coefficient λ^2/M^2 while differing at finite momentum and in higher-point or vacuum response. Consequently the phenomenological action cannot generally select one unique ultraviolet theory, even if its classical metric equations are completely known.

38.4 What the microscopic calculations did and did not demonstrate

Increment	Reproducible result or obstruction	Scientific status
QM026	Exact finite Grassmann determinant projection supports source-selected large-N order; independent cells do not transmit a field.	D/N; zero-dimensional model
QM027–31	Conditional TEGR/Einstein fits, optical Kerr family, metric/orbit nonuniqueness, vacuum and induced-coefficient matching limits.	D/C; not micro selection
QM032	Off-shell Dirac TT bubble with correct universal Weyl log; mixed physical kernel still missing.	D/N; partial loop

Increment	Reproducible result or obstruction	Scientific status
QM033	Six-field J^2 kinetic claim fails ghost-sign audit; full orbit dynamics not classified.	D/N; classification correction
QM034	Exact causal finite-system shared determinant response and soft relative mode; soft coefficient vanishes, vacuum composite remains gapped.	D/N; reduced quantum system
QM035–36	Exact wave and Kerr single-copy checks; WKB/bridge constrains transport rather than source forces.	D/N; phenomenological/kinematic
QM037–38	Full metric-to-Clifford relation, spin connection, Fierz constraints, conformal transport; inverse matching is nonunique.	D/N/C; no derived wave
QM039	One-dimensional interacting bond model transmits a causal cross-species response but develops a Peierls-type large-system instability.	D/N; restricted lattice
QM040-A	A specified 3+1 Euclidean auxiliary model has a nonzero self-consistent isotropic saddle and positive complete $Q = 0$ Hessian.	D/N; *homogeneous* stability only
QM041	Full finite-momentum 16-component one-loop Euclidean Model A kernel; no detected isolated subthreshold pole in benchmarks, while species couplings approach equality at strong ordering.	D/N; sampled channel/regime

QM040–QM041 together. For the explicitly fixed-background Model A, a nonzero stable uniform auxiliary frame, loop-induced gradient stiffness and increasingly similar fermionic kinetic couplings are real calculated effects. Yet the shear restoring-to-stiffness ratio tends to Λ_c^2, no independently isolated subthreshold low-energy field was detected in the tested range, and its fixed background is not the desired fundamental QGD ontology. Do not infer the full metric-free spinor-pregeometry response from the external-metric fermion determinant or from Model A's Euclidean kernel.

38.5 A finite scientific acceptance criterion rather than an endless search

The *phenomenological stage* is conditionally closed at the level of the adopted effective GR dynamics, physical-state/boundary inputs and regular signed-sector representation. It is neither a microscopic derivation nor an independent new classical prediction beyond GR. Its quantum counterterms, quantum tensor slot, and cosmological parameters remain effective/model-dependent.

The *microscopic stage* is open. A candidate must pass *all of the following within one consistent model and controlled approximation*:

1. A well-defined quantum state and nondegenerate *physical* collective phase, accounting for gauge/constraint directions, not just a formal auxiliary field or local condensate.
2. A stable, causal collective response in the source-free exterior with a source-visible residue or well-defined constraint-mediated long-range field, and no wrong-sign physical modes; no microscopic graviton is pre-assumed.
3. Common physical Clifford propagation and correct connection/WKB conserved-current transport for at least independent fermionic probes and the photon sector, including absence of unacceptable chirality/species splitting.
4. A controlled homogeneous vacuum compatible with the same state and source-response theory, without assuming a predicted numerical value for Λ.

5. Correct low-energy matching to the effective QGD action and the full founding master metric, with the regular Gram lift distinguished from microscopic selection.

Failure of any essential check *rejects the specific candidate or approximation*, not the exact metric representation. Passing them would supply serious evidence for a derived $\mathcal{F}_{\mathrm{QM}}$; no known calculation in this chapter passes all five. A research programme may therefore *stop here and report an open origin problem*, rather than launching an unlimited sequence of new candidate Hamiltonians.

38.6 Consolidated findings and final statement

Phenomenological QGD:	conditionally closed effective representation and dynamics,
inverse matching:	mathematically specified constraints, no unique inversion,
microscopic QGD:	collective origin and universal physical wave not yet derived.

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The value of $\mathcal{F}_{\mathrm{QM}}$ would be to *establish the proposed quantum-matter origin of the effective geometry*, potentially calculate presently calibrated coefficients, and provide independent physically distinguishable predictions. None of those outcomes should be counted as demonstrated merely because $\Gamma_{\mathrm{eff}}[g(\Sigma)]$ reproduces Einsteinian solutions. Our established achievements are compatible with QGD's emergent-metric interpretation; they do not yet single it out among theories with the same gravitational phenomenology.

Editorial decision. Retain Chapter 6 as a completed, audited *research-stage synthesis*. Keep the original Chapter 1 spinor block, corrected WKB/current bridge, Option C and the founding master metric intact. Do not promote the simplified fixed-reference signed-sector metric, an external-metric matter determinant or a speculative extra-sector kinetic operator to a new microscopic law. The unsolved $\mathcal{F}_{\mathrm{QM}}$ is stated explicitly, with a bounded acceptance criterion. This concludes the present research stage without implying that another proposed model must automatically follow.

39 QM043: Coarse-Graining, Quantum Statistical Stability, and Collective Basis Selection

Research status and purpose. This section strengthens the inverse-matching correspondence of QM042 by explicitly identifying the operations which could connect many microscopic quantum constituents to a macroscopic QGD collective field. Coarse-graining, statistical concentration, and the formation of four independent basis covectors are *different* assertions. They must not be conflated with a proof of gravity or treated as additional axioms of the founding metric. All conclusions here are conditional on the microscopic state, correlations, gauge treatment, and regulator.

39.1 From local spinor bilinears to a coarse-grained collective observable [C]

For a proposed microscopic spinor theory and its microscopic connection D_μ, consider the already-motivated Hermitian derivative bilinear, with regulator-defined point splitting understood,

$$\widehat{\mathcal{E}}^{a(i)}{}_{\mu}(y) = \frac{i\hbar}{2}\,\overline{\psi}_i(y)\gamma^a \overleftrightarrow{D_\mu}\psi_i(y). \tag{315}$$

It is a candidate *collective operator*, not a tetrad by definition. For a normalized microscopic state $\widehat{\varrho}$, a coarse-graining prescription $\mathcal{C}_\ell$ may be specified by

$$\mathfrak{B}^a{}_{\mu}(x;\ell) = \frac{Z_\ell}{N}\sum_{i=1}^{N} w_i \int \mathrm{d}\nu_\ell(y \mid x)\,\langle \widehat{\mathcal{E}}^{a(i)}{}_{\mu}(y)\rangle_{\widehat{\varrho}}. \tag{316}$$

Here Z_ℓ fixes dimensions, the measure and smearing rule $\mathrm{d}\nu_\ell(y \mid x)$ are *part of the microscopic definition*, and the state is held explicit. On a fixed-background lattice the averaging cells are unambiguous; in a claimed metric-free theory they must instead be constructed relationally or with a justified regulator. Choosing the desired QGD metric to define the kernel and then declaring that kernel to have generated the metric would be circular. Causal time evolution should be defined by a closed-time-path influence functional; Eq. (316) is not a replacement for a retarded response.

The microscopic field may be decomposed in a chosen gauge/relational description into a mean and its connected fluctuations; *both* enter the effective CTP action,

$$e^{i\Gamma^{\mathrm{CTP}}_{\ell,\mathrm{coll}}[B_+,B_-]} \text{ is defined by a regulated microscopic CTP path integral with } \mathcal{C}_\ell\widehat{\mathcal{E}} \text{ constrained to } B_\pm. \tag{317}$$

This is a schematic definition: the microscopic measure, sources and constraint Jacobians must be specified for a concrete model; the exact CTP normalization $Z_\varrho[J,J]=1$ remains mandatory.

39.2 What the law of large numbers does and does not establish [D/C]

For a *single Hermitian, suitably smeared* component X_i, measured in a regulated quantum state, set $\overline{X}_N = N^{-1}\sum_i X_i$. With the symmetrized connected covariance $C_{ij} = \frac{1}{2}\langle\{\delta X_i, \delta X_j\}\rangle_\varrho$, the exact variance identity is

$$\boxed{\mathrm{Var}_\varrho(\overline{X}_N) = \frac{1}{N^2}\sum_{i,j=1}^{N} C_{ij}.} \tag{318}$$

For identical uncorrelated contributions $C_{ii}=v$ and $C_{i\neq j}=0$, this is v/N, with root-mean-square fluctuations of order $N^{-1/2}$. For identical off-diagonal covariance c it is

$$\boxed{\mathrm{Var}_\varrho(\overline{X}_N) = \frac{v}{N} + \frac{N-1}{N}c.} \tag{319}$$

Thus a sufficient concentration condition is $N^{-2}\sum_{ij} C_{ij} \to 0$ as the relevant thermodynamic/coarse-graining limit is taken. For weak spatial correlations, an analogous statement involves an effective number of *independent correlation volumes*, not necessarily the number of fermions in the region. Nonzero long-range correlations can invalidate the naive law-of-large-numbers estimate; critical collective modes are an especially important exception. This argument concerns a measured regulated component; simultaneous quantum statistics of noncommuting frame components requires a further prescription.

No spontaneous frame from averaging alone. If symmetry forces $\langle \widehat{\mathcal{E}}^a{}_\mu \rangle = 0$, concentration can make the mean more sharply zero. A physical frame phase needs dynamical selection or appropriate gauge-invariant/relational order. QM026's finite-N vanishing mean, nonzero determinant composite, and noncommuting source/large-N limits already show why an expectation value of a gauge-dependent frame cannot by itself serve as an order parameter. Nor does statistical concentration supply a propagation equation or massless wave.

39.3 Basis vectors and the effective Dirac operator [D/C]

Assume — this is a *physical phase condition, not a statistical theorem* — that the regulated, renormalized coarse-grained $\mathfrak{B}^a{}_\mu$ admits a nonsingular Lorentzian coframe $e^a{}_\mu$ after the necessary normalization and gauge/relational identifications:

$$\boxed{\det(e^a{}_\mu) \neq 0, \qquad g^{\text{eff}}_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu.} \tag{320}$$

A basis needs four independent directions; averaging parallel or algebraically constrained spinor currents cannot supply these by sheer multiplicity. Fierz identities constrain bilinears derived from a single Dirac spinor; multi-spinor collective observables may have greater rank but still require an explicit test. In a locally Lorentz-symmetric model $\langle e^a{}_\mu \rangle$ in an unfixed gauge is not an observable, so phase tests must be gauge-invariant or relational.

If the selected phase induces a common first-order kinetic operator for physical probes,

$$\mathcal{D}^{\text{eff}}_i = i\hbar\Gamma^\mu_i(\partial_\mu + \Omega_{\mu,i}) - m_i + \cdots, \qquad \{\Gamma^\mu_i, \Gamma^\nu_i\} \overset{\text{IR}}{=} 2g^{\mu\nu}_{\text{eff}}\mathbf{1}, \tag{321}$$

then the effective metric is recoverable from the *full Clifford anticommutator*, not just a matrix trace. The connection and measure must also yield the conserved Dirac current and the original QGD WKB transport bridge. The classical identities apply *after* the coframe exists: they do not dynamically produce it. Universal photon propagation and absence of unwanted axial/chiral cones remain independent tests. The coframe appearing in this subsection is the *physical emergent coframe*; it is not to be silently identified with the QGD reference $T^a{}_\mu$.

39.4 Matching the complete master metric without losing its quantum slot [D/C]

Retain exactly the founding equation, $s_s = -\varepsilon_s$:

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_s \varepsilon_s \sigma^{(s)}_a \sigma^{(s)}_b - \lambda\mathcal{Q}_{ab}\right].} \tag{322}$$

For an independently generated effective metric and an invertible reference coframe, set

$$M_{ab} = (T^{-1})^\mu{}_a\, g^{\text{eff}}_{\mu\nu} (T^{-1})^\nu{}_b, \qquad \sum_s \varepsilon_s \sigma^{(s)}_a \sigma^{(s)}_b = \eta_{ab} - M_{ab} - \lambda\mathcal{Q}_{ab}. \tag{323}$$

This is a *representation/matching condition*, not a newly derived microscopic Euler equation. Its regular eight-sector Gram-lift specialization with $\lambda\mathcal{Q} = 0$ algebraically represents the target metric,

with nonuniqueness of sector representatives. Identifying $\mathcal{Q}_{ab}$ with a renormalized connected quantum covariance remains a *hypothesis*, requiring a specified operator definition, symmetry, and no double counting of matter-loop contributions. The metric-only pullback Euler equation remains the previously chosen Option C,

$$2s_s \Pi^{ab} \sigma_b^{(s)} = \lambda[(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi]^a, \tag{324}$$

provided the relevant microscopic sector actions have no additional independent variations; otherwise they must be included explicitly.

39.5 A refined inverse matching correspondence, not a unique inverse [C/O]

The result of including averaging, correlations and basis selection is a *testable refinement* of QM042:

$$\begin{aligned} \mathscr{I}^{(\ell)}_{\mathrm{QGD}}[\mathcal{F}_{\mathrm{phen}}] = \{ & (S_{\mathrm{micro}}, \widehat{\varrho}, \mathcal{C}_\ell, \mathcal{O}) : \\ & \Gamma^{\mathrm{CTP}}_{\ell,\mathrm{coll}} = \mathcal{C}^{\mathrm{CTP}}_\ell[S_{\mathrm{micro}}, \widehat{\varrho}; \mathcal{O}], \\ & \text{a nondegenerate, physical collective phase exists,} \\ & \mathrm{Match}_{\mathrm{IR}}(\mathcal{D}_{\mathrm{coll}}, \mathcal{D}_{\mathrm{phen}}) = 1 \}. \end{aligned} \tag{325}$$

The matching data $\mathcal{D}$ include (i) the Clifford principal symbol of independent probes, (ii) compatible spin transport and the Chapter 1 WKB/current bridge, (iii) causal source-to-probe response and physical collective spectra, (iv) vacuum/stability and cosmological zero-mode conditions, and (v) coefficients and nonlocal form factors of the adopted QGD effective action. Agreement is *physical/observational and infrared*, allowing field redefinitions and integration of heavy/gauge modes; it is not literal equality of differently parameterized actions. The map generally returns a *family of microscopic theories*, not a unique action. For every candidate the averaging scale must lie in a justified window where microscopic short-distance structure is suppressed but the sought collective wavelengths remain resolved.

39.6 Proof-status ledger and bounded conclusion

Mathematical statement	Proof status / limitation
Regulated connected-covariance and variance identities	Exact
Statistical concentration of the macroscopic mean	Conditional on sufficiently weak correlations
Four independent collective basis covectors	Requires a selected, nondegenerate physical phase
Curved-Dirac / Clifford and WKB transport bridge	Exact conditional identities; no phase selection
Regular QGD signed-sector Gram lift	Established; does not identify unique sectors or $\mathcal{Q}$
Universal causal microscopic gravitational wave	Open; not established by averaging or QM039–QM041

Concluding boundary. Coarse-graining supplies the candidate macroscopic observable; correlation estimates supply conditions under which it behaves classically; basis independence and the Dirac/Clifford bridge show how a *selected* quantum collective phase could define a common propagation geometry. None of these steps alone proves that the state chooses a nondegenerate phase, that the field supports a healthy long-range causal wave, or that photons and fermions see the same geometry. Consequently $\mathcal{F}_{\mathrm{phen}}$ remains *conditionally closed*; $\mathscr{I}^{(\ell)}_{\mathrm{QGD}}$ is now a more constructive inverse-matching prescription; and a derived $\mathcal{F}_{\mathrm{QM}}$ remains *open*. This is the bounded conclusion of Chapter 6, not a mandate for an indefinite sequence of microscopic trial models.

40 QN001–QN005: Analytic quantum action, natural-invariant pullback, and inverse-matching data (4 November 2026)

Scope and provenance. This section consolidates the action-only research QN001 (Dirac curvature-squared heat kernel), QN002A (analytic eight-chapter coefficient audit), QN003 (quadratic massive Dirac/Hartree spectral response), QN004 (common-regulator Euclidean form factors and Gaussian mass elimination), and QN005 (natural-invariant pullback and spherical reduction). It adds an *explicit effective matching target* to the Chapter 6 $\mathcal{F}_{\rm phen} \to \mathcal{F}_{\rm QM}$ programme. QN001–QN005 are conditional matter-effective calculations: they do not derive a physical collective gravitational phase, the historical $\mathcal{Q}_{ab}$, an independent graviton spectrum, or the microscopic $\mathcal{F}_{\rm QM}$. No earlier theorem or historical result in this chapter is superseded. The main text uses $c = \hbar = 1$ except when $\hbar$ marks formal loop order.

Status notation. **[D]**: derived from the displayed action within its domain; **[V]**: independently checked algebraically or numerically; **[C]**: conditional on regulator, model, state, classical branch or renormalization prescription; **[O]**: not derived. A numerical check tests an analytic relation; decimal benchmark values do not replace physical constants or fix renormalized couplings.

40.1 The full founding metric and its restricted quantum calculation [D/C]

The *unchanged complete master metric* is

$$\boxed{g_{\mu\nu}(\boldsymbol{\sigma}, T, \mathcal{Q}) = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N_s} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad s_s = -\varepsilon_s.} \tag{326}$$

The $\sigma^{(s)}$ are macroscopic gravitational one-forms motivated by Chapter 1's spinor structure, but are *not* generally exact WKB phase gradients; their labels are signed gravitational sectors, not automatically spinor modes. For smooth nondegenerate Lorentzian internal form $h_{ab} = \eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}$ and invertible T, Chapter 2's local factorization $h = L^{\mathsf{T}} \eta L$ gives $e^c{}_\mu = L^c{}_a T^a{}_\mu$ and the standard Clifford-compatible Dirac matrices. This validates the *kinematic* curved-Dirac coupling, not a microscopic origin of the frame. The explicitly computed QN001–QN005 metric-loop results use fixed T and the *declared* $\lambda\mathcal{Q} = 0$ specialization; the complete metric is not replaced by that specialization.

40.2 Canonical Lorentzian action and exact classical sector [D/C]

Preserve the accepted Chapter 4/5 baseline:

$$\boxed{\begin{aligned}\Gamma^{\rm ren}_{\rm QGD}[\boldsymbol{\sigma}, \psi_c, A_c; \varrho, \mu] = {} & \frac{1}{16\pi G_N} \int_{\mathcal{M}} \mathrm{d}^4x \sqrt{-g_\Sigma}\,(R_\Sigma - 2\Lambda_{\rm ren}) + S_{\rm GHY}[g_\Sigma] \\ & + S^{\rm cl}_{\rm F1}[\psi_c, A_c; g_\Sigma] + \hbar \Gamma^{(1)}_{\rm m,ren}[g_\Sigma, \psi_c, A_c; \varrho, \mu] + O(\hbar^2), \qquad g_\Sigma = g(\boldsymbol{\sigma}, T, \mathcal{Q}).\end{aligned}} \tag{327}$$

In the Chapter 4 conventions, the previously abbreviated classical terms are

$$\begin{aligned} S_{\rm GHY}[g] &= \frac{1}{8\pi G_N} \int_{\partial\mathcal{M}} \mathrm{d}^3y\, \epsilon \sqrt{|h|}\, K, \\ S^{\rm cl}_{\rm F1} &= \int_{\mathcal{M}} \mathrm{d}^4x \sqrt{-g} \left[\frac{i}{2} \bar\psi_c \gamma^\mu \overleftrightarrow{\nabla}_\mu \psi_c - m_0 \bar\psi_c \psi_c + G_4 (\bar\psi_c \psi_c)^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}\right]. \end{aligned} \tag{328}$$

Here ∇_μ includes the Clifford spin connection and the specified $U(1)$ coupling; G_4 is a contact-matter coupling *different from* G_N. Boundary conditions and matter-source double counting must

be specified. The one-loop functional may be organized (not uniquely split without a subtraction prescription) as

$$\boxed{\begin{aligned}\Gamma^{(1)}_{\mathrm{m,ren}} = &\int \mathrm{d}^4x\sqrt{-g}[\alpha_R(\mu)R^2 + \alpha_C(\mu)C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \alpha_E(\mu)E_4] \\ &+ \Gamma^{(1)}_{\mathrm{nonlocal}}[g;\mu] + \Gamma^{(1)}_{\mathrm{state}}[g;\varrho,\mu] + \Gamma^{(1)}_{\mathrm{vertices}}[g,\psi_c,A_c;\varrho,\mu] + \Gamma^{(1)}_{\mathrm{boundary}} + \ldots, \\ E_4 = &R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2.\end{aligned}} \tag{329}$$

The α_i are *renormalized effective-action parameters*, not QGD predictions merely because covariance permits their operators. In four dimensions, a constant-coefficient Euler term is a bulk topological density up to boundary effects. Quantum state dependence and real-time dynamics require an in–in/Schwinger–Keldysh prescription; an in–out vacuum determinant is not a causal black-hole-evaporation action.

40.3 Analytically determined local Dirac/Hartree contributions [D/V/C]

For *one* parity-even Euclidean four-component Dirac field of constant Hartree mass $M_v > 0$, with the Chapter 5 proper-time cutoff Λ_{PT}, define

$$\boxed{\begin{aligned}&x_v = \frac{M_v^2}{\Lambda_{\mathrm{PT}}^2}, \qquad I_0 = \Gamma(0,x_v), \qquad I_1 = \Lambda_{\mathrm{PT}}^2 e^{-x_v} - M_v^2\Gamma(0,x_v), \\ &I_2 = \frac{1}{2}[(\Lambda_{\mathrm{PT}}^4 - M_v^2\Lambda_{\mathrm{PT}}^2)e^{-x_v} + M_v^4\Gamma(0,x_v)], \quad \Gamma(0,x) = \int_x^\infty \frac{\mathrm{d}t}{t}e^{-t}.\end{aligned}} \tag{330}$$

At the stationary massive Hartree vacuum the proper-time gap equation gives

$$\boxed{\frac{M_v - m_0}{2G_4^{\mathrm{PT}}} = \frac{M_v I_1}{4\pi^2}, \qquad U_v = \frac{I_2 + M_v(M_v - m_0)I_1}{8\pi^2}.} \tag{331}$$

The curvature-linear Dirac contribution is $C_R(M_v)R_E = -I_1R_E/(96\pi^2)$ and contributes to the *already matched* Einstein coefficient. For a single species in the declared convention,

$$\boxed{G_N^{-1} = G_{\mathrm{bare}}^{-1} + \frac{I_1}{6\pi} + \delta_{\mathrm{other}},} \tag{332}$$

where δ_{other} denotes other independent matching/counterterm contributions; this is *not* a prediction of G_N. Likewise, U_v shifts the renormalized vacuum energy but does *not* fix Λ_{ren}, which requires an independent vacuum-energy condition.

With the Chapter 5 proper-time determinant sign convention, the traced spinor local fourth heat coefficient is

$$\boxed{a_4 = \frac{11}{360}E_{4,E} - \frac{1}{20}C_E^2 - \frac{1}{30}\Box_E R_E.} \tag{333}$$

Accordingly, the calculated direct fermion loop through local curvature-squared order is

$$\boxed{\Delta\Gamma^{(4)}_{E,D} = \frac{I_0}{32\pi^2}\int \mathrm{d}^4x\sqrt{g_E}\left[\frac{11}{360}E_{4,E} - \frac{1}{20}C_E^2 - \frac{1}{30}\Box_E R_E\right] + \ldots.} \tag{334}$$

At the stable stationary Hartree branch $K_v = U''(M_v) > 0$, an independent curvature–mass mixing $v = C_R'(M_v) = M_vI_0/(48\pi^2)$ yields

$$\boxed{\begin{aligned}K_v = \frac{m_0 I_1}{4\pi^2(M_v - m_0)} + \frac{M_v^2 I_0}{2\pi^2}, \qquad &\Delta\Gamma^{(4)}_{E,\mathrm{relax}} = -\frac{v^2}{2K_v}\int \mathrm{d}^4x\sqrt{g_E}\,R_E^2 + \ldots, \\ -\frac{v^2}{2K_v} = -\frac{M_v^2 I_0^2(M_v - m_0)}{1152\pi^2[m_0 I_1 + 2M_v^2 I_0(M_v - m_0)]}&.\end{aligned}} \tag{335}$$

This is one condensate-relaxation contribution, not another independent fermion loop, and has to be assigned a consistent Hartree/large-N versus $\hbar$ order. Its physical R^2 coefficient also includes independent finite counterterms. *Do not add* U_v or $C_R R$ again after using them to define $\Lambda_{\rm ren}$ and G_N.

40.4 QN004: one-regulator quadratic quantum action and exact form factors [D/V/C]

For a weak Euclidean metric perturbation $h_{\mu\nu}$ around flat space, write $z = k_E^2$ and

$$\boxed{\begin{aligned} f(\xi) &= \int_0^1 \mathrm{d}y\, e^{-\xi y(1-y)}, \\ \mathcal{B}_C(\xi) &= \frac{3\xi f(\xi) + 12 f(\xi) - \xi - 12}{6\xi^2}, \\ \mathcal{B}_R(\xi) &= \frac{3\xi f(\xi) + 30 f(\xi) + 2\xi - 30}{36\xi^2}. \end{aligned}} \tag{336}$$

The *same* covariant Euclidean proper-time regulator is used for all four terms below:

$$\boxed{\begin{aligned} \mathcal{F}_C^\Lambda(z) &= \frac{1}{32\pi^2}\int_{\Lambda_{\rm PT}^{-2}}^\infty \frac{\mathrm{d}s}{s} e^{-M_v^2 s}\mathcal{B}_C(sz), \\ \mathcal{F}_R^\Lambda(z) &= \frac{1}{32\pi^2}\int_{\Lambda_{\rm PT}^{-2}}^\infty \frac{\mathrm{d}s}{s} e^{-M_v^2 s}\mathcal{B}_R(sz), \\ \mathcal{V}_\Lambda(z) &= \frac{M_v}{8\pi^2}\int_{\Lambda_{\rm PT}^{-2}}^\infty \frac{\mathrm{d}s}{s} e^{-M_v^2 s}\frac{1-f(sz)}{sz}, \\ K_\Lambda(z) &= K_v + \frac{1}{8\pi^2}\int_0^1 \mathrm{d}y \left[(4M_v^2+z)\Gamma\left(0, \frac{M_v^2 + y(1-y)z}{\Lambda_{\rm PT}^2}\right) - 4M_v^2 I_0\right]. \end{aligned}} \tag{337}$$

After Gaussian elimination of the same Hartree mass fluctuation, the *calculated quadratic Euclidean curvature response*, retaining independent finite local matching data, is

$$\boxed{\begin{aligned} \Gamma^{(2)}_{E,\rm reduced}[h] = \Gamma^{(2)}_{E,\rm EH+matter,ren}[h] + \int \frac{\mathrm{d}^4 k_E}{(2\pi)^4}\Big\{ & C^{(1)}(-k)\,\mathfrak{F}_2(z)\,C^{(1)}(k) \\ & + R_E^{(1)}(-k)\,\mathfrak{F}_0(z)\,R_E^{(1)}(k)\Big\} + \Gamma^{(2)}_{E,\rm topological/boundary} + \Gamma^{(2)}_{E,\rm remaining}, \\ \mathfrak{F}_2(z) &= \alpha_C^{\rm fin} + \mathcal{F}_C^\Lambda(z), \\ \mathfrak{F}_0(z) &= \alpha_R^{\rm fin} + \mathcal{F}_R^\Lambda(z) - \frac{\mathcal{V}_\Lambda(z)^2}{2K_\Lambda(z)}. \end{aligned}} \tag{338}$$

Here $C^{(1)}\mathfrak{F}_2 C^{(1)}$ denotes full Weyl-tensor contraction. The metric seagull/contact structure is included by taking the *second variation of the complete covariant heat trace*; dropping these contacts would violate the stress Ward identity. For the specified stable branch $M_v > m_0 \geq 0$, $K_\Lambda(z) > 0$ at real Euclidean $z \geq 0$, so the particular gapped scalar Hartree channel does not generate a massless $1/z$ long-range pole. This says nothing universal about other physical QGD sectors or timelike poles. The Euclidean functions are *not* the complete causal Lorentzian $\Gamma_{\rm state}$ or exact curved Schwarzschild action.

The local consistency checks are

$$\boxed{\mathcal{F}_C^\Lambda(0) = -\frac{I_0}{640\pi^2}, \quad \mathcal{F}_R^\Lambda(0) = 0, \quad \mathcal{V}_\Lambda(0) = \frac{M_v I_0}{48\pi^2}, \quad K_\Lambda(0) = K_v.} \tag{339}$$

Thus $\mathfrak{F}_0(0) = \alpha_R^{\rm fin} - v^2/(2K_v)$ includes the *same* local R_E^2 term in Eq. (335), not a second contribution. In a *separate massless free conformal Dirac regime*, the universal weak-curvature logarithmic coefficient is

$$\boxed{\Gamma^{(2)}_{E,D,\rm nonloc} \supset \frac{1}{640\pi^2}\int \mathrm{d}^4x\, C^{(1)}_{\mu\nu\rho\sigma}\log\left(\frac{-\partial_E^2}{\mu^2}\right) C^{(1)\mu\nu\rho\sigma}, \qquad a_D = \frac{11}{360}, \quad c_D = \frac{1}{20}.} \tag{340}$$

It must *not* be appended independently to the massive Hartree response: the massless and gapped infrared limits, states, and subtraction conventions differ.

40.5 Chapter 2's exact natural-invariant variation of the quantum action [D/C]

Define the *complete* renormalized metric residual and its frame projection by

$$\mathcal{E}_g^{\mu\nu} = \frac{1}{\sqrt{-g}}\frac{\delta\Gamma^{\rm ren}_{\rm QGD}}{\delta g_{\mu\nu}}, \qquad \Pi^{ab} = T^a{}_\mu T^b{}_\nu \mathcal{E}_g^{\mu\nu}. \tag{341}$$

For fixed reference T, differentiable specified $\mathcal{Q}[\boldsymbol{\sigma}]$ and *no independent direct sector action*, Chapter 2's preferred Option C is unchanged but now carries all quantum terms in Π:

$$\boxed{2s_s\Pi^{ab}\sigma_b^{(s)} = \lambda\big[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi\big]^a, \quad s = 1,\dots,N_s.} \tag{342}$$

Option F is the corresponding geometric chain rule $(D_{\sigma^{(s)}}g)^*_{\sqrt{-g}}(\Pi_{\mathcal{G}} + \Pi_{\mathcal{T}} + \Pi_{\mathcal{Q}}) = 0$ with the sector conventions already fixed in Chapters 4–6. The historical balance $\mathcal{S}_{(s)} = \mathcal{G}_{(s)} + \mathcal{T}_{(s)} + \mathcal{Q}_{(s)} = 0$ remains a decomposition of this *single* Euler derivative; the computed matter-loop terms must not be counted again as the historical $\mathcal{Q}_{ab}$ operator. In the declared metric-only branch, Eq. (342) reduces to $\Pi^{ab}\sigma_b^{(s)} = 0$. At rank-deficient configurations, these projected equations need not imply all ten metric equations.

For metric-only $\Gamma_g[g(\boldsymbol{\sigma})]$, $L = D_{\boldsymbol{\sigma}}g$, and admissible variations a, b, the *exact off-shell* Hessian identity is

$$\boxed{D^2\Gamma_\sigma[a,b] = D^2\Gamma_g[La, Lb] + \langle D\Gamma_g, D^2g[a,b]\rangle.} \tag{343}$$

The second term drops *only if the complete metric residual vanishes*, not merely because the projected sector equations vanish. Then $H_\sigma = L^* H_g L$. Consequently QN004 changes the metric Hessian coefficients, but it does not eliminate metric-preserving sector redundancies. If T is independently varied, or if a separately derived sector functional exists, additional Euler terms appear and this conclusion must be revisited.

40.6 QN005: eight signed sectors, rank, and off-shell channels [D/V/C]

At a flat Cartesian background choose eight nonzero covectors in four cancelling signed pairs:

$$g_{\mu\nu} = \eta_{\mu\nu} + \sum_{a=0}^{3}\big(\sigma_\mu^{(a,+)}\sigma_\nu^{(a,+)} - \sigma_\mu^{(a,-)}\sigma_\nu^{(a,-)}\big), \qquad \sigma_0^{(a,+)} = \sigma_0^{(a,-)} = v\,\mathrm{d}x^a,\ v \neq 0. \tag{344}$$

For $a_\mu^{(a,\pm)} = \delta\sigma_\mu^{(a,\pm)}$, the metric Jacobian reads

$$h_{\mu\nu} = (La)_{\mu\nu} = v\sum_{a=0}^{3}\Big[\delta^a_\mu(a_\nu^{(a,+)} - a_\nu^{(a,-)}) + \delta^a_\nu(a_\mu^{(a,+)} - a_\mu^{(a,-)})\Big]. \tag{345}$$

This map has rank ten and kernel dimension $32-10=22$; the nonzero background is essential. For $z=k_E^2>0$, define $h_{\rm TT}=P^{(2)}h$ and $\phi=\theta^{\mu\nu}h_{\mu\nu}/\sqrt{3}$, $\theta_{\mu\nu}=\delta_{\mu\nu}-k_\mu k_\nu/z$. The QN004 quadratic metric action becomes

$$\boxed{\begin{aligned}&\Gamma_E^{(2)}[a]=\int\frac{\mathrm{d}^4k_E}{(2\pi)^4}\left[\mathcal{A}_2(z)\|P^{(2)}La\|^2+\mathcal{A}_0(z)|\phi(La)|^2\right],\\&\mathcal{A}_2(z)=\frac{z}{64\pi G_N}+\frac{z^2}{2}\mathfrak{F}_2(z),\\&\mathcal{A}_0(z)=-\frac{z}{32\pi G_N}+3z^2\mathfrak{F}_0(z).\end{aligned}} \tag{346}$$

For a *complete metric-stationary* flat background with its vacuum stress and $\Lambda_{\rm ren}$ matched away, and generic nonzero z with $\mathcal{A}_{2,0}\neq0$, the ten-metric-component Hessian has rank $5+1=6$ and four diffeomorphism-null directions. Its 32-sector pullback has *the same* rank six with nullity $22+4=26$. The five Euclidean spin-two off-shell components and one transverse scalar are *not* six physical radiative polarizations. At an identically zero-sector background, $L=0$ and this ordinary quadratic natural-invariant response does not arise from a metric-only action.

40.7 QN005: independently varied spherical amplitudes and weak-static quantum action [D/V/C]

On Chapter 2's one-positive-mass-sector spherical branch, vary $A(r)$ and $B(r)$ *independently before imposing the classical null-sector equation*:

$$\sigma=A(r)\,\mathrm{d}(ct)+B(r)\,\mathrm{d}r,\quad X=A^2,\quad Y=B^2,\quad F=1-X,\quad D=1+Y-X, \tag{347}$$

so $g_{tt}=-F$, $g_{tr}=AB$, $g_{rr}=1+Y$, $g_{\theta\theta}=r^2$. Chapter 2's two-function metric Jacobian is nondegenerate where $ABF\neq0$. In the *weak-static* curvature expansion around flat space, the radial curvature combinations are

$$\boxed{R^{(1)}[X,Y]=X''+\frac{2X'}{r}+\frac{2Y'}{r}+\frac{2Y}{r^2},\qquad \mathscr{W}[X,Y]=X''-\frac{X'+Y'}{r}+\frac{2Y}{r^2},\qquad C^{(1)2}=\frac{\mathscr{W}^2}{3}.} \tag{348}$$

For the local quadratic terms at fixed α_R,α_C, set $U=\alpha_R R^{(1)}$ and $V=\alpha_C\mathscr{W}/3$. Direct higher-derivative radial variation yields

$$\boxed{\begin{aligned}&\frac{1}{8\pi}\frac{\delta\Delta\Gamma^{(2)}_{E,\rm loc}}{\delta X}=[r^2(U+V)]''-[r(2U-V)]',\\&\frac{1}{8\pi}\frac{\delta\Delta\Gamma^{(2)}_{E,\rm loc}}{\delta Y}=-[r(2U-V)]'+2(U+V).\end{aligned}} \tag{349}$$

On the classical exterior $X=Y=C/r$, $C=2G_NM/c^2$, one obtains $R^{(1)}=0$, $\mathscr{W}=6C/r^3$, and *both* local radial bulk Euler derivatives vanish. This is a weak-field direct check of the fact that Ricci-flat Einstein backgrounds have no bulk forcing from constant-coefficient R^2 and Weyl-squared terms. It does *not* exclude additional solution branches, boundaries, nonlocal quantum response or state-dependent quantum stress. The classical null relation $A^2=B^2$ was obtained by classical independent variation; it must not be imposed as a QN004 quantum constraint.

The full *weak-static* QN004 nonlocal quadratic correction can also be reduced without guessing local gradient operators. Define the spherical transforms, with j_ℓ the spherical Bessel functions,

$$\widetilde{X}(q)=4\pi\int_0^\infty\mathrm{d}r\,r^2X(r)j_0(qr),\qquad \widetilde{P}(q)=\frac{4\pi}{3}\int_0^\infty\mathrm{d}r\,r^2Y(r)[j_0(qr)+j_2(qr)]. \tag{350}$$

Then, per unit Euclidean time,

$$\boxed{\frac{\Delta\Gamma^{(2),\mathrm{curvature}}_{E,\mathrm{QN004}}}{\Delta\tau} = \frac{1}{2\pi^2}\int_0^\infty \mathrm{d}q\, q^6\Big[\frac{\mathfrak{F}_2(q^2)}{3}\big|\widetilde{X}(q)+\widetilde{P}(q)\big|^2 + \mathfrak{F}_0(q^2)\big|-\widetilde{X}(q)+2\widetilde{P}(q)\big|^2\Big].} \tag{351}$$

It assumes sufficiently regular/regulated decaying profiles and a weak static Euclidean perturbation. The unregulated point-source C/r profile needs distributional source matching; a near-horizon strong-field calculation requires the separate *curved-background, causal* state-dependent theory. No universal QGD-specific $1/r^3$ tail follows solely from the massive QN004 form factors.

40.8 State benchmarks, remaining independent inputs, and Chapter 6 inverse map [C/O]

The curved-spacetime QFT benchmarks must be matched to their *own* states and field content. For one free massless conformal four-component Dirac field in the specified spatially flat FLRW conformal state and anomaly-renormalization convention, Chapter 8 gives

$$\boxed{\begin{aligned}\rho_q &= \frac{11}{960\pi^2}H^4+\frac{C_\varrho}{a^4},\\ p_q &= -\frac{11}{960\pi^2}H^4-\frac{11}{720\pi^2}H^2\dot{H}+\frac{C_\varrho}{3a^4}; \qquad C_\varrho\big|_{\text{conformal thermal}} = \frac{7\pi^2}{60}T_0^4.\end{aligned}} \tag{352}$$

This is a *stress one-point function in a specified background*, not a unique action for arbitrary Lorentzian states. Chapter 5's Boulware far-field correction is likewise a *separate massless, state-dependent semiclassical benchmark*, not the massive Hartree curvature form factor analytically continued to the horizon. Black-hole Hawking/evaporation dynamics requires a regular relevant state and a causal in–in effective action.

Action ingredient	**Status after QN001–QN005**
Full master metric and classical EH–GHY–F1 action	Specified; complete $\mathcal{Q}_{ab}$ operator and independent physical sector selection open.
Vacuum and Einstein coefficients	Explicit regulated I_2, I_1 and Hartree U_v; $G_N, \Lambda_{\mathrm{ren}}$ remain independent physical renormalization data.
Dirac C^2, Euler and local Hartree R^2	Explicit in the stated proper-time model; finite physical $\alpha_R, \alpha_C, \alpha_E$ not fixed.
Massive Euclidean quadratic nonlocal action	Four common-regulator integral kernels and their once-counted Gaussian Schur complement explicit (QN004).
Natural-invariant and spherical pullback	Exact chain rule and Hessian; explicit weak-static radial/local and Fourier–Bessel actions (QN005).
General Lorentzian quantum action	Real-time CTP states, complete curved-background higher-curvature terms, higher loops and physical boundary matching remain open.
Microscopic $\mathcal{F}_{\mathrm{QM}}$	Not derived by calculating any metric-based Dirac determinant.

Operational consequence for inverse matching. The Chapter 6 acceptance set $\mathscr{I}^{(\ell)}_{\mathrm{QGD}}[\mathcal{F}_{\mathrm{phen}}]$ may now include the *calculated and renormalization-qualified* data

$$\begin{aligned}\mathcal{D}_{\mathrm{action}} = \{&\mathcal{F}^\Lambda_C, \mathcal{F}^\Lambda_R, \mathcal{V}_\Lambda, K_\Lambda,\\ &\text{Ward/contact constraints, natural-invariant rank,}\\ &\text{specified-state stress}\}.\end{aligned}$$

This shrinks the family of admissible microscopic models only when the benchmark matter content, regulator/matching, causal state and physical observables are the same. It does not invert the coarse-graining map uniquely; fitting all coefficients is *calibration*, not an independent QGD prediction. Leave some independent semiclassical/gravitational observables as out-of-sample tests.

Bounded conclusion. The metric-only Dirac/Hartree quantum action is now explicitly computable through *quadratic curvature order near flat Euclidean space* in its chosen model, and its natural-invariant variations are controlled by Chapter 2's exact pullback. It gives additional physics beyond a *bare* Einstein–Hilbert action, but no demonstrated new observable beyond ordinary semiclassical GR with the same quantum matter. The physical gravitational collective field, the independent metric derivative structure $\mathcal{Q}_{ab}$ and universal causal microscopic selection remain precisely the Chapter 6 open problem. *Do not promote these conditional effective results into a microscopic theory of gravity.*

41 Q052–Q054: Observable quantum natural invariants and the dark-energy gate (5 November 2026)

Scope and evidence. This section consolidates the new Q052 spherical/cosmological reconstruction, Q053 Schwarzschild/Kerr tests and Q054 cosmological action analysis. It does not replace the full master metric or QN001–QN005 kernels. All formulas are for the declared fixed-reference, metric-only specialization $\lambda\mathcal{Q}_{ab} = 0$ unless another assumption is explicitly stated. The late-time dark-energy tests examine the *existing* renormalized action before considering any additional sector-sensitive proposal.

41.1 The action determines which quantum terms are physical

Write the complete founding master metric and the controlled action as

$$g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \tag{353}$$

$$\Gamma^{\rm ren}_{\rm QGD}[\Sigma] = \Gamma^{\rm cl}_{\rm grav+matter}[g(\Sigma)] + \hbar\Gamma^{(1)}_{\rm quantum}[g(\Sigma); \varrho, \mu] + O(\hbar^2). \tag{354}$$

The field equation is the Euler pullback $(D_\Sigma g)^* \delta\Gamma/\delta g = 0$. On regular full-rank branches it is equivalent to the corresponding quantum-corrected metric Euler equation. The 22 metric-preserving directions of the regular eight-sector representation do not define measurable sector vacuum energies in this action. Consequently, no time-dependent signed-sector representative alone generates a dark-energy stress tensor. Nor does the independently undetermined historical tensor $\mathcal{Q}_{ab}$ receive a unique value from the known one-loop matter response: $q^{\rm back}_{\mu\nu}$ is the already calculated metric deformation, not a second term to add to the action (Q051).

41.2 Spherical quantum behavior and its curvature check

For $\sigma = A(r)\, d(ct) + B(r)\, dr$, $\mu = G_N M/c^2$ and $\ell_P^2 = G_N\hbar/c^3$, Chapter 7's specified massless-Dirac Boulware-state far-field stress gives

$$\boxed{A^2 = \frac{2\mu}{r} + \frac{4\mu\ell_P^2}{15\pi r^3} + \cdots, \quad B^2 = \frac{2\mu}{r} + \frac{6\mu\ell_P^2}{15\pi r^3} + \cdots, \quad B^2 - A^2 = \frac{2\mu\ell_P^2}{15\pi r^3} + \cdots.} \tag{355}$$

With $X = A^2, Y = B^2$, the Chapter 6 weak-field identity $R^{(1)} = X'' + 2X'/r + 2Y'/r + 2Y/r^2$ yields

$$\boxed{\delta R^{(1)} = \frac{(6\times 4 - 4\times 6)\mu\ell_P^2}{15\pi r^5} = 0.} \tag{356}$$

This agrees with the absence of a linear-in-background-curvature conformal trace anomaly; the nonzero $O(\hbar\mu^2/r^6)$ trace and subleading metric coefficients are not determined by this leading check. The state is singular near the Schwarzschild horizon, so this result is not a horizon expansion.

41.3 Kerr: a rotating quantum source, not yet a rotating corrected solution

Local bulk variations of R^2, $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ and E_4 vanish on a four-dimensional Ricci-flat Einstein background with constant couplings and usual boundary assumptions. For one conformal four-component Dirac field, the anomaly on Ricci-flat Kerr gives a *trace source*

$$\boxed{\delta R = -\frac{7\ell_P^2}{720\pi} R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}_{\rm Kerr} + O(\hbar^2).} \tag{357}$$

Its far-field angular structure is $\delta R \propto -\mu^2 \ell_P^2 r^{-6}[1 - 21a^2 \cos^2\theta/r^2 + \cdots]$, with Kerr spin length $a = J/(Mc)$. This trace alone cannot reconstruct a rotating metric, the quantum angular-momentum response, or natural-invariant amplitudes. A quantum state and full conserved $\langle T_{\mu\nu} \rangle$ are needed.

41.4 Cosmological natural invariants and the dark-energy verdict

The physical FLRW combinations $\Xi_\sigma = a^2 - 1$ and $\Upsilon_\sigma = 1 - N^2$ reconstruct $H^2 = \dot{\Xi}_\sigma^2/[4(1-\Upsilon_\sigma)(1+\Xi_\sigma)^2]$. The conformal Dirac state of Chapter 8 contributes a known anomaly, giving the controlled radiation-era expression $\Xi_\sigma = w + \alpha K w/[2(1+w)] + O(\alpha^2)$, $w = 2\sqrt{K}(\tau - \tau_*)$, $\alpha = 11G_N/(360\pi)$ in $c = \hbar = 1$ units. Its relative late-time size is $\alpha H_0^2 \simeq 1.39 \times 10^{-124}$ for $H_0 = 68.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Q054 result (conditional no-self-acceleration gate). In the *regular metric-only theory* with $\Lambda_{\mathrm{ren}} = 0$, no classical matter/vacuum source and no nonanalytic or extra sector term, the local constant-coefficient R^2, C^2 and Euler-density operators yield zero bulk Euler correction on constant-curvature FLRW. In particular $f(R) = R + \beta R^2$ obeys $Rf_R - 2f = -R$ at constant R; only $R = 0$ follows. The conformal anomaly supplies a formal second solution $H^2 = 1/\alpha$, but $Ht_P = \sqrt{360\pi/11} \simeq 10.14$, outside one-loop low-curvature control. Thus the *calculated* local quantum terms do not produce observed late-time accelerating expansion from $\Lambda_{\mathrm{ren}} = 0$.

The existing action *does* admit $\Lambda_{\mathrm{ren}} > 0$ with $w = -1$, but this is the ordinary renormalized cosmological-constant contribution, not a prediction of its magnitude. New dynamical dark energy would require either a genuinely distinct conserved quantum state contribution or an independently declared, representation-sensitive sector action with its own stability and observable-selection tests. The archival temporal-only σ_t kinetic claim and unnormalized quartic potential are not such a derived action. Chapter 8 contains the full Q054 proof and explicit conditional model comparisons.

42 Q061: final quantum-action, symmetry, and energy-status consolidation

Which action is being quantized? The operational gravitational theory is the renormalized metric-mediated action $\Gamma^{\rm ren}_{\rm met}[g(\Sigma)]$ of Chapter 4 on a regular, complete, $\lambda\mathcal{Q}=0$ field-map branch. Its matter-induced local R^2, C^2, E_4 coefficients are consistently renormalized independent EFT inputs; QN004 determines a restricted Euclidean one-loop nonlocal response in a declared proper-time scheme. The same local loop contribution must not be counted both in $c_i(\mu)$ and in the remaining nonlocal/state action.

Physical quantum response. On a metric-shell background the Hessian obeys $H_\Sigma = L^* H_g L$ with $L = D_\Sigma g$. Nonzero quantum corrections alter the observable metric combinations of the natural invariants; local metric-preserving $O(p,q)$ changes do not automatically carry quantum energy. The historical $\mathcal{Q}_{ab}$ in Axiom I has no unique off-shell value from this functional, and a calculated state-dependent $q^{\rm back}_{\mu\nu}$ is not an extra operator (Q051–Q052). Chapter 7 reconstructs explicit Boulware-state spherical backreaction; Chapter 8 reconstructs the conformal anomaly in FLRW, both within their domains.

Causality. A time-dependent quantum source requires the closed-time-path (in–in) response for retarded metric backreaction. QN004 Euclidean form factors are not, by themselves, such a kernel. Finite higher-derivative couplings are treated as perturbative EFT corrections; no extra propagating high-frequency gravitational mode is certified by truncating the derivative expansion.

Dark energy. Signed-sector symmetry, universal metric coupling, and boundary conditions cannot select a metric-null dark-energy amplitude. Q055–Q058 show that a direct positive-kinetic sector action can be internally consistent and support acceleration, but the physical selector and energy scale are new dynamical postulates, not outcomes of Axiom I or II. The simplest shift-symmetric condensate faces a strong early-radiation constraint, whereas a bounded hilltop remains a conditional quintessence-type extension. Chapter 8 contains the detailed tests. Negative static Boulware vacuum energy in Chapter 7 must not be conflated with homogeneous vacuum-like $w=-1$ energy. The original QM research archive is preserved without changing its proof status.